

*A Series of Books in the Mathematical Sciences*  
Victor Klee, Editor

# LINEAR PROGRAMMING

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'I don't want to bore you,' Harvey said, 'but you should understand that these heaps of wire can practically think—linear programming—which means that instead of going through all the alternatives they have a hunch which is the right one.'

*From Billion-Dollar Brain by Len Deighton.  
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# I BASIC THEORY

# Introduction

In this short chapter, we shall explain what is meant by linear programming and sketch a history of this subject.

## A DIET PROBLEM

Polly wonders how much money she must spend on food in order to get all the energy (2,000 kcal), protein (55 g), and calcium (800 mg) that she needs every day. (For iron and vitamins, she will depend on pills. Nutritionists would disapprove, but the introductory example ought to be simple.) She chooses six foods that seem to be cheap sources of the nutrients; her data are collected in Table 1.1.

**Table 1.1 Nutritive Value per Serving**

| Food            | Serving size | Energy (kcal) | Protein (g) | Calcium (mg) | Price per serving (cents) |
|-----------------|--------------|---------------|-------------|--------------|---------------------------|
| Oatmeal         | 28 g         | 110           | 4           | 2            | 3                         |
| Chicken         | 100 g        | 205           | 32          | 12           | 24                        |
| Eggs            | 2 large      | 160           | 13          | 54           | 13                        |
| Whole milk      | 237 cc       | 160           | 8           | 285          | 9                         |
| Cherry pie      | 170 g        | 420           | 4           | 22           | 20                        |
| Pork with beans | 260 g        | 260           | 14          | 80           | 19                        |



Then she begins to think about her menu. For example, 10 servings of pork with beans would take care of all her needs for only (?) \$1.90 per day. On the other hand, 10 servings of pork with beans is a lot of pork with beans—she would not be able to stomach more than 2 servings a day. She decides to impose servings-per-day limits on all six foods:

|                 |                             |
|-----------------|-----------------------------|
| Oatmeal         | at most 4 servings per day  |
| Chicken         | at most 3 servings per day  |
| Eggs            | at most 2 servings per day  |
| Milk            | at most 8 servings per day  |
| Cherry pie      | at most 2 servings per day  |
| Pork with beans | at most 2 servings per day. |

Now, another look at the data shows Polly that 8 servings of milk and 2 servings of cherry pie every day will satisfy the requirements nicely and at a cost of only \$1.12. In fact, she could cut down a little on the pie or the milk or perhaps try a different combination. But so many combinations seem promising that one could go on and on, looking for the best one. Trial and error is not particularly helpful here. To be systematic, we may speculate about some as yet unspecified menu consisting of  $x_1$  servings of oatmeal,  $x_2$  servings of chicken,  $x_3$  servings of eggs, and so on. In order to stay below the upper limits, that menu must satisfy

$$\begin{aligned} 0 &\leq x_1 \leq 4 \\ 0 &\leq x_2 \leq 3 \\ 0 &\leq x_3 \leq 2 \\ 0 &\leq x_4 \leq 8 \\ 0 &\leq x_5 \leq 2 \\ 0 &\leq x_6 \leq 2. \end{aligned} \quad (1.1)$$

And, of course, there are the requirements for energy, protein, and calcium; they lead to the inequalities

$$\begin{aligned} 110x_1 + 205x_2 + 160x_3 + 160x_4 + 420x_5 + 260x_6 &\geq 2,000 \\ 4x_1 + 32x_2 + 13x_3 + 8x_4 + 4x_5 + 14x_6 &\geq 55 \\ 2x_1 + 12x_2 + 54x_3 + 285x_4 + 22x_5 + 80x_6 &\geq 800. \end{aligned} \quad (1.2)$$

If some numbers  $x_1, x_2, \dots, x_6$  satisfy inequalities (1.1) and (1.2), then they describe a satisfactory menu; such a menu will cost, in cents per day,

$$3x_1 + 24x_2 + 13x_3 + 9x_4 + 20x_5 + 19x_6. \quad (1.3)$$

In designing the most economical menu, Polly wants to find numbers  $x_1, x_2, \dots, x_6$  that satisfy (1.1) and (1.2), and make (1.3) as small as possible. As a mathematician

would put it, she wants to

$$\begin{aligned} \text{minimize} \quad & 3x_1 + 24x_2 + 13x_3 + 9x_4 + 20x_5 + 19x_6 \\ \text{subject to} \quad & 0 \leq x_1 \leq 4 \\ & 0 \leq x_2 \leq 3 \\ & 0 \leq x_3 \leq 2 \\ & 0 \leq x_4 \leq 8 \\ & 0 \leq x_5 \leq 2 \\ & 0 \leq x_6 \leq 2 \end{aligned} \quad (1.4)$$

$$110x_1 + 205x_2 + 160x_3 + 160x_4 + 420x_5 + 260x_6 \geq 2000$$

$$4x_1 + 32x_2 + 13x_3 + 8x_4 + 4x_5 + 14x_6 \geq 55$$

$$2x_1 + 12x_2 + 54x_3 + 285x_4 + 22x_5 + 80x_6 \geq 800.$$

Her problem is known as a *diet problem*.

## LINEAR PROGRAMMING

Problems of this kind are called “linear programming problems,” or “LP problems” for short; linear programming is the branch of applied mathematics concerned with these problems. Here are other examples:

$$\begin{aligned} \text{maximize} \quad & 5x_1 + 4x_2 + 3x_3 \\ \text{subject to} \quad & 2x_1 + 3x_2 + x_3 \leq 5 \\ & 4x_1 + x_2 + 2x_3 \leq 11 \\ & 3x_1 + 4x_2 + 2x_3 \leq 8 \\ & x_1, x_2, x_3 \geq 0 \end{aligned} \quad (1.5)$$

(with “ $x_1, x_2, x_3 \geq 0$ ” used as shorthand for “ $x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$ ”) or

$$\begin{aligned} \text{minimize} \quad & 3x_1 - x_2 \\ \text{subject to} \quad & -x_1 + 6x_2 - x_3 + x_4 \geq -3 \\ & 7x_2 + 2x_4 = 5 \\ & x_1 + x_2 + x_3 = 1 \\ & x_3 + x_4 \leq 2 \\ & x_2, x_3 \geq 0. \end{aligned} \quad (1.6)$$

In general, if  $c_1, c_2, \dots, c_n$  are real numbers, then the function  $f$  of real variables  $x_1, x_2, \dots, x_n$  defined by

$$f(x_1, x_2, \dots, x_n) = c_1x_1 + c_2x_2 + \cdots + c_nx_n = \sum_{j=1}^n c_jx_j$$

is called a *linear function*. If  $f$  is a linear function and if  $b$  is a real number, then the equation

$$f(x_1, x_2, \dots, x_n) = b$$

is called a *linear equation* and the inequalities

$$f(x_1, x_2, \dots, x_n) \leq b$$

$$f(x_1, x_2, \dots, x_n) \geq b$$

are called *linear inequalities*. Linear equations and linear inequalities are both referred to as *linear constraints*. Finally, a *linear programming problem* is the problem of maximizing (or minimizing) a linear function subject to a finite number of linear constraints. We shall usually attach different subscripts  $i$  to different constraints and different subscripts  $j$  to different variables. For simplicity of exposition, we shall restrict ourselves in Chapters 1–7 to LP problems of the following form:

$$\begin{aligned} &\text{maximize} && \sum_{j=1}^n c_j x_j \\ &\text{subject to} && \sum_{j=1}^n a_{ij} x_j \leq b_i \quad (i = 1, 2, \dots, m) \\ &&& x_j \geq 0 \quad (j = 1, 2, \dots, n). \end{aligned} \quad (1.7)$$

These problems will be referred to as LP problems in the *standard form*. (The reader should be warned that the terminology is far from unified; several authors prefer the terms *canonical* or *symmetric* form, and others reserve these adjectives for altogether different problems.) For example, (1.5) is a problem in the standard form (with  $n = 3$ ,  $m = 3$ ,  $a_{11} = 2$ ,  $a_{12} = 3$ , and so on). What distinguishes the problems in the standard form from the rest? First, all of their constraints are linear *inequalities*. Secondly, the last  $n$  of the  $m + n$  constraints in (1.7) are very special: they simply stipulate that none of the  $n$  variables may assume negative values. Such constraints are called *nonnegativity constraints*. (Note that problem (1.6) differs from the standard form on both counts: two of its constraints are linear equations and the variables  $x_1, x_4$  may assume negative values.)

The linear function that is to be maximized or minimized in an LP problem is called the *objective function* of that problem. For example, the function  $z$  of variables  $x_1, x_2, x_3, x_4, x_5, x_6$  defined by

$$z(x_1, x_2, \dots, x_6) = 3x_1 + 24x_2 + 13x_3 + 9x_4 + 20x_5 + 19x_6$$

is the objective function of Polly's diet problem (1.4). Numbers  $x_1, x_2, \dots, x_n$  that satisfy all the constraints of an LP problem are said to constitute a *feasible solution* of that problem. For instance, we have observed that

$$x_1 = 0, \quad x_2 = 0, \quad x_3 = 0, \quad x_4 = 8, \quad x_5 = 2, \quad x_6 = 0$$

is a feasible solution of (1.4). Finally, a feasible solution that maximizes the objective function (or minimizes it, depending on the form of the problem) is called an *optimal solution*; the corresponding value of the objective function is called the *optimal value* of the problem. As it turns out, the unique optimal solution of (1.4) is

$$x_1 = 4, \quad x_2 = 0, \quad x_3 = 0, \quad x_4 = 4.5, \quad x_5 = 2, \quad x_6 = 0$$

or simply  $(4, 0, 0, 4.5, 2, 0)$ . Accordingly, the optimal value of (1.4) is 92.5. Not every LP problem has a unique optimal solution; some problems have many different optimal solutions and others have no optimal solutions at all. The latter may occur for one of two radically different reasons: either there are no feasible solutions at all or there are, in a sense, too many of them. The first case may be illustrated on the problem

$$\begin{aligned} &\text{maximize} && 3x_1 - x_2 \\ &\text{subject to} && x_1 + x_2 \leq 2 \\ &&& -2x_1 - 2x_2 \leq -10 \\ &&& x_1, x_2 \geq 0 \end{aligned} \quad (1.8)$$

which has no feasible solutions at all. Such problems are called *infeasible*. On the other hand, even though the problem

$$\begin{aligned} &\text{maximize} && x_1 - x_2 \\ &\text{subject to} && -2x_1 + x_2 \leq -1 \\ &&& -x_1 - 2x_2 \leq -2 \\ &&& x_1, x_2 \geq 0 \end{aligned} \quad (1.9)$$

does have feasible solutions, none of them is optimal: for every number  $M$  there is a feasible solution  $x_1, x_2$  such that  $x_1 - x_2 > M$ . In a sense, (1.9) has such an abundance of feasible solutions that none of them can aspire to be the best. Problems with this property are called *unbounded*. As we shall prove later (Theorem 3.4), every linear programming problem belongs to one of the three categories noted here: it has an optimal solution, is infeasible, or is unbounded.

## HISTORY OF LINEAR PROGRAMMING

As mathematical disciplines go, linear programming is quite young. It started in 1947 when G. B. Dantzig designed the "simplex method" for solving linear programming formulations of U.S. Air Force planning problems. What followed was an exciting period of rapid development in this new field. It soon became clear that a surprisingly wide range of apparently unrelated

problems in production management could be stated in linear programming terms and, most importantly, solved by the simplex method. Such problems, if noticed at all, had traditionally been tackled by a hit-or-miss approach guided only by experience and intuition. The use of linear programming often brought about a considerable increase in the efficiency of the whole operation. (Until then, expansion of the efficiency frontier usually came from technological innovations. This new way to increase efficiency—*under existing technological conditions*—by improvements in organization and planning, made many managers appreciate the practical importance of mathematics. At least, it made them aware of the advantage of stating their decision problems in clear-cut and well-defined terms.) As the popularity of linear programming theory increased, applications in new areas occurred, many of them far from obvious. In turn, these applications stimulated further theoretical research by pointing out the need for solving problems that would have otherwise seemed uninteresting. In this fascinating interplay between theory and applications, a new branch of applied mathematics established itself.

As calculus developed from the seventeenth century's need to solve problems of mechanics, linear programming developed from the twentieth century's need to solve problems of management. Yet other profound influences stimulated the evolution of the new field from its very inception. Economics was one of them: as early as 1947, T. C. Koopmans began pointing out that linear programming provided an excellent framework for the analysis of classical economic theories, such as the renowned system proposed in 1874 by L. Walras. On the other hand, linear programming brought together previously known theorems of pure mathematics concerning such diverse topics as the geometry of convex sets, extremal problems of combinatorial nature, and the theory of two-person games. Finally, it was fortunate and perhaps even inevitable that linear programming developed concurrently with modern computer technology: without electronic computers, present-day large-scale linear programming would be unthinkable.

Scientific fields are rarely born overnight; with the advantage of hindsight, one can often track down the sources that paved the way for the decisive breakthrough. The field of linear programming is no exception. At the core of its mathematical theory is the study of systems of linear inequalities; such systems were investigated by Fourier as far back as 1826. Since then, quite a few other mathematicians have considered the subject, although none of them has devised an algorithm whose efficiency has come close to that of the simplex method. Nevertheless, some of them proved various special cases of a fundamental theorem that is now called the *duality theorem* of linear programming. On the applied side, L. V. Kantorovich pointed out the practical significance of a restricted class of LP problems, and proposed a rudimentary algorithm for their solution as early as 1939. Regrettably, this effort remained neglected in the U.S.S.R. and unknown elsewhere until long after linear programming became an elegant theory through the independent work of Dantzig and others.

In the 1970s, linear programming came twice to public attention. On October 14, 1975, the Royal Swedish Academy of Sciences awarded the Nobel Prize in economic science to L. V. Kantorovich and T. C. Koopmans "for their contributions to the theory of optimum allocation of resources." (As the reader may know, there is no Nobel Prize in mathematics. Apparently the Academy regarded the work of G. B. Dantzig, who is universally recognized as the father of linear programming, as being too mathematical.) The second event was even more dramatic. Ever since the invention of the simplex method, mathematicians had been looking for a *theoretically* satisfactory algorithm to solve LP problems. (A word of explanation is in order: theoretical criteria for judging the efficiency of algorithms are quite different from practical ones. Thus, an algorithm like the simplex method, which is eminently satisfactory in practical applications, may be found theoretically unsatisfactory. The converse is also true: theoretically satisfactory algorithms may be thoroughly useless in practice. We shall return to this distinction in

Chapter 4.) The breakthrough came in 1979 when L. G. Khachian published a description of such an algorithm (based on earlier works by Shor, and by Judin and Nemirovskii). Newspapers around the world published reports of this result, some of them full of hilarious misinterpretations. We shall present the algorithm in the appendix.

For a thorough survey of the history of linear programming, the reader is referred to Chapter 2 of Dantzig's monograph (1963). References to many applications of linear programming may be found in Riley and Gass (1958). Some of the more recent applications are referenced in Gass (1975).

## PROBLEMS

Answers to problems marked with the symbol  $\triangle$  are found at the back of the book.

### 1.1 Which of the problems below are in the standard form?

- a. Maximize  $3x_1 - 5x_2$   
 subject to  $4x_1 + 5x_2 \geq 3$   
 $6x_1 - 6x_2 = 7$   
 $x_1 + 8x_2 \leq 20$   
 $x_1, x_2 \geq 0$ .
- b. Minimize  $3x_1 + x_2 + 4x_3 + x_4 + 5x_5$   
 subject to  $9x_1 + 2x_2 + 6x_3 + 5x_4 + 3x_5 \leq 5$   
 $8x_1 + 9x_2 + 7x_3 + 9x_4 + 3x_5 \leq 2$   
 $x_1, x_2, x_3, x_4 \geq 0$ .
- c. Maximize  $8x_1 - 4x_2$   
 subject to  $3x_1 + x_2 \leq 7$   
 $9x_1 + 5x_2 \leq -2$   
 $x_1, x_2 \geq 0$ .

### 1.2 State in the standard form:

- minimize  $-8x_1 + 9x_2 + 2x_3 - 6x_4 - 5x_5$   
 subject to  $6x_1 + 6x_2 - 10x_3 + 2x_4 - 8x_5 \geq 3$   
 $x_1, x_2, x_3, x_4, x_5 \geq 0$ .

### 1.3 Prove that (1.8) is infeasible and (1.9) is unbounded.

### $\triangle$ 1.4 Find necessary and sufficient conditions for the numbers $s$ and $t$ to make the LP problem

- maximize  $x_1 + x_2$   
 subject to  $sx_1 + tx_2 \leq 1$   
 $x_1, x_2 \geq 0$

- a. have an optimal solution,  
 b. be infeasible,  
 c. be unbounded.

△ 1.5 Prove or disprove: If problem (1.7) is unbounded, then there is a subscript  $k$  such that the problem

$$\begin{aligned} &\text{maximize} && x_k \\ &\text{subject to} && \sum_{j=1}^n a_{ij}x_j \leq b_i \quad (i = 1, 2, \dots, m) \\ & && x_j \geq 0 \quad (j = 1, 2, \dots, n) \end{aligned}$$

is unbounded.

△ 1.6 [Adapted from Greene et al. (1959).] A meat packing plant produces 480 hams, 400 pork bellies, and 230 picnic hams every day; each of these products can be sold either fresh or smoked. The total number of hams, bellies, and picnics that can be smoked during a normal working day is 420; in addition, up to 250 products can be smoked on overtime at a higher cost. The net profits are as follows:

|         | Fresh | Smoked on regular time | Smoked on overtime |
|---------|-------|------------------------|--------------------|
| Hams    | \$8   | \$14                   | \$11               |
| Bellies | \$4   | \$12                   | \$7                |
| Picnics | \$4   | \$13                   | \$9                |

For example, the following schedule yields a total net profit of \$9,965:

|         | Fresh | Smoked | Smoked (overtime) |
|---------|-------|--------|-------------------|
| Hams    | 165   | 280    | 35                |
| Bellies | 295   | 70     | 35                |
| Picnics | 55    | 70     | 105               |

The objective is to find the schedule that maximizes the total net profit. Formulate as an LP problem in the standard form.

1.7 [Adapted from Charnes et al. (1952).] An oil refinery produces four types of raw gasoline: alkylate, catalytic-cracked, straight-run, and isopentane. Two important characteristics of each gasoline are its performance number PN (indicating antiknock properties) and its vapor pressure RVP (indicating volatility). These two characteristics, together with the production levels in barrels per day, are as follows:

|                   | PN  | RVP | Barrels produced |
|-------------------|-----|-----|------------------|
| Alkylate          | 107 | 5   | 3,814            |
| Catalytic-cracked | 93  | 8   | 2,666            |
| Straight-run      | 87  | 4   | 4,016            |
| Isopentane        | 108 | 21  | 1,300            |

These gasolines can be sold either raw, at \$4.83 per barrel, or blended into aviation gasolines (Avgas A and/or Avgas B). Quality standards impose certain requirements on the aviation gasolines; these requirements, together with the selling prices, are as follows:

|         | PN           | RVP       | Price per barrel |
|---------|--------------|-----------|------------------|
| Avgas A | at least 100 | at most 7 | \$6.45           |
| Avgas B | at least 91  | at most 7 | \$5.91           |

The PN and RVP of each mixture are simply weighted averages of the PNs and RVPs of its constituents. For example, the refinery could adopt the following strategy:

- Blend 2,666 barrels of alkylate and 2,666 barrels of catalytic into 5,332 barrels of Avgas A with

$$\text{PN} = \frac{(2,666 \times 107) + (2,666 \times 93)}{5,332} = 100$$

$$\text{RVP} = \frac{(2,666 \times 5) + (2,666 \times 8)}{5,332} = 6.5.$$

- Blend 1,148 barrels of alkylate, 4,016 barrels of straight-run, and 1,024 barrels of isopentane into 6,188 barrels of Avgas B with

$$\text{PN} = \frac{(1,148 \times 107) + (4,016 \times 87) + (1,024 \times 108)}{6,188} = 94.2$$

$$\text{RVP} = \frac{(1,148 \times 5) + (4,016 \times 4) + (1,024 \times 21)}{6,188} = 7.$$

Sell 276 barrels of isopentane raw.

This sample plan yields a total profit of

$$(5,332 \times 6.45) + (6,188 \times 5.91) + (276 \times 4.83) = \$72,296.$$

The refinery aims for the plan that yields the largest possible profit. Formulate as an LP problem in the standard form.

1.8 An electronics company has a contract to deliver 20,000 radios within the next four weeks. The client is willing to pay \$20 for each radio delivered by the end of the first week, \$18 for those delivered by the end of the second week, \$16 by the end of the third week, and \$14 by the end of the fourth week. Since each worker can assemble only 50 radios per week, the company cannot meet the order with its present labor force of 40; hence it must hire and train temporary help. Any of the experienced workers can be taken off the assembly line to instruct a class of three trainees; after one week of instruction, each of the trainees can either proceed to the assembly line or instruct additional new classes.

At present, the company has no other contracts; hence some workers may become idle once the delivery is completed. All of them, whether permanent or temporary, must be kept on the payroll till the end of the fourth week. The weekly wages of a worker, whether assembling, instructing, or being idle, are \$200; the weekly wages of a trainee are \$100. The production costs, excluding the worker's wages, are \$5 per radio.

For example, the company could adopt the following program.

First week: 10 assemblers, 30 instructors, 90 trainees  
Workers' wages: \$8,000

Trainees' wages: \$9,000  
 Profit from 500 radios: \$7,500  
 Net loss: \$9,500

Second week: 120 assemblers, 10 instructors, 30 trainees  
 Workers' wages: \$26,000  
 Trainees' wages: \$3,000  
 Profit from 6,000 radios: \$78,000  
 Net profit: \$49,000

Third week: 160 assemblers  
 Workers' wages: \$32,000  
 Profit from 8,000 radios: \$88,000  
 Net profit: \$56,000

Fourth week: 110 assemblers, 50 idle  
 Workers' wages: \$32,000  
 Profit from 5,500 radios: \$49,500  
 Net profit: \$17,500

This program, leading to a total net profit of \$113,000, is one of many possible programs. The company's aim is to maximize the total net profit. Formulate as an LP problem (not necessarily in the standard form).

- △ 1.9 [S. Masuda (1970); see also V. Chvátal (1983).] The *bicycle problem* involves  $n$  people who have to travel a distance of ten miles, and have one single-seat bicycle at their disposal. The data are specified by the walking speed  $w_j$  and the bicycling speed  $b_j$  of each person  $j$  ( $j = 1, 2, \dots, n$ ); the task is to minimize the arrival time of the last person. (Can you solve the case of  $n = 3$  and  $w_1 = 4$ ,  $w_2 = w_3 = 2$ ,  $b_1 = 16$ ,  $b_2 = b_3 = 12$ ?) Show that the optimal value of the LP problem

$$\begin{aligned} & \text{minimize} && t \\ & \text{subject to} && t - x_j - x'_j - y_j - y'_j \geq 0 \quad (j = 1, 2, \dots, n) \\ & && t - \sum_{j=1}^n y_j - \sum_{j=1}^n y'_j \geq 0 \\ & && w_j x_j - w_j x'_j + b_j y_j - b_j y'_j = 10 \quad (j = 1, 2, \dots, n) \\ & && \sum_{j=1}^n b_j y_j - \sum_{j=1}^n b_j y'_j \leq 10 \\ & && x_j, x'_j, y_j, y'_j \geq 0 \quad (j = 1, 2, \dots, n) \end{aligned}$$

provides a lower bound on the optimal value of the bicycle problem.

## How the Simplex Method Works

In this chapter, we shall learn to solve LP problems in the standard form by the simplex method. A rigorous analysis of the details will be deferred to Chapter 3.

### FIRST EXAMPLE

We shall illustrate the simplex method on the following example:

$$\begin{aligned} & \text{maximize} && 5x_1 + 4x_2 + 3x_3 \\ & \text{subject to} && 2x_1 + 3x_2 + x_3 \leq 5 \\ & && 4x_1 + x_2 + 2x_3 \leq 11 \\ & && 3x_1 + 4x_2 + 2x_3 \leq 8 \\ & && x_1, x_2, x_3 \geq 0. \end{aligned} \tag{2.1}$$

A preliminary step of the method consists of introducing so-called slack variables.

In order to motivate this concept, let us consider the first of our constraints,

$$2x_1 + 3x_2 + x_3 \leq 5. \quad (2.2)$$

For every feasible solution  $x_1, x_2, x_3$ , the value of the left-hand side of (2.1) is at most the value of the right-hand side; often, there may be a slack between the two values. We shall denote the slack by  $x_4$ . That is, we shall define  $x_4 = 5 - 2x_1 - 3x_2 - x_3$ ; with this notation, inequality (2.2) may now be written as  $x_4 \geq 0$ . In an analogous way, the next two constraints give rise to variables  $x_5$  and  $x_6$ . Finally, following a time-honored convention, we shall denote the objective function  $5x_1 + 4x_2 + 3x_3$  by  $z$ . To summarize: for every choice of numbers  $x_1, x_2$ , and  $x_3$ , we shall define numbers  $x_4, x_5, x_6$ , and  $z$  by the formulas

$$\begin{aligned} x_4 &= 5 - 2x_1 - 3x_2 - x_3 \\ x_5 &= 11 - 4x_1 - x_2 - 2x_3 \\ x_6 &= 8 - 3x_1 - 4x_2 - 2x_3 \\ z &= 5x_1 + 4x_2 + 3x_3. \end{aligned} \quad (2.3)$$

With this notation, our problem may be restated as

$$\text{maximize } z \text{ subject to } x_1, x_2, x_3, x_4, x_5, x_6 \geq 0. \quad (2.4)$$

The new variables  $x_4, x_5, x_6$  defined by (2.3) are called *slack variables*; the old variables  $x_1, x_2, x_3$  are usually referred to as the *decision variables*. It is crucial to note that the equations in (2.3) spell out an equivalence between (2.1) and (2.4). More precisely:

- Every feasible solution  $x_1, x_2, x_3$  of (2.1) can be extended, in the unique way determined by (2.3), into a feasible solution  $x_1, x_2, \dots, x_6$  of (2.4).
- Every feasible solution  $x_1, x_2, \dots, x_6$  of (2.4) can be restricted, simply by deleting the slack variables, into a feasible solution  $x_1, x_2, x_3$  of (2.1).
- This correspondence between feasible solutions of (2.1) and feasible solutions of (2.4) carries optimal solutions of (2.1) onto optimal solutions of (2.4), and vice versa.

The grand strategy of the simplex method is that of *successive improvements*: having found some feasible solution  $x_1, x_2, \dots, x_6$  of (2.4), we shall try to proceed to another feasible solution  $\bar{x}_1, \bar{x}_2, \dots, \bar{x}_6$ , which is better in the sense that

$$5\bar{x}_1 + 4\bar{x}_2 + 3\bar{x}_3 > 5x_1 + 4x_2 + 3x_3.$$

Repeating this process a finite number of times, we shall eventually arrive at an optimal solution.

To begin with, we need some feasible solution  $x_1, x_2, \dots, x_6$ . Finding one in our example presents no difficulty: setting the decision variables  $x_1, x_2, x_3$  at zero, we

evaluate the slack variables  $x_4, x_5, x_6$  from (2.3). Hence our initial solution,

$$x_1 = 0, \quad x_2 = 0, \quad x_3 = 0, \quad x_4 = 5, \quad x_5 = 11, \quad x_6 = 8 \quad (2.5)$$

yields  $z = 0$ .

In the spirit of the grand strategy sketched above, we should now look for a feasible solution that yields a higher value of  $z$ . Finding such a solution is not difficult. For example, if we keep  $x_2 = x_3 = 0$  and increase the value of  $x_1$ , we obtain  $z = 5x_1 > 0$ . Thus, if we keep  $x_2 = x_3 = 0$  and set  $x_1 = 1$ , we obtain  $z = 5$  (and  $x_4 = 3, x_5 = 7, x_6 = 5$ ). Better yet, if we keep  $x_2 = x_3 = 0$  and set  $x_1 = 2$ , we obtain  $z = 10$  (and  $x_4 = 1, x_5 = 3, x_6 = 2$ ). However, if we keep  $x_2 = x_3 = 0$  and set  $x_1 = 3$ , we obtain  $z = 15$  and  $x_4 = x_5 = x_6 = -1$ ; this won't do, since feasibility requires  $x_i \geq 0$  for every  $i$ . The moral is that we cannot increase  $x_1$  too much. The question is: *Just how much can we increase  $x_1$  (keeping  $x_2 = x_3 = 0$  at the same time) and still maintain feasibility ( $x_4, x_5, x_6 \geq 0$ )?*

The condition  $x_4 = 5 - 2x_1 - 3x_2 - x_3 \geq 0$  implies  $x_1 \leq \frac{5}{2}$ ; similarly,  $x_5 \geq 0$  implies  $x_1 \leq \frac{11}{4}$  and  $x_6 \geq 0$  implies  $x_1 \leq \frac{8}{3}$ . Of these three bounds, the first is the most stringent. Increasing  $x_1$  up to that bound we obtain our next solution,

$$x_1 = \frac{5}{2}, \quad x_2 = 0, \quad x_3 = 0, \quad x_4 = 0, \quad x_5 = 1, \quad x_6 = \frac{1}{2}. \quad (2.6)$$

Note that this solution yields  $z = \frac{25}{2}$ , which is indeed an improvement over  $z = 0$ .

Next, we should look for a feasible solution that is even better than (2.6). However, this task seems a little more difficult. What made the first iteration so easy? We had at our disposal not only the feasible solution (2.5), but also the system of linear equations (2.3), which guided us in our quest for an improved feasible solution. If we wish to continue in a similar way, we should manufacture a new system of linear equations that relates to (2.6) much as system (2.3) relates to (2.5).

What properties should the new system have? Note that (2.3) expresses the variables that assume positive values in (2.5) in terms of the variables that assume zero values in (2.5). Similarly, the new system should express those variables that assume positive values in (2.6) in terms of the variables that assume zero values in (2.6): in short, it should express  $x_1, x_5, x_6$  (as well as  $z$ ) in terms of  $x_2, x_3$ , and  $x_4$ . In particular, the variable  $x_1$ , which just changed its value from zero to positive should change its position from the right-hand side to the left-hand side of the system of equations. Similarly, the variable  $x_4$ , which just changed its value from positive to zero, should move from the left-hand side to the right-hand side.

To construct the new system, we shall begin with the newcomer to the left-hand side, namely, the variable  $x_1$ . The desired formula for  $x_1$  in terms of  $x_2, x_3, x_4$  is obtained easily from the first equation in (2.3):

$$x_1 = \frac{5}{2} - \frac{3}{2}x_2 - \frac{1}{2}x_3 - \frac{1}{2}x_4. \quad (2.7)$$

Next, in order to express  $x_5$ ,  $x_6$ , and  $z$  in terms of  $x_2$ ,  $x_3$ ,  $x_4$ , we simply substitute from (2.7) into the corresponding rows of (2.3):

$$\begin{aligned}x_5 &= 11 - 4\left(\frac{5}{2} - \frac{3}{2}x_2 - \frac{1}{2}x_3 - \frac{1}{2}x_4\right) - x_2 - 2x_3 \\&= 1 + 5x_2 + 2x_4, \\x_6 &= 8 - 3\left(\frac{5}{2} - \frac{3}{2}x_2 - \frac{1}{2}x_3 - \frac{1}{2}x_4\right) - 4x_2 - 2x_3 \\&= \frac{1}{2} + \frac{1}{2}x_2 - \frac{1}{2}x_3 + \frac{3}{2}x_4, \\z &= 5\left(\frac{5}{2} - \frac{3}{2}x_2 - \frac{1}{2}x_3 - \frac{1}{2}x_4\right) + 4x_2 + 3x_3 \\&= \frac{25}{2} - \frac{7}{2}x_2 + \frac{1}{2}x_3 - \frac{5}{2}x_4.\end{aligned}$$

Hence our new system reads

$$\begin{aligned}x_1 &= \frac{5}{2} - \frac{3}{2}x_2 - \frac{1}{2}x_3 - \frac{1}{2}x_4 \\x_5 &= 1 + 5x_2 + 2x_4 \\x_6 &= \frac{1}{2} + \frac{1}{2}x_2 - \frac{1}{2}x_3 + \frac{3}{2}x_4 \\z &= \frac{25}{2} - \frac{7}{2}x_2 + \frac{1}{2}x_3 - \frac{5}{2}x_4.\end{aligned}\tag{2.8}$$

As we did in the first iteration, we shall now try to increase the value of  $z$  by increasing the value of a suitably chosen right-hand side variable, while at the same time keeping the remaining right-hand side variables fixed at zero. Note that increases in the values of  $x_2$  or  $x_4$  would bring about *decreases* in the value of  $z$ , which is very much against our intentions. Thus, we have no choice: the right-hand side variable to increase its value is necessarily  $x_3$ . How much can we increase  $x_3$ ? The answer can be read directly from system (2.8): with  $x_2 = x_4 = 0$ , the constraint  $x_1 \geq 0$  implies  $x_3 \leq 5$ , the constraint  $x_5 \geq 0$  imposes no restriction at all, and the constraint  $x_6 \geq 0$  implies  $x_3 \leq 1$ . Hence,  $x_3 = 1$  is the best we can do; our new solution is

$$x_1 = 2, \quad x_2 = 0, \quad x_3 = 1, \quad x_4 = 0, \quad x_5 = 1, \quad x_6 = 0.\tag{2.9}$$

(Note that the value of  $z$  just increased from 12.5 to 13.)

As we have learned, getting just the improved solution isn't good enough; we also want a system of linear equations to go with (2.9). In this system, the positive-valued variables  $x_1$ ,  $x_3$ ,  $x_5$  will appear on the left, whereas the zero-valued variables  $x_2$ ,  $x_4$ ,  $x_6$

will appear on the right. To construct the system, we begin again with the newcomer to the left-hand side, namely, the variable  $x_3$ . From the third equation in (2.8), we have  $x_3 = 1 + x_2 + 3x_4 - 2x_6$ ; substituting for  $x_3$  into the remaining equations in (2.8), we obtain

$$\begin{aligned}x_3 &= 1 + x_2 + 3x_4 - 2x_6 \\x_1 &= 2 - 2x_2 - 2x_4 + x_6 \\x_5 &= 1 + 5x_2 + 2x_4 \\z &= 13 - 3x_2 - x_4 - x_6.\end{aligned}\tag{2.10}$$

Now it's time for the third iteration. First of all, from the right-hand side of (2.10) we have to choose a variable whose increase brings about an increase of the objective function. However, there is no such variable: indeed, if we increase any of the right-hand side variables  $x_2$ ,  $x_4$ ,  $x_6$ , we will make the value of  $z$  *decrease*. Thus, it seems that we have come to a standstill. In fact, the very presence of this standstill indicates that we are done; we have solved our problem; the solution described by the last table is optimal. Why? The answer lies hidden in the last row of (2.10):

$$z = 13 - 3x_2 - x_4 - x_6.\tag{2.11}$$

Our last solution (2.9) yields  $z = 13$ ; proving that this solution is optimal amounts to proving that every feasible solution satisfies the inequality  $z \leq 13$ . Since every feasible solution  $x_1, x_2, \dots, x_6$  satisfies, among other relations, the inequalities  $x_2 \geq 0$ ,  $x_4 \geq 0$ , and  $x_6 \geq 0$ , the desired inequality  $z \leq 13$  follows directly from (2.11).

## DICTIONARIES

In general, given a problem

$$\begin{aligned}\text{maximize} \quad & \sum_{j=1}^n c_j x_j \\ \text{subject to} \quad & \sum_{j=1}^n a_{ij} x_j \leq b_i \quad (i = 1, 2, \dots, m) \\ & x_j \geq 0 \quad (j = 1, 2, \dots, n)\end{aligned}\tag{2.12}$$

we first introduce the slack variables  $x_{n+1}, x_{n+2}, \dots, x_{n+m}$  and denote the objective function by  $z$ . That is, we define

$$\begin{aligned}x_{n+i} &= b_i - \sum_{j=1}^n a_{ij} x_j \quad (i = 1, 2, \dots, m) \\ z &= \sum_{j=1}^n c_j x_j.\end{aligned}\tag{2.13}$$



In the framework of the simplex method, each feasible solution  $x_1, x_2, \dots, x_n$  of (2.12) is represented by  $n + m$  nonnegative numbers  $x_1, x_2, \dots, x_{n+m}$ , with  $x_{n+1}, x_{n+2}, \dots, x_{n+m}$  defined by (2.13). In each iteration, the simplex method moves from some feasible solution  $x_1, x_2, \dots, x_{n+m}$  to another feasible solution  $\bar{x}_1, \bar{x}_2, \dots, \bar{x}_{n+m}$ , which is better than the previous one in the sense that

$$\sum_{j=1}^n c_j \bar{x}_j > \sum_{j=1}^n c_j x_j.$$

(Actually, the last statement is not quite correct: the inequality is not always strict. This point and other subtleties will be discussed in Chapter 3.)

As we have seen, it is convenient to associate a system of linear equations with each of the feasible solutions: such systems make it easier to find the improved feasible solutions. They do so by translating any choice of values of the right-hand side variables into the corresponding values of the left-hand side variables and of the objective function. Following J. E. Strum (1972), we shall refer to these systems as *dictionaries*. Thus, every dictionary associated with (2.12) will be a system of linear equations in the variables  $x_1, x_2, \dots, x_{n+m}$  and  $z$ . However, not every system of linear equations in these variables constitutes a dictionary. To begin with, we have defined  $x_{n+1}, x_{n+2}, \dots, x_{n+m}$  and  $z$  in terms of  $x_1, x_2, \dots, x_n$ , and so the  $n + m + 1$  variables are heavily interdependent. This interdependence must be captured by every dictionary associated with (2.12): the translations must be correct. More precisely, we shall insist that:

Every solution of the set of equations comprising a dictionary must be also a solution of (2.13), and vice versa. (2.14)

For example, for every choice of numbers  $x_1, x_2, \dots, x_6$  and  $z$ , the following three statements are equivalent:

- $x_1, x_2, \dots, x_6, z$  constitute a solution of (2.3),
- $x_1, x_2, \dots, x_6, z$  constitute a solution of (2.8),
- $x_1, x_2, \dots, x_6, z$  constitute a solution of (2.10).

In that sense, the three dictionaries (2.3), (2.8), and (2.10) contain the same information concerning the interdependence among the seven variables. Nevertheless, each of the three dictionaries presents this information in its very own way. The form of (2.3) suggests that we are free to choose the numerical values of  $x_1, x_2$ , and  $x_3$  at will, whereupon the values of  $x_4, x_5, x_6$ , and  $z$  are determined: in this dictionary, the decision variables  $x_1, x_2, x_3$  act as independent variables, while  $z$  and the slack variables  $x_4, x_5, x_6$  are dependent on them. Dictionary (2.8) presents  $x_2, x_3, x_4$  as independent and  $x_1, x_5, x_6, z$  as dependent. In dictionary (2.10), the independent variables are  $x_2, x_4, x_6$  and the dependent ones are  $x_3, x_1, x_5, z$ . In general:

The equations of every dictionary must express  $m$  of the variables  $x_1,$

$x_2, \dots, x_{n+m}$  and the objective function  $z$  in terms of the remaining  $n$  (2.15) variables.

The properties (2.14) and (2.15) are the defining properties of dictionaries.

In addition to these two properties, dictionaries (2.3), (2.8), and (2.10) have the following property:

Setting the right-hand side variables at zero and evaluating the left-hand side variables, we arrive at a *feasible* solution.

Dictionaries with this additional property will be called *feasible dictionaries*. Hence, every feasible dictionary describes a feasible solution. However, not every feasible solution is described by a feasible dictionary; for instance, no dictionary describes the feasible solution  $x_1 = 1, x_2 = 0, x_3 = 1, x_4 = 2, x_5 = 5, x_6 = 3$  of (2.1). Feasible solutions that can be described by dictionaries are called *basic*. The characteristic feature of the simplex method is the fact that it works exclusively with basic feasible solutions and ignores all other feasible solutions.

## SECOND EXAMPLE

We shall complete our preview of the simplex method by applying it to another LP problem:

$$\begin{aligned} \text{maximize} \quad & 5x_1 + 5x_2 + 3x_3 \\ \text{subject to} \quad & x_1 + 3x_2 + x_3 \leq 3 \\ & -x_1 + 3x_3 \leq 2 \\ & 2x_1 - x_2 + 2x_3 \leq 4 \\ & 2x_1 + 3x_2 - x_3 \leq 2 \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

In this case, the initial feasible dictionary reads

$$\begin{aligned} x_4 &= 3 - x_1 - 3x_2 - x_3 \\ x_5 &= 2 + x_1 - 3x_3 \\ x_6 &= 4 - 2x_1 + x_2 - 2x_3 \\ x_7 &= 2 - 2x_1 - 3x_2 + x_3 \\ \hline z &= 5x_1 + 5x_2 + 3x_3. \end{aligned} \tag{2.16}$$

(Even though the order of the equations in a dictionary is quite irrelevant, we shall make a habit of writing the formula for  $z$  last and separating it from the rest of the table by a solid line. Of course, that does *not* mean that the last equation is the sum of the previous ones.) This feasible dictionary describes the feasible solution

$$x_1 = 0, \quad x_2 = 0, \quad x_3 = 0, \quad x_4 = 3, \quad x_5 = 2, \quad x_6 = 4, \quad x_7 = 2.$$



However, there is no need to write this solution down, as we just did: the solution is implicit in the dictionary.

In the first iteration, we shall attempt to increase the value of  $z$  by making one of the right-hand side variables positive. At this moment, any of the three variables  $x_1, x_2, x_3$  would do. In small examples, it is common practice to choose the variable that, in the formula for  $z$ , has the largest coefficient: the increase in that variable will make  $z$  increase at the fastest rate (but not necessarily to the highest level). In our case, this rule leaves us a choice between  $x_1$  and  $x_2$ ; choosing arbitrarily, we decide to make  $x_1$  positive. As the value of  $x_1$  increases, so does the value of  $x_5$ . However, the values of  $x_4, x_6$ , and  $x_7$  decrease, and none of them is allowed to become negative. Of the three constraints  $x_4 \geq 0, x_6 \geq 0, x_7 \geq 0$  that impose upper bounds on the increment of  $x_1$ , the last constraint  $x_7 \geq 0$  is the most stringent: it implies  $x_1 \leq 1$ . In the improved feasible solution, we shall have  $x_1 = 1$  and  $x_7 = 0$ . Without writing the new solution down, we shall now construct the new dictionary. All we need to know is that  $x_1$  just made its way from the right-hand side to the left, whereas  $x_7$  went in the opposite direction. From the fourth equation in (2.16), we have

$$x_1 = 1 - \frac{3}{2}x_2 + \frac{1}{2}x_3 - \frac{1}{2}x_7. \quad (2.17)$$

Substituting from (2.17) into the remaining equations of (2.16), we arrive at the desired dictionary

$$\begin{aligned} x_1 &= 1 - \frac{3}{2}x_2 + \frac{1}{2}x_3 - \frac{1}{2}x_7 \\ x_4 &= 2 - \frac{3}{2}x_2 - \frac{3}{2}x_3 + \frac{1}{2}x_7 \\ x_5 &= 3 - \frac{3}{2}x_2 - \frac{5}{2}x_3 - \frac{1}{2}x_7 \\ x_6 &= 2 + 4x_2 - 3x_3 + x_7 \\ \hline z &= 5 - \frac{5}{2}x_2 + \frac{11}{2}x_3 - \frac{5}{2}x_7. \end{aligned} \quad (2.18)$$

The construction of (2.18) completes the first iteration of the simplex method.

### Digression on Terminology

The variables  $x_j$  that appear on the left-hand side of a dictionary are called *basic*; the variables  $x_j$  that appear on the right-hand side are *nonbasic*. The basic variables are said to constitute a *basis*. Of course, the basis changes with each iteration: for example, in the first iteration,  $x_1$  entered the basis whereas  $x_7$  left it. In each iteration,

we first choose the nonbasic variable that is to enter the basis and then we find out which basic variable must leave the basis. The choice of the *entering* variable is motivated by our desire to increase the value of  $z$ ; the determination of the *leaving* variable is based on the requirement that all variables must assume nonnegative values. The leaving variable is that basic variable whose nonnegativity imposes the most stringent upper bound on the increment of the entering variable. The formula for the leaving variable appears in the *pivot row* of the dictionary; the computational process of constructing the new dictionary is referred to as *pivoting*.

### Back to the Second Example

In our example, the variable to enter the basis during the second iteration is quite unequivocally  $x_3$ . This is the only nonbasic variable in (2.18) whose coefficient in the last row is positive. Of the four basic variables,  $x_6$  imposes the most stringent upper bound on the increase of  $x_3$ , and, therefore, has to leave the basis. Pivoting, we arrive at our third dictionary,

$$\begin{aligned} x_3 &= \frac{2}{3} + \frac{4}{3}x_2 + \frac{1}{3}x_7 - \frac{1}{3}x_6 \\ x_1 &= \frac{4}{3} - \frac{5}{6}x_2 - \frac{1}{3}x_7 - \frac{1}{6}x_6 \\ x_4 &= 1 - \frac{7}{2}x_2 + \frac{1}{2}x_6 \\ x_5 &= \frac{4}{3} - \frac{29}{6}x_2 - \frac{4}{3}x_7 + \frac{5}{6}x_6 \\ \hline z &= \frac{26}{3} + \frac{29}{6}x_2 - \frac{2}{3}x_7 - \frac{11}{6}x_6. \end{aligned} \quad (2.19)$$

In the third iteration, the entering variable is  $x_2$  and the leaving variable is  $x_5$ . Pivoting yields the dictionary

$$\begin{aligned} x_2 &= \frac{8}{29} - \frac{8}{29}x_7 + \frac{5}{29}x_6 - \frac{6}{29}x_5 \\ x_3 &= \frac{30}{29} - \frac{1}{29}x_7 - \frac{3}{29}x_6 - \frac{8}{29}x_5 \\ x_1 &= \frac{32}{29} - \frac{3}{29}x_7 - \frac{9}{29}x_6 + \frac{5}{29}x_5 \\ x_4 &= \frac{1}{29} + \frac{28}{29}x_7 - \frac{3}{29}x_6 + \frac{21}{29}x_5 \\ \hline z &= 10 - 2x_7 - x_6 - x_5. \end{aligned} \quad (2.20)$$

At this point, no nonbasic variable can enter the basis without making the value of  $z$  decrease. Hence, the last dictionary describes an optimal solution of our example. That solution is

$$x_1 = \frac{32}{29}, \quad x_2 = \frac{8}{29}, \quad x_3 = \frac{30}{29}$$

and it yields  $z = 10$ .

## FURTHER REMARKS

The reader may have noticed that, having first carefully laid down the definition of a dictionary, we then proceeded to refer to (2.18), (2.19), and (2.20) as dictionaries, without bothering to verify that they do indeed have property (2.14). Such carelessness can be easily justified. Take, for example, system (2.18). Since (2.18) arises from (2.16) by arithmetical operations (namely, pivoting with  $x_1$  entering and  $x_7$  leaving), every solution of (2.16) must be also a solution of (2.18). The converse is also true, since (2.16) can be obtained from (2.18) by pivoting with  $x_7$  entering and  $x_1$  leaving. Hence, every solution of (2.18) is a solution of (2.16), and vice versa. Similar arguments show that every solution of (2.19) is a solution of (2.18), and vice versa; and that every solution of (2.20) is a solution of (2.19), and vice versa.

Another point of concern is the question of the *uniqueness*, as opposed to the *existence*, of optimal solutions. This question will be of no great interest to us; nevertheless, it is easy to deal with and so we will get it out of the way now. Note that in each of our two examples, we not only found an optimal solution, but we also collected the evidence to prove that there is only one optimal solution. For instance, the final dictionary for our first problem reads

$$\begin{aligned} x_3 &= 1 + x_2 + 3x_4 - 2x_6 \\ x_1 &= 2 - 2x_2 - 2x_4 + x_6 \\ x_5 &= 1 + 5x_2 + 2x_4 \\ z &= 13 - 3x_2 - x_4 - x_6. \end{aligned}$$

The last row shows that every feasible solution with  $z = 13$  satisfies  $x_2 = x_4 = x_6 = 0$ ; the rest of the dictionary shows that every such solution satisfies  $x_3 = 1$ ,  $x_1 = 2$ ,  $x_5 = 1$ ; therefore, there is just one optimal solution. A similar argument applies to the second problem.

Of course, there are LP problems with more than just one optimal solution; having solved

such problems by the simplex method, we can effectively describe all the optimal solutions. For example, consider the following dictionary:

$$\begin{aligned} x_4 &= 3 + x_2 - 2x_5 + 7x_3 \\ x_1 &= 1 - 5x_2 + 6x_5 - 8x_3 \\ x_6 &= 4 + 9x_2 + 2x_5 - x_3 \\ z &= 8 - x_3. \end{aligned}$$

The last row shows that every optimal solution satisfies  $x_3 = 0$  (but not necessarily  $x_2 = 0$  or  $x_5 = 0$ ). For such solutions, the rest of the dictionary implies

$$\begin{aligned} x_4 &= 3 + x_2 - 2x_5 \\ x_1 &= 1 - 5x_2 + 6x_5 \\ x_6 &= 4 + 9x_2 + 2x_5. \end{aligned} \tag{2.21}$$

We conclude that every optimal solution arises by the substitution formulas (2.21) from some  $x_2$  and  $x_5$  such that

$$\begin{aligned} -x_2 + 2x_5 &\leq 3 \\ 5x_2 - 6x_5 &\leq 1 \\ -9x_2 - 2x_5 &\leq 4 \\ x_2, x_5 &\geq 0. \end{aligned}$$

(In fact, the inequality  $-9x_2 - 2x_5 \leq 4$  is clearly redundant; its validity is forced by  $x_2 \geq 0$  and  $x_5 \geq 0$ .)

There are a few other rough spots we deliberately failed to point out in our overview of the simplex method. We shall discuss them in Chapter 3.

## TABLEAU FORMAT

The simplex method is often introduced in a format differing from ours. To outline the more popular *tableau format*, we shall return to the first example of this chapter. To begin, let us write down the equations of the first dictionary in a slightly modified form:

$$\begin{aligned} 2x_1 + 3x_2 + x_3 + x_4 &= 5 \\ 4x_1 + x_2 + 2x_3 + x_5 &= 11 \\ 3x_1 + 4x_2 + 2x_3 + x_6 &= 8 \\ -z + 5x_1 + 4x_2 + 3x_3 &= 0. \end{aligned}$$

Recording just the coefficients at the  $x_i$ 's, together with the right-hand sides, we obtain our first tableau:

|   |   |   |   |   |   |    |
|---|---|---|---|---|---|----|
| 2 | 3 | 1 | 1 | 0 | 0 | 5  |
| 4 | 1 | 2 | 0 | 1 | 0 | 11 |
| 3 | 4 | 2 | 0 | 0 | 1 | 8  |
| 5 | 4 | 3 | 0 | 0 | 0 | 0. |

In a similar way, the equations of the second dictionary,

$$\begin{array}{rcl} x_1 + \frac{3}{2}x_2 + \frac{1}{2}x_3 + \frac{1}{2}x_4 & = & \frac{5}{2} \\ -5x_2 & -2x_4 + x_5 & = 1 \\ -\frac{1}{2}x_2 + \frac{1}{2}x_3 - \frac{3}{2}x_4 & + x_6 & = \frac{1}{2} \\ \hline -z - \frac{7}{2}x_2 + \frac{1}{2}x_3 - \frac{5}{2}x_4 & = & -\frac{25}{2} \end{array}$$

give rise to a second tableau:

$$\begin{array}{cccccc} 1 & \frac{3}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 & \frac{5}{2} \\ 0 & -5 & 0 & -2 & 1 & 0 & 1 \\ 0 & -\frac{1}{2} & \frac{1}{2} & -\frac{3}{2} & 0 & 1 & \frac{1}{2} \\ \hline 0 & -\frac{7}{2} & \frac{1}{2} & -\frac{5}{2} & 0 & 0 & -\frac{25}{2} \end{array}$$

It is a routine matter to translate the pivoting rules, previously derived in terms of dictionaries, into the language of tableaux. The following steps describe the procedure; the reader should have no trouble verifying its correctness. (At any rate, the procedure is not important for our exposition since we do not use the tableau format.)

**Step 1.** Examine all numbers in the last row (except the one farthest right, which equals the current value of  $-z$ ). If all of them are negative or zero, stop: the tableau describes an optimal solution. Otherwise find the largest of these numbers; the column in which it appears is called the *pivot column* and corresponds to the entering variable.

For example, the pivot column in our first tableau is the first one:

|   |   |   |   |   |   |    |
|---|---|---|---|---|---|----|
| 2 | 3 | 1 | 1 | 0 | 0 | 5  |
| 4 | 1 | 2 | 0 | 1 | 0 | 11 |
| 3 | 4 | 2 | 0 | 0 | 1 | 8  |
| 5 | 4 | 3 | 0 | 0 | 0 | 0  |

**Step 2.** For each row whose entry  $r$  in the pivot column is positive, look up the entry  $s$  in the rightmost column. The row with the smallest ratio  $\frac{s}{r}$  is called the *pivot row* and corresponds to the leaving variable. (If all the entries in the pivot column are negative or zero, then the problem is unbounded; more on that in Chapter 3.)

In our example, the pivot row is the first row (with  $\frac{s}{r} = \frac{5}{2}$ ):

|   |   |   |   |   |   |    |
|---|---|---|---|---|---|----|
| 2 | 3 | 1 | 1 | 0 | 0 | 5  |
| 4 | 1 | 2 | 0 | 1 | 0 | 11 |
| 3 | 4 | 2 | 0 | 0 | 1 | 8  |
| 5 | 4 | 3 | 0 | 0 | 0 | 0  |

**Step 3.** Divide every entry in the pivot row by the *pivot number*, found in the intersection of the pivot row with the pivot column:

$$\begin{array}{cccccc} 1 & \frac{3}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 & \frac{5}{2} \\ 4 & 1 & 2 & 0 & 1 & 0 & 11 \\ 3 & 4 & 2 & 0 & 0 & 1 & 8 \\ \hline 5 & 4 & 3 & 0 & 0 & 0 & 0 \end{array}$$

**Step 4.** From every remaining row, subtract a suitable multiple of the new pivot row. This operation is designed to make every entry in the pivot column (except for the pivot number) become zero; hence, the “suitable multiple” results when the new pivot row is multiplied by the entry appearing in the pivot column and in the row in question. (In our example, step 4 results in the second tableau.)

A tableau is nothing but a cryptic recording of a dictionary with all the variables collected on the left-hand side and the symbols for these variables omitted. We shall continue to use dictionaries instead, since they are more explicit. (Of course, nothing prevents the reader tired of writing the same symbols  $x_1, x_2, \dots$  over and over again from using the tableau shorthand.) □

## A WARNING

There is often more than one way of describing a particular algorithm; descriptions aimed at clarifying underlying concepts are often quite different from those that suggest efficient computer implementations. The simplex method is no exception. Dictionaries may provide a convenient tool for explaining its basic principles. However, in implementing the method for computer solutions of large problems, considerations of computational efficiency and numerical accuracy overshadow such didactic niceties. We shall begin to study efficient implementations of the simplex method in Chapters 7 and 8.

## PROBLEMS

△ 2.1 Solve the following problems by the simplex method:

- a. maximize  $3x_1 + 2x_2 + 4x_3$   
 subject to  $x_1 + x_2 + 2x_3 \leq 4$   
 $2x_1 + 3x_3 \leq 5$   
 $2x_1 + x_2 + 3x_3 \leq 7$   
 $x_1, x_2, x_3 \geq 0$
- b. maximize  $5x_1 + 6x_2 + 9x_3 + 8x_4$   
 subject to  $x_1 + 2x_2 + 3x_3 + x_4 \leq 5$   
 $x_1 + x_2 + 2x_3 + 3x_4 \leq 3$   
 $x_1, x_2, x_3, x_4 \geq 0$
- c. maximize  $2x_1 + x_2$   
 subject to  $2x_1 + 3x_2 \leq 3$   
 $x_1 + 5x_2 \leq 1$   
 $2x_1 + x_2 \leq 4$   
 $4x_1 + x_2 \leq 5$   
 $x_1, x_2 \geq 0$ .

2.2 Use the simplex method to describe *all* the optimal solutions of the following problem:

- maximize  $2x_1 + 3x_2 + 5x_3 + 4x_4$   
 subject to  $x_1 + 2x_2 + 3x_3 + x_4 \leq 5$   
 $x_1 + x_2 + 2x_3 + 3x_4 \leq 3$   
 $x_1, x_2, x_3, x_4 \geq 0$ .

## Pitfalls and How to Avoid Them

The examples illustrating the simplex method in the preceding chapter were purposely smooth. They did not point out the dangers that can occur. The purpose of the present chapter, therefore, is to rigorously analyze the method by scrutinizing its every step.

### THREE KINDS OF PITFALLS

Three kinds of pitfalls can occur in the simplex method.

- (i) **INITIALIZATION.** We might not be able to start: How do we get hold of a feasible dictionary?
- (ii) **ITERATION.** We might get stuck in some iteration: Can we always choose an entering variable, find the leaving variable, and construct the next feasible dictionary by pivoting?
- (iii) **TERMINATION.** We might not be able to finish: Can the simplex method construct an endless sequence of dictionaries without ever reaching an optimal solution?

In the preceding chapter, INITIALIZATION never came up. Given a problem

$$\begin{aligned} &\text{maximize} && \sum_{j=1}^n c_j x_j \\ &\text{subject to} && \sum_{j=1}^n a_{ij} x_j \leq b_i \quad (i = 1, 2, \dots, m) \\ &&& x_j \geq 0 \quad (j = 1, 2, \dots, n) \end{aligned} \quad (3.1)$$

we constructed the initial feasible dictionary by simply writing down the formulas defining the slack variables and the objective function,

$$\begin{aligned} x_{n+i} &= b_i - \sum_{j=1}^n a_{ij} x_j \quad (i = 1, 2, \dots, m) \\ z &= \sum_{j=1}^n c_j x_j. \end{aligned}$$

In general, this dictionary is feasible if and only if each right-hand side,  $b_i$ , in (3.1) is nonnegative. This is the case if and only if

$$x_1 = 0, \quad x_2 = 0, \quad \dots, \quad x_n = 0$$

is a feasible solution of (3.1). Since the set of zero values is sometimes called the “origin,” problems (3.1) with each right-hand side  $b_i$  nonnegative are referred to as problems with a *feasible origin*. For the moment, we shall avoid the pitfalls of INITIALIZATION by default: we shall restrict ourselves to problems with a feasible origin. Problems with an infeasible origin are discussed on pages 39–42.

### Iteration

Given some feasible dictionary, we have to select an entering variable, to find a leaving variable, and to construct the next feasible dictionary by pivoting.

**Choosing an entering variable.** The entering variable is a *nonbasic variable*  $x_j$  with a *positive coefficient*  $\bar{c}_j$  in the last row of the current dictionary. This rule is ambiguous in the sense that it may provide more than one candidate for entering the basis, or no candidate at all. The latter alternative implies that the current dictionary describes an optimal solution, at which point the method may terminate. More precisely, consider the last row of our current dictionary,

$$z = z^* + \sum_{j \in N} \bar{c}_j x_j$$

with  $N$  standing for the set of subscripts  $j$  of nonbasic variables  $x_j$ . Our current solution, with  $x_j = 0$  whenever  $j \in N$ , gives the objective function the numerical value of  $z^*$ . If  $\bar{c}_j \leq 0$  whenever  $j \in N$ , then every feasible solution, with  $x_j \geq 0$

whenever  $j \in N$ , gives the objective function a numerical value of at most  $z^*$ ; hence the current solution is optimal. On the other hand, if there is more than one candidate for entering the basis, then any of these candidates may serve. (In hand calculations involving small problems, it is customary to choose the candidate  $x_j$  that has the largest coefficient  $\bar{c}_j$ . In most computer implementations of the simplex method, however, this practice is abandoned. More on this subject in Chapter 7.)

**Finding the leaving variable.** The leaving variable is *that basic variable whose nonnegativity imposes the most stringent upper bound on the increase of the entering variable*. Again, this rule is ambiguous in the sense that it may provide more than one candidate for leaving the basis, or no candidate at all. The latter alternative is illustrated on the dictionary

$$\begin{aligned} x_2 &= 5 + 2x_3 - x_4 - 3x_1 \\ x_5 &= 7 - 3x_4 - 4x_1 \\ z &= 5 + x_3 - x_4 - x_1. \end{aligned}$$

The entering variable is  $x_3$ , but neither of the two basic variables  $x_2, x_5$  imposes an upper bound on its increase. Therefore, we can make  $x_3$  as large as we wish (maintaining  $x_1 = x_4 = 0$ ) and still retain feasibility: setting  $x_3 = t$  for any positive  $t$ , we obtain a feasible solution with  $x_1 = 0, x_2 = 5 + 2t, x_4 = 0, x_5 = 7$ , and  $z = 5 + t$ . Since  $t$  can be made arbitrarily large,  $z$  can be made arbitrarily large. We conclude that the problem is *unbounded*: for every number  $M$ , there is a feasible solution  $x_1, x_2, \dots, x_5$  such that  $x_3 - x_4 - x_1 > M$ . The same conclusion can be reached in general: if there is no candidate for leaving the basis, then we can make the value of the entering variable, and therefore also the value of the objective function, as large as we wish. In that case, the problem is unbounded. On the other hand, if there is more than one candidate for leaving the basis, then any of these candidates may serve. Once the entering and leaving variables have been selected, pivoting is a straightforward matter.

**Degeneracy.** The presence of more than one candidate for leaving the basis has interesting consequences. For illustration, consider the dictionary

$$\begin{aligned} x_4 &= 1 - 2x_3 \\ x_5 &= 3 - 2x_1 + 4x_2 - 6x_3 \\ x_6 &= 2 + x_1 - 3x_2 - 4x_3 \\ z &= 2x_1 - x_2 + 8x_3. \end{aligned}$$

Having chosen  $x_3$  to enter the basis, we find that each of the three basic variables  $x_4, x_5, x_6$  limits the increase of  $x_3$  to  $\frac{1}{2}$ . Hence each of these three variables is a candidate for leaving the basis. We arbitrarily choose  $x_4$ . Pivoting as usual, we obtain the dictionary

$$\begin{array}{rcl}
 x_3 & = & 0.5 \qquad \qquad - 0.5x_4 \\
 x_5 & = & - 2x_1 + 4x_2 + 3x_4 \\
 x_6 & = & \qquad x_1 - 3x_2 + 2x_4 \\
 \hline
 z & = & 4 + 2x_1 - x_2 - 4x_4.
 \end{array}$$

This dictionary differs from all the dictionaries we have encountered so far in one important respect: along with the nonbasic variables, the basic variables  $x_5$  and  $x_6$  have value zero in the associated solution. Basic solutions with one or more basic variables at zero are called *degenerate*.

Although harmless in its own right, degeneracy may have annoying side effects. These are illustrated on the next iteration in our example. There,  $x_1$  enters the basis and  $x_5$  leaves; because of degeneracy, the constraint  $x_5 \geq 0$  limits the increment of  $x_1$  to zero. Hence the value of  $x_1$  will remain unchanged, and so will the values of the remaining variables and the value of the objective function  $z$ . This is annoying, for the motivation behind the simplex method is a desire to increase the value of  $z$  in each iteration. In this particular iteration, that desire remains unfulfilled: pivoting changes the dictionary into

$$\begin{array}{rcl}
 x_1 & = & 2x_2 + 1.5x_4 - 0.5x_5 \\
 x_3 & = & 0.5 \qquad \qquad - 0.5x_4 \\
 x_6 & = & - x_2 + 3.5x_4 - 0.5x_5 \\
 \hline
 z & = & 4 + 3x_2 - x_4 - x_5
 \end{array}$$

but it does not affect the associated solution at all. Simplex iterations that do not change the basic solution are called *degenerate*. (As the reader may verify, the next iteration is degenerate again, but the one after that turns out to be nondegenerate and brings us to the optimal solution.)

In a sense, degeneracy is something of an accident: a basic variable may vanish only if the results of successive pivot operations just happen to cancel each other out. And yet degeneracy abounds in LP problems arising from practical applications. It has been said that nearly all such problems yield degenerate basic feasible solutions at some stage of the simplex method. Whenever that happens, the simplex method may stall by going through a few (and sometimes quite a few) degenerate iterations in a row. Typically, such a block of degenerate iterations ends with a breakthrough represented by a nondegenerate iteration; an example of the atypical case is presented next.

### Termination: Cycling

Can the simplex method go through an endless sequence of iterations without ever finding an optimal solution? Yes, it can. To justify this claim, let us consider the initial dictionary

$$\begin{array}{rcl}
 x_5 & = & - 0.5x_1 + 5.5x_2 + 2.5x_3 - 9x_4 \\
 x_6 & = & - 0.5x_1 + 1.5x_2 + 0.5x_3 - x_4 \\
 x_7 & = & 1 - x_1 \\
 \hline
 z & = & 10x_1 - 57x_2 - 9x_3 - 24x_4
 \end{array}$$

and let us agree on the following:

- (i) The entering variable will always be the nonbasic variable that has the largest coefficient in the  $z$ -row of the dictionary.
- (ii) If two or more basic variables compete for leaving the basis, then the candidate with the smallest subscript will be made to leave.

Now the sequence of dictionaries constructed in the first six iterations goes as follows.

After the first iteration:

$$\begin{array}{rcl}
 x_1 & = & 11x_2 + 5x_3 - 18x_4 - 2x_5 \\
 x_6 & = & - 4x_2 - 2x_3 + 8x_4 + x_5 \\
 x_7 & = & 1 - 11x_2 - 5x_3 + 18x_4 + 2x_5 \\
 \hline
 z & = & 53x_2 + 41x_3 - 204x_4 - 20x_5.
 \end{array}$$

After the second iteration:

$$\begin{array}{rcl}
 x_2 & = & - 0.5x_3 + 2x_4 + 0.25x_5 - 0.25x_6 \\
 x_1 & = & - 0.5x_3 + 4x_4 + 0.75x_5 - 2.75x_6 \\
 x_7 & = & 1 + 0.5x_3 - 4x_4 - 0.75x_5 - 13.25x_6 \\
 \hline
 z & = & 14.5x_3 - 98x_4 - 6.75x_5 - 13.25x_6.
 \end{array}$$

After the third iteration:

$$\begin{array}{rcl}
 x_3 & = & 8x_4 + 1.5x_5 - 5.5x_6 - 2x_1 \\
 x_2 & = & - 2x_4 - 0.5x_5 + 2.5x_6 + x_1 \\
 x_7 & = & 1 \qquad \qquad \qquad - x_1 \\
 \hline
 z & = & 18x_4 + 15x_5 - 93x_6 - 29x_1.
 \end{array}$$

After the fourth iteration:

$$\begin{array}{rcl}
 x_4 & = & - 0.25x_5 + 1.25x_6 + 0.5x_1 - 0.5x_2 \\
 x_3 & = & - 0.5x_5 + 4.5x_6 + 2x_1 - 4x_2 \\
 x_7 & = & 1 \qquad \qquad \qquad - x_1 \\
 \hline
 z & = & 10.5x_5 - 70.5x_6 - 20x_1 - 9x_2.
 \end{array}$$

After the fifth iteration:

$$\begin{array}{rcl} x_5 & = & 9x_6 + 4x_1 - 8x_2 - 2x_3 \\ x_4 & = & -x_6 - 0.5x_1 + 1.5x_2 + 0.5x_3 \\ x_7 & = & 1 - x_1 \\ \hline z & = & 24x_6 + 22x_1 - 93x_2 - 21x_3. \end{array}$$

After the sixth iteration:

$$\begin{array}{rcl} x_6 & = & -0.5x_1 + 1.5x_2 + 0.5x_3 - x_4 \\ x_5 & = & -0.5x_1 + 5.5x_2 + 2.5x_3 - 9x_4 \\ x_7 & = & 1 - x_1 \\ \hline z & = & 10x_1 - 57x_2 - 9x_3 - 24x_4. \end{array}$$

Since the dictionary constructed after the sixth iteration is identical with the initial dictionary, the method will go through the same six iterations again and again without ever finding the optimal solution (which, as we shall see later, has  $z = 1$ ). This phenomenon is known as cycling. More precisely, we say that the simplex method *cycles* if one dictionary appears in two different iterations (and so the sequence of iterations leading from the dictionary to itself can be repeated over and over without end). Note that cycling can occur only in the presence of degeneracy: since the value of the objective function increases with each nondegenerate iteration and remains unchanged after each degenerate one, all the iterations in the sequence leading from a dictionary to itself must be degenerate. Cycling is one reason why the simplex method may fail to terminate; the following theorem shows that it is the only reason.

**THEOREM 3.1.** If the simplex method fails to terminate, then it must cycle.

**PROOF.** To begin, note that there are only finitely many ways of choosing  $m$  basic variables from all the  $n + m$  variables. Thus, if the simplex method fails to terminate, then some basis must appear in two different iterations. Now it only remains to be proved that any two dictionaries with the same basis must be identical. (This fact becomes trivial as soon as one describes dictionaries in terms of matrices, as we shall do in Chapter 7. Nevertheless, we can and shall present an easy proof from scratch right now.) Consider two dictionaries

$$\begin{array}{rcl} x_i & = & b_i - \sum_{j \notin B} a_{ij}x_j \quad (i \in B) \\ \hline z & = & v + \sum_{j \notin B} c_jx_j \end{array} \quad (3.2)$$

and

$$\begin{array}{rcl} x_i & = & b_i^* - \sum_{j \notin B} a_{ij}^*x_j \quad (i \in B) \\ \hline z & = & v^* + \sum_{j \notin B} c_j^*x_j \end{array} \quad (3.3)$$

with the same set of basic variables  $x_i$  ( $i \in B$ ). It is a defining property of dictionaries that every solution  $x_1, x_2, \dots, x_{n+m}, z$  of (3.2) is a solution of (3.3) and vice versa. In particular, if  $x_k$  is a nonbasic variable and if  $t$  is a number, then the numbers

$$x_k = t, \quad x_j = 0 \quad (j \notin B \text{ and } j \neq k), \quad x_i = b_i - a_{ik}t \quad (i \in B), \quad z = v + c_k t,$$

constituting a solution of (3.2), must satisfy (3.3). Hence,

$$b_i - a_{ik}t = b_i^* - a_{ik}^*t \quad \text{for all } i \in B, \quad \text{and} \quad v + c_k t = v^* + c_k^* t.$$

Since these identities must hold for all numbers  $t$ , we have

$$b_i = b_i^*, \quad a_{ik} = a_{ik}^* \quad \text{for all } i \in B, \quad \text{and} \quad v = v^*, \quad c_k = c_k^*.$$

Since  $x_k$  was an arbitrary nonbasic variable, the two dictionaries are identical.

Cycling is a rare phenomenon. In fact, constructing an LP problem on which the simplex method may cycle is difficult. [Our example is adapted from K. T. Marshall and J. W. Suurballe (1969). The first example of this size was constructed by E. M. L. Beale (1955) and the first example ever was constructed by A. J. Hoffman (1953). Incidentally, Marshall and Suurballe (1969) proved that if the simplex method cycles off-optimum on a problem that has an optimal solution, then the dictionaries must involve at least six variables and at least three equations.] P. Wolfe (1963) and T. C. T. Kotiah and D. I. Steinberg (1978) reported having come across practical problems that cycled (in 25 and 18 iterations, respectively) but such reports are scarce. For this reason, the remote possibility of cycling is disregarded in most computer implementations of the simplex method.

There are ways of preventing the occurrence of cycling altogether. The classic *perturbation method* and *lexicographic method* avoid cycling by a judicious choice of the leaving variable in each simplex iteration; the more recent *smallest-subscript rule* does so by an easy choice of *both* the entering and the leaving variables. The former alternative maintains the freedom of choice among different candidates for entering the basis, but it requires extra computations to choose the leaving variable; the latter alternative requires no extra work at all, but it gives up the multitude of choices for the entering variable. We shall explain the details of both.

**The perturbation method and the lexicographic method.** The perturbation and the lexicographic methods are closely related. The perturbation method, suggested first by A. Orden and developed independently by A. Charnes (1952), provides an intuitive motivation for the lexicographic method of G. B. Dantzig, A. Orden, and P. Wolfe (1955). The lexicographic method can be seen as an implementation of the perturbation method.

The starting point relies on the observations that cycling can be stamped out by stamping out degeneracy and that degeneracy itself is something of an accident. To elaborate on the second observation, consider a degenerate dictionary. The basic variables currently at zero would most likely assume small nonzero values if the initial right-hand sides,  $b_i$ , were changed slightly; at the same time, if these changes were truly microscopic, then the problem could be considered unchanged for all practical purposes. One way of exploiting these observations is to add a small positive  $\varepsilon$  to each  $b_i$ , and then to apply the simplex method to the resulting problem. This trick (with  $\varepsilon = 10^{-6}$  or so) is actually used in some computer implementations of the simplex method; it helps to reduce the number of degenerate iterations. Nevertheless, it does not constitute a reliable safeguard against cycling: for instance, if the simplex method is applied to the problem

$$\begin{array}{ll} \text{maximize} & 10x_1 - 57x_2 - 9x_3 - 24x_4 + 100x_5 \\ \text{subject to} & x_5 \leq 1 + \varepsilon \\ & 0.5x_1 - 5.5x_2 - 2.5x_3 + 9x_4 + x_5 \leq 1 + \varepsilon \\ & 0.5x_1 - 1.5x_2 - 0.5x_3 + x_4 + x_5 \leq 1 + \varepsilon \\ & x_1 + x_5 \leq 2 + \varepsilon \\ & x_1, x_2, \dots, x_5 \geq 0 \end{array}$$

then the degenerate dictionary

$$\begin{array}{rcll} x_5 = & 1 + \varepsilon & & - x_6 \\ x_7 = & & -0.5x_1 + 5.5x_2 + 2.5x_3 - 9x_4 + & x_6 \\ x_8 = & & -0.5x_1 + 1.5x_2 + 0.5x_3 - & x_4 + x_6 \\ x_9 = & 1 & - x_1 & + x_6 \\ z = & 100 + 100\varepsilon + 10x_1 - 57x_2 - 9x_3 - 24x_4 - & 100x_6 \end{array}$$

is obtained after the first iteration and, as the reader may verify, the simplex method cycles in the next six iterations. (The cycle is essentially the same as that of the preceding example.)

What went wrong here was that the small amounts  $\varepsilon$  added to the right-hand sides cancelled each other out in the first iteration. To guarantee that such cancellations will never take place (and therefore all the dictionaries will remain nondegenerate), we shall perturb the different right-hand sides  $b_1, b_2, \dots, b_m$  by radically different amounts  $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_m$ . More precisely, we shall choose a very small  $\varepsilon_1$  and then make each  $\varepsilon_{i+1}$  much smaller than the preceding  $\varepsilon_i$ : in symbols,

$$0 < \varepsilon_m < \varepsilon_{m-1} < \dots < \varepsilon_2 < \varepsilon_1 < 1. \quad (3.4)$$

Then we shall apply the simplex method to the perturbed problem

$$\text{maximize} \quad \sum_{j=1}^n c_j x_j$$

$$\begin{array}{ll} \text{subject to} & \sum_{j=1}^n a_{ij} x_j \leq b_i + \varepsilon_i \quad (i = 1, 2, \dots, m) \\ & x_j \geq 0 \quad (j = 1, 2, \dots, n). \end{array}$$

This is the *perturbation method*. (The perturbation method is usually presented with  $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_m$  equal to the powers  $\varepsilon, \varepsilon^2, \dots, \varepsilon^m$  of the same small  $\varepsilon$ . Our version makes the subsequent analysis a little more transparent.) For illustration, let us return to our first example on which the simplex method cycled. There, the initial dictionary reads

$$\begin{array}{rcll} x_5 = \varepsilon_1 & & -0.5x_1 + 5.5x_2 + 2.5x_3 - & 9x_4 \\ x_6 = \varepsilon_2 & & -0.5x_1 + 1.5x_2 + 0.5x_3 - & x_4 \\ x_7 = 1 + \varepsilon_3 & - & x_1 & \\ z = & & 10x_1 - 57x_2 - 9x_3 - & 24x_4. \end{array}$$

Again, the entering variable is  $x_1$ . The constraints  $x_5 \geq 0$ ,  $x_6 \geq 0$ , and  $x_7 \geq 0$  limit the increase of  $x_1$  to  $2\varepsilon_1$ ,  $2\varepsilon_2$ , and  $1 + \varepsilon_3$ , respectively. Since  $2\varepsilon_2 < 2\varepsilon_1 < 1 + \varepsilon_3$ , the leaving variable is  $x_6$ , and the next dictionary reads

$$\begin{array}{rcll} x_1 = 2\varepsilon_2 & & + 3x_2 + x_3 - 2x_4 - & 2x_6 \\ x_5 = \varepsilon_1 - \varepsilon_2 & & + 4x_2 + 2x_3 - 8x_4 + & x_6 \\ x_7 = 1 - 2\varepsilon_2 + \varepsilon_3 & - & 3x_2 - x_3 + 2x_4 + & 2x_6 \\ z = & 20\varepsilon_2 & - 27x_2 + x_3 - 44x_4 - & 20x_6. \end{array}$$

Now the only candidate for the entering variable is  $x_3$  and the only candidate for the leaving variable is  $x_7$ . The resulting dictionary,

$$\begin{array}{rcll} x_3 = 1 - 2\varepsilon_2 + \varepsilon_3 & & - 3x_2 + 2x_4 + 2x_6 - & x_7 \\ x_1 = 1 + \varepsilon_3 & & & - x_7 \\ x_5 = 2 + \varepsilon_1 - 5\varepsilon_2 + 2\varepsilon_3 & - & 2x_2 - 4x_4 + 5x_6 - & 2x_7 \\ z = 1 + 18\varepsilon_2 + \varepsilon_3 & & - 30x_2 - 42x_4 - 18x_6 - & x_7 \end{array}$$

is the optimal dictionary for the perturbed problem. It may be converted into the optimal dictionary for the original problem by simply disregarding all the terms involving  $\varepsilon_1, \varepsilon_2, \varepsilon_3$ .

How should we choose the numerical values of  $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_m$ ? The simplest answer is that we do not have to do that at all: rather than committing ourselves to definite values of  $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_m$ , we may just think of these symbols as representing indefinite quantities, which satisfy (3.4). After several iterations of the simplex method, these symbols spread throughout the various rows of the dictionary, but they remain confined to the absolute terms in each of the  $m+1$  rows; the coefficients at the nonbasic variables in the dictionary are unaffected by the perturbation. Now when it comes to finding the leaving variable, each of the constraints  $x_i \geq 0$  for a nonbasic  $x_i$  limits the increase of the entering  $x_j$  to a quantity such as  $2\varepsilon_1, 2\varepsilon_2, 1 + \varepsilon_3$ , or, more generally,

$$r = r_0 + r_1\varepsilon_1 + \dots + r_m\varepsilon_m, \quad s = s_0 + s_1\varepsilon_1 + \dots + s_m\varepsilon_m \quad (3.5)$$

and so on. As we are about to explain, assumption (3.4) allows us to compare the numerical values of such quantities without referring to the precise values of  $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_m$ . If  $r$  and  $s$  in (3.5) are distinct, then there is the smallest subscript  $k$  such that  $r_k \neq s_k$ . It is customary to say that  $r$  is *lexicographically smaller* than  $s$  if  $r_k < s_k$ . (The choice of the term *lexicographically* is explained by observing that, for instance,  $2 + 21\varepsilon_1 + 19\varepsilon_2 + 20\varepsilon_3$  is lexicographically smaller than  $2 + 21\varepsilon_1 + 20\varepsilon_2 + 20\varepsilon_3 + 15\varepsilon_4 + 14\varepsilon_5$  for the same reason that "bust" comes before "button"



in a dictionary.) It is easy to prove that  $r$  is lexicographically smaller than  $s$  if and only if  $r$  is numerically smaller than  $s$  for all values of  $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_m$  that satisfy (3.4). This statement has to be made precise by specifying just what is meant by the symbol  $\ll$  in (3.4); we leave the details for problem 3.7.

The *lexicographic method* is that implementation of the perturbation method in which  $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_m$  are treated as symbols, and quantities such as  $r$  and  $s$  in (3.5) are compared by the lexicographic rule. Note that it is always possible to choose the leaving variable by the lexicographic rule: in every finite set of expressions such as  $r$  and  $s$  in (3.5), there is always one that is lexicographically smaller than or equal to all the others. Even though this fact may be taken for granted intuitively, rigor requires that it be proved; we leave the details for problem 3.6. Another fine point concerns the behavior of the objective function  $z$ . The value of  $z$ , equal to some expression  $v_0 + v_1\varepsilon_1 + \dots + v_m\varepsilon_m$ , remains unchanged in each degenerate iteration and increases, in the lexicographic sense, with each nondegenerate one. (In our example, the increase from 0 to  $20\varepsilon_2$  in the first iteration was followed by the increase from  $20\varepsilon_2$  to  $1 + 18\varepsilon_2 + \varepsilon_3$  in the second iteration.) It is intuitively obvious that the total of two or more lexicographic increases is a lexicographic increase; a rigorous proof of this fact follows from the result of problem 3.5. Now it follows that, even in the generalized context of the lexicographic method, cycling is possible only in the presence of degeneracy. Finally, note that the only function of the terms involving  $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_m$  is to guide us toward the appropriate choice of a leaving variable whenever two or more candidates present themselves in the original problem. If, at any moment, these terms are deleted, then the dictionary for the perturbed problem reduces to a dictionary for the original problem.

**THEOREM 3.2.** The simplex method terminates as long as the leaving variable is selected by the lexicographic rule in each iteration.

**PROOF.** In view of the preceding remarks, we need merely prove that no degenerate dictionary will be constructed. (If all dictionaries are nondegenerate, then all iterations are nondegenerate. In that case, cycling cannot occur and the desired conclusion follows from Theorem 3.1.) Thus, we need only consider an arbitrary row

$$x_k = (r_0 + r_1\varepsilon_1 + \dots + r_m\varepsilon_m) - \sum_{j \notin B} d_j x_j \quad (3.6)$$

of an arbitrary dictionary and to prove that at least one of the  $m+1$  numbers  $r_0, r_1, \dots, r_m$  is distinct from zero. (Actually, we shall prove that at least one of the  $m$  numbers  $r_1, r_2, \dots, r_m$  is distinct from zero.) Writing  $d_k = 1$  and  $d_i = 0$  for all basic variables  $x_i$  distinct from  $x_k$ , we record (3.6) as

$$\sum_{j=1}^{n+m} d_j x_j = r_0 + \sum_{i=1}^m r_i \varepsilon_i. \quad (3.7)$$

Since this equation has been obtained by algebraic manipulations from the definitions of the slack variables,

$$x_{n+i} = b_i + \varepsilon_i - \sum_{j=1}^n a_{ij} x_j \quad (i = 1, 2, \dots, m) \quad (3.8)$$

it must hold for all choices of numbers  $x_1, x_2, \dots, x_{n+m}$  and  $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_m$  that satisfy (3.8). Hence, the equation

$$\sum_{j=1}^n d_j x_j + \sum_{i=1}^m d_{n+i} (b_i + \varepsilon_i - \sum_{j=1}^n a_{ij} x_j) = r_0 + \sum_{i=1}^m r_i \varepsilon_i$$

which is obtained by substituting from (3.8) into (3.7), must hold for all choices of numbers  $x_1, x_2, \dots, x_n$  and  $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_m$ . Writing this identity as

$$\sum_{j=1}^n (d_j - \sum_{i=1}^m d_{n+i} a_{ij}) x_j + \sum_{i=1}^m (d_{n+i} - r_i) \varepsilon_i = r_0 - \sum_{i=1}^m d_{n+i} b_i$$

we observe that the coefficient at each  $x_j$ , the coefficient at each  $\varepsilon_i$ , and the right-hand side must equal zero. Thus

$$\begin{aligned} d_{n+i} &= r_i & \text{for all } i &= 1, 2, \dots, m \\ d_j &= \sum_{i=1}^m d_{n+i} a_{ij} & \text{for all } j &= 1, 2, \dots, n. \end{aligned} \quad (3.9)$$

If all the numbers  $r_1, r_2, \dots, r_m$  were equal to zero, then (3.9) would imply  $d_{n+i} = 0$  for all  $i = 1, 2, \dots, m$  and  $d_j = 0$  for all  $j = 1, 2, \dots, n$ , contradicting the fact that  $d_k = 1$ . ■

With the hindsight provided by Theorem 3.2, it becomes easy to prove that every LP problem in the standard form can be perturbed by adding suitable small numbers  $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_m$  to the right-hand sides  $b_1, b_2, \dots, b_m$  in such a way that the simplex method applied to the perturbed problem will terminate. In fact, the numbers  $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_m$  may be chosen as the powers  $\varepsilon, \varepsilon^2, \dots, \varepsilon^m$  of any sufficiently small positive  $\varepsilon$ . We leave the details for problem 3.8.

As we have observed, the terms involving  $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_m$  are needed only when a tie has to be broken between two or more candidates for leaving the basis. Thus we might just as well wait until such a need arises, and only then introduce an ad hoc perturbation. This idea was developed by P. Wolfe (1963); its lexicographic counterpart comes from G. B. Dantzig (1960).

**Smallest-subscript rule.** This term will refer to breaking ties in the choice of the entering and leaving variables by always choosing the candidate  $x_k$  that has the smallest subscript  $k$ . The motivation for this elegant concept is provided by the following result.

**THEOREM 3.3.** [R. G. Bland (1977).] The simplex method terminates as long as the entering and leaving variables are selected by the smallest-subscript rule in each iteration.

**PROOF.** By virtue of Theorem 3.1, we need only show that cycling is impossible when the smallest-subscript rule is used. We shall do this by deriving a contradiction from the assumption that the smallest-subscript rule leads from some dictionary  $D_0$  to itself in a sequence of degenerate iterations. For definiteness, let us say that this sequence of iterations produces dictionaries  $D_1, D_2, \dots, D_k$  such that  $D_k = D_0$ . A variable will be called *fickle* if it is nonbasic in some of these dictionaries and basic in others. Among all the fickle variables, let  $x_i$  have the largest subscript. In the sequence  $D_0, D_1, \dots, D_k$ , there is a dictionary  $D$  with  $x_i$  leaving (basic in  $D$  but nonbasic in the next dictionary), and some other fickle variable  $x_s$  entering (nonbasic in  $D$  but basic in the

next dictionary). Further along in the sequence  $D_0, D_1, \dots, D_k, D_1, D_2, \dots, D_k$ , there must be a dictionary  $D^*$  with  $x_r$  entering. Let us record  $D$  as

$$\begin{aligned} x_i &= b_i - \sum_{j \in B} a_{ij} x_j \quad (i \in B) \\ z &= v + \sum_{j \in B} c_j x_j. \end{aligned}$$

Since all the iterations leading from  $D$  to  $D^*$  are degenerate, the objective function  $z$  must have the same value  $v$  in both dictionaries. Thus, the last row of  $D^*$  may be recorded as

$$z = v + \sum_{j=1}^{n+m} c_j^* x_j$$

with  $c_j^* = 0$  whenever  $x_j$  is basic in  $D^*$ . Since this equation has been obtained from  $D$  by algebraic manipulations, it must be satisfied by every solution of  $D$ . In particular, it must be satisfied by  $x_s = y, x_j = 0$  ( $j \notin B$  but  $j \neq s$ ),  $x_i = b_i - a_{is}y$  ( $i \in B$ ) and  $z = v + c_s y$  for every choice of  $y$ . Thus we have

$$v + c_s y = v + c_s^* y + \sum_{i \in B} c_i^* (b_i - a_{is} y)$$

and, after simplification,

$$(c_s - c_s^* + \sum_{i \in B} c_i^* a_{is}) y = \sum_{i \in B} c_i^* b_i$$

for every choice of  $y$ . Since the right-hand side of the last equation is a constant independent of  $y$ , we conclude that

$$c_s - c_s^* + \sum_{i \in B} c_i^* a_{is} = 0. \quad (3.10)$$

The rest is easy. Since  $x_s$  is entering in  $D$ , we have  $c_s > 0$ . Since  $x_s$  is not entering in  $D^*$  and yet  $s < t$ , we have  $c_s^* \leq 0$ . Hence (3.10) implies that

$$c_r^* a_{rs} < 0 \quad \text{for some } r \in B. \quad (3.11)$$

Since  $r \in B$ , the variable  $x_r$  is basic in  $D$ ; since  $c_r^* \neq 0$ , the same variable is nonbasic in  $D^*$ . Hence,  $x_r$  is fickle and we have  $r \leq t$ . Actually,  $x_r$  is different from  $x_t$ : since  $x_t$  is leaving in  $D$ , we have  $a_{ts} > 0$  and so  $c_r^* a_{ts} > 0$ . Now  $r < t$  and yet  $x_r$  is not entering in  $D^*$ . Thus, we cannot have  $c_r^* > 0$ . From (3.11), we conclude that

$$a_{rs} > 0.$$

Since all the iterations leading from  $D$  to  $D^*$  are degenerate, the two dictionaries describe the same solution. In particular, the value of  $x_r$  is zero in both dictionaries ( $x_r$  is nonbasic in  $D^*$ ) and so  $b_r = 0$ . Hence  $x_r$  was a candidate for leaving the basis of  $D$ —yet we picked  $x_t$ , even though  $r < t$ . This contradiction completes the proof. ■

One further point: termination of the simplex method can be guaranteed even without abiding by the smallest-subscript rule in every single iteration. We might resort to the smallest-subscript rule, for instance, only when the last fifty or so iterations were degenerate, and abandon it after the next nondegenerate iteration in favor of any other way of choosing the entering and leaving variables. Although cycling might conceivably take place in this case, each block of consecutive degenerate iterations would be followed by a nondegenerate iteration, and so each dictionary would be constructed only a finite number of times.

## Initialization

The only remaining point that needs to be explained is getting hold of the initial feasible dictionary in a problem

$$\begin{aligned} \text{maximize} \quad & \sum_{j=1}^n c_j x_j \\ \text{subject to} \quad & \sum_{j=1}^n a_{ij} x_j \leq b_i \quad (i = 1, 2, \dots, m) \\ & x_j \geq 0 \quad (j = 1, 2, \dots, n) \end{aligned}$$

with an infeasible origin. The trouble with an infeasible origin is twofold. First, it may not be clear that our problem has any feasible solutions at all. Second, even if a feasible solution is apparent, a feasible dictionary may not be. One way of getting around both obstacles uses a so-called *auxiliary problem*,

$$\begin{aligned} \text{minimize} \quad & x_0 \\ \text{subject to} \quad & \sum_{j=1}^n a_{ij} x_j - x_0 \leq b_i \quad (i = 1, 2, \dots, m) \\ & x_j \geq 0 \quad (j = 0, 1, \dots, n). \end{aligned}$$

A feasible solution of the auxiliary problem is readily available: it suffices to set the value of each  $x_j$  with  $1 \leq j \leq n$  at zero and make the value of  $x_0$  sufficiently large. Furthermore, it is easy to see that the original problem has a feasible solution *if and only if* the auxiliary problem has a feasible solution with  $x_0 = 0$ . To put it differently, the original problem has a feasible solution if and only if the optimum value of the auxiliary problem is zero. Hence our plan is to solve the auxiliary problem first; the technical details are illustrated on the problem

$$\begin{aligned} \text{maximize} \quad & x_1 - x_2 + x_3 \\ \text{subject to} \quad & 2x_1 - x_2 + 2x_3 \leq 4 \\ & 2x_1 - 3x_2 + x_3 \leq -5 \\ & -x_1 + x_2 - 2x_3 \leq -1 \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

To avoid unnecessary confusion, we write the auxiliary problem in its maximization form:

$$\begin{aligned} \text{maximize} \quad & -x_0 \\ \text{subject to} \quad & 2x_1 - x_2 + 2x_3 - x_0 \leq 4 \\ & 2x_1 - 3x_2 + x_3 - x_0 \leq -5 \\ & -x_1 + x_2 - 2x_3 - x_0 \leq -1 \\ & x_0, x_1, x_2, x_3 \geq 0. \end{aligned}$$

Writing down the formulas defining the slack variables  $x_4, x_5, x_6$  and the objective function  $w$ , we obtain the dictionary

$$\begin{array}{rcl} x_4 & = & 4 - 2x_1 + x_2 - 2x_3 + x_0 \\ x_5 & = & -5 - 2x_1 + 3x_2 - x_3 + x_0 \\ x_6 & = & -1 + x_1 - x_2 + 2x_3 + x_0 \\ w & = & -x_0 \end{array}$$

which is infeasible. Nevertheless, this infeasible dictionary can be transformed into a feasible one by a single pivot, with  $x_0$  entering and  $x_5$  leaving the basis:

$$\begin{array}{rcl} x_0 & = & 5 + 2x_1 - 3x_2 + x_3 + x_5 \\ x_4 & = & 9 - 2x_2 - x_3 + x_5 \\ x_6 & = & 4 + 3x_1 - 4x_2 + 3x_3 + x_5 \\ w & = & -5 - 2x_1 + 3x_2 - x_3 - x_5. \end{array}$$

In general, the auxiliary problem may be written as

$$\begin{array}{ll} \text{maximize} & -x_0 \\ \text{subject to} & \sum_{j=1}^n a_{ij}x_j - x_0 \leq b_i \quad (i = 1, 2, \dots, m) \\ & x_j \geq 0 \quad (j = 0, 1, \dots, n). \end{array}$$

Writing down the formulas defining the slack variables  $x_{n+1}, x_{n+2}, \dots, x_{n+m}$  and the objective function  $w$  gives us the dictionary

$$\begin{array}{rcl} x_{n+i} & = & b_i - \sum_{j=1}^n a_{ij}x_j + x_0 \quad (i = 1, 2, \dots, m) \\ w & = & -x_0 \end{array}$$

which is infeasible. Nevertheless, this infeasible dictionary can be transformed into a feasible one by a single pivot, with  $x_0$  entering and the "most infeasible"  $x_{n+i}$  leaving the basis. More precisely, the leaving variable is that  $x_{n+k}$  whose negative value,  $b_k$ , has the largest magnitude among all the negative numbers  $b_i$ . After pivoting, the variable  $x_0$  assumes the positive value of  $-b_k$ , whereas each basic  $x_{n+i}$  assumes the nonnegative value of  $b_i - b_k$ . Now we are set to solve the auxiliary problem by the simplex method. In our illustrative example, the computations go as follows.

After the first iteration, with  $x_2$  entering and  $x_6$  leaving:

$$\begin{array}{rcl} x_2 & = & 1 + 0.75x_1 + 0.75x_3 + 0.25x_5 - 0.25x_6 \\ x_0 & = & 2 - 0.25x_1 - 1.25x_3 + 0.25x_5 + 0.75x_6 \\ x_4 & = & 7 - 1.5x_1 - 2.5x_3 + 0.5x_5 + 0.5x_6 \\ w & = & -2 + 0.25x_1 + 1.25x_3 - 0.25x_5 - 0.75x_6. \end{array}$$

After the second iteration, with  $x_3$  entering and  $x_0$  leaving:

$$\begin{array}{rcl} x_3 & = & 1.6 - 0.2x_1 + 0.2x_5 + 0.6x_6 - 0.8x_0 \\ x_2 & = & 2.2 + 0.6x_1 + 0.4x_5 + 0.2x_6 - 0.6x_0 \\ x_4 & = & 3 - x_1 - x_6 + 2x_0 \\ w & = & -x_0. \end{array} \quad (3.12)$$

The last dictionary (3.12) is optimal. Since the optimal value of the auxiliary problem is zero, dictionary (3.12) points out a feasible solution of the original problem:  $x_1 = 0, x_2 = 2.2, x_3 = 1.6$ . Furthermore, (3.12) can be easily converted into the desired feasible dictionary of the original problem. To obtain the first three rows of the desired dictionary, we simply copy down the first three rows of (3.12), omitting all the terms involving  $x_0$ :

$$\begin{array}{rcl} x_3 & = & 1.6 - 0.2x_1 + 0.2x_5 + 0.6x_6 \\ x_2 & = & 2.2 + 0.6x_1 + 0.4x_5 + 0.2x_6 \\ x_4 & = & 3 - x_1 - x_6. \end{array} \quad (3.13)$$

To obtain the last row, we have to express the original objective function

$$z = x_1 - x_2 + x_3 \quad (3.14)$$

in terms of the nonbasic variables  $x_1, x_5, x_6$ . For this purpose, we simply substitute from (3.13) into (3.14), obtaining

$$\begin{aligned} z &= x_1 - (2.2 + 0.6x_1 + 0.4x_5 + 0.2x_6) + (1.6 - 0.2x_1 + 0.2x_5 + 0.6x_6) \\ &= -0.6 + 0.2x_1 - 0.2x_5 + 0.4x_6. \end{aligned}$$

In short, the desired dictionary reads

$$\begin{array}{rcl} x_3 & = & 1.6 - 0.2x_1 + 0.2x_5 + 0.6x_6 \\ x_2 & = & 2.2 + 0.6x_1 + 0.4x_5 + 0.2x_6 \\ x_4 & = & 3 - x_1 - x_6 \\ z & = & -0.6 + 0.2x_1 - 0.2x_5 + 0.4x_6. \end{array}$$

Clearly, the same procedure will transform an optimal dictionary of the auxiliary problem into a feasible dictionary of the original problem whenever  $x_0$  is nonbasic in the former.

Now, let us review the general situation. We have learned how to construct the auxiliary problem and its first feasible dictionary. In the process of solving the auxiliary problem, we may encounter a dictionary where  $x_0$  competes with other variables for leaving the basis. If and when that happens, it is only natural to choose  $x_0$  as the actual leaving variable; immediately after pivoting, we obtain a dictionary where

$$x_0 \text{ is nonbasic, and so the value of } w \text{ is zero.} \quad (3.15)$$

Clearly, a feasible dictionary with this property is optimal. However, we may also reach the optimum of the auxiliary problem while  $x_0$  is still basic. Thus, we may obtain an optimal dictionary where

$$x_0 \text{ is basic and the value of } w \text{ is nonzero} \quad (3.16)$$

or, conceivably, an optimal dictionary where

$$x_0 \text{ is basic and the value of } w \text{ is zero.} \quad (3.17)$$

Let us examine case (3.17). Since the next-to-last dictionary was not yet optimal, the value of  $w = -x_0$  must have changed from some negative level to zero in the last iteration. To put it differently, the value of the basic variable  $x_0$  must have dropped from some positive level to zero in the last iteration. But then  $x_0$  was a candidate for leaving the basis; yet, contrary to our policy, we did not pick it. This contradiction shows that (3.17) cannot occur. Hence the optimal dictionary of the auxiliary problem has either property (3.15) or property (3.16). In the former case, we construct a feasible dictionary of the original problem as illustrated previously and proceed to solve the original problem by the simplex method; in the latter case, we simply conclude that the original problem is infeasible.

This strategy is known as the *two-phase simplex method*. In the *first phase*, we set up and solve the auxiliary problem; if the optimal dictionary turns out to have property (3.15) then we proceed to the *second phase*, solving the original problem itself. We shall return to the two-phase simplex method in Chapter 8.

## THE FUNDAMENTAL THEOREM OF LINEAR PROGRAMMING

This name is given to the following result.

**THEOREM 3.4.** Every LP problem in the standard form has the following three properties:

- (i) If it has no optimal solution, then it is either infeasible or unbounded.
- (ii) If it has a feasible solution, then it has a basic feasible solution.
- (iii) If it has an optimal solution, then it has a basic optimal solution.

**PROOF.** The first phase of the two-phase simplex method either discovers that the problem is infeasible or else it delivers a basic feasible solution. The second phase of the two-phase simplex method either discovers that the problem is unbounded or else it delivers a basic optimal solution. ■

Note that the first property is not shared by problems whose constraints may include *strict* linear inequalities  $\sum a_j x_j < b$ . To take a trivial example, the problem

$$\text{maximize } x \quad \text{subject to } x < 0$$

is neither infeasible nor unbounded and yet it has no optimal solution. The remaining two properties (ii) and (iii) tell us that, when looking for feasible or optimal solutions of an LP problem in the standard form, we may confine our search to a *finite* set. These two properties, easy to establish from scratch, are often used to motivate the simplex method. Our exposition has followed the reverse pattern, with an emphasis placed on actually solving the problem—and the fundamental theorem of linear programming obtained as an effortless afterthought. □

## PROBLEMS

$$\begin{aligned} \triangle 3.1 \quad & \text{Maximize} && x_1 + 3x_2 - x_3 \\ & \text{subject to} && 2x_1 + 2x_2 - x_3 \leq 10 \\ & && 3x_1 - 2x_2 + x_3 \leq 10 \\ & && x_1 - 3x_2 + x_3 \leq 10 \\ & && x_1, x_2, x_3 \geq 0. \end{aligned}$$

- 3.2 In the tableau format, a natural tie-breaking rule for the choice of the pivot row favors the rows that appear higher up in the tableau. Show that in the following example (constructed by H. W. Kuhn), this tie-breaking rule leads to cycling:

$$\begin{aligned} & \text{maximize} && 2x_1 + 3x_2 - x_3 - 12x_4 \\ & \text{subject to} && -2x_1 - 9x_2 + x_3 + 9x_4 \leq 0 \\ & && \frac{1}{3}x_1 + x_2 - \frac{1}{3}x_3 - 2x_4 \leq 0 \\ & && x_1, x_2, x_3, x_4 \geq 0. \end{aligned}$$

- 3.3 Solve problem 3.2 by the perturbation technique.
- 3.4 Arrange the following expressions in a sequence from lexicographically smallest to lexicographically largest:

$$\begin{aligned} & 3 - \varepsilon_1 \\ & 3 \\ & 2 + 10\varepsilon_1 \\ & 3 - 4\varepsilon_1 + \varepsilon_2 \\ & \varepsilon_2 + 3\varepsilon_3 \\ & 3 + 4\varepsilon_1 + \varepsilon_3 \\ & 3 - 4\varepsilon_1 + \varepsilon_2 + \varepsilon_3. \end{aligned}$$

- 3.5 Prove: If  $r = r_0 + r_1\varepsilon_1 + \cdots + r_m\varepsilon_m$  is lexicographically smaller than  $s = s_0 + s_1\varepsilon_1 + \cdots + s_m\varepsilon_m$  and if  $s$  is lexicographically smaller than  $t = t_0 + t_1\varepsilon_1 + \cdots + t_m\varepsilon_m$ , then  $r$  is lexicographically smaller than  $t$ .
- 3.6 Use the result of problem 3.5 to prove that, in every finite set of distinct expressions, such as  $r$  and  $s$  in (3.5), there is an expression that is lexicographically smaller than all the others.

- 3.7 Prove that for every pair of expressions in (3.5) there is a positive number  $\delta$  such that the following two statements are equivalent: (i)  $r$  is lexicographically smaller than  $s$ ; (ii) for every choice of numbers  $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_m$  such that
- $$0 < \varepsilon_1 < \delta \quad \text{and} \quad 0 < \varepsilon_i < \delta \varepsilon_{i-1} \quad \text{for all } i = 2, 3, \dots, m$$
- $r$  is numerically smaller than  $s$ .

- 3.8 Use Theorem 3.2 and the result of problem 3.7 to prove the following. For every LP problem

$$\begin{aligned} &\text{maximize} && \sum_{j=1}^n c_j x_j \\ &\text{subject to} && \sum_{j=1}^n a_{ij} x_j \leq b_i \quad (i = 1, 2, \dots, m) \\ &&& x_j \geq 0 \quad (j = 1, 2, \dots, n) \end{aligned}$$

there is a positive number  $\delta$  such that the simplex method used to

$$\begin{aligned} &\text{maximize} && \sum_{j=1}^n c_j x_j \\ &\text{subject to} && \sum_{j=1}^n a_{ij} x_j \leq b_i + \varepsilon^i \quad (i = 1, 2, \dots, m) \\ &&& x_j \geq 0 \quad (j = 1, 2, \dots, n) \end{aligned}$$

terminates whenever  $0 < \varepsilon < \delta$ .

- △ 3.9 Solve the following problems by the two-phase simplex method:

- a. maximize  $3x_1 + x_2$   
subject to  $x_1 - x_2 \leq -1$   
 $-x_1 - x_2 \leq -3$   
 $2x_1 + x_2 \leq 4$   
 $x_1, x_2 \geq 0$
- b. maximize  $3x_1 + x_2$   
subject to  $x_1 - x_2 \leq -1$   
 $-x_1 - x_2 \leq -3$   
 $2x_1 + x_2 \leq 2$   
 $x_1, x_2 \geq 0$
- c. maximize  $3x_1 + x_2$   
subject to  $x_1 - x_2 \leq -1$   
 $-x_1 - x_2 \leq -3$   
 $2x_1 - x_2 \leq 2$   
 $x_1, x_2 \geq 0$ .

- 3.10 Prove or disprove: A feasible dictionary whose last row reads  $z = z^* + \sum \bar{c}_j x_j$  describes an optimal solution if and only if  $\bar{c}_j \leq 0$  for all  $j$ .

## How Fast Is the Simplex Method?

The subject of this chapter is the number of iterations in the simplex method. We shall also comment on the distinction between theoretically satisfactory and practically satisfactory algorithms, with a particular regard to linear programming.

### TYPICAL NUMBER OF ITERATIONS

For practical problems of the form

$$\begin{aligned} &\text{maximize} && \sum_{j=1}^n c_j x_j \\ &\text{subject to} && \sum_{j=1}^n a_{ij} x_j \leq b_i \quad (i = 1, 2, \dots, m) \\ &&& x_j \geq 0 \quad (j = 1, 2, \dots, n) \end{aligned} \tag{4.1}$$

with  $m < 50$  and  $m + n < 200$ , Dantzig (1963, p. 160) reported the number of iterations as being usually less than  $3m/2$  and only rarely going to  $3m$ . This observation agrees with empirical findings obtained more recently for much larger problems: the

typical number of iterations increases proportionally to  $m$  (with the proportionality constant in the range suggested by Dantzig) and only very slowly with  $n$ . (It is sometimes said that, for a fixed  $m$ , the typical number of iterations is proportional to the logarithm of  $n$ .) Theoretical explanations of this phenomenon were proposed by G. B. Dantzig (1980), K.-H. Borgwardt (1982) and S. Smale (1982). It is this remarkable efficiency of the simplex method that accounts for its staggering success. At the current level of computer technology, typical problems with about 100 constraints and variables are solved in a few seconds; even problems with several thousands of constraints can be handled successfully. (To attain this level of efficiency, the simplex method has to be implemented properly, so that the time *per iteration* is reduced as much as possible. Consequently, the format of dictionaries has to be abandoned in favor of less time-consuming ways of organizing the necessary computations. We shall begin to study this matter in Chapter 7.) For problems with some particular structure amenable to specialized versions of the simplex method (such as the network simplex method of Chapter 19 or generalized upper bounding of Chapter 25), this limit can be pushed even further.

Monte Carlo simulation studies of the number of iterations were pioneered by H. W. Kuhn and R. E. Quandt (1963), who solved a number of problems (4.1) with  $c_j = 1$  for all  $j$ ,  $b_i = 10,000$  for all  $i$ , and each  $a_{ij}$  selected at random from the set of positive integers between 1 and 1,000. A small part of these experiments has been reproduced, on a slightly larger scale, with the results exhibited in Table 4.1. (Each entry in the table represents the average number of iterations over 100 problems.) In each simplex iteration, the entering variable was that nonbasic variable that had the largest coefficient in the  $z$ -row of the dictionary. We shall refer to this selection rule as the *largest-coefficient rule*.

**TABLE 4.1 Average Number of Iterations Required by the Largest-Coefficient Rule**

| $m \backslash n$ | 10   | 20   | 30   | 40   | 50   |
|------------------|------|------|------|------|------|
| 10               | 9.40 | 14.2 | 17.4 | 19.4 | 20.2 |
| 20               |      | 25.2 | 30.7 | 38.0 | 41.5 |
| 30               |      |      | 44.4 | 52.7 | 62.9 |
| 40               |      |      |      | 67.6 | 78.7 |
| 50               |      |      |      |      | 95.2 |

Source: D. Avis and V. Chvátal (1978).

The production management problems solved in practice are very much different from such randomly generated examples. Typically, most of their coefficients  $a_{ij}$  are zeros, the remaining nonzero coefficients occur in clusters that are very far from random, and the range of distinct numerical values of the coefficients is often very small. In spite of these differences, the Monte Carlo simulation results are in striking agreement with the empirical observations quoted above: for instance, if  $n = 50$ , then the average number of iterations is about  $2m$ .

## PROBLEMS REQUIRING AN UNUSUALLY LARGE NUMBER OF ITERATIONS

From a purist point of view, it would be even more reassuring to have a proof that, for *every* problem (4.1), the simplex method would require no more than, say,  $10mn$  iterations to find an optimal solution. However, there is no such proof. Worse than that, there are examples of LP problems that make the simplex method go through an enormous number of iterations. V. Klee and G. J. Minty (1972) have shown that in the process of solving the problem

$$\begin{aligned}
 &\text{maximize} && \sum_{j=1}^n 10^{n-j} x_j \\
 &\text{subject to} && \left( 2 \sum_{j=1}^{i-1} 10^{i-j} x_j \right) + x_i \leq 100^{i-1} \quad (i = 1, 2, \dots, n) \\
 &&& x_j \geq 0 \quad (j = 1, 2, \dots, n)
 \end{aligned} \tag{4.2}$$

the simplex method goes through  $2^n - 1$  iterations. (A proof is outlined in problems 4.2 and 4.3.) This number is quite frightening. For example, at the rate of 100 iterations per second (a reasonably generous estimate), problem (4.2) with  $n = 50$  would take more than 300,000 years to solve! (The empirical and simulation results just quoted do *not* contradict this result. They simply suggest that problems requiring large numbers of iterations must be rare. For this reason, the Klee–Minty examples (4.2) and other similar examples are sometimes referred to as “pathological.”)

As our starting point for further discussion, we choose the Klee–Minty problem with  $n = 3$ ,

$$\begin{aligned}
 &\text{maximize} && 100x_1 + 10x_2 + x_3 \\
 &\text{subject to} && x_1 \leq 1 \\
 &&& 20x_1 + x_2 \leq 100 \\
 &&& 200x_1 + 20x_2 + x_3 \leq 10,000 \\
 &&& x_1, x_2, x_3 \geq 0.
 \end{aligned} \tag{4.3}$$

Using the largest-coefficient rule, we construct the following sequence of dictionaries. The initial dictionary:

$$\begin{array}{rcl} x_4 & = & 1 - x_1 \\ x_5 & = & 100 - 20x_1 - x_2 \\ x_6 & = & 10,000 - 200x_1 - 20x_2 - x_3 \\ \hline z & = & 100x_1 + 10x_2 + x_3. \end{array}$$

After the first iteration:

$$\begin{array}{rcl} x_1 & = & 1 - x_4 \\ x_5 & = & 80 + 20x_4 - x_2 \\ x_6 & = & 9,800 + 200x_4 - 20x_2 - x_3 \\ \hline z & = & 100 - 100x_4 + 10x_2 + x_3. \end{array}$$

After the second iteration:

$$\begin{array}{rcl} x_1 & = & 1 - x_4 \\ x_2 & = & 80 + 20x_4 - x_5 \\ x_6 & = & 8,200 - 200x_4 + 20x_5 - x_3 \\ \hline z & = & 900 + 100x_4 - 10x_5 + x_3. \end{array}$$

After the third iteration:

$$\begin{array}{rcl} x_4 & = & 1 - x_1 \\ x_2 & = & 100 - 20x_1 - x_5 \\ x_6 & = & 8,000 + 200x_1 + 20x_5 - x_3 \\ \hline z & = & 1,000 - 100x_1 - 10x_5 + x_3. \end{array}$$

After the fourth iteration:

$$\begin{array}{rcl} x_4 & = & 1 - x_1 \\ x_2 & = & 100 - 20x_1 - x_5 \\ x_3 & = & 8,000 + 200x_1 + 20x_5 - x_6 \\ \hline z & = & 9,000 + 100x_1 + 10x_5 - x_6. \end{array}$$

After the fifth iteration:

$$\begin{array}{rcl} x_1 & = & 1 - x_4 \\ x_2 & = & 80 + 20x_4 - x_5 \\ x_3 & = & 8,200 - 200x_4 + 20x_5 - x_6 \\ \hline z & = & 9,100 - 100x_4 + 10x_5 - x_6. \end{array}$$

After the sixth iteration:

$$\begin{array}{rcl} x_1 & = & 1 - x_4 \\ x_5 & = & 80 + 20x_4 - x_2 \\ x_3 & = & 9,800 + 200x_4 - 20x_2 - x_6 \\ \hline z & = & 9,900 + 100x_4 - 10x_2 - x_6. \end{array}$$

After the seventh iteration:

$$\begin{array}{rcl} x_4 & = & 1 - x_1 \\ x_5 & = & 100 - 20x_1 - x_2 \\ x_3 & = & 10,000 - 200x_1 - 20x_2 - x_6 \\ \hline z & = & 10,000 - 100x_1 - 10x_2 - x_6. \end{array}$$

In the first iteration, we were led to an unfortunate choice of the entering variable: had we made  $x_3$  rather than  $x_1$  enter the basis, we would have pivoted directly to the final dictionary. In view of this blunder, it is natural to question the expediency of the largest-coefficient rule: perhaps the simplex method would *always* go through only a small number of iterations if it were directed by some other rule. In fact, the largest-coefficient rule is not quite natural. More specifically, it ranks the potential candidates for entering the basis according to their coefficients in the last row of the dictionary: variables with larger coefficients appear to be more promising. But appearances are misleading and the ranking order is easily upset by changes in the scale on which each candidate is measured. For instance, the substitution

$$\bar{x}_1 = x_1, \quad \bar{x}_2 = 0.01x_2, \quad \bar{x}_3 = 0.0001x_3$$

converts the Klee-Minty problem (4.3) into the form

$$\begin{array}{lll} \text{maximize} & 100\bar{x}_1 + 1,000\bar{x}_2 + 10,000\bar{x}_3 & \\ \text{subject to} & \bar{x}_1 & \leq 1 \\ & 20\bar{x}_1 + 100\bar{x}_2 & \leq 100 \\ & 200\bar{x}_1 + 2,000\bar{x}_2 + 10,000\bar{x}_3 & \leq 10,000 \\ & \bar{x}_1, \bar{x}_2, \bar{x}_3 & \geq 0. \end{array}$$

In the first dictionary associated with this new version of (4.3), the nonbasic variable  $\bar{x}_3$  appears most attractive, and so the simplex method reaches the optimal solution in only one iteration.

## ALTERNATIVE PIVOTING RULES

Thus we are led to ranking the candidates  $x_j$  for entering the basis according to criteria that are independent of changes of scale. One criterion of this kind is the increase in the objective function obtained when  $x_j$  actually enters the basis. The



resulting rule (always choose that candidate whose entrance into the basis brings about the largest increase in the objective function) is referred to as the *largest-increase* rule. On the Klee–Minty examples (4.2), the largest-increase rule leads the simplex method to the optimal solution in only one iteration, as opposed to the  $2^n - 1$  iterations required by the previously used largest-coefficient rule. However, the new rule does not always lead to a small number of iterations: R. G. Jeroslow (1973) constructed LP problems that are to the largest-increase rule what the Klee–Minty problems are to the largest-coefficient rule. (More precisely, the number of iterations required by the largest-increase rule grows exponentially with  $m$  and  $n$ .) Again, these examples exploit the myopia inherent in the simplex method. It is conceivable that every easily implemented rule for choosing the entering variable can be tricked in a similar way into requiring very large numbers of iterations.

Which of the two rules is better? On problems arising from applications, the number of iterations required by the largest increase is *usually* smaller than the number of iterations required by the largest coefficient. Simulation experiments lead to a similar outcome (see Table 4.2).

**Table 4.2 Average Numbers of Iterations  
Required by the Largest-Increase Rule**

| $m \backslash n$ | 10   | 20   | 30   | 40   | 50   |
|------------------|------|------|------|------|------|
| 10               | 7.02 | 9.17 | 10.8 | 12.1 | 12.6 |
| 20               |      | 16.2 | 20.2 | 24.2 | 27.3 |
| 30               |      |      | 28.7 | 34.5 | 39.4 |
| 40               |      |      |      | 43.3 | 39.9 |
| 50               |      |      |      |      | 58.9 |

Source: D. Avis and V. Chvátal (1978).

Nevertheless, as the largest-coefficient rule takes less time to execute than the largest increase, it is the former that usually wins in terms of total computing time. More generally, the number of iterations is a poor criterion for assessing the efficiency of a rule for choosing the entering variable. It is the total computing time that counts, and rules that tend to reduce the number of iterations often take too much time to execute. In this light, even the largest-coefficient rule is found too time-consuming and therefore rarely, if ever, used in practice. The choice of entering variables in efficient implementations of the simplex method is influenced by the logistics of handling large problems on a computer; this matter will be studied in Chapter 7.

A systematic rule that always leads to an unambiguous choice of the entering variable, and to an unambiguous choice of the leaving variable in case of a tie, is called a *pivoting rule*. The largest-coefficient rule and the largest-increase rule, amended by unambiguous instructions for tie-breaking, are two examples of pivoting rules; the smallest-subscript rule of Chapter 3 is another.

## EFFICIENCY OF ALGORITHMS IN THEORY AND PRACTICE

As noted in Chapter 1, the theoretical and the practical criteria for judging the efficiency of algorithms are radically different. From the theoretical point of view, an algorithm is satisfactory if its running time increases only slowly with the size of the problem. This is a vague definition as it stands; we are going to make it precise.

Let us consider a fixed class of problems (such as linear programming problems in the standard form) and a fixed algorithm (such as the simplex method) for solving problems in this class. A fair interpretation of the “size of the problem” is the time required to transmit the data. To put it differently, the size of a problem is the number of times you have to hit the keyboard of your typewriter in order to write down the data. For instance, the size of the Klee–Minty problems (4.2) is roughly  $n^3/3$  when  $n$  gets very large (each of the  $i - j + 1$  digits in each coefficient  $2 \cdot 10^{i-j}$  has to be written down). A fair interpretation of “running time” is the total number of elementary steps (such as adding up, multiplying, or comparing two one-digit numbers; executing a “go to” instruction in a computer program; and so on) that have to be executed. (Thus it is implicitly assumed that each elementary step requires one unit of time.) Now for each  $s$ , there may be many (but only finitely many) different problems of size  $s$  in our class, and our algorithm may require different amounts of time  $t_1, t_2, \dots, t_M$  for different problems  $P_1, P_2, \dots, P_M$  of this size. Only the largest of these numbers  $t_i$  matters in the theoretical context. Of course, this largest  $t_i$  depends on  $s$ ; we shall denote it by  $t(s)$ . Thus, our algorithm solves every problem of size  $s$  within  $t(s)$  units of time and actually uses up these  $t(s)$  units of time in the worst case. Finally, the algorithm is considered satisfactory if  $t(s)$  grows only slowly with  $s$ . More precisely, the algorithm is satisfactory if there is a polynomial  $p$  such that  $t(s) \leq p(s)$  for all  $s$ .

This definition, proposed by J. Edmonds (1965), is one of the most fruitful and stimulating concepts in theoretical computer science. [Those wishing for more information on this subject are referred to Garey and Johnson (1979).] Nevertheless, even though this concept does reflect to some extent the reasons why practitioners are satisfied by some algorithms and unsatisfied by others, it fails to capture these reasons fully. Two of the features that make it unrealistic from a practical point of view are:

- (i) The worst-case criterion.
- (ii) The asymptotic point of view.

The inadequacy of the worst-case criterion is demonstrated most dramatically on the case of the simplex method itself: even eminently useful algorithms may be labeled unsatisfactory on the basis of a few isolated examples of a kind that might never come up in practice. The average running time ( $\sum t_i/M$ ) might provide a more realistic criterion than the worst running time ( $\max t_i$ ); unfortunately, a rigorous analysis of the average performance is often much more



difficult than an analysis of the worst performance. The inadequacy of the second feature may manifest itself even when the running time depends only on the size of the problem, so that the average performance and the worst performance coincide. The point is that the actual values  $t(s)$ , with  $s$  restricted to a finite range, do not matter at all; the only thing that counts is the rate of growth of  $t(s)$  as  $s$  increases beyond every bound. Thus, a hypothetical algorithm with a running time  $t = 10^{s/1,000,000,000}$  (rounded up to the nearest integer) would be found theoretically unsatisfactory even though  $t(s) \leq 10$  whenever  $s \leq 10^9$ ; on the other hand, an algorithm with a running time  $t(s) = 10^{1,000}s$  would be found theoretically satisfactory even though  $t(s) \geq 10^{1,000}$  for all  $s$ . Theorists judge algorithms by their worst performance on problems of sizes outside the range of practical interest, whereas practitioners judge algorithms by their typical performance on problems whose sizes are limited to a finite range. (In all fairness, it should be admitted that the theoretical definition is not all that bad. As it turns out, the polynomials bounding the running time of theoretically satisfactory algorithms often assume reasonably small values for reasonably small values of  $s$  and, on the other hand, even the typical running time of theoretically unsatisfactory algorithms will often get out of hand already for small values of  $s$ .)

For many years, while practitioners were trying to reduce the typical running time of the simplex method by yet another 10% or 20%, theorists were trying to answer a fundamental question: Is there a theoretically satisfactory algorithm for solving linear programming problems? Eventually, L. G. Khachian (1979) provided an affirmative answer by presenting such an algorithm. This "ellipsoid method" is surprisingly simple and elegant; we shall describe its details in the appendix. Will this beautiful gem of pure mathematics ever become a serious challenger of the simplex method's supremacy in solving practical LP problems? That remains to be seen; at the time of this writing, it seems very likely that the answer is no.

## PROBLEMS

△ 4.1 Compare the performance of the three pivoting rules discussed in this chapter on the following examples:

- a. maximize  $4x_1 + 5x_2$   
subject to  $2x_1 + x_2 \leq 9$   
 $x_1 \leq 4$   
 $x_2 \leq 3$   
 $x_1, x_2 \geq 0$
- b. maximize  $2x_1 + x_2$   
subject to  $3x_1 + x_2 \leq 3$   
 $x_1, x_2 \geq 0$
- c. maximize  $3x_1 + 5x_2$   
subject to  $x_1 + 2x_2 \leq 5$   
 $x_1 \leq 3$   
 $x_2 \leq 2$   
 $x_1, x_2 \geq 0$

4.2 In the Klee–Minty problem (4.2), denote the slack variables by  $s_1, s_2, \dots, s_n$  rather than by  $x_{n+1}, x_{n+2}, \dots, x_{2n}$ . Prove that in every feasible dictionary, precisely one of the two variables  $x_i, s_i$  is basic.

4.3 Use the result of problem 4.2 and induction on  $n$  to prove that, when the simplex method with the largest coefficient rule is applied to (4.2), the resulting dictionaries have the following properties:

(i) After  $2^{n-1} - 1$  iterations, the last row reads

$$z = 10 \left( 100^{n-2} - \sum_{j=1}^{n-2} 10^{n-1-j} x_j - s_{n-1} \right) + x_n.$$

(ii) After  $2^{n-1}$  iterations, the last row reads

$$z = 90 \cdot 100^{n-2} + 10 \left( \sum_{j=1}^{n-2} 10^{n-1-j} x_j + s_{n-1} \right) - s_n.$$

(iii) After  $2^n - 1$  iterations, the last row reads

$$z = 100^{n-1} - \sum_{j=1}^{n-1} 10^{n-j} x_j - s_n.$$

(iv) After each iteration, all the coefficients in the last row are integers.

## The Duality Theorem

Every maximization LP problem in the standard form gives rise to a minimization LP problem called the dual problem. The two problems are linked in an interesting way. Every feasible solution in one yields a bound on the optimal value of the other. In fact, if one of the two problems has an optimal solution, then so does the other, and the two optimal values coincide. This fact, known as the Duality Theorem, is the subject of the present chapter. We shall also note that, in managerial applications, the variables featured in the dual problem can be interpreted in a very useful way.

### MOTIVATION: FINDING UPPER BOUNDS ON THE OPTIMAL VALUE

We shall begin this chapter with the following LP problem:

$$\begin{array}{ll} \text{maximize} & 4x_1 + x_2 + 5x_3 + 3x_4 \\ \text{subject to} & x_1 - x_2 - x_3 + 3x_4 \leq 1 \\ & 5x_1 + x_2 + 3x_3 + 8x_4 \leq 55 \\ & -x_1 + 2x_2 + 3x_3 - 5x_4 \leq 3 \\ & x_1, x_2, x_3, x_4 \geq 0. \end{array}$$

Rather than *solving* it, we shall try to get a quick *estimate* of the optimal value  $z^*$  of its objective function. To get a reasonably good lower bound on  $z^*$ , we need only come up with a reasonably good feasible solution. For example, the bound  $z^* \geq 5$  comes from considering the feasible solution  $(0, 0, 1, 0)$ . The feasible solution  $(2, 1, 1, \frac{1}{3})$  shows that  $z^* \geq 15$ . Better yet, the feasible solution  $(3, 0, 2, 0)$  yields  $z^* \geq 22$ . Needless to say, such guesswork is vastly inferior to the systematic attack by the simplex method: even if we were lucky enough to hit on the optimal solution, our guess would provide no *proof* that the solution is indeed optimal.

We shall not pursue this line any further: the subject of this chapter stems from a similar quest for *upper* bounds on  $z^*$ . For example, a glance at the data suggests that  $z^* \leq \frac{275}{3}$ . Indeed, multiplying the second constraint by  $\frac{5}{3}$  we obtain the inequality

$$\frac{25}{3}x_1 + \frac{5}{3}x_2 + 5x_3 + \frac{40}{3}x_4 \leq \frac{275}{3}.$$

Hence every feasible solution  $(x_1, x_2, x_3, x_4)$  satisfies the inequality

$$4x_1 + x_2 + 5x_3 + 3x_4 \leq \frac{25}{3}x_1 + \frac{5}{3}x_2 + 5x_3 + \frac{40}{3}x_4 \leq \frac{275}{3}.$$

In particular, this inequality holds for the optimal solution and so  $z^* \leq \frac{275}{3}$ . With a little inspiration, we can improve this bound considerably. For instance, the sum of the second and third constraints reads

$$4x_1 + 3x_2 + 6x_3 + 3x_4 \leq 58.$$

Therefore,  $z^* \leq 58$ . Rather than searching for further improvements in a haphazard way, we shall now describe the strategy in precise and general terms.

We construct *linear combinations* of the constraints. That is, we multiply the first constraint by some number  $y_1$ , the second by  $y_2$ , the third by  $y_3$ , and then we add them up. (In the first case, we had  $y_1 = 0, y_2 = \frac{5}{3}, y_3 = 0$ ; in the second case, we had  $y_1 = 0, y_2 = y_3 = 1$ .) The resulting inequality reads

$$\begin{aligned} (y_1 + 5y_2 - y_3)x_1 + (-y_1 + y_2 + 2y_3)x_2 + (-y_1 + 3y_2 + 3y_3)x_3 + (3y_1 + 8y_2 - 5y_3)x_4 \\ \leq y_1 + 55y_2 + 3y_3. \end{aligned} \quad (5.1)$$

Of course, each of the three multipliers  $y_i$  must be nonnegative: otherwise the corresponding inequality would reverse its direction. Next, we want to use the left-hand side of (5.1) as an upper bound on  $z = 4x_1 + x_2 + 5x_3 + 3x_4$ . This can be justified only if in (5.1), the coefficient at each  $x_j$  is at least as big as the corresponding coefficient in  $z$ . More explicitly, we want

$$\begin{aligned} y_1 + 5y_2 - y_3 &\geq 4 \\ -y_1 + y_2 + 2y_3 &\geq 1 \\ -y_1 + 3y_2 + 3y_3 &\geq 5 \\ 3y_1 + 8y_2 - 5y_3 &\geq 3. \end{aligned}$$

If the multipliers  $y_i$  are nonnegative and if they satisfy these four inequalities, then we may safely conclude that every feasible solution  $(x_1, x_2, x_3, x_4)$  satisfies the inequality

$$4x_1 + x_2 + 5x_3 + 3x_4 \leq y_1 + 55y_2 + 3y_3.$$

In particular, this inequality is satisfied by the optimal solution; therefore

$$z^* \leq y_1 + 55y_2 + 3y_3.$$

Of course, we want as small an upper bound on  $z^*$  as we can possibly get. Thus, we are led to the following LP problem:

$$\begin{array}{ll} \text{minimize} & y_1 + 55y_2 + 3y_3 \\ \text{subject to} & y_1 + 5y_2 - y_3 \geq 4 \\ & -y_1 + y_2 + 2y_3 \geq 1 \\ & -y_1 + 3y_2 + 3y_3 \geq 5 \\ & 3y_1 + 8y_2 - 5y_3 \geq 3 \\ & y_1, y_2, y_3 \geq 0. \end{array}$$

## THE DUAL PROBLEM

This problem is called the *dual* of the original one; the original problem is called the *primal* problem. In general, the dual of the problem

$$\begin{array}{ll} \text{maximize} & \sum_{j=1}^n c_j x_j \\ \text{subject to} & \sum_{j=1}^n a_{ij} x_j \leq b_i \quad (i = 1, 2, \dots, m) \\ & x_j \geq 0 \quad (j = 1, 2, \dots, n) \end{array} \quad (5.2)$$

is defined to be the problem

$$\begin{array}{ll} \text{minimize} & \sum_{i=1}^m b_i y_i \\ \text{subject to} & \sum_{i=1}^m a_{ij} y_i \geq c_j \quad (j = 1, 2, \dots, n) \\ & y_i \geq 0 \quad (i = 1, 2, \dots, m). \end{array} \quad (5.3)$$

(Note that the dual of a maximization problem is a minimization problem. Furthermore, the  $m$  primal constraints  $\sum a_{ij} x_j \leq b_i$  are in a one-to-one correspondence with the  $m$  dual variables  $y_i$ ; conversely, the  $n$  dual constraints  $\sum a_{ij} y_i \geq c_j$  are in a one-

to-one correspondence with the  $n$  primal variables  $x_j$ . The coefficient at each variable in the objective function, primal or dual, appears in the other problem as the right-hand side of the corresponding constraint.)

As in our example, every feasible solution of the dual yields an upper bound on the optimal value of the primal. More explicitly, for every primal feasible solution  $(x_1, x_2, \dots, x_n)$  and for every dual feasible solution  $(y_1, y_2, \dots, y_m)$  we have

$$\sum_{j=1}^n c_j x_j \leq \sum_{i=1}^m b_i y_i. \quad (5.4)$$

The proof of (5.4), which was illustrated at the beginning of this section, can be written down succinctly as

$$\sum_{j=1}^n c_j x_j \leq \sum_{j=1}^n \left( \sum_{i=1}^m a_{ij} y_i \right) x_j = \sum_{i=1}^m \left( \sum_{j=1}^n a_{ij} x_j \right) y_i \leq \sum_{i=1}^m b_i y_i.$$

Inequality (5.4) is extremely useful: if we happen to stumble across a primal feasible solution  $(x_1^*, x_2^*, \dots, x_n^*)$  and a dual feasible solution  $(y_1^*, y_2^*, \dots, y_m^*)$  such that

$$\sum_{j=1}^n c_j x_j^* = \sum_{i=1}^m b_i y_i^*$$

then we may conclude that both of these solutions are optimal. Indeed, (5.4) implies that every primal feasible solution  $(x_1, x_2, \dots, x_n)$  satisfies

$$\sum_{j=1}^n c_j x_j \leq \sum_{i=1}^m b_i y_i^* = \sum_{j=1}^n c_j x_j^*$$

and that every dual feasible solution  $(y_1, y_2, \dots, y_m)$  satisfies

$$\sum_{i=1}^m b_i y_i \geq \sum_{j=1}^n c_j x_j^* = \sum_{i=1}^m b_i y_i^*.$$

For instance, we have an easy way of showing that the primal feasible solution  $x_1 = 0, x_2 = 14, x_3 = 0, x_4 = 5$  of our original example is optimal: just consider the dual feasible solution  $y_1 = 11, y_2 = 0, y_3 = 6$ . It is not at all obvious, however, that an analogous proof of optimality can be given for *every* LP problem that has an optimal solution; this fact is the central theorem of linear programming.

## THE DUALITY THEOREM AND ITS PROOF

The explicit version of the theorem comes from D. Gale, H. W. Kuhn, and A. W. Tucker (1951); its notions originated in conversations between G. B. Dantzig and J. von Neumann in the fall of 1947.

**THEOREM 5.1 (The Duality Theorem).** If the primal (5.2) has an optimal solution  $(x_1^*, x_2^*, \dots, x_n^*)$ , then the dual (5.3) has an optimal solution  $(y_1^*, y_2^*, \dots, y_m^*)$  such that

$$\sum_{j=1}^n c_j x_j^* = \sum_{i=1}^m b_i y_i^*. \quad (5.5)$$

Before presenting the proof, let us briefly illustrate its crucial point: the optimal solution of the *dual* problem can be read off the  $z$ -row of the final dictionary for the *primal* problem. In the example that we used to motivate the concept of the dual problem, the final dictionary reads

$$\begin{aligned} x_2 &= 14 - 2x_1 - 4x_3 - 5x_5 - 3x_7 \\ x_4 &= 5 - x_1 - x_3 - 2x_5 - x_7 \\ x_6 &= 1 + 5x_1 + 9x_3 + 21x_5 + 11x_7 \\ \hline z &= 29 - x_1 - 2x_3 - 11x_5 - 6x_7. \end{aligned}$$

Note that the slack variables  $x_5, x_6, x_7$  can be matched up with the dual variables  $y_1, y_2, y_3$  in a natural way: for instance,  $x_5$  is the slack variable in the first constraint, whereas  $y_1$  represents the multiplier for the same constraint. By the same logic,  $x_6$  goes with  $y_2$  and  $x_7$  goes with  $y_3$ . In the  $z$ -row of the dictionary, the coefficients at the slack variables are

$$-11 \text{ at } x_5, \quad 0 \text{ at } x_6, \quad -6 \text{ at } x_7.$$

Assigning these values with reversed signs to the corresponding dual variables, we obtain the desired optimal solution of the dual:

$$y_1 = 11, \quad y_2 = 0, \quad y_3 = 6.$$

At first, this may seem like pulling a rabbit out of a hat; however, the following general argument explains the magic.

**PROOF OF THEOREM 5.1.** We need only find a *feasible* solution  $(y_1^*, y_2^*, \dots, y_m^*)$  satisfying (5.5); indeed, such a solution will be *optimal* by virtue of the remarks following (5.4). In order to find that solution, we solve the primal problem by the simplex method; having introduced the slack variables

$$x_{n+i} = b_i - \sum_{j=1}^n a_{ij} x_j \quad (i = 1, 2, \dots, m) \quad (5.6)$$

we eventually arrive at the final dictionary. For the sake of definiteness, let us say that the last row of that dictionary reads

$$z = z^* + \sum_{k=1}^{n+m} \bar{c}_k x_k. \quad (5.7)$$

In (5.7), each  $\bar{c}_k$  is a nonpositive number (in fact,  $\bar{c}_k = 0$  whenever  $x_k$  is a basic variable). In addition,  $z^*$  is the optimal value of the objective function, and so

$$z^* = \sum_{j=1}^n c_j x_j^*. \quad (5.8)$$

Defining

$$y_i^* = -\bar{c}_{n+i} \quad (i = 1, 2, \dots, m) \quad (5.9)$$

we claim that  $(y_1^*, y_2^*, \dots, y_m^*)$  is a dual feasible solution satisfying (5.5); the rest of the proof consists of a straightforward verification of our claim. Substituting  $\sum c_j x_j$  for  $z$  and substituting from (5.6) for the slack variables in (5.7) we obtain the identity

$$\sum_{j=1}^n c_j x_j = z^* + \sum_{j=1}^n \bar{c}_j x_j - \sum_{i=1}^m y_i^* \left( b_i - \sum_{j=1}^n a_{ij} x_j \right)$$

which may be written as

$$\sum_{j=1}^n c_j x_j = \left( z^* - \sum_{i=1}^m b_i y_i^* \right) + \sum_{j=1}^n \left( \bar{c}_j + \sum_{i=1}^m a_{ij} y_i^* \right) x_j.$$

This identity, having been obtained by algebraic manipulations from the definitions of the slack variables and the objective function, must hold for every choice of values of  $x_1, x_2, \dots, x_n$ . Hence we have

$$z^* = \sum_{i=1}^m b_i y_i^* \quad (5.10)$$

and

$$c_j = \bar{c}_j + \sum_{i=1}^m a_{ij} y_i^* \quad (j = 1, 2, \dots, n). \quad (5.11)$$

Since  $\bar{c}_k \leq 0$  for every  $k = 1, 2, \dots, n + m$ , (5.11) and (5.9) imply

$$\sum_{i=1}^m a_{ij} y_i^* \geq c_j \quad (j = 1, 2, \dots, n)$$

$$y_i^* \geq 0 \quad (i = 1, 2, \dots, m).$$

Finally, (5.10) and (5.8) imply (5.5). ■

## RELATIONSHIP BETWEEN THE PRIMAL AND DUAL PROBLEMS

Next, let us point out that the dual of the dual is always the primal problem. Indeed, the dual problem may be written as

$$\begin{aligned} &\text{maximize} && \sum_{i=1}^m (-b_i)y_i \\ &\text{subject to} && \sum_{i=1}^m (-a_{ij})y_i \leq -c_j \quad (j = 1, 2, \dots, n) \\ &&& y_i \geq 0 \quad (i = 1, 2, \dots, m). \end{aligned}$$

The dual of this problem is

$$\begin{aligned} &\text{minimize} && \sum_{j=1}^n (-c_j)x_j \\ &\text{subject to} && \sum_{j=1}^n (-a_{ij})x_j \geq -b_i \quad (i = 1, 2, \dots, m) \\ &&& x_j \geq 0 \quad (j = 1, 2, \dots, n) \end{aligned}$$

which is clearly equivalent to the original problem. A nice corollary to this observation and to the duality theorem is that the primal problem has an optimal solution *if and only if* the dual problem has an optimal solution. Note also that if the primal is unbounded, then the dual must be infeasible [this follows directly from (5.4)]. By the same argument, if the dual is unbounded then the primal must be infeasible. However, both primal and dual may be infeasible at the same time. For example, both the problem

$$\begin{aligned} &\text{maximize} && 2x_1 - x_2 \\ &\text{subject to} && x_1 - x_2 \leq 1 \\ &&& -x_1 + x_2 \leq -2 \\ &&& x_1, x_2 \geq 0 \end{aligned}$$

and its dual are infeasible. These conclusions are summarized in Table 5.1.

**Table 5.1**  
**Primal–Dual**  
**Combinations**

|        |            | Dual       |            |            |
|--------|------------|------------|------------|------------|
|        |            | Optimal    | Infeasible | Unbounded  |
| Primal | Optimal    | Possible   | Impossible | Impossible |
|        | Infeasible | Impossible | Possible   | Possible   |
|        | Unbounded  | Impossible | Possible   | Impossible |

In particular, if the primal problem has a feasible solution *and* if the dual problem has a feasible solution, then both problems have optimal solutions.

Duality has important practical implications. In certain cases, we may find it advantageous to apply the simplex method to the dual of the problem that we are really interested in. (Of course, the optimal solution of the primal problem can then be read directly off the final dictionary for the dual.) For example, if  $m = 99$  and  $n = 9$ , then dictionaries will have 100 rows in the primal problem but only 10 rows in the dual. Since the typical number of simplex iterations is proportional to the number of rows in a dictionary and relatively insensitive to the number of variables, we shall most likely be better off solving the dual problem.

From a theoretical point of view, duality is important because it points out an elegant and succinct way of proving optimality of solutions of LP problems: as we have observed, an optimal solution of the dual problem provides a “certificate of optimality” for an optimal solution of the primal problem, and vice versa. Furthermore, the duality theorem asserts that for *every* optimal solution there is a certificate of optimality. To appreciate the impact of this fact, consider a student who is supposed to solve the problem

$$\begin{aligned} &\text{maximize} && \sum_{j=1}^n c_j x_j \\ &\text{subject to} && \sum_{j=1}^n a_{ij} x_j \leq b_i \quad (i = 1, 2, \dots, m) \\ &&& x_j \geq 0 \quad (j = 1, 2, \dots, n). \end{aligned} \tag{5.12}$$

Applying the simplex method to (5.12), the student finds simultaneously an optimal solution  $x_1^*, x_2^*, \dots, x_n^*$  of (5.12) and an optimal solution  $y_1^*, y_2^*, \dots, y_m^*$  of the dual problem

$$\begin{aligned} &\text{minimize} && \sum_{i=1}^m b_i y_i \\ &\text{subject to} && \sum_{i=1}^m a_{ij} y_i \geq c_j \quad (j = 1, 2, \dots, n) \\ &&& y_i \geq 0 \quad (i = 1, 2, \dots, m). \end{aligned} \tag{5.13}$$

Then he shows *both* solutions to his supervisor. The supervisor has an easy way of checking the correctness of the answer. To check the *feasibility* of the allegedly optimal solution, she has to verify the inequalities

$$\begin{aligned} &\sum_{j=1}^n a_{ij} x_j^* \leq b_i \quad (i = 1, 2, \dots, m) \\ &x_j^* \geq 0 \quad (j = 1, 2, \dots, n). \end{aligned} \tag{5.14}$$

To check its *optimality*, she has to verify the inequalities

$$\sum_{i=1}^m a_{ij}y_i^* \geq c_j \quad (j = 1, 2, \dots, n) \quad (5.15)$$

$$y_i^* \geq 0 \quad (i = 1, 2, \dots, m)$$

and the equation

$$\sum_{j=1}^n c_j x_j^* = \sum_{i=1}^m b_i y_i^*. \quad (5.16)$$

Of course, the computational effort involved in these *verifications* is much smaller than the computational effort required to *solve* (5.12) from scratch by the simplex method.

## COMPLEMENTARY SLACKNESS

Now we shall show how the supervisor can often recover the certificate of optimality  $y_1^*, y_2^*, \dots, y_m^*$  from the optimal solution  $x_1^*, x_2^*, \dots, x_n^*$  alone. The key to the procedure is a convenient way of breaking down equation (5.16) into simple constituents.

**THEOREM 5.2.** Let  $x_1^*, x_2^*, \dots, x_n^*$  be a feasible solution of (5.12) and let  $y_1^*, y_2^*, \dots, y_m^*$  be a feasible solution of (5.13). Necessary and sufficient conditions for simultaneous optimality of  $x_1^*, x_2^*, \dots, x_n^*$  and  $y_1^*, y_2^*, \dots, y_m^*$  are

$$\sum_{i=1}^m a_{ij}y_i^* = c_j \quad \text{or} \quad x_j^* = 0 \quad (\text{or both}) \quad \text{for every } j = 1, 2, \dots, n \quad (5.17)$$

and

$$\sum_{j=1}^n a_{ij}x_j^* = b_i \quad \text{or} \quad y_i^* = 0 \quad (\text{or both}) \quad \text{for every } i = 1, 2, \dots, m. \quad (5.18)$$

**PROOF.** Assumptions (5.14) and (5.15) imply

$$c_j x_j^* \leq \left( \sum_{i=1}^m a_{ij} y_i^* \right) x_j^* \quad (j = 1, 2, \dots, n) \quad (5.19)$$

$$\left( \sum_{j=1}^n a_{ij} x_j^* \right) y_i^* \leq b_i y_i^* \quad (i = 1, 2, \dots, m) \quad (5.20)$$

and so

$$\sum_{j=1}^n c_j x_j^* \leq \sum_{j=1}^n \left( \sum_{i=1}^m a_{ij} y_i^* \right) x_j^* = \sum_{i=1}^m \left( \sum_{j=1}^n a_{ij} x_j^* \right) y_i^* \leq \sum_{i=1}^m b_i y_i^*. \quad (5.21)$$

Hence, (5.21) holds with equalities throughout if and only if equalities hold in (5.19) and (5.20). One way to guarantee the equality  $c_j x_j^* = (\sum_{i=1}^m a_{ij} y_i^*) x_j^*$  is to insist that  $x_j^* = 0$ ; failing that, we must require  $c_j = \sum_{i=1}^m a_{ij} y_i^*$ . Hence, equalities hold in (5.19) if and only if condition (5.17) is satisfied. Similarly, equalities hold in (5.20) if and only if condition (5.18) is satisfied.

To summarize, conditions (5.17) and (5.18) are necessary and sufficient for (5.16) to hold. On the other hand, the Duality Theorem shows that (5.16) is necessary and sufficient for simultaneous optimality of  $x_1^*, x_2^*, \dots, x_n^*$  and  $y_1^*, y_2^*, \dots, y_m^*$ . The proof is completed. ■

Conditions (5.17) and (5.18) gain simplicity as soon as we introduce the slack variables

$$x_{n+i} = b_i - \sum_{j=1}^n a_{ij} x_j \quad (i = 1, 2, \dots, m)$$

$$y_{m+j} = -c_j + \sum_{i=1}^m a_{ij} y_i \quad (j = 1, 2, \dots, n).$$

As we observed once before, the primal slack variables  $x_{n+1}, x_{n+2}, \dots, x_{n+m}$  are naturally matched up with the dual decision variables  $y_1, y_2, \dots, y_m$ : each variable  $x_{n+i}$  denotes the slack in the  $i$ th primal constraint, whereas the corresponding  $y_i$  represents the multiplier at the same constraint. Similarly, each primal decision variable  $x_j$  is matched with the dual slack  $y_{m+j}$ . Conditions (5.17) and (5.18) require that in each of the  $m + n$  matching pairs, at least one variable must have value zero. These conditions are usually called the *complementary slackness* conditions; Theorem 5.2 itself is referred to as the Complementary Slackness Theorem.

It is an easy task to convert Theorem 5.2 into a form in which its applicability becomes evident.

**THEOREM 5.3.** A feasible solution  $x_1^*, x_2^*, \dots, x_n^*$  of (5.12) is optimal if and only if there are numbers  $y_1^*, y_2^*, \dots, y_m^*$  such that

$$\sum_{i=1}^m a_{ij} y_i^* = c_j \quad \text{whenever } x_j^* > 0 \quad (5.22)$$

$$y_i^* = 0 \quad \text{whenever } \sum_{j=1}^n a_{ij} x_j^* < b_i$$

and such that

$$\sum_{i=1}^m a_{ij} y_i^* \geq c_j \quad \text{for all } j = 1, 2, \dots, n \quad (5.23)$$

$$y_i^* \geq 0 \quad \text{for all } i = 1, 2, \dots, m.$$

**PROOF.** If  $x_1^*, x_2^*, \dots, x_n^*$  is optimal, then, by Theorem 5.1, there is an optimal solution  $y_1^*, y_2^*, \dots, y_m^*$  of (5.13). That solution, being feasible, satisfies (5.23). By Theorem 5.2, the two optimal solutions satisfy the complementary slackness conditions (5.22).

Conversely, if  $y_1^*, y_2^*, \dots, y_m^*$  satisfy (5.23), then they constitute a feasible solution of (5.13). If they satisfy (5.22) as well, then, by Theorem 5.2,  $x_1^*, x_2^*, \dots, x_n^*$  is an optimal solution of (5.12) and  $y_1^*, y_2^*, \dots, y_m^*$  is an optimal solution of (5.13). ■

Theorem 5.3 is often useful in checking the optimality of allegedly optimal solutions when no certificate of optimality is provided. Confronted with an allegedly optimal solution  $x_1^*, x_2^*, \dots, x_n^*$  of (5.12), we first set up the system of linear equations (5.22) and solve for  $y_1^*, y_2^*, \dots, y_m^*$ . If the solution  $y_1^*, y_2^*, \dots, y_m^*$  is unique, then we are in business:  $x_1^*, x_2^*, \dots, x_n^*$  is optimal if and only if (5.23) holds. We shall illustrate this situation with two examples.

First, let us consider the claim that

$$x_1^* = 2, \quad x_2^* = 4, \quad x_3^* = 0, \quad x_4^* = 0, \quad x_5^* = 7, \quad x_6^* = 0$$

is an optimal solution of the problem

$$\begin{aligned} \text{maximize} \quad & 18x_1 - 7x_2 + 12x_3 + 5x_4 + 8x_6 \\ \text{subject to} \quad & 2x_1 - 6x_2 + 2x_3 + 7x_4 + 3x_5 + 8x_6 \leq 1 \\ & -3x_1 - x_2 + 4x_3 - 3x_4 + x_5 + 2x_6 \leq -2 \\ & 8x_1 - 3x_2 + 5x_3 - 2x_4 + 2x_6 \leq 4 \\ & 4x_1 + 8x_3 + 7x_4 - x_5 + 3x_6 \leq 1 \\ & 5x_1 + 2x_2 - 3x_3 + 6x_4 - 2x_5 - x_6 \leq 5 \\ & x_1, x_2, x_3, x_4, x_5, x_6 \geq 0. \end{aligned}$$

In this case, (5.22) reads

$$\begin{aligned} 2y_1^* - 3y_2^* + 8y_3^* + 4y_4^* + 5y_5^* &= 18 \\ -6y_1^* - y_2^* - 3y_3^* + 2y_5^* &= -7 \\ 3y_1^* + y_2^* - y_4^* - 2y_5^* &= 0 \\ y_2^* &= 0 \\ y_5^* &= 0. \end{aligned}$$

Since its solution  $(\frac{1}{3}, 0, \frac{5}{3}, 1, 0)$  satisfies (5.23), the proposed solution  $x_1^*, x_2^*, \dots, x_6^*$  is optimal.

Second, let us consider the claim that

$$x_1^* = 0, \quad x_2^* = 2, \quad x_3^* = 0, \quad x_4^* = 7, \quad x_5^* = 0$$

is an optimal solution of the problem

$$\begin{aligned} \text{maximize} \quad & 8x_1 - 9x_2 + 12x_3 + 4x_4 + 11x_5 \\ \text{subject to} \quad & 2x_1 - 3x_2 + 4x_3 + x_4 + 3x_5 \leq 1 \\ & x_1 + 7x_2 + 3x_3 - 2x_4 + x_5 \leq 1 \\ & 5x_1 + 4x_2 - 6x_3 + 2x_4 + 3x_5 \leq 22 \\ & x_1, x_2, x_3, x_4, x_5 \geq 0. \end{aligned}$$

Here (5.22) becomes

$$\begin{aligned} -3y_1^* + 7y_2^* + 4y_3^* &= -9 \\ y_1^* - 2y_2^* + 2y_3^* &= 4 \\ y_2^* &= 0. \end{aligned}$$

Since its unique solution  $(3.4, 0, 0.3)$  violates (5.23), the proposed solution  $x_1^*, x_2^*, \dots, x_5^*$  is not optimal.

Of course, this straightforward strategy for verifying optimality of allegedly optimal solutions is applicable only if the system of equations (5.22) has a unique solution. The following result points out conditions under which this is always the case.

**THEOREM 5.4.** If  $x_1^*, x_2^*, \dots, x_n^*$  is a nondegenerate basic feasible solution of (5.12), then (5.22) has a unique solution.

The proof of this theorem is postponed until the end of Chapter 7, where it will become an exercise (problem 7.3).

## ECONOMIC SIGNIFICANCE OF DUAL VARIABLES

For many LP problems

$$\begin{aligned} \text{maximize} \quad & \sum_{j=1}^n c_j x_j \\ \text{subject to} \quad & \sum_{j=1}^n a_{ij} x_j \leq b_i \quad (i = 1, 2, \dots, m) \\ & x_j \geq 0 \quad (j = 1, 2, \dots, n) \end{aligned} \tag{5.24}$$

arising in applications, the variables  $y_1, y_2, \dots, y_m$  in the dual problem can be given a meaningful interpretation. An indication of the way these dual variables should be interpreted follows from a heuristic argument occasionally used in elementary

**PROOF.** If  $x_1^*, x_2^*, \dots, x_n^*$  is optimal, then, by Theorem 5.1, there is an optimal solution  $y_1^*, y_2^*, \dots, y_m^*$  of (5.13). That solution, being feasible, satisfies (5.23). By Theorem 5.2, the two optimal solutions satisfy the complementary slackness conditions (5.22).

Conversely, if  $y_1^*, y_2^*, \dots, y_m^*$  satisfy (5.23), then they constitute a feasible solution of (5.13). If they satisfy (5.22) as well, then, by Theorem 5.2,  $x_1^*, x_2^*, \dots, x_n^*$  is an optimal solution of (5.12) and  $y_1^*, y_2^*, \dots, y_m^*$  is an optimal solution of (5.13). ■

Theorem 5.3 is often useful in checking the optimality of allegedly optimal solutions when no certificate of optimality is provided. Confronted with an allegedly optimal solution  $x_1^*, x_2^*, \dots, x_n^*$  of (5.12), we first set up the system of linear equations (5.22) and solve for  $y_1^*, y_2^*, \dots, y_m^*$ . If the solution  $y_1^*, y_2^*, \dots, y_m^*$  is *unique*, then we are in business:  $x_1^*, x_2^*, \dots, x_n^*$  is optimal if and only if (5.23) holds. We shall illustrate this situation with two examples.

First, let us consider the claim that

$$x_1^* = 2, \quad x_2^* = 4, \quad x_3^* = 0, \quad x_4^* = 0, \quad x_5^* = 7, \quad x_6^* = 0$$

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$$\begin{aligned} \text{maximize} \quad & \sum_{j=1}^n c_j x_j \\ \text{subject to} \quad & \sum_{j=1}^n a_{ij} x_j \leq b_i \quad (i = 1, 2, \dots, m) \\ & x_j \geq 0 \quad (j = 1, 2, \dots, n) \end{aligned} \tag{5.24}$$

arising in applications, the variables  $y_1, y_2, \dots, y_m$  in the dual problem can be given a meaningful interpretation. An indication of the way these dual variables should be interpreted follows from a heuristic argument occasionally used in elementary



physics and known as "dimension analysis." For instance, suppose that (5.24) is the problem of maximizing profit in a furniture manufacturing firm. Each  $x_j$  measures the level of the output of the  $j$ th product (such as desks or chairs), and each  $b_i$  specifies the available amount of the  $i$ th resource (such as wood or metal). Note that *each  $a_{ij}$  is expressed in units of resource  $i$  per unit of product  $j$*  (in fact, each  $a_{ij}$  is the amount of resource  $i$  required in making a unit of product  $j$ ) and that *each  $c_j$  is in dollars per unit of product  $j$*  (in fact, each  $c_j$  is the net profit brought in by a unit of product  $j$ ). To make the left-hand side,  $\sum a_{ij}y_i$ , of each dual constraint commensurate with the right-hand side  $c_j$ , we must express each  $y_i$  in dollars per unit of resource  $i$ . Thus, one is led to suspect that each  $y_i$  measures the unit worth of the  $i$ th resource. The following theorem will validate this suspicion.

**THEOREM 5.5.** If (5.24) has at least one nondegenerate basic optimal solution, then there is a positive  $\varepsilon$  with the following property: If  $|t_i| \leq \varepsilon$  for all  $i = 1, 2, \dots, m$ , then the problem

$$\begin{aligned} &\text{maximize} && \sum_{j=1}^n c_j x_j \\ &\text{subject to} && \sum_{j=1}^n a_{ij} x_j \leq b_i + t_i \quad (i = 1, 2, \dots, m) \\ &&& x_j \geq 0 \quad (j = 1, 2, \dots, n) \end{aligned} \quad (5.25)$$

has an optimal solution and its optimal value equals

$$z^* + \sum_{i=1}^m y_i^* t_i$$

with  $z^*$  standing for the optimal value of (5.24) and with  $y_1^*, y_2^*, \dots, y_m^*$  standing for the optimal solution of its dual.

We postpone the proof of this theorem also until the end of Chapter 7, where it will become an exercise (problem 7.4). At this moment, let us note only that the uniqueness of  $y_1^*, y_2^*, \dots, y_m^*$  is guaranteed by Theorems 5.4 and 5.2.

Theorem 5.5 reveals the effects of small variations in the supplies of the resources on the total net profit of the firm. With each extra unit of resource  $i$ , the profit increases by  $y_i^*$  dollars. Hence,  $y_i^*$  specifies the maximum amount that the firm should be willing to pay, over and above the present trading price, for each extra unit of resource  $i$ . For this reason,  $y_i^*$  is often called the *marginal value* of the  $i$ th resource, the adjective "marginal" referring to the difference between the trading price and the actual worth

of the resource. Another term commonly used for  $y_i^*$  in this context is the *shadow price* of the  $i$ th resource.

To illustrate these findings, imagine a forester who has 100 acres of hardwood timber. Felling the hardwood and letting the area regenerate would cost \$10 per acre in immediate resources and bring a subsequent return of \$50 per acre. An alternative course of action is to fell the hardwood and plant the area with pine; that would cost \$50 per acre with a subsequent return of \$120 per acre. Hence, the *net* profits resulting from the two treatments are \$40 and \$70 per acre, respectively. Unfortunately, the more profitable second treatment cannot be applied to the entire area since only \$4,000 is available to meet the immediate costs. Clearly, the forester's problem is to

$$\begin{aligned} &\text{maximize} && 40x_1 + 70x_2 \\ &\text{subject to} && x_1 + x_2 \leq 100 \\ &&& 10x_1 + 50x_2 \leq 4,000 \\ &&& x_1, x_2 \geq 0. \end{aligned}$$

Its optimal solution is  $x_1^* = 25$  and  $x_2^* = 75$ . Hence, the forester should fell the hardwood throughout the entire area, letting 25 acres regenerate and planting the remaining 75 acres with pine. According to this program, the initial investment of \$4,000 yields the ultimate net profit of \$6,250.

Evidently, the forester's initial capital represents a valuable resource. In fact, the forester might be well advised to increase the level of this resource by taking out a short-term loan; the resulting extra profit might make up even for a drastic interest rate. For example, suppose that she could borrow \$100 now and pay back \$180 later; should she do that? On the other hand, she might be tempted to divert some of her \$4,000 to other lucrative enterprises. For example, suppose she could invest \$100 now and collect \$180 later; should she do that? According to Theorem 5.5, the answers lie hidden in the optimal solution

$$y_1^* = 32.5, \quad y_2^* = 0.75$$

of the dual problem: the forester should take out (limited) loans if and only if the interest is lower than 75 cents per dollar and she should make (small) investments if and only if the profit is greater than 75 cents per dollar.

These claims, whose validity is guaranteed by Theorem 5.5, are easy to justify directly. Having borrowed  $t$  dollars, the forester aims to

$$\begin{aligned} &\text{maximize} && 40x_1 + 70x_2 \\ &\text{subject to} && x_1 + x_2 \leq 100 \\ &&& 10x_1 + 50x_2 \leq 4,000 + t \\ &&& x_1, x_2 \geq 0. \end{aligned} \quad (5.26)$$

Every feasible solution  $x_1, x_2$  of this problem satisfies the inequalities

$$\begin{aligned} 40x_1 + 70x_2 &= 32.5(x_1 + x_2) + 0.75(10x_1 + 50x_2) \\ &\leq 3,250 + 0.75(4,000 + t) = 6,250 + 0.75t \end{aligned} \quad (5.27)$$

and so the extra profit will never exceed  $0.75t$ . In fact, if  $t \leq 1,000$ , then the forester can realize the additional profit of  $0.75t$  by letting

$$x_1 = 25 - 0.025t, \quad x_2 = 75 + 0.025t. \quad (5.28)$$

Investments in other enterprises give rise to negative values of  $t$  in (5.26); as a result of such investments, the net profit from the original enterprise diminishes. If  $-t$  dollars are diverted to alternative investments ( $-t$  is positive!) then, by (5.27), the profit from the hardwood felling enterprise will drop by  $0.75(-t)$  or even more. In fact, if  $-t \leq 3,000$ , then the drop can be limited to only  $0.75(-t)$  by choosing  $x_1$  and  $x_2$  according to (5.28).

It should perhaps be emphasized that Theorem 5.5 deals with *small* changes  $t_i$  in the resource levels; its conclusions may fail when the  $t_i$ 's are large. For instance, our forester has no use for loans exceeding \$1,000 and, should she wish to invest all of her \$4,000 in another enterprise, she would be ill advised to demand only 75 cents of profit on each dollar. (A part of Theorem 5.5 can be salvaged even if the  $t_i$ 's are large; see problem 5.9.)

Now suppose that a previously unavailable opportunity arises for the forester to engage in an activity such as, say, felling the hardwood and planting the area with conifer. For a quick assessment of this activity, the forester may appeal to the marginal values of her resources: \$32.5 per acre of hardwood and \$0.75 per dollar of capital. If the new activity requires  $a$  dollars per acre, then the resources consumed by this activity per acre are valued at  $$(32.5 + 0.75a)$  and the activity is worth considering if and only if its net profit per acre exceeds this figure. Further examples of this kind are presented in problems 5.6 and 5.7.

In closing, let us mention that models of economy often fall into the realm of linear programming. In particular, many theorems concerning economic equilibria may be deduced from the Duality Theorem and the Complementary Slackness Theorem. Their discussion exceeds the scope of this text; the interested reader is referred to D. Gale (1960) and R. Dorfman, P. A. Samuelson, and R. M. Solow (1958).

## PROBLEMS

5.1 Illustrate Theorem 5.1 on each of the three LP problems in problem 2.1.

$$\begin{aligned} \triangle 5.2 \quad & \text{Maximize} && -x_1 - 2x_2 \\ & \text{subject to} && -3x_1 + x_2 \leq -1 \\ & && x_1 - x_2 \leq 1 \\ & && -2x_1 + 7x_2 \leq 6 \\ & && 9x_1 - 4x_2 \leq 6 \\ & && -5x_1 + 2x_2 \leq -3 \\ & && 7x_1 - 3x_2 \leq 6 \\ & && x_1, x_2 \geq 0. \end{aligned}$$

$\triangle 5.3$  For each of the two problems below, use Theorem 5.3 to check the optimality of the proposed solution.

$$\begin{aligned} \text{a. Maximize} & \quad 7x_1 + 6x_2 + 5x_3 - 2x_4 + 3x_5 \\ \text{subject to} & \quad x_1 + 3x_2 + 5x_3 - 2x_4 + 2x_5 \leq 4 \\ & \quad 4x_1 + 2x_2 - 2x_3 + x_4 + x_5 \leq 3 \\ & \quad 2x_1 + 4x_2 + 4x_3 - 2x_4 + 5x_5 \leq 5 \\ & \quad 3x_1 + x_2 + 2x_3 - x_4 - 2x_5 \leq 1 \\ & \quad x_1, x_2, x_3, x_4, x_5 \geq 0. \end{aligned}$$

$$\text{Proposed solution: } x_1^* = 0, \quad x_2^* = \frac{4}{3}, \quad x_3^* = \frac{2}{3}, \quad x_4^* = \frac{5}{3}, \quad x_5^* = 0.$$

$$\begin{aligned} \text{b. Maximize} & \quad 4x_1 + 5x_2 + x_3 + 3x_4 - 5x_5 + 8x_6 \\ \text{subject to} & \quad x_1 - 4x_3 + 3x_4 + x_5 + x_6 \leq 1 \\ & \quad 5x_1 + 3x_2 + x_3 - 5x_5 + 3x_6 \leq 4 \\ & \quad 4x_1 + 5x_2 - 3x_3 + 3x_4 - 4x_5 + x_6 \leq 4 \\ & \quad -x_2 + 2x_4 + x_5 - 5x_6 \leq 5 \\ & \quad -2x_1 + x_2 + x_3 + x_4 + 2x_5 + 2x_6 \leq 7 \\ & \quad 2x_1 - 3x_2 + 2x_3 - x_4 + 4x_5 + 5x_6 \leq 5 \\ & \quad x_1, x_2, x_3, x_4, x_5, x_6 \geq 0. \end{aligned}$$

$$\text{Proposed solution: } x_1 = 0, \quad x_2 = 0, \quad x_3 = \frac{5}{2}, \quad x_4 = \frac{7}{2}, \quad x_5 = 0, \quad x_6 = \frac{1}{2}.$$

$\triangle 5.4$  In problem 1.6, one of the possible strategies is as follows:

- Smoke all 400 bellies on regular time.
- Smoke 20 picnics on regular time and 210 on overtime.
- Smoke 40 hams on overtime and sell 440 fresh.

Use Theorem 5.3 to find out whether this strategy is optimal or not.

5.5 In problem 1.7, one of the possible strategies is as follows:

- Blend 3,754 barrels of alkylate, 2,666 barrels of catalytic, 920 barrels of straight-run, and 543 barrels of isopentane into 7,883 barrels of Avgas A.
- Blend 60 barrels of alkylate, 3,096 barrels of straight-run, and 672 barrels of isopentane into 3,828 barrels of Avgas B.
- Sell 85 barrels of isopentane raw.

Use Theorem 5.3 to find out whether this strategy is optimal or not.

- 5.6 In the optimal solution to problem 1.6, all the bellies and picnics are smoked. However, sufficiently drastic changes in market prices might provide an incentive to change this policy. Assume that the market price of fresh bellies increases by  $x$  dollars, while all the other prices remain fixed at their original levels. How large would  $x$  have to be in order to make it profitable for the plant to sell fresh bellies? Ask and answer a similar question for picnics. How would the sales of small amounts of fresh bellies and picnics affect the rest of the operation? What precisely do "small amounts" mean in this context?
- 5.7 In the optimal solution to problem 1.7, 85 barrels of isopentane are sold raw at \$4.83 per barrel. Find the break-even selling prices for raw alkylate, catalytic, and straight-run. Next, assume there is a demand for Avgas C with PN at least 80 and RVP at most 7. Find the break-even selling price of this gasoline.
- 5.8 Can you interpret the complementary slackness conditions in economic terms?
- 5.9 Let  $z^*$  be the optimal value of (5.24) and let  $y_1^*, y_2^*, \dots, y_m^*$  be any optimal solution of the dual problem. Prove that
- $$\sum_{j=1}^n c_j x_j \leq z^* + \sum_{i=1}^m y_i^* t_i$$
- for every feasible solution  $x_1, x_2, \dots, x_n$  of (5.25).
- 5.10 Construct an example showing that the conclusion of Theorem 5.5 may fail if the hypothesis that (5.24) has a nondegenerate basic optimal solution is omitted.

## Gaussian Elimination and Matrices

This chapter reviews facts about systems of linear equations and matrices that will be needed later. We shall begin with Gaussian elimination and discuss ways of enhancing its accuracy and speed. Then we shall introduce the notion of matrix multiplication and review Gaussian elimination in terms of matrices. Finally, we shall establish a few fundamental facts concerning nonsingular matrices.

### GAUSSIAN ELIMINATION

Perhaps the most popular method for solving systems of linear equations is known as *Gaussian elimination*. This method consists of a successive elimination of variables followed by a back substitution. We shall illustrate it on the system

$$\begin{aligned} 5x_1 + 4x_2 + 3x_3 + 9x_4 &= 49 \\ 3x_1 + 2x_2 + x_3 + 2x_4 &= 19 \\ -14x_1 - 8x_2 - 7x_3 + 5x_4 &= -68 \\ 12x_1 + 6x_2 - 25x_3 + 10x_4 &= -38. \end{aligned} \tag{6.1}$$

The first equation in (6.1) may be converted into the formula

x\_1 = 9.8 - 0.8x\_2 - 0.6x\_3 - 1.8x\_4. (6.2)

Substituting from (6.2) into the remaining equations in (6.1) we obtain

3(9.8 - 0.8x\_2 - 0.6x\_3 - 1.8x\_4) + 2x\_2 + x\_3 + 2x\_4 = 19
-14(9.8 - 0.8x\_2 - 0.6x\_3 - 1.8x\_4) - 8x\_2 - 7x\_3 + 5x\_4 = -68
12(9.8 - 0.8x\_2 - 0.6x\_3 - 1.8x\_4) + 6x\_2 - 25x\_3 + 10x\_4 = -38

and, after simplification,

-0.4x\_2 - 0.8x\_3 - 3.4x\_4 = -10.4
3.2x\_2 + 1.4x\_3 + 30.2x\_4 = 69.2
-3.6x\_2 - 32.2x\_3 - 11.6x\_4 = -155.6. (6.3)

Clearly, numbers x\_1, x\_2, x\_3, x\_4 satisfy (6.1) if and only if they satisfy (6.2), (6.3). In this sense, system (6.1) is equivalent to system (6.2), (6.3). Since variable x\_1 does not appear in (6.3), we say that it has been eliminated. Just as we eliminated x\_1 from (6.1), we may now eliminate x\_2 from (6.3). The first equation in (6.3) may be written as

x\_2 = 26 - 2x\_3 - 8.5x\_4. (6.4)

Substituting from (6.4) into the remaining two equations in (6.3) we obtain, after simplification, the system

-5x\_3 + 3x\_4 = -14
-25x\_3 + 19x\_4 = -62. (6.5)

To eliminate x\_3 from (6.5), we write the first equation of (6.5) in the form

x\_3 = 2.8 + 0.6x\_4. (6.6)

Substituting from (6.6) into the remaining equation in (6.5) we obtain

4x\_4 = 8. (6.7)

Now, (6.7) yields

x\_4 = 2. (6.8)

Substituting from (6.8) back into (6.6) we find

x\_3 = 4. (6.9)

Substituting from (6.9) and (6.8) back into (6.4) we find

x\_2 = 1. (6.10)

Finally, substituting from (6.10), (6.9), and (6.8) back into (6.2) we obtain

x\_1 = 3. (6.11)

The same strategy applies in general: we eliminate x\_1, x\_2, ... in this order and then find the values of x\_n, x\_{n-1}, ..., x\_1 by back substitution. The elimination process merits a comment. When the variables x\_1, x\_2, ..., x\_{k-1} have been eliminated, we are confronted with a system of n - k + 1 equations in the remaining n - k + 1 variables x\_k, x\_{k+1}, ..., x\_n. Among these n - k + 1 equations, we choose an equation

sum\_{j=k}^n a\_j x\_j = b (6.12)

such that a\_k != 0; this number a\_k is called the pivot. Substituting from

x\_k = b/a\_k - sum\_{j=k+1}^n (a\_j/a\_k) x\_j

into the remaining n - k equations, we eliminate x\_k and obtain a system of n - k equations in n - k variables. (However, it may happen that each of the n - k + 1 equations of (6.12) has a\_k = 0. In other words, the system of n - k + 1 equations may involve only the n - k variables x\_{k+1}, x\_{k+2}, ..., x\_n. This circumstance indicates that the original system of n equations in n variables either has no solutions at all or it has infinitely many of them. Since this case is of no importance in our present discussion, we shall not elaborate on the details until Theorem 6.1, which will follow later.)

It is natural to ask how much work is required to solve a system of n equations in n variables by this method. For our example, Table 6.1 gives a detailed account of the work involved.

Table 6.1 Operations Count

Table with 4 columns: To calculate, Additions, Multiplications, Divisions. Rows include equations (6.2) through (6.11) and a Total row.

Note: Additions and subtractions are essentially the same operation; therefore we count both under the heading Additions.

In general, the work required to eliminate x\_k amounts to n - k + 1 divisions, followed by (n - k)(n - k + 1) multiplications and (n - k)(n - k + 1) additions. Then, evaluating x\_k by back substitution takes n - k multiplications and n - k additions. The totals are:

Additions: sum\_{k=1}^n (n - k)(n - k + 1) + sum\_{k=1}^n (n - k) = n(n - 1)(2n + 5)/6.

Multiplications: sum\_{k=1}^n (n - k)(n - k + 1) + sum\_{k=1}^n (n - k) = n(n - 1)(2n + 5)/6.

Divisions: sum\_{k=1}^n (n - k + 1) = n(n + 1)/2.

Hence, there are about n^3/3 additions, n^3/3 multiplications, and n^2/2 divisions to carry out. Since the additions are relatively easy and the divisions are relatively few, the main computational

burden comes from the multiplications. (Incidentally, note that the method applied to  $2n$  equations in  $2n$  variables takes roughly  $8n^3/3$  multiplications, *eight times* the amount required by  $n$  equations in  $n$  variables.)

Similar methods for solving *small* systems of linear equations have been in use for centuries. An ancient Chinese text, "Nine Chapters on the Mathematical Art," includes examples of three equations in three variables solved by an elimination procedure. This book dates from the Han dynasty of 206 B.C.–A.D. 220 (the period in which the *I-Ching* was written), but some of its contents may be as old as 1100 B.C. Other examples appear in various writings scattered throughout the ages. However, a systematic study of the method as a tool for solving *arbitrary* systems of linear equations was first undertaken by C. F. Gauss (1777–1855) and A. M. Legendre (1752–1833). Additional information on the history of elimination methods may be found in Eves (1964), Smith (1953), and Struik (1967).

Two obvious criteria for assessing the performance of any numerical algorithm are (i) the accuracy of results and (ii) the speed of execution. We shall analyze Gaussian elimination from both points of view, beginning with numerical accuracy.

### Accuracy of Gaussian Elimination

Numerical computations are usually done in *floating-point arithmetic*. That is, each nonzero number is represented in the form  $x \cdot 10^k$  such that  $k$  is an integer and  $0.1 \leq |x| < 1$ . The number  $x$  is stored only to a *d-digit precision* (or "*d* significant digits") for some fixed positive integer  $d$ ; that is, only the first  $d$  digits of  $x$  following the decimal point are stored. Since the same convention applies to results of arithmetical operations, floating-point computations are subject to rounding errors. For instance, if  $d = 3$ , then the result of

$$(0.523 \cdot 10^2) \times (-0.324 \cdot 10^{-5}) = -0.169452 \cdot 10^{-3}$$

is stored as  $-0.169 \cdot 10^{-3}$ . In long chain calculations, these rounding errors may accumulate to such an extent that the final result is useless.

It would be a grave misconception to assume that only long chain calculations lead to numerically inaccurate results: even small systems of linear equations may cause trouble. To illustrate, we shall solve the system

$$\begin{aligned} 0.0001x_1 + x_2 &= 1 \\ 0.5x_1 + 0.5x_2 &= 1 \end{aligned} \quad (6.13)$$

rounding the intermediate results to three significant digits. Using the first equation to eliminate  $x_1$  we obtain

$$\begin{aligned} x_1 &= 10,000 - 10,000x_2 \\ -5,000x_2 &= -5,000 \end{aligned}$$

whereupon back substitution yields  $x_2 = 1$  and  $x_1 = 0$ . But the true solution of (6.13), rounded to three significant digits, is  $x_1 = 1$  and  $x_2 = 1$ . What went wrong? The pivot 0.0001 was too small relative to the remaining coefficients. In our first outline of Gaussian elimination, we observed that *zero* pivots *must* be avoided since division by zero is impossible. Now it seems reasonable to infer that *small* pivots *should* be avoided since division by small numbers is likely to produce serious rounding errors. This precaution may be implemented in various ways. For instance, while eliminating  $x_1, x_2, \dots$  in this order, we might always choose the pivot that has the largest magnitude among all the available candidates. Or, going a step further, we might also change the order in which the variables are eliminated, so that the pivot magnitude in each iteration is as large as possible. The latter strategy is known as *complete pivoting*, whereas the former, with the order of elimination fixed, is known as *partial pivoting*. Our discussion will be restricted to partial pivoting since this strategy is quite adequate in practice. In our example,

partial pivoting tells us to eliminate  $x_1$  from the second equation. Rounding the intermediate results to three significant digits, we obtain

$$\begin{aligned} x_1 &= 2 - x_2 \\ x_2 &= 1 \end{aligned}$$

whereupon back substitution yields  $x_2 = 1$  and  $x_1 = 1$ . Thus, all is well—at least in this case.

Investigation into the numerical accuracy of elimination methods has an interesting history. An early paper by H. Hotelling (1943) established a discouraging fact: Even when  $n = 100$ , the computed solution  $\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n$  of

$$\sum_{j=1}^n a_{ij}x_j = b_i \quad (i = 1, 2, \dots, n) \quad (6.14)$$

may differ from the true solution  $x_1, x_2, \dots, x_n$  to such extent that (unless we work to the precision of more than 60 decimal digits) the computations are meaningless. Although comparing the computed solution  $\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n$  with the true solution  $x_1, x_2, \dots, x_n$  seems to be the natural way of assessing the accuracy of the computations, this point of view, nowadays known as the *forward error analysis*, fell out of favor around the 1950s. The computed solution may suddenly look much better, if it turns out to be the precise solution of a system

$$\sum_{j=1}^n \tilde{a}_{ij}\tilde{x}_j = b_i \quad (i = 1, 2, \dots, n) \quad (6.15)$$

whose coefficients  $\tilde{a}_{ij}$  differ from the original coefficients  $a_{ij}$  only slightly. This new point of view, known as the *backward error analysis*, makes perfect sense in applications where the coefficients  $a_{ij}$  are often obtained by imprecise measurements and/or estimations. The idea of backward error analysis, implicit in papers by J. von Neumann and H. H. Goldstine (1947) and by A. M. Turing (1948), was put forth by W. Givens (1954) and championed by J. H. Wilkinson (1963). Wilkinson subjected elimination methods to backward error analysis and found that Gaussian elimination with partial pivoting yields remarkably accurate results. His arguments guarantee the existence of a system (6.15) that is very close to (6.14). Roughly speaking (the precise statement is a little more intricate), each difference  $\tilde{a}_{ij} - a_{ij}$  amounts to at most  $n$  of the individual errors made in the process of solving (6.14).

Of course, this guarantee loses much of its appeal when the individual rounding errors are large compared with  $a_{ij}$ . This situation may happen when the various coefficients differ from each other so widely that the individual rounding errors resulting from handling large  $a_{ij}$  become overwhelming when compared with small  $a_{ij}$ . To illustrate, let us solve the system

$$\begin{aligned} x_1 + 10,000x_2 &= 10,000 \\ 0.5x_1 + 0.5x_2 &= 1 \end{aligned} \quad (6.16)$$

by Gaussian elimination with partial pivoting, working to three significant digits. Eliminating  $x_1$  from the first equation, we obtain

$$\begin{aligned} x_1 &= 10,000 - 10,000x_2 \\ -5,000x_2 &= -5,000 \end{aligned}$$

whereupon back substitution yields  $x_2 = 1, x_1 = 0$ . This result is unsatisfactory even from the point of view of backward error analysis since every system

$$\begin{aligned} \tilde{a}_{11}\tilde{x}_1 + \tilde{a}_{12}\tilde{x}_2 &= 10,000 \\ \tilde{a}_{21}\tilde{x}_1 + \tilde{a}_{22}\tilde{x}_2 &= 1 \end{aligned}$$

satisfied by  $\tilde{x}_1 = 0$ ,  $\tilde{x}_2 = 1$  must have  $\tilde{a}_{22} = 1$ , and therefore differ significantly from (6.16). Fortunately, there is often a way of avoiding such blunders. Individual rounding errors tend to become less significant when the various nonzero coefficients  $a_{ij}$  have roughly the same magnitude. Systems with this property are called "well-scaled." Many systems

$$\sum_{j=1}^n a_{ij}x_j = b_i \quad (i = 1, 2, \dots, n) \quad (6.17)$$

that arise in practice are well-scaled; other systems can often be made well-scaled by simple transformations. Having chosen positive numbers  $r_1, r_2, \dots, r_m$  and  $s_1, s_2, \dots, s_n$ , we multiply each row  $i$  in (6.17) by  $r_i$  and substitute  $s_j x'_j$  for each  $x_j$ . The resulting system reads  $\sum r_i a_{ij}(s_j x'_j) = r_i b_i$ , or

$$\sum_{j=1}^n (r_i a_{ij} s_j) x'_j = r_i b_i \quad (i = 1, 2, \dots, n). \quad (6.18)$$

The transformation of (6.17) into (6.18) is called *scaling*, with the numbers  $r_i$  and  $s_j$  referred to as *scaling factors*. A suitable choice of these scaling factors helps to smooth out discrepancies among the magnitudes of the various nonzero coefficients and often yields a well-scaled system (6.18). For instance, if (6.17) reads

$$\begin{aligned} 0.5x_1 + 0.002x_2 + 2x_3 + 40x_4 &= 12 \\ 30x_1 + 0.4x_2 + 500x_3 + 1000x_4 &= 600 \\ 2x_1 + 0.01x_2 + 30x_3 + 300x_4 &= 90 \\ 0.1x_1 + 0.005x_2 + 4x_3 + 20x_4 &= 14 \end{aligned}$$

and if we choose

$$\begin{aligned} r_1 &= 10, & r_2 &= 0.1, & r_3 &= 1, & r_4 &= 10 \\ s_1 &= 1, & s_2 &= 100, & s_3 &= 0.1, & s_4 &= 0.01 \end{aligned}$$

then (6.18) assumes the well-scaled form

$$\begin{aligned} 5x'_1 + 2x'_2 + 2x'_3 + 4x'_4 &= 120 \\ 3x'_1 + 4x'_2 + 5x'_3 + x'_4 &= 60 \\ 2x'_1 + x'_2 + 3x'_3 + 3x'_4 &= 90 \\ x'_1 + 5x'_2 + 4x'_3 + 2x'_4 &= 140. \end{aligned}$$

Incidentally, note the advantage of choosing powers of 10 for the scaling factors. Since we are working in the decimal notation, each evaluation of  $r_i a_{ij} s_j$  amounts to a mere shift of the decimal point. For the same reason, the scaling factors in machine computations (working in binary notation) are chosen to be powers of 2. Given any system (6.17), one can always find scaling factors that are powers of 10 and yield a system (6.18) that is nearly well-scaled or, more precisely, *equilibrated* in the sense that

$$0.1 < \max_j |r_i a_{ij} s_j| \leq 1 \quad \text{for all } i$$

and

$$0.1 < \max_i |r_i a_{ij} s_j| \leq 1 \quad \text{for all } j.$$

(In the binary system, powers of 10 would be replaced by powers of 2, and the lower bounds 0.1 would be replaced by 0.5.) The procedure is easy: first find scaling factors  $r_i$  such that

$$0.1 < \max_j |r_i a_{ij}| \leq 1$$

for all  $i$ , and then complete your choice by suitable values of  $s_j$ . For instance, this procedure applied to (6.16) yields first  $r_1 = 10^{-4}$ ,  $r_2 = 1$  and then  $s_1 = 1$ ,  $s_2 = 1$ . Next, the resulting system (6.13) is solved by Gaussian elimination with partial pivoting. As observed previously, the resulting computed solution is  $\tilde{x}_1 = 1$ ,  $\tilde{x}_2 = 1$ . Note that this result is satisfactory even from the point of view of backward error analysis:  $\tilde{x}_1, \tilde{x}_2$  constitute the precise solution of the system

$$\begin{aligned} \tilde{x}_1 + 9,999\tilde{x}_2 &= 10,000 \\ 0.5\tilde{x}_1 + 0.5\tilde{x}_2 &= 1 \end{aligned}$$

whose coefficients differ relatively little from those of (6.16).

Scaling is a common practice whose theoretical properties are not completely understood. There is often more than one way of scaling a system (6.17) into an equilibrated system (6.18), and it is not clear, a priori, which choice of scaling factors yields the best results. For instance, setting  $r_1 = 1$ ,  $r_2 = 1$ ,  $s_1 = 1$ ,  $s_2 = 10^{-4}$  in (6.16), we obtain the equilibrated system

$$\begin{aligned} x_1 + x'_2 &= 10,000 \\ 0.5x_1 + 0.00005x'_2 &= 1. \end{aligned}$$

Now Gaussian elimination with partial pivoting, working to three significant digits, delivers  $x'_2 = 10,000$  and  $x_1 = 0$ . Thus we obtain once again the unsatisfactory solution  $x_1 = 0$ ,  $x_2 = 1$ . Fortunately, examples such as this illustrate the exception rather than the rule. In practice, Gaussian elimination with partial pivoting, preceded by some form of scaling (such as equilibration), produces results whose numerical accuracy is quite adequate. For more information on this subject, the reader is referred to G. Forsythe and C. B. Moler (1967).

### Speed of Gaussian Elimination

Now we turn to an analysis of the speed of Gaussian elimination. As we have already observed, Gaussian elimination requires about  $n^3/3$  multiplications to solve a system

$$\sum_{j=1}^n a_{ij}x_j = b_i \quad (i = 1, 2, \dots, n). \quad (6.19)$$

However, it may happen that many of these  $n^3/3$  multiplications are multiplications by zero, in which case the number of genuine multiplications is considerably reduced. Typically (the atypical case is illustrated in problem 6.2), such reductions occur only if many of the coefficients  $a_{ij}$  in (6.19) are zeros. Systems with this property are called *sparse*, whereas systems with most of the coefficients  $a_{ij}$  nonzero are called *dense*. Large systems (6.19) encountered in practice are often very sparse: even when  $n$  runs to thousands or more, each variable  $x_j$  may occur with a nonzero coefficient  $a_{ij}$  in only five or ten equations. However, sparsity alone does not guarantee noticeable reductions in the running time of Gaussian elimination. For instance, consider the sparse system

$$\begin{aligned} x_1 + x_2 + x_3 + x_4 + x_5 + x_6 &= 4 \\ x_1 + 6x_2 &= 5 \\ x_1 + 6x_3 &= 5 \\ x_1 + 6x_4 &= 5 \\ x_1 + 6x_5 &= 5 \\ x_1 + 6x_6 &= 5. \end{aligned} \quad (6.20)$$

Eliminating  $x_1$  from the first equation, we obtain the completely dense system

$$\begin{aligned} 5x_2 - x_3 - x_4 - x_5 - x_6 &= 1 \\ -x_2 + 5x_3 - x_4 - x_5 - x_6 &= 1 \\ -x_2 - x_3 + 5x_4 - x_5 - x_6 &= 1 \\ -x_2 - x_3 - x_4 + 5x_5 - x_6 &= 1 \\ -x_2 - x_3 - x_4 - x_5 + 5x_6 &= 1 \end{aligned}$$

and no time savings can be claimed in the subsequent iterations of Gaussian elimination. This example illustrates a phenomenon known as *fill-in*: when a few variables have been eliminated, the remaining system may become considerably denser than the original one.

Attempts to curb fill-in rely on freedom in the choice of pivots. The variables  $x_1, x_2, \dots, x_n$  can be eliminated in an arbitrary order, and each can be eliminated from an arbitrary equation in which it appears with a nonzero coefficient; a judicious choice of the next variable to be eliminated, and of the equation from which to eliminate it, may reduce fill-in considerably. An early proposal along these lines was made by H. M. Markowitz (1957). It goes as follows: let  $p_i$  stand for the number of nonzero coefficients in the  $i$ th equation, and let  $q_j$  stand for the number of equations involving  $x_j$ . In the first iteration, when a variable  $x_j$  has to be eliminated from an  $i$ th equation, choose  $i$  and  $j$  so that the product  $(p_i - 1)(q_j - 1)$  is as small as possible (and, of course,  $a_{ij} \neq 0$ ). In each subsequent iteration, follow the same rule, with the  $p$ 's and  $q$ 's updated. For example, in (6.20) we have

$$p_1 = 6, \quad p_2 = 2, \quad p_3 = 2, \quad p_4 = 2, \quad p_5 = 2, \quad p_6 = 2$$

and

$$q_1 = 6, \quad q_2 = 2, \quad q_3 = 2, \quad q_4 = 2, \quad q_5 = 2, \quad q_6 = 2.$$

Since the product  $(p_i - 1)(q_j - 1)$  attains the minimum value 1 if and only if  $2 \leq i \leq 6$  and  $2 \leq j \leq 6$ , Markowitz's rule directs us to eliminate a variable  $x_j$  with  $2 \leq j \leq 6$  from the  $j$ th equation (we have to have  $a_{ij} \neq 0$ !). Breaking the tie arbitrarily, we eliminate  $x_2$  from the second equation,

$$x_2 = \frac{5}{6} - \frac{1}{6}x_1$$

and obtain the system

$$\begin{aligned} \frac{5}{6}x_1 + x_3 + x_4 + x_5 + x_6 &= \frac{19}{6} \\ x_1 + 6x_3 &= 5 \\ x_1 + 6x_4 &= 5 \\ x_1 + 6x_5 &= 5 \\ x_1 + 6x_6 &= 5. \end{aligned}$$

This is a significant improvement over the elimination of  $x_1$  from the first equation in (6.20); now we have created no new nonzero coefficients at all. (Furthermore, this phenomenon persists throughout the remaining iterations as long as the pivots are selected by Markowitz's rule.)

We shall return to ways of curbing fill-in by judicious choice of pivots later in this chapter, after Gaussian elimination has been described in terms of matrices.

## Summary of Elimination Methods

Gaussian elimination offers a great deal of flexibility in the choice of pivots; this flexibility may be exploited to enhance the accuracy as well as the speed of the method. When the choice of a pivot dictated by considerations of accuracy (partial pivoting) disagrees with a choice suggested by considerations of speed (fill-in reduction), it is usually the latter that gets preference. Considerations of accuracy take over only in extreme situations, such as when a choice of another equation (and the same variable) would increase the size of the pivot a thousand times or even more. (Of course, when Gaussian elimination is applied to completely dense systems, partial pivoting may be given full rein since all hopes of enhancing speed vanish.)

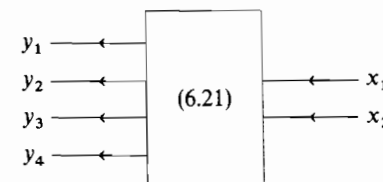
In many textbooks, Gaussian elimination is presented together with a variant, in which each  $x_k$  is eliminated not only from the remaining system of equations in  $x_{k+1}, x_{k+2}, \dots, x_n$  but also from the substitution formulas for  $x_1, x_2, \dots, x_{k-1}$ . This method appeared in a posthumous edition of a textbook written by a German geometer, W. Jordan, and became known as *Gauss-Jordan elimination*. It is inferior to Gaussian elimination on at least three counts: it requires  $n^3/2$  multiplications on completely dense systems, it is liable to suffer from more fill-in on sparse systems, and its numerical accuracy tends to be worse. For these reasons, we shall not discuss it.

## MATRIX MULTIPLICATION

The system of linear equations

$$\begin{aligned} y_1 &= 4x_1 + 3x_2 \\ y_2 &= -x_1 + 4x_2 \\ y_3 &= -3x_1 + 2x_2 \\ y_4 &= 2x_1 + 5x_2 \end{aligned} \tag{6.21}$$

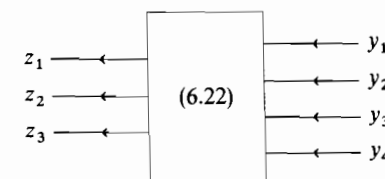
may be thought of as a device transforming inputs  $x_1, x_2$  into outputs  $y_1, y_2, y_3, y_4$ :



Similarly, the system

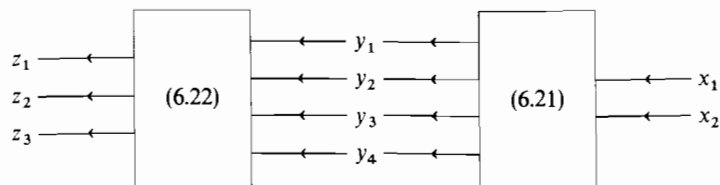
$$\begin{aligned} z_1 &= 5y_1 - 2y_2 + 3y_3 + y_4 \\ z_2 &= y_1 + 2y_2 - y_3 + 3y_4 \\ z_3 &= -y_1 - 3y_2 + 3y_3 + 2y_4 \end{aligned} \tag{6.22}$$

may be thought of as a device transforming inputs  $y_1, y_2, y_3, y_4$  into outputs  $z_1, z_2, z_3$ :





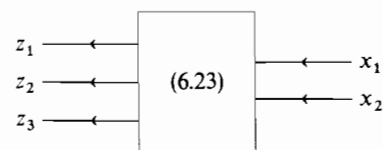
plugging the outputs of (6.21) into the inputs of (6.22),



amounts to substituting for  $y_1, y_2, y_3, y_4$  from (6.21) into (6.22): after simplification, we obtain

$$\begin{aligned} z_1 &= 15x_1 + 18x_2 \\ z_2 &= 11x_1 + 24x_2 \\ z_3 &= -6x_1 + x_2. \end{aligned} \quad (6.23)$$

Thus, the concatenation of the two devices (6.21) and (6.22) may be replaced by a single device:



More generally, consider a device that transforms inputs  $x_1, x_2, \dots, x_n$  into outputs

$$y_k = \sum_{j=1}^n b_{kj}x_j \quad (k = 1, 2, \dots, p) \quad (6.24)$$

and another device that transforms inputs  $y_1, y_2, \dots, y_p$  into outputs

$$z_i = \sum_{k=1}^p a_{ik}y_k \quad (i = 1, 2, \dots, m). \quad (6.25)$$

Plugging the outputs of the first device into the inputs of the second amounts to substituting for  $y_1, y_2, \dots, y_p$  from (6.24) into (6.25):

$$z_i = \sum_{k=1}^p a_{ik} \left( \sum_{j=1}^n b_{kj}x_j \right) = \sum_{j=1}^n \left( \sum_{k=1}^p a_{ik}b_{kj} \right) x_j.$$

Again, the concatenation of the two devices may be replaced by a single device

$$z_i = \sum_{j=1}^n c_{ij}x_j \quad (i = 1, 2, \dots, m)$$

defined by

$$c_{ij} = \sum_{k=1}^p a_{ik}b_{kj}. \quad (6.26)$$

It is customary to represent systems of linear functions by rectangular arrays of the corresponding coefficients. Thus,

$$\mathbf{A} = \begin{bmatrix} 5 & -2 & 3 & 1 \\ 1 & 2 & -1 & 3 \\ -1 & -3 & 3 & 2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 4 & 3 \\ -1 & 4 \\ -3 & 2 \\ 2 & 5 \end{bmatrix}, \quad \text{and} \quad \mathbf{C} = \begin{bmatrix} 15 & 18 \\ 11 & 24 \\ -6 & 1 \end{bmatrix}$$

represent (6.22), (6.21), and (6.23), respectively. Such arrays are called *matrices*. [The term *matrix* first appeared in a memoir published by A. Cayley in 1858. For historical details, see T. Hawkins (1974).]

A matrix with  $m$  rows and  $n$  columns is referred to as a “matrix of size  $m \times n$ ” or simply as an “ $m \times n$  matrix.” It is convenient to denote the entry in the  $i$ th row and in the  $j$ th column of a matrix  $\mathbf{A}$  by  $a_{ij}$ , the corresponding entry in a matrix  $\mathbf{B}$  by  $b_{ij}$ , and so on. This convention is often stressed in phrases such as “let  $\mathbf{A} = (a_{ij})$  be an  $m \times n$  matrix.” The *product*  $\mathbf{AB}$  of an  $m \times p$  matrix  $\mathbf{A}$  and a  $p \times n$  matrix  $\mathbf{B}$  is the  $m \times n$  matrix  $\mathbf{C}$  defined by (6.26). This definition is motivated by the fact that the matrix  $\mathbf{C} = \mathbf{AB}$  represents the device obtained by concatenating the devices represented by  $\mathbf{A}$  and  $\mathbf{B}$ . (Note that the product  $\mathbf{AB}$  is defined only if the number of rows in  $\mathbf{B}$  matches the number of columns in  $\mathbf{A}$ .) The terminology derives from the special case  $m = p = n = 1$ , when all three of the matrices  $\mathbf{A}$ ,  $\mathbf{B}$ , and  $\mathbf{AB}$  reduce to single numbers, and the number  $\mathbf{AB}$  is the ordinary product of the numbers  $\mathbf{A}$  and  $\mathbf{B}$ . However, the familiar properties of number multiplication do not always extend to the more general context of matrix multiplication. For example, if  $\mathbf{A}$  and  $\mathbf{B}$  are matrices, then we do not necessarily have  $\mathbf{AB} = \mathbf{BA}$ : to begin with, one or both of these products may be undefined; and even if both  $\mathbf{AB}$  and  $\mathbf{BA}$  are defined (which is the case if and only if  $\mathbf{A}$  and  $\mathbf{B}$  are square matrices of the same size), they may be not equal to each other. For instance, if

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

then

$$\mathbf{AB} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{BA} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

Nevertheless, just as  $(ab)c = a(bc)$  for any choice of numbers  $a, b, c$ , so we have

$$(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC}) \quad (6.27)$$

for any choice of matrices  $\mathbf{A}, \mathbf{B}, \mathbf{C}$  such that the products  $\mathbf{AB}$  and  $\mathbf{BC}$  are defined. This property of matrix multiplication is known as its *associativity*. In view of our motivation of matrix multiplication, (6.27) is intuitively obvious: either of its two sides represents the device obtained by concatenating the three devices represented by  $\mathbf{A}, \mathbf{B}, \mathbf{C}$ . Verifying (6.27) directly from the definition of matrix multiplication is easy; we omit the routine details. Once (6.27) is established, we may delete the parentheses on either side and write simply  $\mathbf{ABC}$  without any ambiguity.

In number multiplication, the number “one” plays a special part: we have  $1 \cdot x = x \cdot 1 = x$  for all numbers  $x$ . In matrix multiplication, this part is played by so-called *identity matrices*, square matrices with ones on the diagonal and zeros off the diagonal. For instance, we have

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 5 & -2 & 3 & 1 \\ 1 & 2 & -1 & 3 \\ -1 & -3 & 4 & 2 \end{bmatrix} = \begin{bmatrix} 5 & -2 & 3 & 1 \\ 1 & 2 & -1 & 3 \\ -1 & -3 & 4 & 2 \end{bmatrix}$$

and

$$\begin{bmatrix} 5 & -2 & 3 & 1 \\ 1 & 2 & -1 & 3 \\ -1 & -3 & 4 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & -2 & 3 & 1 \\ 1 & 2 & -1 & 3 \\ -1 & -3 & 4 & 2 \end{bmatrix}.$$



More generally, with  $I_k$  standing for the identity matrix of size  $k \times k$ , we have

$$I_m A = A I_n = A$$

for every  $m \times n$  matrix  $A$ . This fact, apart from being easy to verify mechanically, is also intuitively obvious: the identity matrix  $I_k$  represents a transformation device that transfers the  $k$  inputs directly into the  $k$  outputs without a change. When there is no danger of confusion, we shall sometimes denote the identity matrix  $I_k$  simply by  $I$ .

A matrix having only one row is called a *row vector*, and a matrix having only one column is called a *column vector*. The entries in a (row or column) vector are referred to as its *components*, and their number is the vector's *length*. Note that an  $m \times n$  matrix may be multiplied by a column vector of length  $n$  on the right, and by a row vector of length  $m$  on the left. (The following observation concerning matrix products  $AB$  is often handy: if  $\mathbf{a}$  is the  $i$ th row of  $A$  and if  $\mathbf{b}$  is the  $j$ th column of  $B$ , then  $\mathbf{aB}$  is the  $i$ th row of  $AB$  and  $\mathbf{Ab}$  is the  $j$ th column of  $AB$ .)

The formalism of matrices provides a succinct way of recording systems of linear equations. For instance, the system

$$\begin{aligned} 4x_1 + 3x_2 + 9x_3 &= 3 \\ 2x_1 - x_2 + 2x_3 &= 1 \\ 8x_1 + 7x_2 - 5x_3 &= 4 \end{aligned}$$

may be recorded as  $A\mathbf{x} = \mathbf{b}$  with

$$A = \begin{bmatrix} 4 & 3 & 9 \\ 2 & -1 & 2 \\ 8 & 7 & -5 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \text{and } \mathbf{b} = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}.$$

More generally, every system of  $n$  linear equations in  $n$  variables may be recorded as  $A\mathbf{x} = \mathbf{b}$  such that  $A$  is an  $n \times n$  matrix and  $\mathbf{x}, \mathbf{b}$  are column vectors of length  $n$ .

SPARSE AND SPECIALLY STRUCTURED MATRICES

As we have already observed, matrices  $A$  appearing in large systems  $A\mathbf{x} = \mathbf{b}$  found in practice are *sparse*: most of their entries  $a_{ij}$  are zeros. Such matrices may be multiplied fast and stored compactly. For instance, the matrix

$$\begin{bmatrix} 3 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 5 & 0 & 0 & 6 & 0 & 0 & 0 \\ 0 & 0 & 0 & 8 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 9 & 0 & 4 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 5 & 6 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 4 & 0 & 0 & 7 \\ 0 & 0 & 0 & 0 & 0 & 0 & 6 & 5 & 0 & 7 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 6 & 0 & 7 & 0 \end{bmatrix}$$

may be stored in the following "packed form":

|            |                                                              |
|------------|--------------------------------------------------------------|
| column 1:  | 3 in row 1.                                                  |
| column 2:  | 2 in row 1, 9 in row 4.                                      |
| column 3:  | 2 in row 2.                                                  |
| column 4:  | 5 in row 2, 8 in row 3, 4 in row 4.                          |
| column 5:  | 2 in row 3, 5 in row 5.                                      |
| column 6:  | 6 in row 5, 2 in row 6.                                      |
| column 7:  | 6 in row 2, 4 in row 5, 4 in row 7, 6 in row 8, 6 in row 10. |
| column 8:  | 5 in row 8, 2 in row 9.                                      |
| column 9:  | 3 in row 9, 7 in row 10.                                     |
| column 10: | 7 in row 7, 7 in row 8.                                      |

The actual technical details involving memory locations and so on would lead us astray; the interested reader is referred to R. P. Tewarson (1973) and J. de Buchet (1971).

We shall sometimes emphasize sparsity by using a suggestive notation. For example, the sparse matrix shown previously might be recorded as

$$\begin{bmatrix} 3 & 2 & & & & & & & & \\ & & 2 & 5 & & & 6 & & & \\ & & & 8 & 2 & & & & & \\ & 9 & & 4 & & & & & & \\ & & & & 5 & 6 & 4 & & & \\ & & & & & 2 & & & & \\ & & & & & & 4 & & & 7 \\ & & & & & & 6 & 5 & & 7 \\ & & & & & & & 2 & 3 & \\ & & & & & & 6 & & 7 & \end{bmatrix}$$

and, more generally, some or all of the zero entries omitted in a presentation of a sparse matrix.

Three special kinds of matrices will play an important part in our discussions: permutation matrices, eta matrices, and triangular matrices. A *permutation matrix*, such as

$$\begin{bmatrix} & & 1 \\ 1 & & \\ & 1 & \end{bmatrix}$$

arises from the identity matrix by permuting its rows and/or columns. To put it differently, a permutation matrix has precisely one nonzero entry in each row and precisely one nonzero entry in each column; all of these nonzero entries are ones. An *eta matrix*, such as

$$\begin{bmatrix} 1 & 3 & \\ & 1 & 5 \\ & & 4 \\ & 2 & 1 \end{bmatrix}$$

differs from the identity matrix in only one column, referred to as its *eta column*. A matrix  $A = (a_{ij})$  is called *lower triangular* if  $a_{ij} = 0$  whenever  $j > i$  and *upper triangular* if  $a_{ij} = 0$  whenever  $j < i$ . Thus,

$$\begin{bmatrix} 3 & & & \\ 5 & 6 & & \\ 0 & 9 & 8 & \\ 1 & 7 & 1 & 2 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 4 & 1 & 2 & 0 \\ & 3 & 6 & 4 \\ & & 1 & 9 \\ & & & 8 \end{bmatrix}$$

are lower and upper triangular, respectively.

## THE ALGEBRA OF MATRICES

Matrices can be not only multiplied; they can also be added. If  $\mathbf{A} = (a_{ij})$  and  $\mathbf{B} = (b_{ij})$  are matrices of the same size, then their *sum*  $\mathbf{A} + \mathbf{B}$  is the matrix  $\mathbf{C} = (c_{ij})$ , defined by  $c_{ij} = a_{ij} + b_{ij}$ . This operation generalizes ordinary number addition and obeys several familiar laws of ordinary number arithmetic. For instance,

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$$

$$\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C}$$

and

$$(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}$$

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$$

whenever either side is defined. Finally, matrices may be multiplied by ordinary numbers: if  $t$  is a number and if  $\mathbf{A} = (a_{ij})$  is a matrix, then  $t\mathbf{A}$  denotes the matrix  $\mathbf{B} = (b_{ij})$  defined by  $b_{ij} = ta_{ij}$ . Again, we have

$$t(\mathbf{A} + \mathbf{B}) = t\mathbf{A} + t\mathbf{B}$$

$$(s + t)\mathbf{A} = s\mathbf{A} + t\mathbf{A}$$

and so on.

For every  $m \times n$  matrix  $\mathbf{A} = (a_{ij})$ , the  $n \times m$  matrix  $(a_{ji})$  is called the *transpose* of  $\mathbf{A}$  and is denoted by  $\mathbf{A}^T$ . In particular, the transpose of every row vector is a column vector and vice versa. Relying on this convention, we shall often record the column vector with components  $x_1, x_2, \dots, x_n$  as  $[x_1, x_2, \dots, x_n]^T$ .

## GAUSSIAN ELIMINATION IN TERMS OF MATRICES

Suppose that we have eliminated  $x_1, x_2, x_3$  from a system of six linear equations in  $x_1, x_2, \dots, x_6$ . Now we are left with a system of three linear equations in  $x_4, x_5, x_6$  and with back substitution formulas for  $x_1, x_2, x_3$ . For definiteness, let us write the formulas as

$$\begin{aligned} x_1 &= d_1 - c_{12}x_2 - c_{13}x_3 - c_{14}x_4 - c_{15}x_5 - c_{16}x_6 \\ x_2 &= d_2 - c_{23}x_3 - c_{24}x_4 - c_{25}x_5 - c_{26}x_6 \\ x_3 &= d_3 - c_{34}x_4 - c_{35}x_5 - c_{36}x_6 \end{aligned} \quad (6.28)$$

and the remaining system as

$$\begin{aligned} c_{44}x_4 + c_{45}x_5 + c_{46}x_6 &= d_4 \\ c_{54}x_4 + c_{55}x_5 + c_{56}x_6 &= d_5 \\ c_{64}x_4 + c_{65}x_5 + c_{66}x_6 &= d_6. \end{aligned} \quad (6.29)$$

Now we may convert the first of the last three equations into

$$x_4 = \frac{d_4}{c_{44}} - \frac{c_{45}}{c_{44}}x_5 - \frac{c_{46}}{c_{44}}x_6 \quad (6.30)$$

and, substituting from (6.30) into (6.29), eliminate  $x_4$ :

$$\begin{aligned} \left(c_{55} - c_{54} \frac{c_{45}}{c_{44}}\right)x_5 + \left(c_{56} - c_{54} \frac{c_{46}}{c_{44}}\right)x_6 &= d_5 - c_{54} \frac{d_4}{c_{44}} \\ \left(c_{65} - c_{64} \frac{c_{45}}{c_{44}}\right)x_5 + \left(c_{66} - c_{64} \frac{c_{46}}{c_{44}}\right)x_6 &= d_6 - c_{64} \frac{d_4}{c_{44}}. \end{aligned} \quad (6.31)$$

In matrix notation, the intermediate results (6.28), (6.29) *before* the elimination of  $x_4$  may be recorded as

$$\begin{bmatrix} 1 & c_{12} & c_{13} & c_{14} & c_{15} & c_{16} \\ & 1 & c_{23} & c_{24} & c_{25} & c_{26} \\ & & 1 & c_{34} & c_{35} & c_{36} \\ & & & c_{44} & c_{45} & c_{46} \\ & & & c_{54} & c_{55} & c_{56} \\ & & & c_{64} & c_{65} & c_{66} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \\ d_5 \\ d_6 \end{bmatrix} \quad (6.32)$$

whereas the intermediate results (6.28), (6.30), (6.31) *after* the elimination of  $x_4$  may be recorded as

$$\begin{bmatrix} 1 & c_{12} & c_{13} & c_{14} & c_{15} & c_{16} \\ & 1 & c_{23} & c_{24} & c_{25} & c_{26} \\ & & 1 & c_{34} & c_{35} & c_{36} \\ & & & 1 & \frac{c_{45}}{c_{44}} & \frac{c_{46}}{c_{44}} \\ & & & & c_{55} - c_{54} \frac{c_{45}}{c_{44}} & c_{56} - c_{54} \frac{c_{46}}{c_{44}} \\ & & & & c_{65} - c_{64} \frac{c_{45}}{c_{44}} & c_{66} - c_{64} \frac{c_{46}}{c_{44}} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ \frac{d_4}{c_{44}} \\ d_5 - c_{54} \frac{d_4}{c_{44}} \\ d_6 - c_{64} \frac{d_4}{c_{44}} \end{bmatrix} \quad (6.33)$$

Note that the transformation of (6.32) into (6.33) amounts to multiplying both sides of (6.32) by the matrix

$$\mathbf{L} = \begin{bmatrix} 1 & & & & & \\ & 1 & & & & \\ & & 1 & & & \\ & & & \frac{1}{c_{44}} & & \\ & & & -\frac{c_{54}}{c_{44}} & 1 & \\ & & & -\frac{c_{64}}{c_{44}} & & 1 \end{bmatrix}$$

on the left. Of course, if  $c_{44} = 0$ , then this transformation cannot be carried out and we are forced to use a different row of (6.29) in order to eliminate  $x_4$ . For instance, if we use the last row of (6.29), then we obtain

$$x_4 = \frac{d_6}{c_{64}} - \frac{c_{65}}{c_{64}}x_5 - \frac{c_{66}}{c_{64}}x_6$$

and

$$\begin{aligned} \left(c_{55} - c_{54}\frac{c_{65}}{c_{64}}\right)x_5 + \left(c_{56} - c_{54}\frac{c_{66}}{c_{64}}\right)x_6 &= d_5 - c_{54}\frac{d_6}{c_{64}} \\ \left(c_{45} - c_{44}\frac{c_{65}}{c_{64}}\right)x_5 + \left(c_{46} - c_{44}\frac{c_{66}}{c_{64}}\right)x_6 &= d_4 - c_{44}\frac{d_6}{c_{64}}. \end{aligned}$$

In matrix terms again, this operation may be viewed as passing from (6.32) to

$$\begin{bmatrix} 1 & c_{12} & c_{13} & c_{14} & c_{15} & c_{16} \\ & 1 & c_{23} & c_{24} & c_{25} & c_{26} \\ & & 1 & c_{34} & c_{35} & c_{36} \\ & & & c_{64} & c_{65} & c_{66} \\ & & & c_{54} & c_{55} & c_{56} \\ & & & c_{44} & c_{45} & c_{46} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ d_6 \\ d_5 \\ d_4 \end{bmatrix} \quad (6.34)$$

and then to

$$\begin{bmatrix} 1 & c_{12} & c_{13} & c_{14} & c_{15} & c_{16} \\ & 1 & c_{23} & c_{24} & c_{25} & c_{26} \\ & & 1 & c_{34} & c_{35} & c_{36} \\ & & & 1 & \frac{c_{65}}{c_{64}} & \frac{c_{66}}{c_{64}} \\ & & & & c_{55} - c_{54}\frac{c_{65}}{c_{64}} & c_{56} - c_{54}\frac{c_{66}}{c_{64}} \\ & & & & c_{45} - c_{44}\frac{c_{65}}{c_{64}} & c_{46} - c_{44}\frac{c_{66}}{c_{64}} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ \frac{d_6}{c_{64}} \\ d_5 - c_{54}\frac{d_6}{c_{64}} \\ d_4 - c_{44}\frac{d_6}{c_{64}} \end{bmatrix} \quad (6.35)$$

Note that the passage from (6.32) to (6.34) amounts to multiplying both sides of (6.32) by the matrix

$$P = \begin{bmatrix} 1 & & & & & \\ & 1 & & & & \\ & & 1 & & & \\ & & & 1 & & \\ & & & & 1 & \\ & & & & & 1 \end{bmatrix}$$

on the left and that the passage from (6.34) to (6.35) amounts to multiplying both sides of (6.34) by the matrix

$$L = \begin{bmatrix} 1 & & & & & \\ & 1 & & & & \\ & & 1 & & & \\ & & & \frac{1}{c_{64}} & & \\ & & & -\frac{c_{54}}{c_{64}} & 1 & \\ & & & -\frac{c_{44}}{c_{64}} & & 1 \end{bmatrix}$$

on the left.

More generally, having eliminated  $x_1, x_2, \dots, x_{k-1}$  in this order from a system of  $n$  linear equations in  $x_1, x_2, \dots, x_n$ , we may record our intermediate results in matrix notation as

$$Cx = d. \quad (6.36)$$

The first  $k-1$  rows of this system provide substitution formulas for  $x_1, x_2, \dots, x_{k-1}$ , whereas the last  $n-k+1$  rows constitute the residual system of equations in  $x_k, x_{k+1}, \dots, x_n$ . Hence, the matrix  $C = (c_{ij})$  has a rather special structure. In its first  $k-1$  columns, all the entries on the diagonal are ones, whereas all the entries below the diagonal are zeros. The subsequent elimination of  $x_k$  yields a similar system,

$$\bar{C}x = \bar{d} \quad (6.37)$$

in the first  $k$  columns of  $\bar{C}$ , all the entries on the diagonal are ones whereas all the entries below the diagonal are zeros. As in our example, the transformation of (6.36) into (6.37) amounts to multiplying both sides of (6.36) on the left first by a permutation matrix  $P = P_k$  and then by a lower triangular eta matrix  $L = L_k$ . The permutation matrix  $P$  is obtained from the identity matrix by interchanging its row  $k$  with some row  $r$  such that  $r \geq k$  and  $c_{rk} \neq 0$ . (We shall comment later on the case  $c_{ik} = 0$  for all  $i = k, k+1, \dots, n$ .) Thus, the matrix  $T = PC$  arises from  $C$  by interchanging its rows  $k$  and  $r$ . In particular, writing  $T = (t_{ij})$  we have  $t_{kk} = c_{rk} \neq 0$ . The lower triangular eta matrix  $L_k$  differs from the identity matrix only in its  $k$ th column whose entries  $l_{ik}$  are defined by

$$l_{ik} = \begin{cases} 0 & \text{if } i < k \\ \frac{1}{t_{kk}} & \text{if } i = k \\ -\frac{t_{ik}}{t_{kk}} & \text{if } i > k \end{cases}$$

Thus, the new system (6.37) reads  $L_k P_k Cx = L_k P_k d$ . The elimination process applied to a system  $Ax = b$  may be seen as moving through a sequence of systems

$$\begin{aligned} Ax &= b \\ L_1 P_1 Ax &= L_1 P_1 b \\ L_2 P_2 L_1 P_1 Ax &= L_2 P_2 L_1 P_1 b \\ &\vdots \\ L_n P_n L_{n-1} P_{n-1} \cdots L_1 P_1 Ax &= L_n P_n L_{n-1} P_{n-1} \cdots L_1 P_1 b. \end{aligned}$$

Since the last system in this sequence consists entirely of substitution formulas, the matrix

$$U = L_n P_n L_{n-1} P_{n-1} \cdots L_1 P_1 A$$

is upper triangular and all of its diagonal entries are ones.

### Triangular Factorization

In Gaussian elimination, as described at the beginning of this chapter, the matrices  $P_1, L_1, \dots, P_n, L_n$  are not stored explicitly; instead,

$$C = L_{k-1} P_{k-1} \cdots L_1 P_1 A \quad \text{and} \quad d = L_{k-1} P_{k-1} \cdots L_1 P_1 b$$

are updated into

$$\bar{C} = L_k P_k C \quad \text{and} \quad \bar{d} = L_k P_k d$$

in each iteration. Nevertheless, the description given just now suggests a way of computing  $P_1, L_1, \dots, P_n, L_n$  and  $U = L_n P_n \cdots L_1 P_1 A$  explicitly and independently of the right-hand side  $b$  in the original system. That is, in each iteration, having just computed  $P_k$  and  $L_k$ , we merely have to update  $C$  into  $\bar{C} = L_k P_k C$ . Thus the total amount of work involved in computing  $P_1, L_1, \dots, P_n, L_n$  and  $U$  is roughly equal to the total amount of work involved in solving

$$Ax = b \tag{6.38}$$

by Gaussian elimination. Since  $U$  and the eta matrices  $L_1, L_2, \dots, L_n$  are triangular, we shall refer to the matrices  $P_1, L_1, \dots, P_n, L_n$  and  $U$  as a *triangular factorization* of  $A$ . (Strictly speaking, this term is a misnomer, as it is  $U$  rather than  $A$  that is being factorized. Nevertheless, we shall let it stand. A closely related concept is the *LU-decomposition* of  $A$ ; see problem 6.11.)

Once the triangular factorization of  $A$  is available, system (6.38) may be solved fast for an arbitrary right-hand side  $b$ . We first compute

$$v = L_n P_n L_{n-1} P_{n-1} \cdots L_1 P_1 b$$

and then solve the system

$$Ux = v.$$

To see that this procedure yields the correct solution  $x$  of (6.38), recall that  $Ux = v$  is precisely the set of substitution formulas that would have been obtained by Gaussian elimination. In particular, as  $U = (u_{ij})$  is an upper triangular matrix with  $u_{ii} = 1$  for all  $i$ , solving the system  $Ux = v$  amounts to a simple back substitution. We obtain the values of  $x_n, x_{n-1}, \dots, x_1$  in this order by setting

$$x_i = v_i - \sum_{j=i+1}^n u_{ij} x_j$$

in each iteration. Thus, once  $v$  is known, the total amount of work involved in solving  $Ux = v$  comes to only  $n(n-1)/2$  multiplications and the same number of additions; if the matrix  $U$  is sparse, then this number reduces even further. Similarly, the long chain calculation of  $v$  may be implemented quickly: we first compute  $P_1 b$ , then  $L_1(P_1 b)$ , then  $P_2(L_1 P_1 b)$ , then  $L_2(P_2 L_1 P_1 b)$ , and so on. Each multiplication of a  $P_k$  by a column vector on the right is done without any arithmetic; each multiplication of an  $L_k$  by a column vector on the right takes only  $n-k+1$  multiplications and  $n-k$  additions. Thus, the total amount of work involved in computing  $v$  comes to only  $n(n+1)/2$  multiplications and  $n(n-1)/2$  additions; again, if the  $k$ th column of each  $L_k$  is very sparse, then these numbers are reduced much further.

The practical impact of these observations is clear: when faced with the task of solving several different systems

$$Ax = b, \quad Ax = b', \quad Ax = b'', \quad \dots \tag{6.39}$$

with the same left-hand side  $Ax$ , one is well-advised to first “preprocess” the matrix  $A$  by finding its triangular factorization. The preprocessing step takes about as much work as solving just one of the systems (6.39) by Gaussian elimination: about  $n^3/3$  multiplications and  $n^3/3$  additions when  $A$  is completely dense and possibly much less when  $A$  is sparse. As soon as this step is executed, each of the systems (6.39) may be solved relatively quickly: the number of multiplications and the number of additions are roughly equal to the total number of nonzero entries in the matrix  $U$  and in the eta columns of  $L_1, L_2, \dots, L_n$ . Even in the completely dense case,  $n^2$  multiplications and  $n^2 - n$  additions suffice.

Similarly, every system  $yA = c$ , with  $y$  and  $c$  standing for row rather than column vectors, may be solved by first solving the triangular system  $wU = c$  and then evaluating  $y = wL_n P_n \cdots L_1 P_1$ ; now  $yA = wU = c$ . Again, no more than  $n^2/2$  multiplications and additions are required in solving  $wU = c$  and roughly the same number of operations suffice to evaluate  $y = ((wL_n)P_n) \cdots L_1 P_1$ .

### Computing Accurate and Sparse Triangular Factorizations

The tricks used to enhance the accuracy and speed of Gaussian elimination in solving systems of linear equations apply just as well in computing triangular factorizations. In particular, partial pivoting amounts to a judicious choice of the permutation matrix  $P_k$  in each iteration, so that each off-diagonal entry  $-t_{ik}/t_{kk}$  in the eta column of  $L_k$  has a size of at most one. A more detailed explanation is called for in the case of tricks such as Markowitz’s pivot selection, which change the order in which the variables  $x_1, x_2, \dots, x_n$  are eliminated.

To begin, note that the sequence of pivots selected by Markowitz’s strategy may be determined without carrying out the numerical computations. For an illustration, consider the system

$$\begin{aligned} a_{11}x_1 & & & & & + a_{16}x_6 & = b_1 \\ a_{21}x_1 + a_{22}x_2 & & & + a_{24}x_4 + a_{25}x_5 + a_{26}x_6 & = b_2 \\ a_{31}x_1 & & + a_{33}x_3 & & & & = b_3 \\ a_{41}x_1 & & + a_{43}x_3 + a_{44}x_4 + a_{45}x_5 & & & & = b_4 \\ & & & & a_{54}x_4 + a_{55}x_5 & & = b_5 \\ a_{62}x_2 + a_{63}x_3 + a_{64}x_4 & & & & + a_{66}x_6 & = b_6. \end{aligned} \tag{6.40}$$

Markowitz’s selection rule is independent of the actual values of the coefficients  $a_{ij}$  shown in (6.40): as long as all of these coefficients are nonzero, the rule tells us to choose  $a_{16}, a_{33}$  or  $a_{55}$  [all of them having  $(p_i - 1)(q_j - 1) = 2$ ] for the first pivot. Breaking the tie arbitrarily, we choose  $a_{16}$ . Unless lucky cancellations (such as those illustrated in problem 6.2) occur, the system that results from the elimination of  $x_6$  from the first equation in (6.40) will read

$$\begin{aligned} a'_{21}x_1 + a'_{22}x_2 & & + a'_{24}x_4 + a'_{25}x_5 & = b'_2 \\ a'_{31}x_1 & & + a'_{33}x_3 & = b'_3 \\ a'_{41}x_1 & & + a'_{43}x_3 + a'_{44}x_4 + a'_{45}x_5 & = b'_4 \\ & & & a'_{54}x_4 + a'_{55}x_5 & = b'_5 \\ a'_{61}x_1 + a'_{62}x_2 + a'_{63}x_3 + a'_{64}x_4 & & & = b'_6 \end{aligned}$$

(with the nonzero coefficient  $a'_{61} = -a_{11}a_{66}/a_{16}$  created by fill-in). In the second iteration, Markowitz’s rule suggests  $a'_{33}$  or  $a'_{55}$  [both of them having  $(p_i - 1)(q_j - 1) = 2$  again] as the

pivot; breaking the tie arbitrarily, we choose  $a'_{33}$ . Again, the resulting system of equations may be recorded symbolically, the next pivot chosen, and so on. As the reader may verify, a continuation of this process may lead to the sequence of pivots

$$a_{16}, a_{33}, a_{55}, a_{22}, a_{41}, a_{64} \quad (6.41)$$

(where we write  $a_{33}$  for  $a'_{33}$  and so on).

Gaussian elimination with a prescribed sequence of pivots yields itself to a neat description in terms of matrices. The trick is to permute the variables and the equations of the system  $\mathbf{Ax} = \mathbf{b}$  in such a way that, in the resulting system  $\tilde{\mathbf{A}}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$ , the prescribed pivots appear on the diagonal in the prescribed order. We leave it to the reader to verify (problem 6.6) that

$$\tilde{\mathbf{A}} = \mathbf{PAQ}, \quad \tilde{\mathbf{x}} = \mathbf{Q}^T \mathbf{x}, \quad \tilde{\mathbf{b}} = \mathbf{Pb}$$

with permutation matrices  $\mathbf{P}$  and  $\mathbf{Q}$  determined by the prescribed sequence of pivots: if the  $k$ th pivot in this sequence is  $a_{ij}$ , then the  $k$ th row of  $\mathbf{P}$  is the  $i$ th row of the identity matrix  $\mathbf{I}$  and the  $k$ th column of  $\mathbf{Q}$  is the  $j$ th column of  $\mathbf{I}$ . For instance, if  $\mathbf{Ax} = \mathbf{b}$  is (6.40), and the prescribed sequence of pivots is (6.41), then  $\tilde{\mathbf{A}}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$  reads

$$\begin{aligned} a_{16}x_6 & & + a_{11}x_1 & = b_1 \\ & a_{33}x_3 & + a_{31}x_1 & = b_3 \\ & & a_{55}x_5 & + a_{54}x_4 = b_5 \\ a_{26}x_6 & + a_{25}x_5 & + a_{22}x_2 & + a_{21}x_1 + a_{24}x_4 = b_2 \\ & a_{43}x_3 + a_{45}x_5 & & + a_{41}x_1 + a_{44}x_4 = b_4 \\ a_{66}x_6 + a_{63}x_3 & & + a_{62}x_2 & + a_{64}x_4 = b_6 \end{aligned}$$

and we have

$$\mathbf{P} = \begin{bmatrix} 1 & & & & & \\ & 1 & & & & \\ & & 1 & & & \\ 1 & & & 1 & & \\ & & & & 1 & \\ & & & & & 1 \end{bmatrix}, \quad \mathbf{Q} = \begin{bmatrix} & & & & 1 & \\ & & & 1 & & \\ & 1 & & & & \\ & & & & & \\ & & 1 & & & \\ 1 & & & & & \end{bmatrix}$$

Now Gaussian elimination applied to  $\mathbf{Ax} = \mathbf{b}$  with the prescribed sequence of pivots is nothing but Gaussian elimination applied to  $\tilde{\mathbf{A}}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$  with the first variable eliminated from the first equation in the first iteration, the second variable eliminated from the second equation in the second iteration, and so on. Thus, it transforms  $\tilde{\mathbf{A}}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$  first into  $\tilde{\mathbf{L}}_1\tilde{\mathbf{A}}\tilde{\mathbf{x}} = \tilde{\mathbf{L}}_1\tilde{\mathbf{b}}$ , then into  $\tilde{\mathbf{L}}_2\tilde{\mathbf{L}}_1\tilde{\mathbf{A}}\tilde{\mathbf{x}} = \tilde{\mathbf{L}}_2\tilde{\mathbf{L}}_1\tilde{\mathbf{b}}$ , and so on until an upper triangular system

$$\tilde{\mathbf{L}}_n\tilde{\mathbf{L}}_{n-1}\cdots\tilde{\mathbf{L}}_1\tilde{\mathbf{A}}\tilde{\mathbf{x}} = \tilde{\mathbf{L}}_n\tilde{\mathbf{L}}_{n-1}\cdots\tilde{\mathbf{L}}_1\tilde{\mathbf{b}}$$

is obtained.

When fill-in in Gaussian elimination is curbed by a suitable choice of a pivot sequence, the resulting triangular factorization

$$\tilde{\mathbf{L}}_n\tilde{\mathbf{L}}_{n-1}\cdots\tilde{\mathbf{L}}_1\tilde{\mathbf{A}} = \tilde{\mathbf{U}}$$

is sparser than the triangular factorization

$$\mathbf{L}_n\mathbf{L}_{n-1}\cdots\mathbf{L}_1\mathbf{A} = \mathbf{U}.$$

(This is the case in our example; as the reader may verify by symbolic computations, fill-in creates four extra nonzeros in the triangular factorization of  $\mathbf{A}$  and only one in the triangular factorization of  $\tilde{\mathbf{A}}$ .) Thus Markowitz's pivot selection may be viewed as producing permutation matrices  $\mathbf{P}$  and  $\mathbf{Q}$  with the aim of reducing the number of nonzeros in the triangular factorization of  $\tilde{\mathbf{A}} = \mathbf{PAQ}$ .

Why is it important to curb the number of nonzeros in the triangular factorization of  $\tilde{\mathbf{A}} = \mathbf{PAQ}$ ? Sparse triangular factorizations are computed faster than dense ones, but that is not the main point. As we have remarked, every system  $\mathbf{Ax} = \mathbf{b}$  may be permuted into  $\tilde{\mathbf{A}}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$  with  $\tilde{\mathbf{x}} = \mathbf{Q}^T \mathbf{x}$  and  $\tilde{\mathbf{b}} = \mathbf{Pb}$ ; similarly (problem 6.6 again), every system  $\mathbf{yA} = \mathbf{c}$  may be permuted into  $\tilde{\mathbf{y}}\tilde{\mathbf{A}} = \tilde{\mathbf{c}}$ , with  $\tilde{\mathbf{y}} = \mathbf{yP}^T$  and  $\tilde{\mathbf{c}} = \mathbf{cQ}$ . Hence,  $\mathbf{Ax} = \mathbf{b}$  may be solved by first permuting  $\mathbf{b}$  into  $\tilde{\mathbf{b}}$ , then solving  $\tilde{\mathbf{A}}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$ , and finally permuting  $\tilde{\mathbf{x}}$  into  $\mathbf{x}$ ; similarly,  $\mathbf{yA} = \mathbf{c}$  may be solved by first permuting  $\mathbf{c}$  into  $\tilde{\mathbf{c}}$ , then solving  $\tilde{\mathbf{y}}\tilde{\mathbf{A}} = \tilde{\mathbf{c}}$ , and finally permuting  $\tilde{\mathbf{y}}$  into  $\mathbf{y}$ . In either case, this roundabout procedure requires only one multiplication and one addition per each nonzero in the triangular factorization of  $\tilde{\mathbf{A}}$  (not counting the diagonal ones in  $\tilde{\mathbf{L}}_1, \tilde{\mathbf{L}}_2, \dots, \tilde{\mathbf{L}}_n$ ). Thus, sparser triangular factorizations of  $\tilde{\mathbf{A}}$  allow faster solutions of systems  $\mathbf{Ax} = \mathbf{b}$  and  $\mathbf{yA} = \mathbf{c}$ .

The early work of Markowitz was followed by intensive research into reducing the number of nonzeros in the triangular factorization of  $\mathbf{PAQ}$  by an appropriate choice of permutation matrices  $\mathbf{P}$  and  $\mathbf{Q}$ . Reducing this number to its absolute minimum seems difficult [in fact, D. J. Rose and R. E. Tarjan (1978) proved that it is difficult when  $\mathbf{Q}$  is required to be  $\mathbf{P}^T$ ]. Nevertheless, algorithms for finding a good (although not necessarily the best) choice of  $\mathbf{P}$  and  $\mathbf{Q}$  may be guided by simple observations on the nature of fill-in in computing the triangular factorization

$$\tilde{\mathbf{L}}_n\tilde{\mathbf{L}}_{n-1}\cdots\tilde{\mathbf{L}}_1\tilde{\mathbf{A}} = \tilde{\mathbf{U}}. \quad (6.42)$$

For instance (problem 6.7), if the  $k$ th column of  $\tilde{\mathbf{A}}$  has no nonzeros above the diagonal, then the pattern of nonzeros in this column is carried intact into the  $k$ th column of  $\tilde{\mathbf{L}}_k$  and, in addition, the  $k$ th column of  $\tilde{\mathbf{U}}$  has no nonzeros at all except the diagonal one. To put it differently, fill-in may occur only in those columns where  $\tilde{\mathbf{A}}$  has nonzero above the diagonal; such columns are called *spikes*.

A particularly successful procedure, designed by E. Hellerman and D. C. Rarick (1971, 1972), aims to permute  $\mathbf{A}$  into an  $\tilde{\mathbf{A}} = \mathbf{PAQ}$ , with the nonzeros above the diagonal confined to small "bumps", as illustrated in Figure 6.1, and with only a few spikes in each bump. On sparse

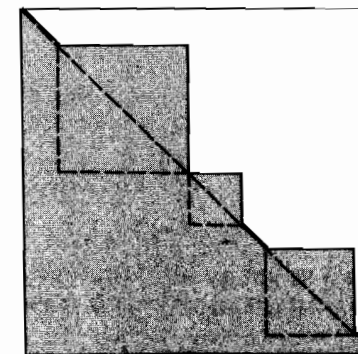


Figure 6.1

matrices  $A$  that are encountered in practice, typically with up to ten nonzeros per column, this procedure yields excellent results. Usually, the number of nonzeros in the triangular factorization of  $\tilde{A}$  (discounting, as usual, the diagonal ones in  $\tilde{L}_1, \tilde{L}_2, \dots, \tilde{L}_n$ ) is within 120% of the number of nonzeros in  $A$ , and the total number of spikes is less than a hundred or so even when  $A$  has several thousand columns. (Of course, these empirical findings hinge on the special structure of sparse matrices encountered in practice, which allows these matrices to be permuted into the near-triangular form. On randomly generated sparse matrices, even the most sophisticated procedures would fail miserably: since the nonzeros in such matrices  $A$  form a patternless mosaic, nearly all columns in any  $PAQ$  are spikes and fill-in is catastrophic.)

The potential conflict between accuracy and speed, discussed earlier in the context of solving systems of linear equations, arises in the same way when triangular factorizations are computed, and may be resolved by the same compromise. More precisely, the  $k$ th pivot in the sequence prescribed by  $P$  and  $Q$  may vanish or at least become dangerously small in the first  $k - 1$  iterations of Gaussian elimination, in which case (6.42) has to be replaced by

$$\tilde{L}_n \tilde{P}_n \cdots \tilde{L}_1 \tilde{P}_1 \tilde{A} = \tilde{U}$$

with  $\tilde{P}_k \neq I$ . The vague phrase "dangerously small" may be interpreted as "more than  $c$  times smaller than another entry in the last  $n - k$  rows of the  $k$ th column of  $\tilde{L}_{k-1} \tilde{P}_{k-1} \cdots \tilde{L}_1 \tilde{P}_1 \tilde{A}$ " for some fixed value of  $c$ . Setting  $c$  equal to 100 or so represents a concession to the rigid rules of partial pivoting (which would require  $c$  to be equal to 1) and yet it forces only a few of the matrices  $\tilde{P}_k$  to differ from  $I$ . As a result, numerical accuracy hardly suffers, while the number of nonzeros in the triangular factorization increases only slightly (typically by a fraction of a percent).

## SINGULAR AND NONSINGULAR MATRICES

Gaussian elimination will fail to deliver a triangular factorization of  $A$  if and only if there will be no available pivots in some iteration. If this happens in the  $k$ th iteration, then the matrix

$$C = L_{k-1} P_{k-1} \cdots L_1 P_1 A$$

has a rather special structure. In its first  $k - 1$  columns, all the entries on the diagonal are ones and all the entries below the diagonal are zeros; furthermore,  $c_{ik} = 0$  for all  $i = k, k + 1, \dots, n$ . For instance, if  $n = 6$  and  $k = 4$ , then

$$C = \begin{bmatrix} 1 & c_{12} & c_{13} & c_{14} & c_{15} & c_{16} \\ & 1 & c_{23} & c_{24} & c_{25} & c_{26} \\ & & 1 & c_{34} & c_{35} & c_{36} \\ & & & & c_{45} & c_{46} \\ & & & & c_{55} & c_{56} \\ & & & & c_{65} & c_{66} \end{bmatrix}$$

In this case, defining

$$w_3 = c_{34}$$

$$w_2 = c_{24} - c_{23}w_3$$

$$w_1 = c_{14} - c_{12}w_2 - c_{13}w_3$$

and  $w_4 = -1, w_5 = 0, w_6 = 0$ , we obtain

$$C \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \\ w_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

In the last equation, all the components of the right-hand side vector are zeros; such vectors are called *zero vectors* and are denoted by  $\mathbf{0}$  when there is no danger of confusion between row and column vectors of different lengths. Our example with  $n = 6$  and  $k = 4$  illustrates a general procedure for finding a column vector  $\mathbf{w}$  such that  $\mathbf{w} \neq \mathbf{0}$  but  $C\mathbf{w} = \mathbf{0}$ : having defined  $w_{k-1}, \dots, w_1$  in this order by

$$w_i = c_{ik} - \sum_{j=i+1}^{k-1} c_{ij}w_j$$

we set  $w_k$  equal to  $-1$  and  $w_i$  equal to  $0$  whenever  $i > k$ . Now we are in a position to prove a fundamental fact.

**THEOREM 6.1.** Every square matrix  $A$  has precisely one of the following two properties:

- (i) For every right-hand side  $\mathbf{b}$ , the system  $A\mathbf{x} = \mathbf{b}$  has either no solution at all or infinitely many solutions.
- (ii) For every right-hand side  $\mathbf{b}$ , the system  $A\mathbf{x} = \mathbf{b}$  has precisely one solution.

**PROOF.** The procedure discussed above either finds those matrices  $P_1, L_1, \dots, P_{k-1}, L_{k-1}$  and a nonzero vector  $\mathbf{w}$  such that

$$L_{k-1} P_{k-1} \cdots L_1 P_1 A \mathbf{w} = \mathbf{0}$$

or else it delivers a triangular factorization of  $A$ . In the former case, we recall that, for every right-hand side  $\mathbf{b}$ , the two systems

$$A\mathbf{x} = \mathbf{b} \quad \text{and} \quad L_{k-1} P_{k-1} \cdots L_1 P_1 A\mathbf{x} = L_{k-1} P_{k-1} \cdots L_1 P_1 \mathbf{b}$$

have the same set of solutions. In particular, setting  $\mathbf{b}$  equal to  $\mathbf{0}$ , we conclude that  $A\mathbf{w} = \mathbf{0}$ . But then  $A$  has property (i): as soon as  $A\mathbf{x} = \mathbf{b}$  for some  $\mathbf{x}$ , we also have  $A(\mathbf{x} + t\mathbf{w}) = \mathbf{b}$  for every real number  $t$ . On the other hand, if the procedure delivers a triangular factorization  $P_1, L_1 P_2, L_2, \dots, P_n, L_n, U$  of  $A$ , then we recall that, for every right-hand side  $\mathbf{b}$ , the two systems

$$A\mathbf{x} = \mathbf{b} \quad \text{and} \quad U\mathbf{x} = L_n P_n \cdots L_1 P_1 \mathbf{b}$$

have the same set of solutions. Since the triangular system  $U\mathbf{x} = \mathbf{v}$  has precisely one solution for every right-hand side  $\mathbf{v}$ , it follows that  $A$  has property (ii) in this case. ■

Because property (i) is atypical, matrices with this property are called *singular*. Matrices with property (ii) are called *nonsingular*.

### The Inverse of a Matrix

Returning to the similarities between matrix multiplication and number multiplication, we observe that the reciprocal  $a^{-1}$  of a number  $a$  is a number  $b$  such that  $ab = ba = 1$ . By analogy, we might say that a matrix  $\mathbf{B}$  is a reciprocal of a matrix  $\mathbf{A}$  if each of the two products  $\mathbf{AB}$ ,  $\mathbf{BA}$  is an identity matrix. Nowadays, the term *reciprocal* sounds a little archaic; it has been replaced by the term *inverse*. In order for  $\mathbf{B}$  to be an inverse of  $\mathbf{A}$ , both products  $\mathbf{AB}$  and  $\mathbf{BA}$  must be defined; hence  $\mathbf{A}$  and  $\mathbf{B}$  must be square matrices of the same size.

**THEOREM 6.2.** A matrix has an inverse if and only if it is nonsingular.

**PROOF.** To establish the “if” part, consider a nonsingular  $n \times n$  matrix  $\mathbf{A}$ . For each of the  $n$  columns  $\mathbf{b}$  of the  $n \times n$  identity matrix  $\mathbf{I}$ , the system  $\mathbf{Ax} = \mathbf{b}$  has a solution; these  $n$  solutions may be seen as the  $n$  columns of a matrix  $\mathbf{X}$  such that  $\mathbf{AX} = \mathbf{I}$ . We claim that  $\mathbf{XA} = \mathbf{I}$ , and so  $\mathbf{X}$  is an inverse of  $\mathbf{A}$ . Our claim may be justified by observing that  $\mathbf{A}(\mathbf{XA}) = (\mathbf{AX})\mathbf{A} = \mathbf{A}$  and that  $\mathbf{AI} = \mathbf{A}$ : now if  $\mathbf{XA}$  and  $\mathbf{I}$  differed in their  $k$ th columns, then the system  $\mathbf{Ax} = \mathbf{b}$ , with  $\mathbf{b}$  standing for the  $k$ th column of  $\mathbf{A}$ , would have two different solutions (namely, the  $k$ th column of  $\mathbf{XA}$  and the  $k$ th column of  $\mathbf{I}$ ), contradicting the nonsingularity of  $\mathbf{A}$ .

To establish the “only if” part, consider an  $n \times n$  matrix  $\mathbf{A}$  along with its inverse  $\mathbf{B}$  and an arbitrary vector  $\mathbf{b}$ . If  $\mathbf{Ax} = \mathbf{b}$ , then  $\mathbf{x} = \mathbf{BAx} = \mathbf{Bb}$ ; on the other hand,  $\mathbf{A}(\mathbf{Bb}) = \mathbf{b}$ . Hence,  $\mathbf{Bb}$  is the unique solution of  $\mathbf{Ax} = \mathbf{b}$  and  $\mathbf{A}$  is nonsingular. ■

Note that an inverse of  $\mathbf{A}$ , if it exists at all, is unique:  $\mathbf{BA} = \mathbf{I}$  and  $\mathbf{AC} = \mathbf{I}$  imply  $\mathbf{B} = \mathbf{BI} = \mathbf{B}(\mathbf{AC}) = (\mathbf{BA})\mathbf{C} = \mathbf{IC} = \mathbf{C}$ . Hence, we may speak of *the* inverse, rather than *an* inverse, of  $\mathbf{A}$ . Just as the reciprocal of a number  $a$  is denoted by  $a^{-1}$ , the inverse of a matrix  $\mathbf{A}$  is denoted by  $\mathbf{A}^{-1}$ .

Inverses will interest us only as theoretical concepts, never as computational devices. The reasons for this are summarized in Box 6.1.

#### BOX 6.1 Why Inverses Should Not Be Used in Solving Systems of Linear Equations

One might be misled into the belief that the inverse  $\mathbf{A}^{-1}$  is a handy device for solving systems  $\mathbf{Ax} = \mathbf{b}$ : the straight evaluation of  $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$  takes only  $n^2$  multiplications and  $n^2 - n$  additions. However, there are several good reasons for using a triangular factorization of  $\mathbf{A}$  instead.

- (i) Computing  $\mathbf{A}^{-1}$  takes considerably longer (about three times as long in the completely dense case) than computing the triangular factorization.
- (ii) Since extra calculations are involved in computing  $\mathbf{A}^{-1}$ , the results are more likely to suffer from rounding errors.
- (iii) The calculation of  $\mathbf{v} = \mathbf{L}_n\mathbf{P}_n \cdots \mathbf{L}_1\mathbf{P}_1\mathbf{b}$  and the solution of  $\mathbf{Ux} = \mathbf{v}$  are considerably streamlined when the matrix  $\mathbf{U}$  and the eta columns of  $\mathbf{L}_1, \mathbf{L}_2, \dots, \mathbf{L}_n$  are sparse. (As we have seen, this is the case when  $\mathbf{A}$  is sparse and has only a few spikes.) However, similar time-savings cannot be claimed in the straight evaluation of  $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$ : even very sparse matrices tend to have dense inverses.

### PROBLEMS

#### 6.1 Scale the system

$$\begin{aligned} 0.3x_1 + 0.08x_2 + 0.002x_3 + 4x_4 + 0.0002x_5 &= 1.19 \\ 2,000x_1 + 200x_2 + 60x_3 + 20,000x_4 + 3x_5 &= 1,500 \\ 60x_1 + 6x_2 + 0.7x_3 + 500x_4 + 0.04x_5 &= 28 \\ 400x_1 + 40x_2 + 8x_3 + 2,000x_4 + 0.3x_5 &= 210 \\ 5x_1 + 0.2x_2 + 0.06x_3 + 30x_4 + 0.004x_5 &= 2. \end{aligned}$$

#### △ 6.2 Solve the system

$$\begin{aligned} 3x_1 - 9x_2 + 6x_3 + 3x_4 - 3x_5 - 6x_6 &= -3 \\ 2x_1 + x_2 + 4x_3 + 2x_4 - 2x_5 - 4x_6 &= -2 \\ x_1 + 2x_2 + 5x_3 + x_4 - x_5 - 2x_6 &= 4 \\ 3x_1 + 5x_2 + x_3 + 2x_4 - 3x_5 - 6x_6 &= 1 \\ x_1 + 4x_2 - 2x_3 + 3x_4 + 2x_5 - 2x_6 &= 8 \\ 2x_1 + 3x_2 + x_3 - 4x_4 + 5x_5 + x_6 &= 2. \end{aligned}$$

#### △ 6.3 Solve the system

$$\begin{aligned} 4x_1 &+ 3x_4 + 2x_5 &= 14 \\ &2x_2 &+ 3x_5 + 4x_6 &= 18 \\ 3x_1 + 2x_2 &+ 6x_5 &= 19 \\ 3x_1 &+ 2x_3 &= 7 \\ 6x_1 &+ 4x_3 &+ 5x_5 &= 24 \\ &5x_3 + 6x_4 &+ 2x_6 &= 26. \end{aligned}$$

#### △ 6.4 Find a triangular factorization of the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & -2 & 4 \\ 1 & 5 & -1 & 5 \\ 1 & 3 & -3 & 6 \\ -1 & -3 & 3 & -8 \end{bmatrix}.$$

Use this result to solve the systems  $\mathbf{Ax} = \mathbf{b}$ ,  $\mathbf{Ax} = \mathbf{b}'$ ,  $\mathbf{Ax} = \mathbf{b}''$  with

$$\mathbf{b} = \begin{bmatrix} 6 \\ 10 \\ 7 \\ -1 \end{bmatrix}, \quad \mathbf{b}' = \begin{bmatrix} 8 \\ 12 \\ 9 \\ -11 \end{bmatrix}, \quad \mathbf{b}'' = \begin{bmatrix} 1 \\ 4 \\ 0 \\ 0 \end{bmatrix}$$

and the systems  $\mathbf{yA} = \mathbf{c}$ ,  $\mathbf{yA} = \mathbf{c}'$  with

$$\mathbf{c} = [7, 23, -16, 33], \quad \mathbf{c}' = [2, 8, -3, 9].$$

#### 6.5 Prove that $(\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T$ .

6.6 Let  $\mathbf{P}$  and  $\mathbf{Q}$  be permutation matrices of the same size as a square matrix  $\mathbf{A}$ . Prove that  $\tilde{\mathbf{A}}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$  with  $\tilde{\mathbf{A}} = \mathbf{PAQ}$ ,  $\tilde{\mathbf{x}} = \mathbf{Q}^T\mathbf{x}$ , and  $\tilde{\mathbf{b}} = \mathbf{Pb}$  is just another way of recording the system  $\mathbf{Ax} = \mathbf{b}$ , and that  $\tilde{\mathbf{y}}\tilde{\mathbf{A}} = \tilde{\mathbf{c}}$  with  $\tilde{\mathbf{A}} = \mathbf{PAQ}$ ,  $\tilde{\mathbf{y}} = \mathbf{yP}^T$ ,  $\tilde{\mathbf{c}} = \mathbf{cQ}$  is just another way of recording the system  $\mathbf{yA} = \mathbf{c}$ .

6.7 The statement that “in computing the triangular factorization  $\mathbf{L}_n\mathbf{L}_{n-1} \cdots \mathbf{L}_1\mathbf{A} = \mathbf{U}$ , fill-in occurs in the  $j$ th column” means that  $\mathbf{L}_j$  has a nonzero entry  $l_{ij}$  where  $a_{ij} = 0$  or  $\mathbf{U}$  has a nonzero entry  $u_{ij}$  where  $a_{ij} = 0$ . Prove that fill-in is restricted to the spikes of  $\mathbf{A}$ .

- 6.8 Prove that the product of two lower triangular matrices is lower triangular and that the inverse of a nonsingular lower triangular matrix is lower triangular.
- 6.9 Prove that the product  $\mathbf{AB}$  of two square matrices is nonsingular if and only if both  $\mathbf{A}$  and  $\mathbf{B}$  are nonsingular.
- 6.10 Let  $\mathbf{L}_n \mathbf{P}_n \cdots \mathbf{L}_1 \mathbf{P}_1 \mathbf{A} = \mathbf{U}$  be a triangular factorization of  $\mathbf{A}$ ; let matrices  $\mathbf{Q}_n, \mathbf{Q}_{n-1}, \dots, \mathbf{Q}_0$  be defined by  $\mathbf{Q}_n = \mathbf{I}$  and  $\mathbf{Q}_{k-1} = \mathbf{Q}_k \mathbf{P}_k$  ( $k = n, n-1, \dots, 1$ ) and let  $\mathbf{L}_k^*$  stand for the matrix  $\mathbf{Q}_k \mathbf{L}_k \mathbf{Q}_k^T$ . Prove that:
- (i) Each  $\mathbf{Q}_k$  is a permutation matrix agreeing with  $\mathbf{I}$  in the first  $k$  rows and columns.
  - (ii)  $\mathbf{L}_n \mathbf{P}_n \cdots \mathbf{L}_1 \mathbf{P}_1 = \mathbf{L}_n^* \cdots \mathbf{L}_1^* \mathbf{Q}_0$ .
  - (iii) Each  $\mathbf{L}_k^*$  is a lower triangular eta matrix whose eta column is the  $k$ th column.
- 6.11 An *LU-decomposition* of a matrix  $\mathbf{A}$  consists of a lower triangular matrix  $\mathbf{L}$ , an upper triangular matrix  $\mathbf{U}$ , and a permutation matrix  $\mathbf{P}$  such that  $\mathbf{PA} = \mathbf{LU}$ . Use the results of problems 6.8, 6.9, and 6.10 to prove that every nonsingular matrix has an LU-decomposition.
- 6.12 At the end of this chapter, the claim was made that “even very sparse matrices tend to have dense inverses.” Artificially constructed examples do not support empirical claims such as this one very convincingly, but here is one anyway:

$$\text{if } \mathbf{A} = \begin{bmatrix} 1 & & & & 1 \\ 1 & 1 & & & \\ & 1 & 1 & & \\ & & 1 & 1 & \\ & & & 1 & 1 \end{bmatrix} \text{ then } \mathbf{A}^{-1} = \begin{bmatrix} 0.5 & 0.5 & -0.5 & 0.5 & -0.5 \\ -0.5 & 0.5 & 0.5 & -0.5 & 0.5 \\ 0.5 & -0.5 & 0.5 & 0.5 & -0.5 \\ -0.5 & 0.5 & -0.5 & 0.5 & 0.5 \\ 0.5 & -0.5 & 0.5 & -0.5 & 0.5 \end{bmatrix}.$$

Compute a triangular factorization of  $\mathbf{A}$  and compare its density with the density of  $\mathbf{A}^{-1}$ . Generalize this example to arbitrary sizes  $n \times n$  such that  $n$  is odd. (Why is there no generalization with  $n$  even?)

## The Revised Simplex Method

In each iteration of the simplex method, one basic feasible solution is replaced by another. When the old solution is represented by a dictionary, the new solution is easy to find, and only a small part of the dictionary is actually used for that purpose. This part may be reconstructed directly from the original data and the new solution found without any reference to dictionaries. The resulting implementations of the simplex method are known under the generic name of the *revised simplex method*; the implementation of the simplex method that updates a dictionary in each iteration is known as the *standard simplex method*. We shall see that each iteration of the revised simplex method requires solving two systems of linear equations. Typically, these systems are not solved from scratch; instead, some device facilitating their solution is used and updated in each iteration. The device presented in this chapter is essentially the same as the popular “product form of the inverse” developed by G. B. Dantzig and W. Orchard-Hays (1954).

Each iteration of the revised simplex method may or may not take less time than the corresponding iteration of the standard simplex method. The outcome of this comparison depends not only on the particular implementation of the revised simplex method but also on the nature of the data. We shall see that, on the typical large and sparse LP problems solved in applications, the revised simplex method works faster



than the standard simplex method. This is the reason why modern computer programs for solving LP problems always use some form of the revised simplex method.

### MATRIX DESCRIPTION OF DICTIONARIES

Our preliminary task is to develop an understanding of the relationship between dictionaries and the original data. For illustration, we shall consider the dictionary

$$\begin{aligned} x_1 &= 54 - 0.5x_2 - 0.5x_4 - 0.5x_5 + 0.5x_6 \\ x_3 &= 63 - 0.5x_2 - 0.5x_4 + 0.5x_5 - 1.5x_6 \\ x_7 &= 15 + 0.5x_2 - 0.5x_4 + 0.5x_5 + 2.5x_6 \\ z &= 1782 - 2.5x_2 + 1.5x_4 - 3.5x_5 - 8.5x_6 \end{aligned} \quad (7.1)$$

arising from the problem

$$\begin{aligned} \text{maximize} \quad & 19x_1 + 13x_2 + 12x_3 + 17x_4 \\ \text{subject to} \quad & 3x_1 + 2x_2 + x_3 + 2x_4 \leq 225 \\ & x_1 + x_2 + x_3 + x_4 \leq 117 \\ & 4x_1 + 3x_2 + 3x_3 + 4x_4 \leq 420 \\ & x_1, x_2, x_3, x_4 \geq 0 \end{aligned} \quad (7.2)$$

after two iterations of the standard simplex method. To relate the coefficients in (7.1) to the data in (7.2), we first recall that the top three equations in the dictionary are equivalent to the three equations

$$\begin{aligned} 3x_1 + 2x_2 + x_3 + 2x_4 + x_5 &= 225 \\ x_1 + x_2 + x_3 + x_4 + x_6 &= 117 \\ 4x_1 + 3x_2 + 3x_3 + 4x_4 + x_7 &= 420. \end{aligned} \quad (7.3)$$

Hence they arise by solving (7.3) for  $x_1, x_3$ , and  $x_7$ . In matrix terms, this solution may be described quite compactly. First, we record system (7.3) as  $\mathbf{Ax} = \mathbf{b}$  with

$$\mathbf{A} = \begin{bmatrix} 3 & 2 & 1 & 2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 1 & 0 \\ 4 & 3 & 3 & 4 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 225 \\ 117 \\ 420 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \end{bmatrix}.$$

To emphasize the fact that only the basic variables  $x_1, x_3, x_7$  are treated as unknowns, we write  $\mathbf{Ax}$  as  $\mathbf{A}_B \mathbf{x}_B + \mathbf{A}_N \mathbf{x}_N$  with

$$\mathbf{A}_B = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 1 & 0 \\ 4 & 3 & 1 \end{bmatrix}, \quad \mathbf{A}_N = \begin{bmatrix} 2 & 2 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 3 & 4 & 0 & 0 \end{bmatrix}, \quad \mathbf{x}_B = \begin{bmatrix} x_1 \\ x_3 \\ x_7 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} x_2 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix}$$

and then cast the system  $\mathbf{Ax} = \mathbf{b}$  in the form

$$\mathbf{A}_B \mathbf{x}_B = \mathbf{b} - \mathbf{A}_N \mathbf{x}_N. \quad (7.4)$$

Since the square matrix  $\mathbf{A}_B$  happens to be nonsingular, both sides of (7.4) may be multiplied by  $\mathbf{A}_B^{-1}$  on the left. Thus we obtain

$$\mathbf{x}_B = \mathbf{A}_B^{-1} \mathbf{b} - \mathbf{A}_B^{-1} \mathbf{A}_N \mathbf{x}_N \quad (7.5)$$

which is a compact record of the top three equations in (7.1). To obtain the fourth equation, we record the objective function  $z$  as  $\mathbf{cx}$  with

$$\mathbf{c} = [19, 13, 12, 17, 0, 0, 0]$$

or, more suggestively, as  $\mathbf{c}_B \mathbf{x}_B + \mathbf{c}_N \mathbf{x}_N$ , with

$$\mathbf{c}_B = [19, 12, 0] \quad \text{and} \quad \mathbf{c}_N = [13, 17, 0, 0].$$

Substituting for  $\mathbf{x}_B$  from (7.5) we obtain

$$z = \mathbf{c}_B (\mathbf{A}_B^{-1} \mathbf{b} - \mathbf{A}_B^{-1} \mathbf{A}_N \mathbf{x}_N) + \mathbf{c}_N \mathbf{x}_N = \mathbf{c}_B \mathbf{A}_B^{-1} \mathbf{b} + (\mathbf{c}_N - \mathbf{c}_B \mathbf{A}_B^{-1} \mathbf{A}_N) \mathbf{x}_N.$$

Thus, dictionary (7.1) may be recorded in matrix terms as

$$\begin{aligned} \mathbf{x}_B &= \mathbf{A}_B^{-1} \mathbf{b} - \mathbf{A}_B^{-1} \mathbf{A}_N \mathbf{x}_N \\ z &= \mathbf{c}_B \mathbf{A}_B^{-1} \mathbf{b} + (\mathbf{c}_N - \mathbf{c}_B \mathbf{A}_B^{-1} \mathbf{A}_N) \mathbf{x}_N \end{aligned} \quad (7.6)$$

More generally, consider an arbitrary LP problem in the standard form

$$\begin{aligned} \text{maximize} \quad & \sum_{j=1}^n c_j x_j \\ \text{subject to} \quad & \sum_{j=1}^n a_{ij} x_j \leq b_i \quad (i = 1, 2, \dots, m) \\ & x_j \geq 0 \quad (j = 1, 2, \dots, n). \end{aligned}$$

After the introduction of the slack variables  $x_{n+1}, x_{n+2}, \dots, x_{n+m}$ , this problem may be recorded as

$$\begin{aligned} \text{maximize} \quad & \mathbf{cx} \\ \text{subject to} \quad & \mathbf{Ax} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}. \end{aligned}$$

(The matrix  $\mathbf{A}$  has  $m$  rows and  $n + m$  columns, of which the last  $m$  form the identity matrix. The column vector  $\mathbf{x}$  has length  $n + m$  and the column vector  $\mathbf{b}$  has length  $m$ . The row vector  $\mathbf{c}$  has length  $n + m$  and its last  $m$  components are zeros.) Each basic feasible solution  $\mathbf{x}^*$  of this problem partitions  $x_1, x_2, \dots, x_{n+m}$  into  $m$  basic and  $n$  nonbasic variables. As in our example, this partition induces a partition of  $\mathbf{A}$  into  $\mathbf{A}_B$  and  $\mathbf{A}_N$ , a partition of  $\mathbf{x}$  into  $\mathbf{x}_B$  and  $\mathbf{x}_N$ , and a partition of  $\mathbf{c}$  into  $\mathbf{c}_B$  and  $\mathbf{c}_N$ . We propose to show that

matrix  $\mathbf{A}_B$  is nonsingular (7.7)

by showing that the system  $\mathbf{A}_B \mathbf{x}_B = \mathbf{b}$  has precisely one solution. The existence of a solution is evident: since the basic feasible solution  $\mathbf{x}^*$  satisfies  $\mathbf{A} \mathbf{x}^* = \mathbf{b}$  and  $\mathbf{x}_N^* = \mathbf{0}$ , it satisfies  $\mathbf{A}_B \mathbf{x}_B^* = \mathbf{A} \mathbf{x}^* - \mathbf{A}_N \mathbf{x}_N^* = \mathbf{b}$ . To verify that there are no other solutions, consider an arbitrary vector  $\tilde{\mathbf{x}}_B$  such that  $\mathbf{A}_B \tilde{\mathbf{x}}_B = \mathbf{b}$  and set  $\tilde{\mathbf{x}}_N = \mathbf{0}$ . Since the resulting vector  $\tilde{\mathbf{x}}$  satisfies  $\mathbf{A} \tilde{\mathbf{x}} = \mathbf{A}_B \tilde{\mathbf{x}}_B + \mathbf{A}_N \tilde{\mathbf{x}}_N = \mathbf{b}$ , it must satisfy the top  $m$  equations in the dictionary representing  $\mathbf{x}^*$ . But then  $\tilde{\mathbf{x}}_N = \mathbf{0}$  implies  $\tilde{\mathbf{x}}_B = \mathbf{x}_B^*$ . Thus the proof of (7.7) is completed. Now the arguments given above show that the dictionary representing  $\mathbf{x}^*$  has the form (7.6). The matrix  $\mathbf{A}_B$  is called the *basis matrix* or (when there is no danger of confusion with the set of basic variables) simply the *basis*. It is customary to denote the basis matrix by  $\mathbf{B}$  rather than  $\mathbf{A}_B$ . We shall bow to this convention and record the dictionary as

$$\begin{aligned} \mathbf{x}_B &= \mathbf{B}^{-1} \mathbf{b} - \mathbf{B}^{-1} \mathbf{A}_N \mathbf{x}_N \\ z &= \mathbf{c}_B \mathbf{B}^{-1} \mathbf{b} + (\mathbf{c}_N - \mathbf{c}_B \mathbf{B}^{-1} \mathbf{A}_N) \mathbf{x}_N. \end{aligned}$$

Of course,  $\mathbf{B}^{-1} \mathbf{b}$  is nothing but the vector  $\mathbf{x}_B^*$  specifying the current values of the basic variables.

## THE REVISED SIMPLEX METHOD

In each iteration of the simplex method, we first choose the entering variable, then find the leaving variable, and finally update the current basic feasible solution. An examination of the way these tasks are carried out in the standard simplex method will lead us to the alternative, the revised simplex method. For illustration, we shall consider the update of the feasible dictionary (7.1) in the standard simplex method. The corresponding iteration of the revised simplex method begins with

$$\mathbf{x}_B^* = \begin{bmatrix} x_1^* \\ x_3^* \\ x_7^* \end{bmatrix} = \begin{bmatrix} 54 \\ 63 \\ 15 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 1 & 0 \\ 4 & 3 & 1 \end{bmatrix}.$$

The entering variable may be any nonbasic variable with a positive coefficient in the last row of the dictionary. As previously observed, the coefficients in this row form

the vector  $\mathbf{c}_N - \mathbf{c}_B \mathbf{B}^{-1} \mathbf{A}_N$ . If the standard simplex method is used, then this vector is readily available as part of the dictionary; in our example, we have

$$z = \cdots - 2.5x_2 + 1.5x_4 - 3.5x_5 - 8.5x_6. \quad (7.8)$$

If the revised simplex method is used, then the vector  $\mathbf{c}_N - \mathbf{c}_B \mathbf{B}^{-1} \mathbf{A}_N$  is computed in two steps: first we find  $\mathbf{y} = \mathbf{c}_B \mathbf{B}^{-1}$  by solving the system  $\mathbf{y} \mathbf{B} = \mathbf{c}_B$  and then we calculate  $\mathbf{c}_N - \mathbf{y} \mathbf{A}_N$ . In our example, we would first solve the system

$$[y_1, y_2, y_3] \cdot \begin{bmatrix} 3 & 1 & 0 \\ 1 & 1 & 0 \\ 4 & 3 & 1 \end{bmatrix} = [19, 12, 0]$$

to find  $\mathbf{y} = [y_1, y_2, y_3] = [3.5, 8.5, 0]$  and then we would calculate

$$[13, 17, 0, 0] - [3.5, 8.5, 0] \cdot \begin{bmatrix} 2 & 2 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 3 & 4 & 0 & 0 \end{bmatrix} = [-2.5, 1.5, -3.5, -8.5]$$

to find the vector featured in (7.8). As the only positive component of this vector is its second component, the second component  $x_4$  of the vector  $\mathbf{x}_N = [x_2, x_4, x_5, x_6]^T$  enters the basis. Incidentally, note that the components of  $\mathbf{c}_N - \mathbf{y} \mathbf{A}_N$  may be calculated individually; if a nonbasic variable  $x_j$  corresponds to a component  $c_j$  of  $\mathbf{c}_N$  and to a column  $\mathbf{a}$  of  $\mathbf{A}_N$ , then the corresponding component of  $\mathbf{c}_N - \mathbf{y} \mathbf{A}_N$  equals  $c_j - \mathbf{y} \mathbf{a}$ . Thus, the entering variable may be any nonbasic variable  $x_j$  for which  $\mathbf{y} \mathbf{a} < c_j$ . The corresponding column  $\mathbf{a}$  of  $\mathbf{A}$  is called the *entering column*.

To determine the leaving variable, we increase the value  $t$  of the entering variable from zero to some positive level, maintaining the values of the remaining nonbasic variables at their zero levels and adjusting the values of the basic variables so as to preserve the constraints  $\mathbf{A} \mathbf{x} = \mathbf{b}$ . As  $t$  increases, the values of the basic variables change until a variable whose value is the first to drop to zero leaves the basis. To find the leaving variable and the largest admissible value of  $t$ , we have to know how precisely the values of the basic variables change with the changes of  $t$ . If the standard simplex method is used, then this information is readily available as part of the dictionary; in our example, we have

$$\begin{aligned} x_1 &= 54 \cdots - 0.5x_4 \cdots & x_1 &= 54 - 0.5t \\ x_3 &= 63 \cdots - 0.5x_4 \cdots, \quad \text{and so} & x_3 &= 63 - 0.5t. \\ x_7 &= 15 \cdots - 0.5x_4 \cdots & x_7 &= 15 - 0.5t \end{aligned} \quad (7.9)$$

More generally, the top  $m$  equations of the dictionary read  $\mathbf{x}_B = \mathbf{x}_B^* - \mathbf{B}^{-1} \mathbf{A}_N \mathbf{x}_N$ , and so  $\mathbf{x}_B$  changes from  $\mathbf{x}_B^*$  to  $\mathbf{x}_B^* - t \mathbf{d}$ , with  $\mathbf{d}$  standing for the column of  $\mathbf{B}^{-1} \mathbf{A}_N$  that corresponds to the entering variable. Note that  $\mathbf{d} = \mathbf{B}^{-1} \mathbf{a}$ , with  $\mathbf{a}$  standing for the entering column. If the revised simplex method is used, then only  $\mathbf{x}_B^*$  is readily

available, whereas  $\mathbf{d}$  is obtained by solving the system  $\mathbf{Bd} = \mathbf{a}$ . In our example, we would solve the system

$$\begin{bmatrix} 3 & 1 & 0 \\ 1 & 1 & 0 \\ 4 & 3 & 1 \end{bmatrix} \cdot \mathbf{d} = \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix} \quad \text{to find the vector} \quad \mathbf{d} = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$$

featured in (7.9). We find easily that  $t$  can be increased all the way to 30, at which point  $54 - 0.5t = 39$ ,  $63 - 0.5t = 48$ ,  $15 - 0.5t = 0$ , and  $x_7$  leaves the basis.

So far, the revised simplex method has been requiring computations not needed in the standard simplex method. This trend gets reversed at the end of the iteration: whereas the standard simplex method requires a laborious update of the entire dictionary, no such computations are needed in the revised simplex method. In our example, the revised simplex method merely enters the next iteration with

$$\mathbf{x}_B^* = \begin{bmatrix} x_1^* \\ x_3^* \\ x_4^* \end{bmatrix} = \begin{bmatrix} 39 \\ 48 \\ 30 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 1 & 1 \\ 4 & 3 & 4 \end{bmatrix}.$$

Incidentally, note that the order of the columns of  $\mathbf{B}$  is unimportant as long as it matches the order of the components of  $\mathbf{x}_B^*$ : the next iteration could just as well be entered with

$$\mathbf{x}_B^* = \begin{bmatrix} x_3^* \\ x_4^* \\ x_1^* \end{bmatrix} = \begin{bmatrix} 48 \\ 30 \\ 39 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 3 & 4 & 4 \end{bmatrix}.$$

To put it differently, the fact that the variables  $x_1, x_2, \dots, x_{n+m}$  happen to be ordered by their subscripts is just coincidental; the columns of  $\mathbf{B}$  may be presented in any other order. An ordered list of the basic variables that specifies the actual order of the  $m$  columns of  $\mathbf{B}$  is called the *basis heading*. We shall find it convenient to replace the leaving variable by the entering variable in each update of the basis heading: the corresponding update of  $\mathbf{B}$  amounts to a replacement of the *leaving column* by the entering column.

Our development of the revised simplex method is summarized in Box 7.1.

### An Economic Interpretation of the Revised Simplex Method

The revised simplex method is intimately related to two subjects presented in Chapter 5: the Complementary Slackness Theorem and the economic interpretation of dual variables. To illustrate the relationship, we shall consider a hypothetical furniture-manufacturing company.

- A bookcase requires three hours of work, one unit of metal, and four units of wood, and it brings in a net profit of \$19.

### BOX 7.1 An Iteration of the Revised Simplex Method

*Step 1.* Solve the system  $\mathbf{yB} = \mathbf{c}_B$ .

*Step 2.* Choose an entering column. This may be any column  $\mathbf{a}$  of  $\mathbf{A}_N$  such that  $\mathbf{ya}$  is less than the corresponding component of  $\mathbf{c}_N$ . If there is no such column, then the current solution is optimal.

*Step 3.* Solve the system  $\mathbf{Bd} = \mathbf{a}$ .

*Step 4.* Find the largest  $t$  such that  $\mathbf{x}_B^* - t\mathbf{d} \geq \mathbf{0}$ . If there is no such  $t$ , then the problem is unbounded; otherwise, at least one component of  $\mathbf{x}_B^* - t\mathbf{d}$  equals zero and the corresponding variable is leaving the basis.

*Step 5.* Set the value of the entering variable at  $t$  and replace the values  $\mathbf{x}_B^*$  of the basic variables by  $\mathbf{x}_B^* - t\mathbf{d}$ . Replace the leaving column of  $\mathbf{B}$  by the entering column and, in the basis heading, replace the leaving variable by the entering variable.

- A desk requires two hours of work, one unit of metal, and three units of wood, and it brings in a net profit of \$13.
- A chair requires one hour of work, one unit of metal, and three units of wood, and it brings in a net profit of \$12.
- A bedframe requires two hours of work, one unit of metal, and four units of wood, and it brings in a net profit of \$17.
- Only 225 hours of labor, 117 units of metal, and 420 units of wood are available per day.

Note that the problem of maximizing the total net profit of the company, under the assumption that all the furniture can be sold, is nothing but our old example (7.2).

Now suppose that a program of making 54 bookcases and 63 chairs per day has been proposed to the company. To find out if this program is optimal, we may appeal to the Complementary Slackness Theorem (the version presented as Theorem 5.3): a feasible solution  $\mathbf{x}^*$  is optimal if and only if there are numbers  $y_1, y_2, \dots, y_m$  that satisfy a certain system of equations and a certain system of inequalities. In this particular example,  $\mathbf{x}^*$  is the basic feasible solution with

$$\mathbf{x}_B^* = \begin{bmatrix} x_1^* \\ x_3^* \\ x_7^* \end{bmatrix} = \begin{bmatrix} 54 \\ 63 \\ 15 \end{bmatrix}$$

the system of equations is

$$\begin{aligned}
 3y_1 + y_2 + 4y_3 &= 19 \\
 y_1 + y_2 + 3y_3 &= 12 \\
 y_3 &= 0
 \end{aligned} \tag{7.10}$$

and the system of inequalities is

$$\begin{aligned}
 2y_1 + y_2 + 3y_3 &\geq 13 \\
 2y_1 + y_2 + 4y_3 &\geq 17 \\
 y_1 &\geq 0 \\
 y_2 &\geq 0.
 \end{aligned} \tag{7.11}$$

Note that (7.10) is nothing but  $\mathbf{yB} = \mathbf{c}_B$ , that (7.11) is nothing but  $\mathbf{yA}_N \geq \mathbf{c}_N$ , and that this observation generalizes: if  $\mathbf{x}^*$  is a nondegenerate basic feasible solution, then the system of equations featured in the Complementary Slackness Theorem is nothing but  $\mathbf{yB} = \mathbf{c}_B$  and the system of inequalities is nothing but  $\mathbf{yA}_N \geq \mathbf{c}_N$ . But  $\mathbf{yB} = \mathbf{c}_B$  is precisely the system of equations solved in step 1 of an iteration of the revised simplex method and  $\mathbf{yA}_N \geq \mathbf{c}_N$  is the system of inequalities considered in step 2. Thus the first two steps in each iteration of the revised simplex method may be seen as checking the current feasible solution  $\mathbf{x}^*$  for optimality by the Complementary Slackness Theorem. (Actually, this statement is not quite correct: if  $\mathbf{x}^*$  happens to be degenerate, then the system of equations featured in the Complementary Slackness Theorem consists of fewer than  $m$  equations and forms a proper subsystem of  $\mathbf{yB} = \mathbf{c}_B$ .)

Furthermore, these first two steps may be given an economic interpretation along the lines described in Chapter 5. Solving system (7.10), or  $\mathbf{yB} = \mathbf{c}_B$  in general, may be interpreted as assigning temporary *shadow prices* to the *resources* (time, metal, and wood) in such a way that the total shadow price of the resources consumed by each of the three *basic activities* (making bookcases, making chairs, and leaving wood unused) matches the net profit returned by this activity. Thus, the solution  $y_1 = 3.5$ ,  $y_2 = 8.5$ ,  $y_3 = 0$  of (7.10) appraises time at \$3.50/hour, metal at \$8.50/unit, and wood at \$0/unit. Evaluating the left-hand side of (7.11), or  $\mathbf{yA}_N$  in general, may be interpreted as finding the total shadow price of the resources consumed by each of the *nonbasic activities* (making desks, making bedframes, leaving working time unused, and leaving metal unused); this operation is sometimes referred to as *pricing out* the nonbasic activities. If none of these activities pays back more than it consumes (that is, if  $\mathbf{c}_N \leq \mathbf{yA}_N$ ), then the current program is optimal. (A converse of this implication, although guaranteed by the Complementary Slackness Theorem whenever  $\mathbf{x}^*$  is nondegenerate, does not hold in general: a degenerate  $\mathbf{x}^*$  may be optimal even if some of the inequalities in  $\mathbf{c}_N \leq \mathbf{yA}_N$  are violated. In that case, the simplex method will go through a few degenerate iterations without changing  $\mathbf{x}^*$  until it comes up with a basis that yields a vector  $\mathbf{y}$  with  $\mathbf{yA}_N \geq \mathbf{c}_N$ .) In our example, making bedframes does pay back more (\$17) than it consumes (time worth \$7, metal worth \$8.50, and wood worth nothing under the current pricing scheme).

Where the Complementary Slackness Theorem leaves off, the revised simplex method continues: it attempts to construct an improved program by *substituting* the profitable *entering activity* (making bedframes) for a suitable mix of the basic activities. The mix,  $d_i$  units of each basic activity  $i$  per unit of the entering activity, must consume resources at the same rate as the entering activity itself. In our example, this requirement gives rise to the system

$$\begin{aligned}
 3d_1 + d_3 &= 2 \\
 d_1 + d_3 &= 1 \\
 4d_1 + 3d_3 + d_7 &= 4
 \end{aligned} \tag{7.12}$$

and the solution of this system,  $d_1 = 0.5$ ,  $d_3 = 0.5$ ,  $d_7 = 0.5$ , specifies the concentrations  $d_i$  of the constituents  $i$  in the mix: each bedframe will be substituted for half a bookcase plus half a chair plus half a unit of unused wood. [Of course, (7.12) is nothing but the system  $\mathbf{Bd} = \mathbf{a}$  solved in step 3 of the iteration.] Since the substitution raises the company's profit (by \$1.50 per bedframe), the largest admissible number  $t$  of chairs should be substituted for  $0.5t$  bookcases plus  $0.5t$  chairs plus  $0.5t$  units of unused wood; since only 15 units of unused wood are available, the value of  $t$  is limited to 30. (This is step 4 of the iteration, where the largest admissible value of  $t$  is determined.) The resulting improved program calls for 39 bookcases, 48 chairs, and 30 bedframes to be made every day. The three new basic activities are making bookcases, making chairs, and making bedframes; the old basic activity of leaving wood unused has just become nonbasic. (This is step 5, where the substitution is actually carried out.)

Along similar lines, each iteration of the revised simplex method may be interpreted in economic terms of pricing and substitution. The interpretation becomes a little less intuitive when some of the numbers  $y_1, y_2, \dots, y_m$  or  $d_1, d_2, \dots, d_m$  come out negative, but it may be justified even in those cases.

### Eta Factorization of the Basis

The efficiency of the revised simplex method hinges on the ease of implementing steps 1 and 3 of each iteration. Typically, the systems  $\mathbf{yB} = \mathbf{c}_B$  and  $\mathbf{Bd} = \mathbf{a}$  are not solved from scratch; instead, some device is used to facilitate their solutions and is updated at the end of each iteration. Thus our description of the revised simplex method encompasses a whole class of implementations, each depending on the choice of device that facilitates solutions of the two systems. We are about to describe the simplest of these devices, almost the same (see problem 7.13) as the popular "product form of the inverse" developed by G. B. Dantzig and W. Orchard-Hays (1954). A class of devices that are more efficient, but also more complicated, will be presented in Chapter 24.

Let  $\mathbf{B}_k$  denote the basis matrix obtained after  $k$  iterations of the simplex method, so that each  $\mathbf{B}_k$  differs from the preceding  $\mathbf{B}_{k-1}$  in only one column. Consider a fixed

$k$  and say that it is the  $p$ th column in which  $\mathbf{B}_k$  differs from  $\mathbf{B}_{k-1}$ . Now the  $p$ th column of  $\mathbf{B}_k$  is the entering column  $\mathbf{a}$  selected in step 2 of the  $k$ th iteration and appearing as the right-hand side in the system  $\mathbf{B}_{k-1}\mathbf{d} = \mathbf{a}$ , which is solved in step 3 of the same iteration. Hence

$$\mathbf{B}_k = \mathbf{B}_{k-1}\mathbf{E}_k \quad (7.13)$$

with  $\mathbf{E}_k$  standing for the identity matrix whose  $p$ th column is replaced by  $\mathbf{d}$ . (To verify this matrix equation, we need only compare its two sides column by column, keeping in mind that the  $j$ th column of  $\mathbf{B}_{k-1}\mathbf{E}_k$  equals  $\mathbf{B}_{k-1}$  multiplied by the  $j$ th column of  $\mathbf{E}_k$  on the right.) For instance,

$$\begin{bmatrix} 3 & 1 & 2 \\ 1 & 1 & 1 \\ 4 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 1 & 0 \\ 4 & 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0.5 \\ & 1 & 0.5 \\ & & 0.5 \end{bmatrix}$$

in the example just used. The importance of equation (7.13) for the revised simplex method is paramount: no matter what device is used to solve the two systems  $\mathbf{y}\mathbf{B}_{k-1} = \mathbf{c}_B$  and  $\mathbf{B}_{k-1}\mathbf{d} = \mathbf{a}$ , its update invariably relies on the fact that  $\mathbf{B}_k = \mathbf{B}_{k-1}\mathbf{E}_k$  with the eta matrix  $\mathbf{E}_k$  readily available.

When the initial basis consists of the slack variables, we have  $\mathbf{B}_0 = \mathbf{I}$  and successive applications of (7.13) yield  $\mathbf{B}_1 = \mathbf{E}_1$ ,  $\mathbf{B}_2 = \mathbf{E}_1\mathbf{E}_2$ ,  $\mathbf{B}_3 = \mathbf{E}_1\mathbf{E}_2\mathbf{E}_3$ , and so on. Thus we have

$$\mathbf{B}_k = \mathbf{E}_1\mathbf{E}_2 \cdots \mathbf{E}_k.$$

This *eta factorization* of  $\mathbf{B}_k$  suggests a convenient way of solving the two systems of equations: the system  $\mathbf{y}\mathbf{B}_k = \mathbf{c}_B$  may be seen as

$$(((\mathbf{y}\mathbf{E}_1)\mathbf{E}_2) \cdots)\mathbf{E}_k = \mathbf{c}_B$$

and the system  $\mathbf{B}_k\mathbf{d} = \mathbf{a}$  may be seen as

$$\mathbf{E}_1(\mathbf{E}_2(\cdots(\mathbf{E}_k\mathbf{d}))) = \mathbf{a}.$$

For instance,  $\mathbf{y}\mathbf{B}_4 = \mathbf{c}_B$  may be solved by solving the sequence of systems

$$\mathbf{u}\mathbf{E}_4 = \mathbf{c}_B, \quad \mathbf{v}\mathbf{E}_3 = \mathbf{u}, \quad \mathbf{w}\mathbf{E}_2 = \mathbf{v}, \quad \text{and} \quad \mathbf{y}\mathbf{E}_1 = \mathbf{w}$$

(so that  $\mathbf{y}\mathbf{B}_4 = \mathbf{y}\mathbf{E}_1\mathbf{E}_2\mathbf{E}_3\mathbf{E}_4 = \mathbf{w}\mathbf{E}_2\mathbf{E}_3\mathbf{E}_4 = \mathbf{v}\mathbf{E}_3\mathbf{E}_4 = \mathbf{u}\mathbf{E}_4 = \mathbf{c}_B$  as desired) and  $\mathbf{B}_4\mathbf{d} = \mathbf{a}$  may be solved by solving the sequence of systems

$$\mathbf{E}_1\mathbf{u} = \mathbf{a}, \quad \mathbf{E}_2\mathbf{v} = \mathbf{u}, \quad \mathbf{E}_3\mathbf{w} = \mathbf{v}, \quad \text{and} \quad \mathbf{E}_4\mathbf{d} = \mathbf{w}$$

(so that  $\mathbf{B}_4\mathbf{d} = \mathbf{E}_1\mathbf{E}_2\mathbf{E}_3\mathbf{E}_4\mathbf{d} = \mathbf{E}_1\mathbf{E}_2\mathbf{E}_3\mathbf{w} = \mathbf{E}_1\mathbf{E}_2\mathbf{v} = \mathbf{E}_1\mathbf{u} = \mathbf{a}$  as desired). At first, this way of solving  $\mathbf{y}\mathbf{B}_k = \mathbf{c}_B$  and  $\mathbf{B}_k\mathbf{d} = \mathbf{a}$  may seem rather awkward: in order to solve one system of linear equations, we resort to solving  $k$  systems. Note, however, that systems such as  $\mathbf{v}\mathbf{E}_i = \mathbf{u}$  or  $\mathbf{E}_i\mathbf{v} = \mathbf{u}$  are extremely easy to solve: if the eta column of  $\mathbf{E}_i$  has  $s$  nonzero entries, then only  $s - 1$  multiplications,  $s - 1$  additions, and

one division are required. Before discussing the efficiency of this scheme any further, let us illustrate it with an example.

The example is again problem (7.2),

$$\text{maximize } \mathbf{c}\mathbf{x} \quad \text{subject to } \mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0}$$

with

$$\mathbf{A} = \begin{bmatrix} 3 & 2 & 1 & 2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 1 & 0 \\ 4 & 3 & 3 & 4 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 225 \\ 117 \\ 420 \end{bmatrix}, \quad \mathbf{c} = [19, 13, 12, 17, 0, 0, 0].$$

As usual, we let the slack variables form the initial basis, so that  $\mathbf{B}_0 = \mathbf{I}$  and

$$\mathbf{x}_B^* = \begin{bmatrix} x_5^* \\ x_6^* \\ x_7^* \end{bmatrix} = \begin{bmatrix} 225 \\ 117 \\ 420 \end{bmatrix}.$$

The first iteration of the revised simplex method begins.

*Step 1.* The system  $\mathbf{y}\mathbf{B}_0 = \mathbf{c}_B$  reduces to  $\mathbf{y} = [0, 0, 0]$ .

*Step 2.* Since  $c_3 - \mathbf{y} \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} = 12$ , we may let  $x_3$  enter the basis.

*Step 3.* The system  $\mathbf{B}_0\mathbf{d} = \mathbf{a}$  reduces to

$$\mathbf{d} = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}.$$

*Step 4.* The largest  $t$  such that  $225 - t \geq 0$ ,  $117 - t \geq 0$ ,  $420 - 3t \geq 0$  is  $t = 117$ . Since  $117 - t = 0$ , the leaving variable is  $x_6$ .

*Step 5.* Now we have

$$\begin{bmatrix} x_5^* \\ x_3^* \\ x_7^* \end{bmatrix} = \begin{bmatrix} 225 - t \\ t \\ 420 - 3t \end{bmatrix} = \begin{bmatrix} 108 \\ 117 \\ 69 \end{bmatrix} \quad \text{and} \quad \mathbf{B}_1 = \mathbf{E}_1 = \begin{bmatrix} 1 & 1 \\ & 1 \\ & 3 & 1 \end{bmatrix}.$$

The second iteration begins.

*Step 1.* Solving the system  $\mathbf{y}\mathbf{B}_1 = \mathbf{c}_B$ , which reads

$$\mathbf{y} \begin{bmatrix} 1 & 1 \\ & 1 \\ & 3 & 1 \end{bmatrix} = [0, 12, 0], \quad \text{we find } \mathbf{y} = [0, 12, 0].$$

*Step 2.* Since  $c_1 - \mathbf{y} \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix} = 7$ , we may let  $x_1$  enter the basis.

Step 3. Solving the system  $\mathbf{B}_1 \mathbf{d} = \mathbf{a}$ , which reads

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 3 & 1 \end{bmatrix} \cdot \mathbf{d} = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}, \text{ we find } \mathbf{d} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}.$$

Step 4. The largest  $t$  such that  $108 - 2t \geq 0$ ,  $117 - t \geq 0$ ,  $69 - t \geq 0$  is  $t = 54$ . Since  $108 - 2t = 0$ , the leaving variable is  $x_5$ .

Step 5. Now we have

$$\begin{bmatrix} x_1^* \\ x_3^* \\ x_7^* \end{bmatrix} = \begin{bmatrix} t \\ 117 - t \\ 69 - t \end{bmatrix} = \begin{bmatrix} 54 \\ 63 \\ 15 \end{bmatrix} \text{ and } \mathbf{B}_2 = \mathbf{E}_1 \mathbf{E}_2 \text{ with } \mathbf{E}_2 = \begin{bmatrix} 2 & & \\ 1 & 1 & \\ 1 & & 1 \end{bmatrix}.$$

The third iteration begins.

Step 1. We shall solve the system  $\mathbf{yB}_2 = \mathbf{c}_B$  as  $(\mathbf{yE}_1)\mathbf{E}_2 = \mathbf{c}_B$ . Solving the system  $\mathbf{uE}_2 = \mathbf{c}_B$ , which reads

$$\mathbf{u} \begin{bmatrix} 2 & & \\ 1 & 1 & \\ 1 & & 1 \end{bmatrix} = [19, 12, 0], \text{ we find } \mathbf{u} = [3.5, 12, 0].$$

Solving the system  $\mathbf{yE}_1 = \mathbf{u}$ , which reads

$$\mathbf{y} \begin{bmatrix} 1 & 1 & \\ 1 & & \\ 3 & 1 & \end{bmatrix} = [3.5, 12, 0], \text{ we find } \mathbf{y} = [3.5, 8.5, 0].$$

Step 2. Since  $c_4 - \mathbf{y} \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix} = 1.5$ , we may let  $x_4$  enter the basis.

Step 3. We shall solve the system  $\mathbf{B}_2 \mathbf{d} = \mathbf{a}$  as  $\mathbf{E}_1(\mathbf{E}_2 \mathbf{d}) = \mathbf{a}$ . Solving the system  $\mathbf{E}_1 \mathbf{u} = \mathbf{a}$ , which reads

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 3 & 1 \end{bmatrix} \cdot \mathbf{u} = \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}, \text{ we find } \mathbf{u} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Solving the system  $\mathbf{E}_2 \mathbf{d} = \mathbf{u}$ , which reads

$$\begin{bmatrix} 2 & \\ 1 & 1 \\ 1 & & 1 \end{bmatrix} \cdot \mathbf{d} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \text{ we find } \mathbf{d} = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}.$$

Step 4. The largest  $t$  such that  $54 - 0.5t \geq 0$ ,  $63 - 0.5t \geq 0$ ,  $15 - 0.5t \geq 0$  is  $t = 30$ . Since  $15 - 0.5t = 0$ , the leaving variable is  $x_7$ .

Step 5. Now we have

$$\begin{bmatrix} x_1^* \\ x_3^* \\ x_4^* \end{bmatrix} = \begin{bmatrix} 54 - 0.5t \\ 63 - 0.5t \\ t \end{bmatrix} = \begin{bmatrix} 39 \\ 48 \\ 30 \end{bmatrix} \text{ and } \mathbf{B}_3 = \mathbf{E}_1 \mathbf{E}_2 \mathbf{E}_3 \text{ with } \mathbf{E}_3 = \begin{bmatrix} 1 & 0.5 \\ & 1 & 0.5 \\ & & 1 \end{bmatrix}.$$

The fourth iteration begins.

Step 1. We shall solve the system  $\mathbf{yB}_3 = \mathbf{c}_B$  as  $((\mathbf{yE}_1)\mathbf{E}_2)\mathbf{E}_3 = \mathbf{c}_B$ . Solving the system  $\mathbf{uE}_3 = \mathbf{c}_B$ , which reads

$$\mathbf{u} \begin{bmatrix} 1 & 0.5 \\ & 1 & 0.5 \\ & & 1 \end{bmatrix} = [19, 12, 17], \text{ we find } \mathbf{u} = [19, 12, 3].$$

Solving the system  $\mathbf{vE}_2 = \mathbf{u}$ , which reads

$$\mathbf{v} \begin{bmatrix} 2 & \\ 1 & 1 \\ 1 & & 1 \end{bmatrix} = [19, 12, 3], \text{ we find } \mathbf{v} = [2, 12, 3].$$

Solving the system  $\mathbf{yE}_1 = \mathbf{v}$ , which reads

$$\mathbf{y} \begin{bmatrix} 1 & 1 \\ 1 & & \\ 3 & 1 & \end{bmatrix} = [2, 12, 3], \text{ we find } \mathbf{y} = [2, 1, 3].$$

Step 2. Since

$$\mathbf{c}_N - \mathbf{yA}_N = [13, 0, 0, 0] - [2, 1, 3] \cdot \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 3 & 0 & 0 & 1 \end{bmatrix} = [-1, -2, -1, -3]$$

we find no candidate for entering the basis. Hence the current solution is optimal.

Even though  $\mathbf{B}_0 = \mathbf{I}$  whenever the initial basis consists of the slack variables, the case of an arbitrary  $\mathbf{B}_0$  is worth considering. In this case, the identity  $\mathbf{B}_k = \mathbf{E}_1 \mathbf{E}_2 \cdots \mathbf{E}_k$  generalizes into

$$\mathbf{B}_k = \mathbf{B}_0 \mathbf{E}_1 \mathbf{E}_2 \cdots \mathbf{E}_k$$

and the two systems  $\mathbf{yB}_k = \mathbf{c}_B$ ,  $\mathbf{B}_k \mathbf{d} = \mathbf{a}$  may be solved as  $((\mathbf{yB}_0)\mathbf{E}_1) \cdots \mathbf{E}_k = \mathbf{c}_B$  and  $\mathbf{B}_0(\mathbf{E}_1(\cdots(\mathbf{E}_k \mathbf{d}))) = \mathbf{a}$ , respectively. Now a triangular factorization

$$\mathbf{L}_m \mathbf{P}_m \cdots \mathbf{L}_1 \mathbf{P}_1 \mathbf{B}_0 = \mathbf{U}$$

of the initial basis  $\mathbf{B}_0$  may be computed before the first iteration and then used again and again in conjunction with the growing sequence  $\mathbf{E}_1, \mathbf{E}_2, \dots, \mathbf{E}_k$ . Note that

$$\mathbf{U} = \mathbf{U}_m \mathbf{U}_{m-1} \cdots \mathbf{U}_1$$

with each  $U_j$  standing for the eta matrix obtained when the  $j$ th column of  $I$  is replaced by the  $j$ th column of  $U$  (a verification of this claim is left for problem 7.6), and so

$$L_m P_m \cdots L_1 P_1 B_k = U_m U_{m-1} \cdots U_1 E_1 E_2 \cdots E_k.$$

In this notation, the system  $yB_k = c_B$  may be solved by first solving  $((yU_m)U_{m-1}) \cdots E_k = c_B$  and then replacing  $y$  by  $((yL_m P_m) \cdots L_1 P_1)$ . The details of this procedure may be spelled out as follows.

1. Set  $i = k$  and  $y = c_B$ .
2. If  $i \geq 1$ , then set  $v = y$ , replace  $y$  by the solution of  $yE_i = v$ , replace  $i$  by  $i - 1$ , and repeat this step.
3. Set  $j = 1$ .
4. If  $j \leq m$ , then set  $v = y$ , replace  $y$  by the solution of  $yU_j = v$ , replace  $j$  by  $j + 1$ , and repeat this step.
5. Set  $j = m$ .
6. If  $j \geq 1$ , then replace  $y$  by  $yL_j P_j$ , replace  $j$  by  $j - 1$ , and repeat this step.

Similarly, the system  $B_k d = a$  may be solved as  $U_m(U_{m-1}(\cdots(E_k d))) = (L_m P_m(\cdots(L_1 P_1 a)))$ ; the details of this procedure may be spelled out as follows.

1. Set  $j = 1$  and  $d = a$ .
2. If  $j \leq m$ , then replace  $d$  by  $L_j P_j d$ , replace  $j$  by  $j + 1$ , and repeat this step.
3. Set  $j = m$ .
4. If  $j \geq 1$ , then set  $v = d$ , replace  $d$  by the solution of  $U_j d = v$ , replace  $j$  by  $j - 1$ , and repeat this step.
5. Set  $i = 1$ .
6. If  $i \leq k$ , then set  $v = d$ , replace  $d$  by the solution of  $E_i d = v$ , replace  $i$  by  $i + 1$ , and repeat this step.

To store each  $E_i$ , we need only store its eta column and record the position of this column in the matrix. Furthermore, if the eta columns are sufficiently sparse, then they may be stored in the "packed form" mentioned in Chapter 6, so that only the nonzero entries are stored and their positions in the column recorded. The same remark applies to the triangular eta matrices  $L_j$  and  $U_j$ . Each of the permutation matrices  $P_j$ , obtained by interchanging some row of  $I$  with the  $j$ th row, may be represented by a single pointer specifying the interchanged row. A sequential file storing the matrices

$$P_1, L_1, P_2, L_2, \dots, P_m, L_m, U_m, U_{m-1}, \dots, U_1, E_1, E_2, \dots, E_k$$

in this fashion is called the *eta file*. This file is scanned backward, from  $E_k$  to  $P_1$ , in solving the system  $yB_k = c_B$ , and it is solved forward, from  $P_1$  to  $E_k$ , in solving the system  $B_k d = a$ . For this reason, the procedure for solving  $yB_k = c_B$  is sometimes

referred to as the *backward transformation*, or BTRAN, and the procedure for solving  $B_k d = a$  is referred to as the *forward transformation*, or FTRAN. Note that the backward and the forward scans alternate and that each new item  $E_{k+1}$  is added to the open end of the file after the file has been scanned forward all the way to  $E_k$  and before the next scan backward to  $P_1$  begins. (The reader should be warned that the term *eta file* is usually employed in connection with the "product form of the inverse," in which case it refers to a different file; see problem 7.13.)

### Refactorizations

Since the eta file grows with each iteration, BTRAN and FTRAN become progressively more and more laborious; eventually, they could even take longer than solving the two systems  $yB_k = c_B$  and  $B_k d = a$  from scratch. Such counterproductive uses of the eta file may be avoided by discarding the whole file from time to time and treating the current  $B_k$  as a new  $B_0$ : compute a fresh triangular factorization of this matrix, and let a new sequence  $E_1, E_2, E_3, \dots$  grow from that point on. These periodic *refactorizations* of the basis keep the overall time spent on executions of steps 1 and 3 within acceptable limits. (For historical reasons, refactorizations are sometimes referred to as "reinversions.")

How often should the basis be refactorized? If  $T_0$  stands for the time spent on the refactorization, if  $T_k$  stands for the time spent on BTRAN and FTRAN in the  $k$ th iteration after refactorization, and if the basis is refactorized after  $r$  iterations, then the average time per execution of steps 1 and 3, including an appropriate share of the overhead  $T_0$ , comes to

$$T_r^* = \frac{1}{r} \sum_{k=0}^r T_k. \quad (7.14)$$

Obviously,  $r$  should be chosen so as to minimize  $T_r^*$ . A trivial way of doing so relies on the observation that  $T_1^*, T_2^*, T_3^*, \dots$  first decrease (as the overhead  $T_0$  gets distributed over more and more iterations) and then they begin to grow (as the length of the eta file begins to take over). Thus, we need only keep track of the cumulative total  $T_0 + T_1 + \cdots + T_k$  and refactorize as soon as this quantity divided by  $k$  stops decreasing. (A rigorous proof of this claim, relying only on the natural assumption that  $T_1 \leq T_2 \leq T_3 \leq \cdots$  is left for problem 7.7.)

In solving large sparse problems arising from applications, the basis is refactorized quite frequently, often after every twenty iterations or so. An exact analysis of the reasons behind these frequent refactorizations is both impossible and unnecessary: impossible, since the relevant statistics vary unpredictably from one problem to the next and unnecessary, since there is no point in a theoretical justification of a policy whose practical success has been firmly established. All the same, the insight provided by an inexact analysis is better than no insight at all. For this reason, we are going to present a few observations concerning the behavior of the large sparse problems encountered in practice.



The matrices featured in these problems are very sparse, typically with at most ten nonzeros per column. Furthermore, they have an implicit special structure allowing each basis matrix to be permuted into a near-triangular form, as discussed in Chapter 6. The resulting triangular factorizations of  $\mathbf{B}_0$  are very sparse, typically with up to  $12m$  nonzeros or so (discounting, as usual, the diagonal ones in  $\mathbf{L}_1, \mathbf{L}_2, \dots, \mathbf{L}_m$ ). If  $\mathbf{B}_0 = \mathbf{I}$ , then the eta columns of  $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3, \dots$  are initially very sparse: in particular, the eta column of  $\mathbf{E}_1$  is nothing but some column of  $\mathbf{A}$ . However, as more and more slack variables are pivoted out of the basis, the density of these eta columns increases until it reaches a steady state that persists after subsequent refactorizations. Typically, the eta columns are 25%–50% dense in this steady state, although densities outside this range do occur.

These statistics permit us to estimate the time required to execute BTRAN and FTRAN in each iteration of the revised simplex method. Taking the time required to execute one multiplication and one addition as our basic unit, we observe that the time required to execute BTRAN equals the length of the eta file; assuming that the eta columns are 50% dense, we may estimate the length of the eta file in the  $k$ th iteration by  $12m + 0.5m(k - 1)$ . The same quantity provides an upper bound on the time required to execute FTRAN. However, many of the multiplications in FTRAN are multiplications by zero, and so considerable time-savings can be claimed in this case. In short, the point is that the right-hand side  $\mathbf{a}$  of the system  $\mathbf{B}_k \mathbf{d} = \mathbf{a}$  will be very sparse, but the right-hand side  $\mathbf{c}_B$  of the system  $\mathbf{y} \mathbf{B}_k = \mathbf{c}_B$  may be completely dense. More precisely, consider the last stage of FTRAN, where systems  $\mathbf{E}_i \mathbf{d} = \mathbf{v}$  are solved. Since the right-hand side  $\mathbf{v}$  is a solution of  $\mathbf{B}_{i-1} \mathbf{v} = \mathbf{a}$ , about  $m/2$  or more of its  $m$  components may be expected to be zeros:  $\mathbf{v}$  would have been the eta column in the  $(i - 1)$ th iteration after the last refactorization if  $\mathbf{a}$  were the entering column then. Thus there is at least an even chance that  $\mathbf{v}$  will have a zero in that position in which the eta column of  $\mathbf{E}_i$  occurs; if that is the case, then the system  $\mathbf{E}_i \mathbf{d} = \mathbf{v}$  reduces to  $\mathbf{d} = \mathbf{v}$ . Since similar, if not better, time-savings occur in the earlier stages of FTRAN, we estimate the time required to execute FTRAN by a half of the time required to execute BTRAN. Thus, we are led to estimate the total time required to execute BTRAN and FTRAN in the  $k$ th iteration by

$$T_k \approx 18m + 0.75m(k - 1).$$

Estimating the time  $T_0$  required to refactorize  $\mathbf{B}_0$  is much trickier: permuting  $\mathbf{B}_0$  into the near-triangular form may take just about as much time as the subsequent computation of the triangular factorization. For the sake of argument, we make an educated guess at

$$T_0 \approx 100m.$$

Substituting these estimations into (7.14), we obtain

$$T_r^* \approx \frac{m}{r} \left( 100 + \sum_{k=1}^r (18 + 0.75(k - 1)) \right) \approx m \left( \frac{100}{r} + 18 + 0.375r \right).$$

Now a routine exercise in calculus (left for problem 7.8) shows that right-hand side is minimized by  $r \approx 16$ , a value within the range of the typical refactorization frequencies used in practice. For future reference, note that refactorizations with this frequency (i) keep the average time required to execute steps 1 and 3 (including the refactorization overhead) down to an estimated  $30m$  or so, and (ii) keep the average length of the eta file down to an estimated  $16m$  or so.

One flaw in this approximate analysis is the assumption that the density of the eta columns has reached its steady state. This steady state is reached only after most of the slack variables

have been pivoted out of the basis, which takes about  $m$  iterations. But the simplex method may terminate after just about  $m$  iterations, in which case the steady state is simply never reached. Nevertheless, the results of the analysis are not completely misleading. True, the density of the eta columns will fall far below the estimated 50% in the early iterations. However,  $\mathbf{B}_0$  will share many columns with  $\mathbf{I}$  in this case, and so the time required to find its triangular factorization and the number of nonzeros in this factorization will be reduced as well. Even though  $30m$  will become a gross overestimate of the average time required to execute steps 1 and 3, the optimal refactorization frequency will remain in the neighborhood of twenty or so.

## THE REVISED SIMPLEX METHOD VERSUS THE STANDARD SIMPLEX METHOD

On the large sparse problems encountered in practice, an iteration of the revised simplex method takes less time than an iteration of the standard simplex method.

To gain an insight into this phenomenon, let us extend the approximate analysis just presented. We have estimated the average time required to execute steps 1 and 3 by  $30m$  or so; the time required to execute step 2 does not exceed the total number of nonzeros in  $\mathbf{A}$ , estimated at no more than  $10n$ . Since step 4 requires up to  $m$  divisions and step 5 requires up to  $m$  multiplications and additions, we are led to estimate the total time per iteration of the revised simplex method at about

$$32m + 10n.$$

On the other hand, the time required to update the dictionary in an iteration of the revised simplex method equals the number of nonzeros in the pivot row times the number of nonzeros in the pivot column. Since each eta column is nothing but a column of the corresponding dictionary, and since we have estimated the density of the eta columns at 50%, we shall estimate the density of the dictionaries by the same figure. Now about  $n/2$  nonzeros may be expected in each row and about  $m/2$  nonzeros in each column of the dictionary; thus, we are led to estimate the total time per iteration of the standard simplex method at

$$mn/4.$$

Note that we may assume  $n \geq m$ , for otherwise it would be easier to solve the problem by solving its dual, as noted in Chapter 5. In fact,  $n$  is considerably bigger than  $m$  in many of the problems encountered in practice; even  $n \geq 2m$  is quite usual. Thus, we are led to conclude that, on the typical sparse problems encountered in practice, the revised simplex method beats the standard simplex method as soon as the number of rows exceeds 100. (Problems with 100 rows are considered small by present-day standards; some people call a problem large only if it has at least 2,000 rows.)

Furthermore, when the simplex method is applied to very large problems, its running time is influenced by another factor, the nature of which is more engineering than mathematical. This factor arises from the way computers are constructed: data may be stored either in central core



memory, which can be accessed fast but whose capacity is relatively small, or in peripheral memory (on disks, drums, or tapes), which has practically unlimited capacity but is slow to access. The time required to retrieve data from peripheral memory may be considerable compared to the time required to execute the various arithmetical operations. When the length of the eta file (estimated before at about  $16m$  on the average) exceeds the capacity of the central memory, the file has to be retrieved from peripheral memory twice in each iteration of the revised simplex method: first in step 1 (BTRAN) and then in step 3 (FTRAN). In addition, when the number of nonzeros in  $\mathbf{A}$  (estimated before at no more than  $10n$ ) exceeds the capacity of central memory, then these nonzeros, or at least some of them, have to be retrieved from the peripheral memory in step 2. Thus we are led to estimate the average number of items retrieved from the peripheral memory in each iteration of the revised simplex method at

$$32m + 10n.$$

On the other hand, even if the dictionaries used in the standard simplex method remain reasonably sparse, new nonzeros in these dictionaries are created in unpredictable places with each iteration. Thus storing the dictionaries in a packed form would create awkward difficulties. If all their entries, including the zeros, are stored explicitly, then the total number of items retrieved from the peripheral memory in each iteration of the standard simplex method comes to

$$mn.$$

We conclude that considerations of the retrieval time favor the revised simplex method even more decisively than considerations of the time required to carry out the various arithmetical operations.

For these reasons, it is the revised simplex method rather than the standard simplex method that is used in computer programs for solving LP problems. The sparsity and the special structure of the data make the revised simplex method superior and constitute the two leitmotifs of large-scale linear programming.

However, it would be wrong to believe that the revised simplex method is always faster than the standard simplex method: an easy argument, left for problem 7.9, shows that an iteration of the revised simplex method takes longer than an iteration of the standard simplex method whenever the bases  $\mathbf{B}_k$  are completely dense and  $n < 2m$ .

Incidentally, it may be a good policy to use the revised simplex method even in solving small LP problems by hand. In addition to its obvious didactic value, this practice makes it possible to spot numerical mistakes immediately by comparing  $\mathbf{yB}$  with  $\mathbf{c}_B$ ,  $\mathbf{Bd}$  with  $\mathbf{a}$ , and  $\mathbf{Bx}_B^*$  with  $\mathbf{b}$ .

## PRICING IN THE REVISED SIMPLEX METHOD

The fact that the components of  $\mathbf{c}_N - \mathbf{yA}_N$  may be computed individually in the revised simplex method has important practical implications. One may discover quite a few candidates for entering the basis by computing only a part of the vector  $\mathbf{c}_N - \mathbf{yA}_N$ . This strategy is known as

*partial pricing*; the generic term *pricing* refers to any method of computing the vector  $\mathbf{c}_N - \mathbf{yA}_N$ , or a part of it, for the purpose of selecting an entering variable. Variations on partial pricing are unlimited. For instance, the heuristic assumption that a current candidate for entering the basis is likely to remain eligible during the next few iterations suggests the following strategy. Compute the first, for instance,  $n/3$  components of  $\mathbf{c}_N - \mathbf{yA}_N$  and set aside perhaps 40 of the most promising candidates for entering the basis. In the subsequent iterations, limit your choice of entering variables to these 40 candidates. When only about 20 of these candidates remain eligible, compute the next  $n/3$  components of  $\mathbf{c}_N - \mathbf{yA}_N$  and repeat the whole process. Another popular alternative is known as *multiple pricing*: having selected, say, 10 promising candidates for entering the basis, consider the restricted problem that involves only these 10 nonbasic variables along with the  $m$  current basic variables. Only after an optimal solution of the restricted problem is found, go on to select the next 10 candidates and to repeat the whole process. The effect of these strategies on the overall number of iterations is hard to predict; the motivation behind their development is the desire to minimize the overall computing time by reducing the time per iteration.

The merits of partial and multiple pricing become even more pronounced when the problem is so large that the matrix  $\mathbf{A}$  has to be stored in peripheral memory and its columns retrieved in step 2 of each iteration. Partial pricing reduces the retrieval time by retrieving fewer columns; multiple pricing transfers the 10 or so columns into central memory and solves the restricted problem there. (Because of its special format, the restricted problem is solved by the *standard* simplex method; each dictionary, having  $m + 1$  rows and only 10 or so columns, is small enough to fit in core.)

Incidentally, judging candidates for entering the basis simply by the magnitude of the components of  $\mathbf{c}_N - \mathbf{yA}_N$  is not the best policy: typically, the number of iterations may be reduced by the use of another criterion. Two such criteria, closely related to each other, have been developed by P. M. J. Harris (1973) and D. Goldfarb and J. K. Reid (1977); they are known as *Devex* and *steepest edge*, respectively. The extra computations involved in these alternatives do pay off: the overall running time decreases in most cases and sometimes it drops to only 50% or less of the original figure.

## ZERO TOLERANCES

The number zero plays a special role in the selection of an entering variable: a nonbasic variable is eligible for entering the basis if and only if the corresponding component of  $\mathbf{c}_N - \mathbf{yA}_N$  is greater than zero. Rounding errors can make zeros or small negative numbers appear as small positive numbers; making the corresponding nonbasic variables enter the basis would be a blunder. To safeguard against this trap, a small positive number  $\varepsilon_1$ , called a *zero tolerance*, is chosen in advance; then a component of  $\mathbf{c}_N - \mathbf{yA}_N$  is considered positive if and only if its computed value exceeds  $\varepsilon_1$ . Another zero tolerance,  $\varepsilon_2$ , safeguards against divisions by extremely small numbers (which tend to produce the most dangerous rounding errors): if some choice of the entering variable leads to an eta matrix whose diagonal element in the eta column is less than  $\varepsilon_2$  in magnitude, then this choice is rejected and another nonbasic variable is made to enter the basis. Yet another zero tolerance,  $\varepsilon_3$ , is used in monitoring the accuracy of computations by comparing  $\mathbf{Bx}_B^*$  with  $\mathbf{b}$ : if these two vectors differ in any component by more than  $\varepsilon_3$ , then the basis is refactored immediately. Setting the actual values of the various zero tolerances is a delicate matter requiring some expertise; in particular, an inappropriate choice of  $\varepsilon_1$  might lead the computations astray. B. A. Murtagh (1981) gives  $\varepsilon_1 = 10^{-5}$ ,  $\varepsilon_2 = 10^{-8}$ ,  $\varepsilon_3 = 10^{-6}$  as typical values for computations with 15 decimal digit precision.

## PROBLEMS

- 7.1 Solve problem 2.1 by the revised simplex method.
- 7.2 Solve problem 1.6 by the revised simplex method and interpret each iteration in economic terms.
- 7.3 Prove Theorem 5.4.
- 7.4 Prove Theorem 5.5.
- △ 7.5 Solve the systems  $yE_1E_2E_3E_4 = [1, 2, 3]$  and  $E_1E_2E_3E_4d = [1, 2, 3]^T$  with
- $$E_1 = \begin{bmatrix} 1 & 3 \\ & 0.5 \\ & 4 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 2 & & \\ 1 & 1 & \\ & 4 & 1 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 1 & & 1 \\ & 1 & 3 \\ & & 1 \end{bmatrix}, \quad E_4 = \begin{bmatrix} -0.5 & & \\ 3 & 1 & \\ & 1 & 1 \end{bmatrix}.$$
- 7.6 Let  $U$  be an  $m \times m$  upper triangular matrix and let  $U_j$  be the eta matrix obtained when the  $j$ th column of  $I$  is replaced by the  $j$ th column of  $U$ . Prove that  $U = U_m U_{m-1} \cdots U_1$ .
- 7.7 Let  $T_0, T_1, T_2, \dots$  be a sequence of numbers such that  $T_{k+1} \geq T_k$  whenever  $k \geq 1$ . Prove that the sequence  $T_1^*, T_2^*, T_3^*, \dots$  defined by
- $$T_r^* = \frac{1}{r} \sum_{k=0}^r T_k$$
- has the following property: if  $T_r^* \leq T_{r+1}^*$ , then  $T_{r+1}^* \leq T_{r+2}^*$ .
- 7.8 Given positive numbers  $a, b, c$ , minimize  $a/r + b + cr$ .
- 7.9 Show that an iteration of the revised simplex method takes longer than an iteration of the standard simplex method whenever the bases  $B_k$  are completely dense and  $n < 2m$ .
- 7.10 What is the optimal refactorization frequency in the revised simplex method when all the bases are completely dense?
- 7.11 Show that, if all the bases are completely dense, an iteration of the revised simplex method takes less time on the average than an iteration of the standard simplex method as long as (i) the basis is refactorized with an appropriate frequency, (ii)  $n > 5m$ , and (iii) only  $n/3$  out of the  $n$  nonbasic variables are priced out in each iteration. (Note, however, that assumption (iii) makes the comparison unfair. First, the number of iterations tends to increase when partial pricing is used. Secondly, (iii) is untenable when there are only a few candidates for entering the basis, which is the typical situation in the last iterations.)
- 7.12 Design a procedure that, given an  $m \times m$  nonsingular matrix  $B_0$ , finds eta matrices  $E_1^*, E_2^*, \dots, E_m^*$  such that
- $$B_0 = E_1^* E_2^* \cdots E_m^*.$$
- (Hint: Assuming that you have found  $E_1^*, E_2^*, \dots, E_j^*$ , whose product agrees with  $B_0$  in the first  $j$  columns, show how to find  $E_{j+1}^*$ .) In the revised simplex method, this eta factorization of  $B_0$  may be updated into
- $$B_k = E_1^* E_2^* \cdots E_{m+k}^*$$
- (by setting  $E_{m+i}^* = E_i$  in the  $i$ th iteration) and used to solve the two systems  $yB_k = c_B$  and  $B_k d = a$ . However, it is better to use a triangular factorization of  $B_0$  instead (as we

have done in the text): triangular factorizations are sparser and easier to compute than eta factorizations.

- 7.13 The device commonly presented in place of our “eta factorization of the basis” is the “product form of the inverse,” which consists of a sequence of eta matrices whose product is  $B_k^{-1}$ . How can this device be used and updated in each iteration? (Hint: Consult problem 7.12.)
- 7.14 Tradition seems to be the reason for clinging to the product form of the inverse: early versions of the revised simplex method used  $B_k^{-1}$ , represented by its individual entries, to solve the systems  $yB_k = c_B$  and  $B_k d = a$ . How can this device be used and updated in each iteration? Why is it unsuitable for solving the large sparse problems typically encountered in practice? (Hint: Consult problem 6.12.)

## General LP Problems: Solutions by the Simplex Method

Many linear programming problems involve explicit upper bounds on individual variables. For instance, one might be confronted with the problem

$$\begin{aligned} &\text{maximize} && \sum_{j=1}^n c_j x_j \\ &\text{subject to} && \sum_{j=1}^n a_{ij} x_j \leq b_i \quad (i = 1, 2, \dots, m) \\ &&& 0 \leq x_j \leq u_j \quad (j = 1, 2, \dots, n) \end{aligned} \quad (8.1)$$

such that each  $u_j$  is a positive number. This problem may be cast in the standard form

$$\begin{aligned} &\text{maximize} && \sum_{j=1}^n c_j x_j \\ &\text{subject to} && \sum_{j=1}^n a_{ij} x_j \leq b_i \quad (i = 1, 2, \dots, m) \\ &&& x_j \leq u_j \quad (j = 1, 2, \dots, n) \\ &&& x_j \geq 0 \quad (j = 1, 2, \dots, n) \end{aligned}$$

and then solved by the simplex method; when the revised simplex method is used, the size of each basis matrix is  $(m + n) \times (m + n)$ . Alternatively, the simplex method can be made to work on (8.1) directly in such a way that, when the revised simplex method is used, the size of each basis matrix is only  $m \times m$ . This technique, suggested by G. B. Dantzig (1955) and known as *upper bounding*, can be applied in the more general context of problems

$$\begin{aligned} &\text{maximize} && \sum_{j=1}^n c_j x_j \\ &\text{subject to} && \sum_{j=1}^n a_{ij} x_j = b_i \quad (i = 1, 2, \dots, m) \\ &&& l_j \leq x_j \leq u_j \quad (j = 1, 2, \dots, n). \end{aligned} \quad (8.2)$$

Here, each  $l_j$  is either a number or the symbol  $-\infty$ , meaning that no lower bound is imposed on  $x_j$ , and each  $u_j$  is either a number or the symbol  $+\infty$ , meaning that no upper bound is imposed on  $x_j$ . [To cast (8.1) in the form of (8.2), we need only add the slack variables  $x_{n+1}, x_{n+2}, \dots, x_{n+m}$ , with  $l_{n+i} = 0$  and  $u_{n+i} = +\infty$  for all  $i$ .] No generality is lost in assuming that  $l_j \leq u_j$  for all  $j$  [otherwise (8.2) is trivially infeasible] but, for technical reasons, it is convenient to admit variables  $x_j$  with  $l_j = u_j$  (even though such variables, whose values are fixed, can be deleted at once and the right-hand side adjusted appropriately).

The modifications that enable the simplex method to handle (8.2) directly are the subject of this chapter. Thus, we shall learn to solve even the most general LP problems, for *every LP problem is easily presentable in the form of (8.2)*. To justify this claim, we need only observe that minimizing  $\sum c_j x_j$  is tantamount to maximizing  $\sum (-c_j) x_j$  and that every inequality constraint can be converted into an equation by the introduction of an appropriate slack variable. (The only inequalities not converted into equations are the explicit bounds on individual variables; each of these is simply recorded as  $x_j \geq l_j$  or  $x_j \leq u_j$ .)

### HOW TO HANDLE EXPLICIT BOUNDS ON INDIVIDUAL VARIABLES

In matrix terms, (8.2) may be recorded as

$$\text{maximize } \mathbf{c}\mathbf{x} \quad \text{subject to } \mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{l} \leq \mathbf{x} \leq \mathbf{u}. \quad (8.3)$$

We shall say that the matrix  $\mathbf{A}$  has  $m$  rows and  $n$  columns; then  $\mathbf{b}$  is a column vector of length  $m$ ,  $\mathbf{c}$  is a row vector of length  $n$ , and  $\mathbf{x}, \mathbf{l}, \mathbf{u}$  are column vectors of length  $n$ . Note that we allow variables  $x_j$  with  $l_j = -\infty$  and  $u_j = +\infty$ . Such variables are called *free* or *unrestricted*, whereas the remaining variables are called *bounded* or

restricted. We shall say that a solution  $\mathbf{x}^*$  of  $\mathbf{Ax} = \mathbf{b}$  is a *basic solution* of (8.3) if the  $n$  components of  $\mathbf{x}$  can be partitioned into  $m$  "basic" and  $n - m$  "nonbasic" variables in such a way that (i) the  $m$  columns of  $\mathbf{A}$  corresponding to the basic variables form a nonsingular matrix and (ii) the value  $x_j^*$  of each bounded nonbasic variable  $x_j$  is  $l_j$  or  $u_j$ . A basic solution  $\mathbf{x}^*$  is called *feasible* if  $\mathbf{l} \leq \mathbf{x}^* \leq \mathbf{u}$ . [This definition is consistent with our original definition of a basic feasible solution in the special case when (8.3) arises from an LP problem in the standard form.] For instance, if

$$\mathbf{A} = \begin{bmatrix} 3 & 1 & 5 & 6 & 9 & 4 & 3 & 4 & 7 & 6 & 4 & 5 \\ 1 & 0 & 9 & 5 & 8 & 1 & 2 & 7 & 8 & 7 & 9 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 72 \\ 62 \end{bmatrix},$$

$$\mathbf{c} = [2, 1, -2, -2, 3, 2, 3, -4, 0, -2, -3, 3]$$

and

$$\mathbf{l} = [-5, -\infty, -4, -2, 2, 0, 0, 3, -\infty, -\infty, -\infty, -\infty]^T$$

$$\mathbf{u} = [+ \infty, 3, -2, 3, 5, 1, + \infty, + \infty, 0, 5, + \infty, + \infty]^T$$

then

$$\mathbf{x}^* = [1, 0, -2, 3, 2, 0, 0, 3, 0, 5, -1, 1]^T$$

is a basic feasible solution; the variables  $x_1, x_2$  are basic, whereas  $x_3, x_4, \dots, x_{12}$  are nonbasic. (Note that, in the present context, a basic solution is not always determined by the choice of basic variables alone. For instance,

$$\mathbf{x}^* = [2, 2, -2, 3, 2, 0, 0, 3, 0, 5, -1, 0]^T$$

and

$$\mathbf{x}^* = [1, 1, -2, 3, 2, 1, 0, 3, 0, 5, -1, 0]^T$$

are other basic feasible solutions with  $x_1, x_2$  basic and  $x_3, x_4, \dots, x_{12}$  nonbasic in our example.)

To solve problem (8.3), the simplex method replaces one basic feasible solution by another in every (nondegenerate) iteration. Again, the idea is to change the value of only one of the nonbasic variables (the entering variable) in such a way that, when the constraints  $\mathbf{Ax} = \mathbf{b}$  are maintained by adjusting the values of the basic variables, feasibility is preserved and the value of the objective function increases. For instance, we might change the value of  $x_7$  in our example from zero to some positive  $t$ . When the values of the basic variables are adjusted to  $x_1 = 1 - 2t$  and  $x_2 = 3t$ , feasibility is preserved (as long as  $t$  is sufficiently small) and the value of the objective function increases from  $-10$  to  $-10 + 2t$ . Alternatively, we might change the value of  $x_{12}$  from  $1$  to  $1 - t$  for some positive  $t$ . When the values of the basic variables are adjusted to  $x_1 = 1 + t$  and  $x_2 = 2t$ , feasibility is preserved (as long as  $t$  is sufficiently small) and the value of the objective function increases from  $-10$  to  $-10 + t$ .

General changes of this kind may be investigated along the lines of Chapter 7: having defined  $\mathbf{B}, \mathbf{A}_N, \mathbf{x}_B, \mathbf{x}_N$  and  $\mathbf{c}_B, \mathbf{c}_N$ , we observe that the constraints  $\mathbf{Ax} = \mathbf{b}$  are equivalent to

$$\mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b} - \mathbf{B}^{-1}\mathbf{A}_N\mathbf{x}_N \quad (8.4)$$

and that every vector  $\mathbf{x}$  satisfying these constraints satisfies

$$\mathbf{c}\mathbf{x} = \mathbf{y}\mathbf{b} + (\mathbf{c}_N - \mathbf{y}\mathbf{A}_N)\mathbf{x}_N \quad (8.5)$$

with  $\mathbf{y} = \mathbf{c}_B\mathbf{B}^{-1}$ . Now consider a basic feasible solution  $\mathbf{x}^*$ ; let  $\mathbf{a}$  stand for the column of  $\mathbf{A}_N$  corresponding to some nonbasic  $x_j$ , and write  $\mathbf{d} = \mathbf{B}^{-1}\mathbf{a}$ . If the value of  $x_j$  changes from  $x_j^*$  to  $x_j^* + t$  but the remaining nonbasic variables stay fixed at their current values, then the right-hand side of (8.4) changes from  $\mathbf{B}^{-1}\mathbf{b} - \mathbf{B}^{-1}\mathbf{A}_N\mathbf{x}_N^*$  (which equals  $\mathbf{x}_B^*$ ) to  $\mathbf{B}^{-1}\mathbf{b} - \mathbf{B}^{-1}\mathbf{A}_N\mathbf{x}_N^* - t\mathbf{B}^{-1}\mathbf{a}$  (which equals  $\mathbf{x}_B^* - t\mathbf{d}$ ). Therefore, to keep the constraints (8.4) satisfied, we must change the values of the basic variables from  $\mathbf{x}_B^*$  to  $\mathbf{x}_B^* - t\mathbf{d}$ . The effect of this change on the value of the objective function can be seen in (8.5): since the component of  $\mathbf{c}_N - \mathbf{y}\mathbf{A}_N$  corresponding to  $x_j$  is  $c_j - \mathbf{y}\mathbf{a}$ , the right-hand side of (8.5) changes from  $\mathbf{y}\mathbf{b} + (\mathbf{c}_N - \mathbf{y}\mathbf{A}_N)\mathbf{x}_N^*$  (which equals  $\mathbf{c}\mathbf{x}^*$ ) to  $\mathbf{y}\mathbf{b} + (\mathbf{c}_N - \mathbf{y}\mathbf{A}_N)\mathbf{x}_N^* + (c_j - \mathbf{y}\mathbf{a})t$  (which equals  $\mathbf{c}\mathbf{x}^* + (c_j - \mathbf{y}\mathbf{a})t$ ). To summarize:

If  $x_j$  changes from  $x_j^*$  to  $x_j^* + t$ , then  $\mathbf{x}_B$  has to change from  $\mathbf{x}_B^*$  to  $\mathbf{x}_B^* - t\mathbf{d}$  and  $\mathbf{c}\mathbf{x}$  changes from  $\mathbf{c}\mathbf{x}^*$  to  $\mathbf{c}\mathbf{x}^* + (c_j - \mathbf{y}\mathbf{a})t$ .

Replacing  $t$  with  $-t$ , we see the following:

If  $x_j$  changes from  $x_j^*$  to  $x_j^* - t$ , then  $\mathbf{x}_B$  has to change from  $\mathbf{x}_B^*$  to  $\mathbf{x}_B^* + t\mathbf{d}$  and  $\mathbf{c}\mathbf{x}$  changes from  $\mathbf{c}\mathbf{x}^*$  to  $\mathbf{c}\mathbf{x}^* - (c_j - \mathbf{y}\mathbf{a})t$ .

We wish to increase the value of the objective function  $\mathbf{c}\mathbf{x}$  by changing the value of the entering nonbasic variable from  $x_j^*$  to  $x_j^* + t$  or  $x_j^* - t$  for some positive  $t$ . The increase to  $x_j^* + t$  is allowed only if  $x_j^* < u_j$  and brings about the desired increase in  $\mathbf{c}\mathbf{x}$  only if  $c_j - \mathbf{y}\mathbf{a} > 0$ ; the decrease to  $x_j^* - t$  is allowed only if  $x_j^* > l_j$  and brings about the desired increase in  $\mathbf{c}\mathbf{x}$  only if  $c_j - \mathbf{y}\mathbf{a} < 0$ . Hence, the entering variable  $x_j$  must satisfy either

$$c_j - \mathbf{y}\mathbf{a} > 0 \quad \text{and} \quad x_j^* < u_j \quad (8.6)$$

or

$$c_j - \mathbf{y}\mathbf{a} < 0 \quad \text{and} \quad x_j^* > l_j. \quad (8.7)$$

The absence of nonbasic variables  $x_j$  satisfying (8.6) or (8.7) implies optimality of our current solution  $\mathbf{x}^*$ . To justify this claim, denote the row vector  $\mathbf{c}_N - \mathbf{y}\mathbf{A}_N$  by  $\bar{\mathbf{c}}_N$ . If no nonbasic variable  $x_j$  satisfies (8.6), then  $x_j^* = u_j$  whenever  $\bar{c}_j > 0$ ; if no nonbasic variable  $x_j$  satisfies (8.7), then  $x_j^* = l_j$  whenever  $\bar{c}_j < 0$ . But then  $\bar{c}_j x_j \leq \bar{c}_j x_j^*$  whenever  $l_j \leq x_j \leq u_j$ . Thus, every feasible solution  $\mathbf{x}$  of (8.3) satisfies  $\bar{\mathbf{c}}_N \mathbf{x}_N \leq \bar{\mathbf{c}}_N \mathbf{x}_N^*$  and, by virtue of (8.5), it satisfies  $\mathbf{c}\mathbf{x} \leq \mathbf{c}\mathbf{x}^*$ . Hence,  $\mathbf{x}^*$  is an optimal solution of (8.3).

As in Chapter 7, the vector  $\mathbf{y} = \mathbf{c}_B \mathbf{B}^{-1}$  may be found by solving the system  $\mathbf{y}\mathbf{B} = \mathbf{c}_B$ . In our example, this system reads

$$\mathbf{y} \begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix} = [2, 1], \quad \text{and so we have} \quad \mathbf{y} = [1, -1].$$

It is easy to check that the only nonbasic variables satisfying (8.6) are  $x_5, x_7$ , and  $x_{11}$ , whereas the only nonbasic variables satisfying (8.7) are  $x_4, x_{10}$ , and  $x_{12}$ . Hence, the entering variable may be any of the nonbasic variables  $x_4, x_5, x_7, x_{10}, x_{11}, x_{12}$ .

It will be convenient to write

$$x_j(t) = x_j^* + t, \quad \mathbf{x}_B(t) = \mathbf{x}_B^* - t\mathbf{d}$$

if the entering variable  $x_j$  satisfies (8.6), and

$$x_j(t) = x_j^* - t, \quad \mathbf{x}_B(t) = \mathbf{x}_B^* + t\mathbf{d}$$

if the entering variable  $x_j$  satisfies (8.7). As in Chapter 7, the vector  $\mathbf{d} = \mathbf{B}^{-1}\mathbf{a}$  featured here may be found by solving the system  $\mathbf{B}\mathbf{d} = \mathbf{a}$ . We intend to replace  $x_j^* = x_j(0)$  by  $x_j(t)$  and  $\mathbf{x}_B^* = \mathbf{x}_B(0)$  by  $\mathbf{x}_B(t)$  for some positive  $t$ . The resulting solution of  $\mathbf{A}\mathbf{x} = \mathbf{b}$  will remain feasible as long as

$$l_j \leq x_j(t) \leq u_j \quad \text{and} \quad \mathbf{l}_B \leq \mathbf{x}_B(t) \leq \mathbf{u}_B. \quad (8.8)$$

Since the value of the objective function increases with  $t$ , we wish to make  $t$  as large as these constraints allow. Let us distinguish among three cases:

- (i) The upper bound on  $t$  imposed by  $\mathbf{l}_B \leq \mathbf{x}_B(t) \leq \mathbf{u}_B$  is stricter than the upper bound (if any) imposed by  $l_j \leq x_j(t) \leq u_j$ .
- (ii) The upper bound on  $t$  imposed by  $l_j \leq x_j(t) \leq u_j$  is at least as strict as the upper bound (if any) imposed by  $\mathbf{l}_B \leq \mathbf{x}_B(t) \leq \mathbf{u}_B$ .
- (iii) Neither  $l_j \leq x_j(t) \leq u_j$  nor  $\mathbf{l}_B \leq \mathbf{x}_B(t) \leq \mathbf{u}_B$  impose any upper bounds on  $t$ .

Each of these three cases will be illustrated on our example.

If the entering variable is  $x_5$ , then we solve the system

$$\begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix} \cdot \mathbf{d} = \begin{bmatrix} 9 \\ 8 \end{bmatrix} \quad \text{and find} \quad \mathbf{d} = \begin{bmatrix} 8 \\ -15 \end{bmatrix}.$$

Now the constraints (8.8) assume the form

$$2 \leq 2 + t \leq 5 \quad \text{and} \quad -5 \leq 1 - 8t, \quad 15t \leq 3$$

which reduces to  $t \leq 0.2$ . This is an instance of (i): when  $t$  is set at its largest admissible value, one of the basic variables reaches its lower or upper bound. This variable leaves the basis (whereas the entering variable enters). In our case, the new basic feasible solution is

$$\mathbf{x}^* = [-0.6, 3, -2, 3, 2.2, 0, 0, 3, 0, 5, -1, 1]^T$$

with the variables  $x_1$  and  $x_5$  basic.

If the entering variable is  $x_4$ , then we solve the system

$$\begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix} \cdot \mathbf{d} = \begin{bmatrix} 6 \\ 5 \end{bmatrix} \quad \text{and find} \quad \mathbf{d} = \begin{bmatrix} 5 \\ -9 \end{bmatrix}.$$

Now the constraints (8.8) assume the form

$$-2 \leq 3 - t \leq 3 \quad \text{and} \quad -5 \leq 1 + 5t, \quad -9t \leq 3$$

which reduces to  $t \leq 5$ . This is an instance of (ii): when  $t$  is set at its largest admissible value, the entering variable switches from one of its bounds to the other, and the basis remains unchanged. (In particular, the entering variable does *not* enter the basis.) In our case, the new basic feasible solution is

$$\mathbf{x}^* = [26, -45, -2, -2, 2, 0, 0, 3, 0, 5, -1, 1]^T$$

with the old basic variables  $x_1, x_2$  remaining basic.

If the entering variable is  $x_{10}$ , then we solve the system

$$\begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix} \cdot \mathbf{d} = \begin{bmatrix} 6 \\ 7 \end{bmatrix} \quad \text{and find} \quad \mathbf{d} = \begin{bmatrix} 7 \\ -15 \end{bmatrix}.$$

Now the constraints (8.8) assume the form

$$5 - t \leq 5 \quad \text{and} \quad -5 \leq 1 + 7t, \quad -15t \leq 3$$

imposing no upper bound on  $t$  whatsoever. This is an instance of (iii): as  $t$  can be made arbitrarily large, the problem is unbounded. In our case, every

$$\mathbf{x}^* = [1 + 7t, -15t, -2, 3, 2, 0, 0, 3, 0, 5 - t, -1, 1]^T$$

with  $t \geq 0$  constitutes a feasible solution with  $\mathbf{c}\mathbf{x}^* = -10 + t$ .

To summarize, when the revised simplex method is made to work directly on (8.3), each of its iterations can be described as in Box 8.1.

**BOX 8.1 An Iteration of the Revised Simplex Method**

**Step 1.** Solve the system  $\mathbf{yB} = \mathbf{c}_B$ .

**Step 2.** Choose an entering variable  $x_j$ . This may be any nonbasic variable  $x_j$  such that, with  $\mathbf{a}$  standing for the corresponding column of  $\mathbf{A}$ , we have either  $\mathbf{ya} < c_j$ ,  $x_j^* < u_j$ , or  $\mathbf{ya} > c_j$ ,  $x_j^* > l_j$ . If there is no such variable then stop; the current solution  $\mathbf{x}^*$  is optimal.

**Step 3.** Solve the system  $\mathbf{Bd} = \mathbf{a}$ .

**Step 4.** Define  $x_j(t) = x_j^* + t$  and  $\mathbf{x}_B(t) = \mathbf{x}_B^* - t\mathbf{d}$  in case  $\mathbf{ya} < c_j$  and  $x_j(t) = x_j^* - t$ ,  $\mathbf{x}_B(t) = \mathbf{x}_B^* + t\mathbf{d}$  in case  $\mathbf{ya} > c_j$ . If the constraints

$$l_j \leq x_j(t) \leq u_j, \quad \mathbf{l}_B \leq \mathbf{x}_B(t) \leq \mathbf{u}_B$$

are satisfied for all positive  $t$  then stop; the problem is unbounded. Otherwise set  $t$  at the largest value allowed by these constraints. If the upper bound imposed on  $t$  by the constraints  $\mathbf{l}_B \leq \mathbf{x}_B(t) \leq \mathbf{u}_B$  is stricter than the upper bound imposed by  $l_j \leq x_j(t) \leq u_j$ , then determine the leaving variable. This may be any basic variable  $x_i$  such that the upper bound imposed on  $t$  by  $l_i \leq x_i(t) \leq u_i$  alone is as strict as the upper bound imposed by all the constraints in  $\mathbf{l}_B \leq \mathbf{x}_B(t) \leq \mathbf{u}_B$ .

**Step 5.** Replace  $x_j^*$  by  $x_j(t)$  and  $\mathbf{x}_B^*$  by  $\mathbf{x}_B(t)$ . If the value of the entering variable  $x_j$  has just switched from one of its bounds to the other, then proceed directly to step 2 of the next iteration. Otherwise, replace the leaving variable  $x_i$  by the entering variable  $x_j$  in the basis heading, and replace the leaving column of  $\mathbf{B}$  by the entering column  $\mathbf{a}$ .

**Degeneracy and Termination**

Our definition of a basic feasible solution admits basic feasible solutions  $\mathbf{x}^*$  with  $x_i^* = l_i$  or  $x_i^* = u_i$  for one or more basic variables  $x_i$ . Such basic feasible solutions are called *degenerate*. When a simplex iteration begins with a degenerate basic feasible solution  $\mathbf{x}^*$ , the constraints  $\mathbf{l}_B \leq \mathbf{x}_B(t) \leq \mathbf{u}_B$  may force  $t$  equal to 0 in step 4. In that case, the entering variable enters the basis and the leaving variable leaves, but the solution  $\mathbf{x}^*$  remains unchanged. Such iterations are called *degenerate*. The value of the objective function remains unchanged after each degenerate iteration and increases with each nondegenerate one. Hence only a sequence of degenerate iterations may create *cycling*, defined as the appearance of the same basic feasible solution with the same set of basic variables in two different iterations. [Note that, in the special case when (8.3) arises from an LP problem in the standard form, this definition of cycling is equivalent to the definition used in Chapter 3.] Theorem 3.1 told us that, on LP problems in the standard form cycling is the only reason why the simplex method may fail to terminate. Now we extend the result to arbitrary LP problems.

**THEOREM 8.1.** If the simplex method, applied to a problem

$$\text{maximize } \mathbf{cx} \quad \text{subject to } \mathbf{Ax} = \mathbf{b}, \quad \mathbf{l} \leq \mathbf{x} \leq \mathbf{u}$$

fails to terminate, then it must cycle.

**PROOF.** Assume that the simplex method fails to terminate and so goes through an endless sequence of iterations. At the beginning of each iteration, ask (i) which variables are basic, (ii) which nonbasic restricted variables are at their lower bounds, and (iii) which nonbasic restricted variables are at their upper bounds. Since there are only a finite number of possible answers to these three questions, one answer must come up in two different iterations. It remains to be proved that the basic feasible solutions  $\mathbf{x}'$  and  $\mathbf{x}''$  featured in these two iterations are also the same. Since

$$\mathbf{x}'_B = \mathbf{B}^{-1}\mathbf{b} - \mathbf{B}^{-1}\mathbf{A}_N\mathbf{x}'_N \quad \text{and} \quad \mathbf{x}''_B = \mathbf{B}^{-1}\mathbf{b} - \mathbf{B}^{-1}\mathbf{A}_N\mathbf{x}''_N$$

proving that  $\mathbf{x}' = \mathbf{x}''$  reduces to proving that  $\mathbf{x}'_N = \mathbf{x}''_N$ . Since the answers to questions (ii) and (iii) were the same in the two iterations, we have  $x'_j = x''_j$  whenever  $x_j$  is a nonbasic restricted variable. Now it only remains to be proved that  $x'_j = x''_j$  whenever  $x_j$  is a nonbasic free variable. But this conclusion follows at once from the observations that (a) if a nonbasic free variable changes its value in some iteration, then it enters the basis in that iteration and (b) a basic free variable never leaves the basis. ■

As noted in Chapter 3, cycling is very rare in practice, and its remote possibility is disregarded in most computer implementations of the simplex method. Furthermore, Theorem 3.3 may be extended as follows.

**THEOREM 8.2.** The simplex method, applied to a problem

$$\text{maximize } \mathbf{cx} \quad \text{subject to } \mathbf{Ax} = \mathbf{b}, \quad \mathbf{l} \leq \mathbf{x} \leq \mathbf{u}$$

terminates as long as the entering and the leaving variables are selected by the smallest-subscript rule in each iteration. ■

The proof follows the proof of Theorem 3.3 almost completely, except that now some of the inequalities in the argument may get reversed, depending on which of the three variables  $x_i$ ,  $x_s$ ,  $x_r$  are at their lower bounds and which of them are at their upper bounds. We leave the tedious details for problem 8.3. □

**The Two-Phase Simplex Method**

Now the only remaining difficulty in applying the simplex method to problems with explicit bounds on individual variables occurs in initialization: How do we get hold of a basic feasible solution to begin with? This difficulty may be overcome by a number of variations on the two-phase simplex method of Chapter 3. The idea remains the same: first the constraints  $\mathbf{Ax} = \mathbf{b}$  are extended by means of *artificial*

variables, making a basic feasible solution readily available, and then the values of these variables are driven to zero by solving an *auxiliary problem*. This strategy can be implemented in a variety of ways; one of them goes as follows.

When confronted with a problem

$$\begin{aligned} &\text{maximize} && \sum_{j=1}^n c_j x_j \\ &\text{subject to} && \sum_{j=1}^n a_{ij} x_j = b_i \quad (i = 1, 2, \dots, m) \\ &&& l_j \leq x_j \leq u_j \quad (j = 1, 2, \dots, n) \end{aligned} \quad (8.9)$$

introduce the artificial variables  $x_{n+1}, x_{n+2}, \dots, x_{n+m}$  and consider the constraints

$$\begin{aligned} &\sum_{j=1}^n a_{ij} x_j + x_{n+i} = b_i \quad (i = 1, 2, \dots, m) \\ &l_j \leq x_j \leq u_j \quad (j = 1, 2, \dots, n+m) \end{aligned} \quad (8.10)$$

with  $l_{n+1}, l_{n+2}, \dots, l_{n+m}$  and  $u_{n+1}, u_{n+2}, \dots, u_{n+m}$  defined as follows. Choose numbers  $\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n$  such that  $\tilde{x}_j = l_j$  or  $\tilde{x}_j = u_j$  for every bounded variable  $x_j$ , and set  $\tilde{x}_{n+i} = b_i - \sum_{j=1}^n a_{ij} \tilde{x}_j$  for all  $i = 1, 2, \dots, m$ . If  $\tilde{x}_{n+i} \geq 0$ , then set  $l_{n+i} = 0$  and  $u_{n+i} = +\infty$ ; if  $\tilde{x}_{n+i} < 0$ , then set  $l_{n+i} = -\infty$  and  $u_{n+i} = 0$ .

Now (8.9) has a feasible solution if and only if (8.10) has a solution with all the artificial variables at zero. In a sense, the degree to which a solution of (8.10) fails to provide a feasible solution of (8.9) can be measured by the sum of the magnitudes of the artificial variables. This sum equals  $\sum w_{n+i} x_{n+i}$ , with  $w_{n+i} = 1$  when  $l_{n+i} = 0$  and  $w_{n+i} = -1$  when  $u_{n+i} = 0$ ; its value is nonnegative for every solution of (8.10), and it equals zero if and only if all the artificial variables equal zero. Thus, the issue of existence of a feasible solution of (8.9) may be settled by solving the auxiliary problem,

$$\text{minimize} \quad \sum_{i=1}^m w_{n+i} x_{n+i} \quad \text{subject to (8.10)}$$

or, in the maximization form,

$$\text{maximize} \quad \sum_{i=1}^m (-w_{n+i}) x_{n+i} \quad \text{subject to (8.10)}.$$

The first phase of the two-phase simplex method consists of solving the auxiliary problem by the simplex method; this phase is initialized by the basic feasible solution  $\tilde{x}$  just described, with the  $m$  artificial variables being basic. At the end of the first phase, we either discover that the optimal value of the auxiliary problem is not zero, and so the original problem (8.9) has no feasible solutions at all, or else we

find an optimal solution  $\mathbf{x}^*$  of the auxiliary problem such that  $x_{n+1}^* = x_{n+2}^* = \dots = x_{n+m}^* = 0$ . In the latter case, the basic feasible solution  $\mathbf{x}^*$  is used to initialize the second phase of the two-phase simplex method. The second phase consists of applying the simplex method to an equivalent version of the original problem (8.9),

$$\begin{aligned} &\text{maximize} && \sum_{j=1}^n c_j x_j \\ &\text{subject to} && \sum_{j=1}^n a_{ij} x_j + x_{n+i} = b_i \quad (i = 1, 2, \dots, m) \\ &&& l_j \leq x_j \leq u_j \quad (j = 1, 2, \dots, n) \\ &&& 0 \leq x_{n+i} \leq 0 \quad (i = 1, 2, \dots, m). \end{aligned} \quad (8.11)$$

The only reason for retaining the artificial variables  $x_{n+i}$  in (8.11) is the fact that some of them may be featured as basic in the initial basic feasible solution  $\mathbf{x}^*$ . Each artificial variable that is nonbasic in  $\mathbf{x}^*$  may be deleted at once; each artificial variable that is basic in  $\mathbf{x}^*$  but becomes nonbasic later may be deleted as soon as it leaves the basis.

It is often the case that at least some of the  $m$  artificial variables  $x_{n+i}$  involved in the auxiliary problem may be omitted from the auxiliary problem from the very start and that the roles these variables play in the solution  $\tilde{x}$  initializing the first phase may be played by suitably chosen variables  $x_j$  of the original problem (8.9). In particular, this is the case for those artificial variables that simply duplicate slack variables in the constraints of the auxiliary problem. For instance, if (8.9) reads

$$\begin{aligned} &\text{maximize} && 3x_1 + x_2 + x_3 + 2x_4 \\ &\text{subject to} && 3x_1 + 2x_2 + 4x_3 + 2x_4 &= 2 \\ &&& 4x_1 + x_2 - 3x_3 + x_4 + x_5 &= -1 \\ &&& x_1 - 3x_2 + x_3 + 3x_4 &+ x_6 &= 3 \\ &&& 2x_1 + x_2 + x_3 + 3x_4 &&+ x_7 &= 5 \\ &&& x_1, x_2, \dots, x_7 &\geq 0 \end{aligned}$$

then the auxiliary problem assumes the form

$$\begin{aligned} &\text{maximize} && &&& -x_8 + x_9 - x_{10} - x_{11} \\ &\text{subject to} && 3x_1 + 2x_2 + 4x_3 + 2x_4 &&& + x_8 &= 2 \\ &&& 4x_1 + x_2 - 3x_3 + x_4 + x_5 &&& + x_9 &= -1 \\ &&& x_1 - 3x_2 + x_3 + 3x_4 &&& + x_6 &+ x_{10} &= 3 \\ &&& 2x_1 + x_2 + x_3 + 3x_4 &&& + x_7 &&+ x_{11} &= 5 \\ &&& x_1, x_2, \dots, x_7 &\geq 0, && x_8 \geq 0, && x_9 \leq 0, && x_{10} \geq 0, && x_{11} \geq 0. \end{aligned}$$



In the constraints of this auxiliary problem, the artificial variables  $x_{10}$  and  $x_{11}$  duplicate the slack variables  $x_6$  and  $x_7$ , respectively. Having made this observation, we may replace the auxiliary problem by

$$\begin{array}{llll} \text{maximize} & & & -x_8 + x_9 \\ \text{subject to} & 3x_1 + 2x_2 + 4x_3 + 2x_4 & + x_8 & = 2 \\ & 4x_1 + x_2 - 3x_3 + x_4 + x_5 & + x_9 & = -1 \\ & x_1 - 3x_2 + x_3 + 3x_4 & + x_6 & = 3 \\ & 2x_1 + x_2 + x_3 + 3x_4 & + x_7 & = 5 \\ & x_1, x_2, \dots, x_7 \geq 0, x_8 \geq 0, x_9 \leq 0 \end{array}$$

which can be initialized by  $\tilde{\mathbf{x}}_B = [\tilde{x}_8, \tilde{x}_9, \tilde{x}_6, \tilde{x}_7]^T = [2, -1, 3, 5]^T$ ,  $\tilde{\mathbf{x}}_N = \mathbf{0}$ . Thus, we bypass two iterations in the first phase: the first iteration with  $x_6$  entering and  $x_{10}$  leaving, the second iteration with  $x_7$  entering and  $x_{11}$  leaving. (In fact, some people prefer not to use the term *artificial variable* at all. They would think of  $x_9$  as the slack variable  $x_5$ , whose bounds  $l_5 = 0, u_5 = +\infty$  have been temporarily changed to  $l_5 = -\infty, u_5 = 0$ , and refer to  $x_8$  as “the slack variable of the equation  $3x_1 + 2x_2 + 4x_3 + 2x_4 = 2$ ,” whose bounds  $l_8 = 0, u_8 = 0$  have been temporarily changed to  $l_8 = 0, u_8 = +\infty$ .)

Incidentally, observe that the original objective function  $\sum c_j x_j$  does not affect the auxiliary problem at all. This is only fair: the issue to be settled in the first phase is that of feasibility, and the objective function is quite irrelevant there. Nevertheless, a complete disregard for the original objective may have unpleasant side effects: if the basic feasible solution delivered at the end of the first phase gives the original objective function an extremely low value, then the second phase may require a relatively large number of iterations to reach an optimal solution. (Of course, these statements are just intuitive observations; even though they tend to be true in general, they are not true under all circumstances.) It seems sensible, therefore, to keep an eye on the original objective function while the artificial variables are being driven out of the basis, so as to avoid unnecessarily sharp drops in its value. This idea is sometimes implemented by maximizing

$$\sum_{j=1}^n c_j x_j - M \sum_{i=1}^m w_{n+i} x_{n+i}$$

rather than  $\sum (-w_{n+i})x_{n+i}$  in the first phase. When the coefficient  $M$  is very large, minimizing  $\sum w_{n+i} x_{n+i}$  is still the primary concern, but sharp drops in the value of  $\sum c_j x_j$  do become undesirable to some extent. This variation on the two-phase method is referred to as the “big M method.” (There is no evidence to support the belief that the “big M” stands for “big mother.”)

Other popular variations involve maximizing

$$r \sum_{j=1}^n c_j x_j - \sum_{i=1}^m s_i w_{n+i} x_{n+i}$$

for suitably chosen “scaling factors”  $r$  and  $s_1, s_2, \dots, s_m$ . The actual choice of these numbers may depend on the current values of the artificial variables  $x_{n+i}$ , and so it may change from one iteration to the next. Then the auxiliary problem is no longer a linear programming problem (the objective function ceases to be linear), but all is well as long as all the artificial variables vanish in the end. The simplest example arises when  $r = 0$  and

$$s_i = \begin{cases} 1 & \text{if } x_{n+i} \neq 0 \\ 0 & \text{if } x_{n+i} = 0 \end{cases}$$

with each artificial variable  $x_{n+i}$  deleted as soon as it leaves the basis.

Finally, the two-phase simplex method may often be implemented without the use of artificial variables. To do so on a problem

$$\text{maximize } \mathbf{c}\mathbf{x} \quad \text{subject to } \mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{l} \leq \mathbf{x} \leq \mathbf{u}$$

with  $\mathbf{A}$  having size  $m \times n$  as usual, we require only a nonsingular matrix  $\mathbf{B}$  formed by some  $m$  columns of  $\mathbf{A}$ . Now we may choose the values  $\mathbf{x}_N^*$  of the nonbasic variables in such a way that  $x_i^* = l_i$  or  $x_i^* = u_i$  for each bounded nonbasic  $x_i$ . Having computed the values  $\mathbf{x}_B^*$  of the basic variables by solving the system  $\mathbf{B}\mathbf{x}_B^* = \mathbf{b} - \mathbf{A}_N \mathbf{x}_N^*$ , we obtain a basic solution  $\mathbf{x}^*$  that, in general, is infeasible: for each basic variable  $x_i$ , we may have  $x_i^* < l_i$  or  $x_i^* > u_i$ . Writing  $i \in P$  in the former case and  $i \in Q$  in the latter, we may measure the “infeasibility” of  $\mathbf{x}^*$  by the quantity

$$\sum_{i \in P} (l_i - x_i^*) + \sum_{i \in Q} (x_i^* - u_i).$$

Since we wish to drive the basic variables inside their bounds, our momentary aim is to increase the value of the objective function

$$\sum_{i \in P} x_i - \sum_{i \in Q} x_i$$

subject to the original constraints  $\mathbf{A}\mathbf{x} = \mathbf{b}, \mathbf{l} \leq \mathbf{x} \leq \mathbf{u}$  with  $l_i$  replaced by  $-\infty$  whenever  $i \in P$  and  $u_i$  replaced by  $+\infty$  whenever  $i \in Q$ . For this purpose, we may use the simplex method as usual. However, as the value of our objective function increases, the value of some  $x_i$  with  $i \in P$  may rise to  $l_i$ , or the value of some  $x_i$  with  $i \in Q$  may drop to  $u_i$ . At this very moment, the objective function must be updated and the original bounds  $l_i, u_i$  restored (otherwise further changes in the value of the entering variable might actually increase the infeasibility of the current solution); the variable  $x_i$  leaves the basis at the same time. Thus the first phase produces a sequence of basic solutions whose infeasibility decreases with each nondegenerate iteration until a basic feasible solution is found or its nonexistence established. We shall not go into the details any further.



### Getting Rid of Artificial Basic Variables

Unless the first phase of the two-phase simplex method discovers that (8.9) is infeasible, it delivers a basic feasible solution of the equivalent problem (8.11). This solution is not necessarily basic in the original problem (8.9), for some of its basic variables may be the artificial variables  $x_{n+i}$ . As previously observed, the presence of artificial variables in the basis creates no problem in the second phase of the two-phase simplex method. Nevertheless, it is only natural to question the role of these variables. Our investigations begin with a procedure which, given a basic feasible solution of (8.11), attempts to drive all artificial variables out of the basis (see Box 8.2). Without loss of generality, let us assume that each artificial basic variable  $x_{n+k}$  appears in the  $k$ th position of the basis heading. Note that each column of  $\mathbf{B}$  is either a column of  $\mathbf{A}$  or a column of  $\mathbf{I}$ .

#### BOX 8.2 Driving Artificial Variables Out of the Basis

*Step 0.* Let  $S$  denote the set of all the subscripts  $i$  such that  $x_{n+i}$  is a basic variable.

*Step 1.* If  $S$  is empty then stop. Otherwise delete some subscript  $k$  from  $S$ .

*Step 2.* Solve the system  $\mathbf{rB} = \mathbf{e}$  with  $\mathbf{e}$  standing for the  $k$ th row of the identity matrix. If  $\mathbf{rA}$  is the zero vector, then return to step 1. Otherwise, there is a nonbasic variable  $x_j$  such that  $\mathbf{ra} \neq 0$  for the corresponding column  $\mathbf{a}$  of  $\mathbf{A}$ . Replace the  $k$ th column of  $\mathbf{B}$  by  $\mathbf{a}$ ; replace  $x_{n+k}$  by  $x_j$  in the basis heading and return to step 1.

To verify that the updated matrix  $\mathbf{B}$  remains nonsingular after each execution of step 2, let  $\mathbf{B}$  stand for the matrix entering step 2 and note that its updated version equals  $\mathbf{BE}$  with  $\mathbf{E}$  standing for the eta matrix whose  $k$ th column is  $\mathbf{B}^{-1}\mathbf{a}$ . Thus we need only verify that  $\mathbf{E}$  is nonsingular. But  $\mathbf{E}$  is nonsingular if and only if the diagonal entry in its eta column is nonzero; now it remains to be observed only that this diagonal entry is  $\mathbf{e}(\mathbf{B}^{-1}\mathbf{a})$  and that  $(\mathbf{eB}^{-1})\mathbf{a} = \mathbf{ra} \neq 0$ .

Upon termination of this procedure, some artificial variables may persist in the basis. This apparent failure is actually a success in its own way: persistence of  $x_{n+k}$  in the basis means that the  $k$ th equation in (8.9) is redundant and may be deleted altogether. More precisely, let  $J$  denote the set of subscripts  $k$  such that  $x_{n+k}$  persists in the basis, and let  $I$  denote the set of subscripts  $1, 2, \dots, m$  that do not belong to  $J$ . We claim that every solution of

$$\sum_{j=1}^n a_{ij}x_j = b_i \quad (i \in I) \quad (8.12)$$

satisfies all the equations

$$\sum_{j=1}^n a_{ij}x_j = b_i \quad (i = 1, 2, \dots, m).$$

To justify this claim, consider an arbitrary subscript  $k$  in  $J$ , along with the vector  $\mathbf{r}$  computed immediately after the deletion of  $k$  from  $S$ . Since  $\mathbf{rA}$  is the zero vector, we have

$$\sum_{i=1}^m r_i a_{ij} = 0 \quad \text{for all } j = 1, 2, \dots, n. \quad (8.13)$$

Since  $\mathbf{rB} = \mathbf{e}$ , we have  $r_k = 1$  and  $r_j = 0$  for all the remaining subscripts  $j$  in  $J$ . Hence (8.13) may

be recorded as

$$a_{kj} = -\sum_{i \in I} r_i a_{ij}$$

and every solution of (8.12) must satisfy

$$\sum_{j=1}^n a_{kj}x_j = \sum_{j=1}^n \left( -\sum_{i \in I} r_i a_{ij} \right) x_j = \sum_{i \in I} \left( -r_i \sum_{j=1}^n a_{ij}x_j \right) = \sum_{i \in I} (-r_i b_i). \quad (8.14)$$

In particular, (8.14) must be satisfied by the feasible solution  $x_1^*, x_2^*, \dots, x_n^*$  of (8.9), and so

$$\sum_{i \in I} (-r_i b_i) = \sum_{j=1}^n a_{kj}x_j^* = b_k.$$

We conclude that every solution of (8.12) must satisfy

$$\sum_{j=1}^n a_{kj}x_j = b_k$$

which is the desired result.

Thus (8.9) and the problem

$$\begin{aligned} &\text{maximize} && \sum_{j=1}^n c_j x_j \\ &\text{subject to} && \sum_{j=1}^n a_{ij}x_j = b_i \quad (i \in I) \\ &&& l_j \leq x_j \leq u_j \quad (j = 1, 2, \dots, n) \end{aligned} \quad (8.15)$$

are equivalent in the sense that they have precisely the same set of feasible solutions. In particular, the feasible solution of (8.9) delivered by the first phase of the two-phase simplex method is a feasible solution of (8.15). We claim that this solution is basic in (8.15), with the appropriate set of basic variables having been delivered by the procedure described earlier. To justify this claim, denote by  $B$  the set of subscripts  $j$  with  $1 \leq j \leq n$  for which  $x_j$  appears in the basis heading upon termination of the procedure. We have to verify only that the system

$$\sum_{j \in B} a_{ij}x_j = b_i \quad (i \in I)$$

has a unique solution. But this claim follows at once from the fact that the system

$$\sum_{j \in B} a_{ij}x_j = b_i \quad (i \in I)$$

$$\sum_{j \in B} a_{ij}x_j + x_{n+i} = b_i \quad (i \in J)$$

has a unique solution.

Our findings may be summarized as follows.

**THEOREM 8.3.** If (8.9) has a feasible solution, then some set  $I$  of subscripts  $1, 2, \dots, m$  has the following two properties: (i) problems (8.9) and (8.15) have precisely the same set of feasible solutions and (ii) problem (8.15) has a basic feasible solution.

Of course, more important than the mere statement of Theorem 8.3 is the fact that we can actually find the set  $I$ , and that we can do so reasonably quickly. If our procedure fails to drive all the artificial variables out of the basis, then we can apply the second phase of the two-phase simplex method to the smaller problem (8.15), where each iteration requires less work than it would have required in the larger problem (8.11).

### Treatment of Free Variables

As the reader may have observed, each free variable  $x_j$  may be eliminated from the problem

$$\text{maximize } \mathbf{c}\mathbf{x} \quad \text{subject to} \quad \mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{l} \leq \mathbf{x} \leq \mathbf{u}$$

by first converting one of the equations into a formula for  $x_j$  and then substituting from this formula into the remaining equations. For illustration, consider the problem

$$\begin{aligned} &\text{maximize} && x_1 \\ &\text{subject to} && x_1 + x_2 + x_3 + x_4 + x_5 + x_6 - x_7 = 10 \\ & && x_1 + 2x_2 && + x_7 = 2 \\ & && && x_3 + 2x_4 && + x_7 = 5 \\ & && && && x_5 + 2x_6 + x_7 = 8 \\ & && && && x_1, x_2, x_3, x_4, x_5, x_6 \geq 0, \quad -\infty \leq x_7 \leq +\infty. \end{aligned}$$

Having converted the first equation into the formula

$$x_7 = -10 + x_1 + x_2 + x_3 + x_4 + x_5 + x_6$$

we substitute into the remaining three equations and obtain an equivalent version of our problem,

$$\begin{aligned} &\text{maximize} && x_1 \\ &\text{subject to} && 2x_1 + 3x_2 + x_3 + x_4 + x_5 + x_6 = 12 \\ & && x_1 + x_2 + 2x_3 + 3x_4 + x_5 + x_6 = 15 \\ & && x_1 + x_2 + x_3 + x_4 + 2x_5 + 3x_6 = 18 \\ & && x_1, x_2, x_3, x_4, x_5, x_6 \geq 0. \end{aligned}$$

More generally, whenever all the free variables are eliminated in this manner, not only do these variables disappear but the number of equations is reduced as well. Now the following strategy suggests itself: first eliminate all the free variables, then solve the new version of the problem by the simplex method, and finally determine the values of the free variables by going back to the substitution formulas. Since the new version is smaller than the original, this strategy has at least two apparent advantages: (i) each of the systems  $\mathbf{yB} = \mathbf{c}_B$  and  $\mathbf{Bd} = \mathbf{a}$ , being smaller in the new version, is easier to solve, (ii) the number of iterations is likely to be smaller in the new version. However, if the original problem is sparse, then the elimination of the

free variables may yield a problem that, although smaller, is also much denser. In that case, the first advantage is lost: relatively small but dense systems may be harder to solve than relatively large but sparse ones. On the other hand, the effects of the second advantage may be felt even without the elimination of free variables: when a free variable has entered the basis, it will never leave again. Thus, a problem with  $k$  free variables in the basis is like a problem with  $k$  equations deleted. For this reason, it is good policy to pivot all the free variables into the basis as soon as possible.

## THE FUNDAMENTAL THEOREM OF LINEAR PROGRAMMING REVIEWED

In Chapter 3, we established a fundamental property of LP problems in the standard form: whenever such a problem has a feasible solution, it has a basic feasible solution. This property is not shared by every problem

$$\text{maximize } \mathbf{c}\mathbf{x} \quad \text{subject to} \quad \mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{l} \leq \mathbf{x} \leq \mathbf{u}. \quad (8.16)$$

For instance, the problem

$$\begin{aligned} &\text{maximize} && x_1 + x_2 + x_3 + x_4 + x_5 \\ &\text{subject to} && 3x_1 + 4x_2 + 2x_3 + 5x_4 + x_5 = 6 \\ & && 2x_1 + x_2 + 3x_3 && - x_5 = -1 \\ & && x_1 + 2x_2 && + 3x_4 + x_5 = 4 \\ & && && x_1, x_2, x_3, x_4, x_5 \geq 0 \end{aligned} \quad (8.17)$$

has feasible solutions ( $x_1^* = x_2^* = x_3^* = 0$ ,  $x_4^* = x_5^* = 1$  is one) but it has no basic feasible solution. To justify this claim, recall that (8.16) can have a basic feasible solution only if some three columns of

$$\mathbf{A} = \begin{bmatrix} 3 & 4 & 2 & 5 & 1 \\ 2 & 1 & 3 & 0 & -1 \\ 1 & 2 & 0 & 3 & 1 \end{bmatrix}$$

form a nonsingular matrix  $\mathbf{B}$ . But we have

$$[3, -2, -5] \cdot \mathbf{A} = [0, 0, 0, 0, 0]$$

and so  $[3, -2, -5] \cdot \mathbf{B} = [0, 0, 0]$  for each matrix  $\mathbf{B}$  formed by three columns of  $\mathbf{A}$ . Now  $\mathbf{B}$  must be singular, for otherwise

$$[3, -2, -5] = [3, -2, -5] \cdot \mathbf{B}\mathbf{B}^{-1} = [0, 0, 0] \cdot \mathbf{B}^{-1} = [0, 0, 0]$$

which is a contradiction.

More generally, (8.16) can have a basic solution only if some  $m$  columns of the  $m \times n$  matrix  $\mathbf{A}$  form a nonsingular matrix. Not all matrices  $\mathbf{A}$  have this property; those that do are said to have *full row rank*. (If  $\mathbf{rA}$  is the zero vector of length  $n$  for some nonzero vector  $\mathbf{r}$  of length  $m$ ,

then, by the argument used in the previous example,  $A$  does not have full row rank. See also problem 8.4.) Thus, Theorem 3.4 can be generalized only to problems (8.16) whose matrices have full row rank.

Furthermore, a little care must be taken, so that the actual impact of Theorem 3.4, as well as its statement, is generalized. An important consequence of Theorem 3.4 is the fact that the search for an optimal solution of an LP problem in the standard form may be restricted to a finite set, namely, the set of basic feasible solutions. Confining the search for an optimal solution of (8.16) to the set of its basic feasible solutions would not hold the same attraction: problems involving free variables may have infinitely many basic feasible solutions. For instance, if

$$A = \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 0 & 3 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

and  $l = [-\infty, 0, 0, -\infty]^T$ ,  $u = [1, +\infty, 5, +\infty]^T$ , then  $[1 - t, t, 0, t]^T$  is a basic feasible solution of (8.16), with  $x_1$  and  $x_2$  basic, whenever  $t \geq 0$ . To remove this blemish, we introduce the notion of a *normal* basic solution: this is any basic solution in which each nonbasic free variable is set at zero. Clearly, every problem (8.16) has only a finitely many normal basic solutions.

**THEOREM 8.4.** If (8.16) has no optimal solution, then it is either infeasible or unbounded. Furthermore, if  $A$  has full row rank, then (8.16) has the following two properties:

- (i) If it has a feasible solution, then it has a normal basic feasible solution.
- (ii) If it has an optimal solution, then it has a normal basic optimal solution.

**PROOF.** The first statement follows easily from an analysis of the two-phase simplex method applied to (8.9). The first phase terminates either by discovering that (8.9) is infeasible or by producing a basic feasible solution of the equivalent problem (8.11). In the latter case, the second phase takes place. This phase terminates either by discovering that (8.11), and therefore also (8.9), is unbounded, or by producing a basic optimal solution of (8.11) that yields an optimal solution of (8.9). The termination of each phase is guaranteed by Theorem 8.2 as long as the smallest-subscript pivoting rule is used.

To prove (i) and (ii), assume that  $A$  has full row rank. If (8.9) has a feasible solution, then the first phase terminates by producing a basic feasible solution  $x_1^*, x_2^*, \dots, x_{n+m}^*$  of (8.11). The procedure used in the proof of Theorem 8.3 will drive all the artificial variables out of the basis; otherwise, it would find a nonzero row vector  $r$  of length  $m$  such that  $rA$  is the zero vector of length  $n$ , contradicting the assumption that  $A$  has full row rank. Hence  $x_1^*, x_2^*, \dots, x_n^*$  is a basic feasible solution of (8.9) in this case. To ensure that this solution is normal, we need only choose the numbers  $\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n$  used to construct (8.10) in such a way that  $\tilde{x}_j = 0$  whenever  $x_j$  is free. Indeed, as soon as the simplex method is initialized by a normal feasible solution, it will construct only normal basic feasible solutions; if a nonbasic free variable actually changes its value in some iteration, then it enters the basis in the same iteration and it remains basic from that point on. Finally, if (8.9) has an optimal solution, then the second phase, initialized by the normal basic feasible solution  $x_1^*, x_2^*, \dots, x_n^*$ , finds a normal basic optimal solution. ■

As observed earlier, the assumption that  $A$  has full row rank cannot be dropped in proving (i) and (ii). Nevertheless, this assumption is satisfied whenever (8.16) arises from a problem in the standard form. In this case, the last  $m$  columns of  $A$  form the identity matrix. Hence Theorem 8.4 does generalize Theorem 3.4.

## PROBLEMS

△ 8.1 Solve the following problems:

a. maximize  $-3x_1 - x_2 - x_3 + 2x_4 - x_5 + x_6 + x_7 - 4x_8$   
 subject to  $x_1 + 3x_3 + x_4 - 5x_5 - 2x_6 + 4x_7 - 6x_8 = 7$   
 $x_2 - 2x_3 - x_4 + 4x_5 + x_6 - 3x_7 + 5x_8 = -3$   
 $0 \leq x_1 \leq 8$   
 $0 \leq x_2 \leq 6$   
 $0 \leq x_3 \leq 4$   
 $0 \leq x_4 \leq 15$   
 $0 \leq x_5 \leq 2$   
 $0 \leq x_6 \leq 10$   
 $0 \leq x_7 \leq 10$   
 $0 \leq x_8 \leq 3$ .

b. maximize  $3x_1 + x_2 + 4x_3 + 2x_4$   
 subject to  $x_1 + 4x_2 + 3x_3 + 3x_4 \leq 2$   
 $x_1 + 3x_2 - x_3 + x_4 \leq -2$   
 $x_1 + 2x_2 + 3x_3 + 2x_4 \leq 3$   
 $x_1 + 3x_2 - 2x_3 + x_4 \leq -3$   
 $x_3, x_4 \geq 0$ .

c. maximize  $5x_1 + 2x_2 - 3x_3 + 3x_4 + 6x_5 + x_6$   
 subject to  $3x_1 + x_2 - 4x_3 + 2x_4 + 5x_5 + x_6 \leq 3$   
 $-5x_1 + 4x_2 + 2x_3 - 3x_4 + 2x_5 + 3x_6 \leq 25$   
 $x_1 + x_2 + 2x_3 + x_4 + x_5 + 2x_6 = 4$   
 $x_1 \geq 0, 2 \leq x_2 \leq 10, x_3 \leq 0, -3 \leq x_4 \leq 3$ .

d. maximize  $8x_1 + 6x_2 - 2x_3 + 6x_4 - 3x_5$   
 subject to  $3x_1 + 2x_2 - x_3 + 3x_4 + 2x_5 = 7$   
 $-x_1 + 2x_2 - 2x_3 = 10$   
 $x_1 - 3x_4 + 2x_5 = 7$   
 $2x_1 - 3x_2 + 2x_3 + 3x_4 = -10$   
 $2x_1 + x_3 - x_5 \leq -6$   
 $x_1, x_2 \geq 0$ .

e. Maximize  $x_1$   
 subject to  $3x_1 + 5x_2 + 3x_3 + 2x_4 + x_5 + 2x_6 + x_7 = 6$   
 $4x_1 + 6x_2 + 5x_3 + 3x_4 + 5x_5 + 5x_6 + 6x_7 = 11$   
 $2x_1 + 4x_2 + x_3 + x_4 - 3x_5 - x_6 - 4x_7 = 1$   
 $x_1 + x_2 + 2x_3 + x_4 + 4x_5 + 3x_6 + 5x_7 = 5$   
 $x_1, x_2, \dots, x_7 \geq 0$ .

8.2 Solve Polly's diet problem from Chapter 1 and illustrate each iteration in economic terms.

8.3 Give a detailed proof of Theorem 8.2.

8.4 Prove that a matrix  $A$  does not have full row rank if and only if  $rA = 0$  for some nonzero row vector  $r$ .

8.5 Illustrate Theorem 8.3 on problem 8.1.e.

## 8.6 Construct an example of a problem

maximize  $\mathbf{c}\mathbf{x}$  subject to  $\mathbf{A}\mathbf{x} = \mathbf{b}$ ,  $\mathbf{l} \leq \mathbf{x} \leq \mathbf{u}$

such that (i) there is a feasible solution, (ii) the  $m \times n$  matrix  $\mathbf{A}$  has a full row rank, (iii) in every basic feasible solution, at least one free variable is nonbasic, and yet (iv) there are at most  $m$  free variables.

## 8.7 Prove the following theorem. If the problem

$$\begin{aligned} &\text{maximize} && \sum_{j=1}^n c_j x_j \\ &\text{subject to} && \sum_{j=1}^n a_{ij} x_j = b_i \quad (i = 1, 2, \dots, m) \\ &&& l_j \leq x_j \leq u_j \quad (j = 1, 2, \dots, n) \end{aligned} \quad (8.18)$$

has at least one nondegenerate basic optimal solution, then there are uniquely defined numbers  $y_1^*, y_2^*, \dots, y_m^*$  and a positive  $\varepsilon$  with the following property: If  $|t_i| \leq \varepsilon$  for all  $i = 1, 2, \dots, m$ , then the problem

$$\begin{aligned} &\text{maximize} && \sum_{j=1}^n c_j x_j \\ &\text{subject to} && \sum_{j=1}^n a_{ij} x_j = b_i + t_i \quad (i = 1, 2, \dots, m) \\ &&& l_j \leq x_j \leq u_j \quad (j = 1, 2, \dots, n) \end{aligned}$$

has an optimal solution and its optimal value equals

$$z^* + \sum_{i=1}^m y_i^* t_i$$

with  $z^*$  standing for the optimal value of (8.18).

8.8 Illustrate the economic significance of the preceding result on the example of problem 1.8.

8.9 State and prove a variation on problem 8.7, involving changes in the bounds  $l_j$  and  $u_j$  rather than changes in the right-hand sides  $b_i$ .

## General LP Problems: Theorems on Duality and Infeasibility

In Chapter 8, we learned to solve general LP problems by the simplex method. The focus of this chapter is more theoretical: we shall extend the duality theorem, proved in Chapter 5 for LP problems in the standard form, to the general context of arbitrary LP problems. Furthermore, we shall prove a theorem characterizing infeasible LP problems: in other words, we shall characterize unsolvable systems of linear inequalities and equations.

### THE DUALITY THEOREM

In Chapter 8, we observed that every LP problem can be converted into the form

$$\text{maximize } \mathbf{c}\mathbf{x} \quad \text{subject to } \mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{l} \leq \mathbf{x} \leq \mathbf{u}.$$

In the present context, it will be convenient to view LP problems from a different angle: (i) the inequalities  $\sum a_{ij}x_j \leq b_i$  will be left alone rather than converted into equations with slack variables and (ii) except for the nonnegativity constraints  $x_j \geq 0$ , explicit bounds on individual variables will be regarded as instances of  $\sum a_{ij}x_j \leq b_i$ . Thus, every LP problem will be presented as

$$\begin{aligned}
& \text{maximize} && \sum_{j=1}^n c_j x_j \\
& \text{subject to} && \sum_{j=1}^n a_{ij} x_j \leq b_i \quad (i \in I) \\
& && \sum_{j=1}^n a_{ij} x_j = b_i \quad (i \in E) \\
& && x_j \geq 0 \quad (j \in R).
\end{aligned} \tag{9.1}$$

Here the set of the constraint subscripts  $1, 2, \dots, m$  splits into subsets  $I$  and  $E$  corresponding to inequalities and equations, respectively. In the set of the variable subscripts  $1, 2, \dots, n$ , the subset  $R$  marks out the variables  $x_j$  that are explicitly restricted to the nonnegative range. Variables  $x_j$  with  $j \notin R$  will be called *free* (even though some of the constraints  $\sum a_{ij} x_j \leq b_i$  may reduce to explicit bounds on these variables) and the corresponding set of variable subscripts  $j$  will be denoted by  $F$ . In particular, (9.1) is in the standard form if and only if  $E = \emptyset$  and  $F = \emptyset$ .

By a *linear combination* of the constraints

$$\begin{aligned}
& \sum_{j=1}^n a_{ij} x_j \leq b_i \quad (i \in I) \\
& \sum_{j=1}^n a_{ij} x_j = b_i \quad (i \in E)
\end{aligned} \tag{9.2}$$

we mean any linear inequality

$$\sum_{j=1}^n \left( \sum_{i=1}^m a_{ij} y_i \right) x_j \leq \sum_{i=1}^m b_i y_i \tag{9.3}$$

such that  $y_1, y_2, \dots, y_m$  are numbers satisfying

$$y_i \geq 0 \quad \text{whenever } i \in I. \tag{9.4}$$

Every linear combination of (9.2) arises by first multiplying both sides of each constraint by the appropriate  $y_i$ , so that

$$\begin{aligned}
& y_i \left( \sum_{j=1}^n a_{ij} x_j \right) \leq b_i y_i \quad (i \in I) \\
& y_i \left( \sum_{j=1}^n a_{ij} x_j \right) = b_i y_i \quad (i \in E)
\end{aligned}$$

is obtained, and then adding up the results, so that

$$\sum_{i=1}^m y_i \left( \sum_{j=1}^n a_{ij} x_j \right) \leq \sum_{i=1}^m b_i y_i$$

is obtained, and finally changing the order of summation in the left-hand side of

the last inequality. Hence, every solution  $x_1, x_2, \dots, x_n$  of the inequalities and equations (9.2) must satisfy all their linear combinations (9.3).

Next, if numbers  $y_1, y_2, \dots, y_m$  are chosen in such a way that

$$\sum_{i=1}^m a_{ij} y_i \geq c_j \quad \text{whenever } j \in R \tag{9.5}$$

and

$$\sum_{i=1}^m a_{ij} y_i = c_j \quad \text{whenever } j \in F \tag{9.6}$$

then every feasible solution  $x_1, x_2, \dots, x_n$  of (9.1) satisfies

$$c_j x_j \leq \left( \sum_{i=1}^m a_{ij} y_i \right) x_j \quad \text{for all } j = 1, 2, \dots, n$$

(with the sign of equality whenever  $j \in F$ ), and so

$$\sum_{j=1}^n c_j x_j \leq \sum_{j=1}^n \left( \sum_{i=1}^m a_{ij} y_i \right) x_j. \tag{9.7}$$

Finally, if numbers  $x_1, x_2, \dots, x_n$  satisfy both (9.3) and (9.7), then they satisfy the inequality

$$\sum_{j=1}^n c_j x_j \leq \sum_{i=1}^m b_i y_i. \tag{9.8}$$

To summarize, the number  $\sum b_i y_i$  provides an upper bound on the optimal value of (9.1) whenever the numbers  $y_1, y_2, \dots, y_m$  satisfy (9.4), (9.5), and (9.6). Naturally, we are interested in finding as good an upper bound as we can. Thus, we are led to

$$\begin{aligned}
& \text{minimize} && \sum_{i=1}^m b_i y_i \\
& \text{subject to} && \sum_{i=1}^m a_{ij} y_i \geq c_j \quad (j \in R) \\
& && \sum_{i=1}^m a_{ij} y_i = c_j \quad (j \in F) \\
& && y_i \geq 0 \quad (i \in I).
\end{aligned} \tag{9.9}$$

This linear programming problem is called the *dual problem* or simply the *dual* of the linear programming problem (9.1); in this context, the original problem (9.1) is sometimes referred to as the *primal problem* or simply the *primal*. [Note that, in the special case when (9.1) is in the standard form, the present definition reduces to that given in Chapter 5.] As we have just observed, every feasible solution  $x_1, x_2, \dots, x_n$  of the primal problem (9.1) and every feasible solution  $y_1, y_2, \dots, y_m$  of the dual

problem (9.9) satisfy the inequality (9.8). This inequality is sometimes referred to as "the weak duality theorem" or "the easy part of the duality theorem." (The more difficult part, which we shall prove, asserts that the gap between the two sides of (9.8) can always be closed by a suitable choice of  $x_1, x_2, \dots, x_n$  and  $y_1, y_2, \dots, y_m$ .)

Incidentally, note that the variables of the dual are in a one-to-one correspondence with the constraints of the primal and that the constraints of the dual are in a one-to-one correspondence with the variables of the primal.

**Table 9.1 Primal-Dual Correspondence**

| In the dual            | In the primal          |
|------------------------|------------------------|
| Restricted variables   | Inequality constraints |
| Free variables         | Equation constraints   |
| Inequality constraints | Restricted variables   |
| Equation constraints   | Free variables         |

Note also that the dual of the dual is the primal: problem (9.9) may be presented as

$$\begin{aligned}
 &\text{maximize} && \sum_{i=1}^m (-b_i)y_i \\
 &\text{subject to} && \sum_{i=1}^m (-a_{ij})y_i \leq -c_j \quad (j \in R) \\
 &&& \sum_{i=1}^m (-a_{ij})y_i = -c_j \quad (j \in F) \\
 &&& y_i \geq 0 \quad (i \in I)
 \end{aligned}$$

and its dual problem

$$\begin{aligned}
 &\text{minimize} && \sum_{j=1}^n (-c_j)x_j \\
 &\text{subject to} && \sum_{j=1}^n (-a_{ij})x_j \geq -b_i \quad (i \in I) \\
 &&& \sum_{j=1}^n (-a_{ij})x_j = -b_i \quad (i \in E) \\
 &&& x_j \geq 0 \quad (j \in R)
 \end{aligned}$$

is just another presentation of (9.1).

Occasionally, students get confused when asked to construct dual problems of LP problems such as

$$\begin{aligned}
 &\text{maximize} && 3x_1 + 2x_2 + 5x_3 \\
 &\text{subject to} && 5x_1 + 3x_2 + x_3 = -8 \\
 &&& 4x_1 + 2x_2 + 8x_3 \leq 23 \\
 &&& 6x_1 + 7x_2 + 3x_3 \geq 1 \\
 &&& x_1 \leq 4, x_3 \geq 0.
 \end{aligned}$$

The routine way of accomplishing this task consists of presenting the maximization problem in the form of (9.1) and then writing (9.9). Thus, our example would be presented as

$$\begin{aligned}
 &\text{maximize} && 3x_1 + 2x_2 + 5x_3 \\
 &\text{subject to} && 5x_1 + 3x_2 + x_3 = -8 \\
 &&& 4x_1 + 2x_2 + 8x_3 \leq 23 \\
 &&& -6x_1 - 7x_2 - 3x_3 \leq -1 \\
 &&& x_1 \leq 4 \\
 &&& x_3 \geq 0
 \end{aligned}$$

and its dual found to be

$$\begin{aligned}
 &\text{minimize} && -8y_1 + 23y_2 - y_3 + 4y_4 \\
 &\text{subject to} && 5y_1 + 4y_2 - 6y_3 + y_4 = 3 \\
 &&& 3y_1 + 2y_2 - 7y_3 = 2 \\
 &&& y_1 + 8y_2 - 3y_3 \geq 5 \\
 &&& y_2, y_3, y_4 \geq 0.
 \end{aligned}$$

Similarly, the routine way of finding the dual of a minimization problem consists of presenting the problem as (9.9) and then writing (9.1). When this convention is observed, the dual of a maximization problem is always a minimization problem and vice versa.

**THEOREM 9.1 (The Duality Theorem).** If a linear programming problem has an optimal solution, then its dual has an optimal solution and the optimal values of the two problems coincide.

**PROOF.** We shall rely on an analysis of the simplex method as presented in Chapter 8. Every LP problem may be presented in the form (9.1), whereupon its dual assumes the form (9.9). If (9.1) has an optimal solution, then the simplex method, applied to its equivalent version

$$\begin{aligned}
&\text{maximize} && \sum_{j=1}^n c_j x_j \\
&&& \sum_{j=1}^n a_{ij} x_j + x_{n+i} = b_i \quad (i \in I) \\
&&& \sum_{j=1}^n a_{ij} x_j = b_i \quad (i \in E) \\
&&& x_j \geq 0 \quad (j \in R) \\
&&& x_{n+i} \geq 0 \quad (i \in I)
\end{aligned}$$

finds an optimal solution  $x_1^*, x_2^*, \dots, x_n^*$ . Consider the vector  $y = [y_1^*, y_2^*, \dots, y_m^*]$  featured in the final iteration of the simplex method. In step 1, this vector is computed by solving the system  $yB = c_B$ ; hence

$$\begin{aligned}
\sum_{i=1}^m a_{ij} y_i^* &= c_j \quad \text{whenever } 1 \leq j \leq n \text{ and } x_j \text{ is basic,} \\
y_i^* &= 0 \quad \text{whenever } i \in I \text{ and } x_{n+i} \text{ is basic.}
\end{aligned}$$

In step 2, no candidate for entering the basis is found; hence

$$\begin{aligned}
\sum_{i=1}^m a_{ij} y_i^* &= c_j \quad \text{whenever } j \in F \text{ and } x_j \text{ is nonbasic} \\
\sum_{i=1}^m a_{ij} y_i^* &\geq c_j \quad \text{whenever } j \in R \text{ and } x_j \text{ is nonbasic} \\
y_i^* &\geq 0 \quad \text{whenever } i \in I \text{ and } x_{n+i} \text{ is nonbasic.}
\end{aligned}$$

Now it follows that

$$\left( \sum_{i=1}^m a_{ij} y_i^* \right) x_j^* = c_j x_j^* \quad \text{for all } j = 1, 2, \dots, n: \quad (9.10)$$

the inequality  $\sum a_{ij} y_i^* \geq c_j$ , valid for all  $j = 1, 2, \dots, n$ , can hold with the sharp inequality sign only if  $j \in R$  and  $x_j$  is nonbasic, in which case  $x_j^* = 0$ . Similarly, it follows that

$$\left( \sum_{j=1}^n a_{ij} x_j^* \right) y_i^* = b_i y_i^* \quad \text{for all } i = 1, 2, \dots, m: \quad (9.11)$$

the inequality  $\sum a_{ij} x_j^* \leq b_i$ , valid for all  $i = 1, 2, \dots, m$ , can hold with the sharp inequality sign only if  $i \in I$  and  $x_{n+i}$  is basic, in which case  $y_i^* = 0$ . Note that (9.10) and (9.11) imply

$$\sum_{j=1}^n c_j x_j^* = \sum_{j=1}^n \left( \sum_{i=1}^m a_{ij} y_i^* \right) x_j^* = \sum_{i=1}^m \left( \sum_{j=1}^n a_{ij} x_j^* \right) y_i^* = \sum_{i=1}^m b_i y_i^*.$$

To summarize,  $y_1^*, y_2^*, \dots, y_m^*$  is a feasible solution of the dual (9.9) and

$$\sum_{i=1}^m b_i y_i^* = \sum_{j=1}^n c_j x_j^*. \quad (9.12)$$

Now it only remains to be shown that every feasible solution  $y_1, y_2, \dots, y_m$  of (9.9) has

$$\sum_{i=1}^m b_i y_i \geq \sum_{i=1}^m b_i y_i^*$$

for then  $y_1^*, y_2^*, \dots, y_m^*$  is an optimal solution of (9.9). But this inequality follows directly from (9.12) and (9.8) with  $x_1, x_2, \dots, x_n$  replaced by  $x_1^*, x_2^*, \dots, x_n^*$ . ■

Our proof shows that the simplex method, applied to an arbitrary LP problem that has an optimal solution, finds an optimal solution of this problem simultaneously with an optimal solution of its dual. Additional results involving arbitrary LP problems and their duals can be developed along the lines of Chapter 5.

## UNSOLVABLE SYSTEMS OF LINEAR INEQUALITIES AND EQUATIONS

In the remainder of this chapter, we shall characterize infeasible linear programs. Now it is convenient to present every LP problem in the form

$$\begin{aligned}
&\text{maximize} && \sum_{j=1}^n c_j x_j \\
&\text{subject to} && \sum_{j=1}^n a_{ij} x_j \leq b_i \quad (i \in I) \\
&&& \sum_{j=1}^n a_{ij} x_j = b_i \quad (i \in E)
\end{aligned}$$

with the nonnegativity constraints  $x_j \geq 0$ , if any, viewed as special instances of  $\sum a_{ij} x_j \leq b_i$ . Since we are concerned only with the existence of feasible solutions, the objective function  $\sum c_j x_j$  is irrelevant; our task amounts to characterizing unsolvable systems

$$\begin{aligned}
&\sum_{j=1}^n a_{ij} x_j \leq b_i \quad (i \in I) \\
&\sum_{j=1}^n a_{ij} x_j = b_i \quad (i \in E).
\end{aligned} \quad (9.13)$$

At the beginning of this chapter, we defined a linear combination of the constraints (9.13) as any linear inequality

$$\sum_{j=1}^n \left( \sum_{i=1}^m a_{ij} y_i \right) x_j \leq \sum_{i=1}^m b_i y_i$$

such that  $y_1, y_2, \dots, y_m$  are numbers satisfying  $y_i \geq 0$  whenever  $i \in I$ . In addition, we observed that every solution of a system of linear constraints must satisfy all linear combinations of this system. This observation may be used to demonstrate the unsolvability of systems such as

$$\begin{aligned} x_1 + 3x_2 + 2x_3 + 4x_4 &\leq 5 \\ 3x_1 + x_2 + 2x_3 + x_4 &\leq 4 \\ 5x_1 + 3x_2 + 3x_3 + 3x_4 &= 9 \\ -x_3 &\leq 0 \\ -x_4 &\leq 0 \end{aligned} \quad (9.14)$$

in a simple and convincing manner. In this particular example, we need only observe that the multipliers

$$y_1 = 1, \quad y_2 = 3, \quad y_3 = -2, \quad y_4 = 2, \quad y_5 = 1$$

yield a linear combination of (9.14) that reads

$$0 \cdot x_1 + 0 \cdot x_2 + 0 \cdot x_3 + 0 \cdot x_4 \leq -1. \quad (9.15)$$

Now it becomes obvious that (9.14) is unsolvable: any solution of this system would have to satisfy the clearly unsolvable inequality (9.15).

More generally, we shall follow H. W. Kuhn (1956) in saying that a system (9.13) is *inconsistent* if there are numbers  $y_1, y_2, \dots, y_m$  such that

$$\left. \begin{aligned} y_i &\geq 0 \quad \text{whenever } i \in I \\ \sum_{i=1}^m a_{ij} y_i &= 0 \quad \text{for all } j = 1, 2, \dots, n \\ \sum_{i=1}^m b_i y_i &< 0 \end{aligned} \right\}. \quad (9.16)$$

Thus, (9.13) is inconsistent if and only if one of its linear combinations reads

$$0 \cdot x_1 + 0 \cdot x_2 + \dots + 0 \cdot x_n \leq c \quad (9.17)$$

for some negative number  $c$ . As soon as the multipliers  $y_1, y_2, \dots, y_m$  satisfying (9.16) are exhibited, it becomes obvious that (9.13) is unsolvable: any solution of this system would have to satisfy the clearly unsatisfiable inequality (9.17). Hence every inconsistent system is unsolvable; now we propose to show the converse.

**THEOREM 9.2.** A system of linear inequalities and equations is unsolvable if and only if it is inconsistent.

**PROOF.** Since the “if” part has just been proved, we have to prove only the “only if” part. For this purpose, we shall appeal to the relationship between system (9.13) and the problem

$$\begin{aligned} &\text{maximize} && \sum_{i=1}^m (-x_{n+i}) \\ &\text{subject to} && \sum_{j=1}^n a_{ij} x_j + w_i x_{n+i} \leq b_i \quad (i \in I) \\ &&& \sum_{j=1}^n a_{ij} x_j + w_i x_{n+i} = b_i \quad (i \in E) \\ &&& x_{n+i} \geq 0 \quad (i = 1, 2, \dots, m) \end{aligned} \quad (9.18)$$

with  $w_i = 1$  if  $b_i \geq 0$  and  $w_i = -1$  if  $b_i < 0$ . Theorem 8.4 guarantees that (9.18) always has an optimal solution; obviously, the optimal value of (9.18) is zero if and only if (9.13) is solvable. In particular, if (9.13) is unsolvable, then the optimal value of (9.18) is negative. In that case, Theorem 9.1 guarantees that the dual of (9.18),

$$\begin{aligned} &\text{minimize} && \sum_{i=1}^m b_i y_i \\ &\text{subject to} && \sum_{i=1}^m a_{ij} y_i = 0 \quad (j = 1, 2, \dots, n) \\ &&& w_i y_i \geq -1 \quad (i = 1, 2, \dots, m) \\ &&& y_i \geq 0 \quad (i \in I) \end{aligned}$$

has an optimal solution and that its optimal value is negative. But then the optimal solution  $y_1, y_2, \dots, y_m$  satisfies (9.16), and so (9.13) is inconsistent. ■

Before proving a supplementary result on unsolvable systems of linear inequalities, let us make note of an immediate corollary of Theorem 8.3.

**THEOREM 9.3.** If a system of  $m$  linear equations has a nonnegative solution, then it has a solution with at most  $m$  variables positive.

**PROOF.** If the system

$$\begin{aligned} \sum_{j=1}^n a_{ij} x_j &= b_i \quad (i = 1, 2, \dots, m) \\ x_j &\geq 0 \quad (j = 1, 2, \dots, n) \end{aligned} \quad (9.19)$$



has a solution, then, by Theorem 8.3, there is some set  $I$  of subscripts  $1, 2, \dots, m$  such that (i) systems (9.19) and

$$\sum_{j=1}^n a_{ij}x_j = b_i \quad (i \in I) \quad (9.20)$$

$$x_j \geq 0 \quad (j = 1, 2, \dots, n)$$

have precisely the same set of solutions and (ii) system (9.20) has a basic feasible solution  $x_1^*, x_2^*, \dots, x_n^*$ . Now it only remains to be observed that  $x_j^* = 0$  whenever  $x_j$  is nonbasic and that at most  $|I|$  variables  $x_j$  are basic. ■

**THEOREM 9.4.** Every unsolvable system of linear inequalities in  $n$  variables contains an unsolvable subsystem of at most  $n + 1$  inequalities.

**PROOF.** If  $Ax \leq b$  is unsolvable, then, by Theorem 9.2, it is inconsistent. Hence, some row vector  $y^*$  satisfies  $y^* \geq 0$ ,  $y^*A = 0$ ,  $y^*b < 0$ . Denote  $y^*b$  by  $c$ , and consider the system  $yA = 0$ ,  $yb = c$  consisting of  $n + 1$  equations. Since  $y^*$  is a nonnegative solution, Theorem 9.3 guarantees the existence of a nonnegative solution  $\bar{y}$  with at most  $n + 1$  positive components  $\bar{y}_i$ . The desired subsystem consists of those inequalities  $\sum a_{ij}x_j \leq b_i$  for which  $\bar{y}_i > 0$ ; since  $\sum \bar{y}_i a_{ij} = 0$  for all  $j$  but  $\sum \bar{y}_i b_i = c < 0$ , this subsystem is inconsistent and therefore unsolvable. ■

## PROBLEMS

### 9.1 Construct the dual problems of

$$\begin{aligned} &\text{maximize} && x_1 - x_2 \\ &\text{subject to} && 2x_1 + 3x_2 - x_3 + x_4 \leq 0 \\ & && 3x_1 + x_2 + 4x_3 - 2x_4 \geq 3 \\ & && -x_1 - x_2 + 2x_3 + x_4 = 1 \\ & && x_2, x_3 \geq 0 \end{aligned}$$

and of

$$\begin{aligned} &\text{minimize} && x_1 - x_2 \\ &\text{subject to} && 2x_1 + 3x_2 - x_3 + x_4 \leq 0 \\ & && 3x_1 + x_2 + 4x_3 - 2x_4 \geq 3 \\ & && -x_1 - x_2 + 2x_3 + x_4 = 1 \\ & && x_2, x_3 \geq 0. \end{aligned}$$

### 9.2 In matrix notation, an LP in the standard form reads

$$\text{maximize } cx \quad \text{subject to } Ax \leq b, \quad x \geq 0$$

and its dual reads

$$\text{minimize } yb \quad \text{subject to } yA \geq c, \quad y \geq 0.$$

What is the dual of: maximize  $cx$  subject to  $Ax \leq b$ ? The dual of: maximize  $cx$  subject to  $Ax = b, x \geq 0$ ?

### 9.3 Construct the dual problem of

$$\begin{aligned} &\text{maximize} && z \\ &\text{subject to} && z - \sum_{i=1}^m a_{ij}x_i \leq 0 \quad (j = 1, 2, \dots, n) \\ & && \sum_{i=1}^m x_i = 1 \\ & && x_i \geq 0 \quad (i = 1, 2, \dots, m). \end{aligned}$$

### 9.4 Maximize

$$\begin{aligned} &x_1 + x_2 + x_3 \\ &\text{subject to} && 2x_1 - 4x_2 + x_3 \leq -1 \\ & && x_1 + 5x_2 + x_3 \leq 16 \\ & && x_1 + x_3 \leq 5 \\ & && 2x_1 + 4x_2 - x_3 \leq 8 \\ & && x_1 - 3x_2 + x_3 \leq 0 \\ & && -4x_1 + 3x_2 \leq 4 \\ & && 4x_1 - 3x_2 + 5x_3 \leq 10 \\ & && x_1 + 2x_2 + x_3 \leq 9. \end{aligned}$$

### 9.5 Generalize Theorems 5.2 and 5.3 to the context of general LP problems. Use your result to find out if

$$x_1^* = 3, \quad x_2^* = -1, \quad x_3^* = 0, \quad x_4^* = 2$$

is an optimal solution of the problem

$$\begin{aligned} &\text{maximize} && 6x_1 + x_2 - x_3 - x_4 \\ &\text{subject to} && x_1 + 2x_2 + x_3 + x_4 \leq 5 \\ & && 3x_1 + x_2 - x_3 \leq 8 \\ & && x_2 + x_3 + x_4 = 1 \\ & && x_3, x_4 \geq 0. \end{aligned}$$

### 9.6 One feasible solution of problem 1.8 calls for 10 assemblers and 30 instructors in the first week, followed by 130 assemblers in each of the remaining three weeks. Use the result of problem 9.5 to find out if this solution is optimal or not.

### 9.7 Illustrate Theorems 9.2 and 9.4 on the system

$$\begin{aligned} &2x + 6y - z \leq -6 \\ &4x - 4y + 3z \leq 0 \\ &3x + y - 4z \leq 3 \\ &7x - 3y - 6z \leq 10 \\ &x - 5y + 2z \leq 2 \\ &7x + 13y - 6z \leq -7 \\ &-6x - 2y + 3z \leq 0. \end{aligned}$$

## Sensitivity Analysis

In linear programming problems arising from applications, the numerical data often represent only rough estimates of quantities that are inherently difficult to measure or predict. Market prices may fluctuate, supplies of raw materials and demands for finished products are often unknown in advance, and production may be affected by a variety of accidental events such as machine breakdowns. In such cases, solving the initial LP problem is only the starting point for further analysis of the situation.

By replacing the original data by more pessimistic or optimistic estimates of the unknown quantities, we may create a number of variations on the original theme. Since each of these new LP problems might possibly represent the actual situation, it is useful to find out how the optimal solutions vary with the changes in data. It may be, for example, that the optimal solution is particularly sensitive to changes in only a small set of parameters; if possible, these parameters should then be estimated with greater accuracy. Other variations on the original theme arise when new variables and/or constraints are introduced. New variables may appear when new products are developed. Similarly, new constraints may result from new production policies. It may also happen that a new constraint actually had to be satisfied from

the beginning but was simply forgotten in the original formulation. Investigations of such changes are referred to as *sensitivity analysis* or *postoptimality analysis*.

Sensitivity analysis could be conducted in a straightforward way by solving each of the modified problems from scratch. However, similar problems are likely to have similar solutions, and so it is often easier to exploit the results obtained in solving the original problem. This alternative strategy is the subject of the present chapter.

### DUAL-FEASIBLE DICTIONARIES

In Chapter 2, we introduced the notion of a *dictionary* associated with a problem

$$\begin{aligned} &\text{maximize} && \sum_{j=1}^n c_j x_j \\ &\text{subject to} && \sum_{j=1}^n a_{ij} x_j \leq b_i \quad (i = 1, 2, \dots, m) \\ &&& x_j \geq 0 \quad (j = 1, 2, \dots, n). \end{aligned} \quad (10.1)$$

The dictionary is any system of equations

$$\begin{aligned} x_r &= \bar{b}_r - \sum_{s \notin B} \bar{a}_{rs} x_s \quad (r \in B) \\ z &= \bar{d} + \sum_{s \notin B} \bar{c}_s x_s \end{aligned} \quad (10.2)$$

such that every solution  $x_1, x_2, \dots, x_{n+m}, z$  of the system

$$\begin{aligned} x_{n+i} &= b_i - \sum_{j=1}^n a_{ij} x_j \quad (i = 1, 2, \dots, m) \\ z &= \sum_{j=1}^n c_j x_j \end{aligned}$$

is a solution of (10.2) and vice versa. Note that (10.2) may be a dictionary even if  $\bar{b}_r < 0$  for some  $r \in B$ . That is, not every dictionary is *feasible* in the sense defined in Chapter 2.

Our immediate objective is to point out a certain one-to-one correspondence between the dictionaries associated with (10.1) and the dictionaries associated with the dual problem. In order to make this correspondence more transparent, we shall change the way in which the variables  $x_k$  are subscripted; instead of denoting the decision variables by  $x_1, x_2, \dots, x_n$  and the slack variables by  $x_{n+1}, x_{n+2}, \dots, x_{n+m}$  as usual, we shall denote the decision variables by  $x_{m+1}, x_{m+2}, \dots, x_{m+n}$  and the slack variables by  $x_1, x_2, \dots, x_m$ . Thus, (10.1) will be recorded as

$$\begin{aligned}
& \text{maximize} && \sum_{j=m+1}^{m+n} c_j x_j \\
& \text{subject to} && \sum_{j=m+1}^{m+n} a_{ij} x_j \leq b_i \quad (i = 1, 2, \dots, m) \\
& && x_j \geq 0 \quad (j = m+1, m+2, \dots, m+n)
\end{aligned} \tag{10.3}$$

and the corresponding initial dictionary will be recorded as

$$\begin{aligned}
x_i &= b_i - \sum_{j=m+1}^{m+n} a_{ij} x_j \quad (i = 1, 2, \dots, m) \\
z &= \sum_{j=m+1}^{m+n} c_j x_j
\end{aligned} \tag{10.4}$$

The dual problem of (10.3) is

$$\begin{aligned}
& \text{minimize} && \sum_{i=1}^m b_i y_i \\
& \text{subject to} && \sum_{i=1}^m a_{ij} y_i \geq c_j \quad (j = m+1, m+2, \dots, m+n) \\
& && y_i \geq 0 \quad (i = 1, 2, \dots, m)
\end{aligned} \tag{10.5}$$

and the corresponding initial dictionary reads

$$\begin{aligned}
y_j &= -c_j + \sum_{i=1}^m a_{ij} y_i \quad (j = m+1, m+2, \dots, m+n) \\
-w &= -\sum_{i=1}^m b_i y_i
\end{aligned} \tag{10.6}$$

Note that dictionaries (10.4) and (10.6) are mirror images of each other in the sense that the actual coefficients appearing in a row of (10.4) are found with reversed signs in a column of (10.6) and vice versa. For instance, if (10.3) is the problem

$$\begin{aligned}
& \text{maximize} && 4x_3 - 13x_4 + 7x_5 \\
& \text{subject to} && 3x_3 + 2x_4 + 5x_5 \leq 5 \\
& && x_3 - 3x_4 + 2x_5 \leq 3 \\
& && x_3, x_4, x_5 \geq 0
\end{aligned}$$

then the dual problem (10.5) reads

$$\text{minimize} \quad 5y_1 + 3y_2$$

$$\begin{aligned}
& \text{subject to} && 3y_1 + y_2 \geq 4 \\
& && 2y_1 - 3y_2 \geq -13 \\
& && 5y_1 + 2y_2 \geq 7 \\
& && y_1, y_2 \geq 0
\end{aligned}$$

and the two dictionaries (10.4) and (10.6) are

$$\begin{aligned}
& && y_3 = -4 + 3y_1 + y_2 \\
x_1 &= 5 - 3x_3 - 2x_4 - 5x_5 && y_4 = 13 + 2y_1 - 3y_2 \\
x_2 &= 3 - x_3 + 3x_4 - 2x_5 && \text{and } y_5 = -7 + 5y_1 + 2y_2 \\
z &= 4x_3 - 13x_4 + 7x_5 && -w = -5y_1 - 3y_2
\end{aligned}$$

respectively. A further examination of this particular example reveals that the primal and the dual dictionaries come in matching pairs. For instance,

$$\begin{aligned}
& && y_2 = 4 - 3y_1 + y_3 \\
x_1 &= -4 + 3x_2 - 11x_4 + x_5 && y_4 = 1 + 11y_1 - 3y_3 \\
x_3 &= 3 - x_2 + 3x_4 - 2x_5 && \text{goes with } y_5 = 1 - y_1 + 2y_3 \\
z &= 12 - 4x_2 - x_4 - x_5 && -w = -12 + 4y_1 - 3y_3 \\
& && y_1 = 1 + 2y_3 - y_5 \\
x_3 &= -5 - 2x_1 + 5x_2 - 19x_4 && y_2 = 1 - 5y_3 + 3y_5 \\
x_5 &= 4 + x_1 - 3x_2 + 11x_4 && \text{goes with } y_4 = 12 + 19y_3 - 11y_5 \\
z &= 8 - x_1 - x_2 - 12x_4 && -w = -8 + 5y_3 - 4y_5
\end{aligned}$$

and so on. In each of these pairs, the primal dictionary is a mirror image of the dual dictionary in the same way that (10.4) is the mirror image of (10.6). This situation is no accident: if

$$\begin{aligned}
x_r &= \bar{b}_r - \sum_{s \in N} \bar{a}_{rs} x_s \quad (r \in B) \\
z &= \bar{d} + \sum_{s \in N} \bar{c}_s x_s
\end{aligned} \tag{10.7}$$

is a primal dictionary arising from (10.4), with  $x_r (r \in B)$  basic and  $x_s (s \in N)$  nonbasic, then

$$\begin{aligned}
y_s &= -\bar{c}_s + \sum_{r \in B} \bar{a}_{rs} y_r \quad (s \in N) \\
-w &= -\bar{d} - \sum_{r \in B} \bar{b}_r y_r
\end{aligned} \tag{10.8}$$

is a dual dictionary arising from (10.6), with  $y_s (s \in N)$  basic and  $y_r (r \in B)$  nonbasic.

A verification of this claim, left for problem 10.1, amounts to tedious plugging and grinding.

Since Chapter 2, we have been referring to dictionaries (10.7) as *feasible* if  $\bar{b}_r \geq 0$  for all  $r \in B$ . Now we shall refer to (10.7) as *dual-feasible* if the corresponding dual dictionary (10.8) is feasible. Thus, (10.7) is dual-feasible if and only if  $\bar{c}_s \leq 0$  for all  $s \in N$ .

## THE DUAL SIMPLEX METHOD

As observed in Chapter 5, every LP problem in the standard form may be solved by applying the simplex method to its dual; that is, an optimal solution of the primal problem may be read off the optimal dictionary for the dual problem. The observations made in the last section imply that this strategy may be implemented without any reference to the dual problem: the sequence of feasible dictionaries created by the simplex method working on the dual problem may be represented by a sequence of dual-feasible dictionaries associated with the primal problem. The resulting algorithm, designed by C. E. Lemke (1954) and known as the *dual simplex method*, constitutes a valuable tool of sensitivity analysis.

An explicit description of the dual simplex method follows mechanically from the fact that a dual variable  $y_k$  is basic in (10.8) if and only if the corresponding primal variable  $x_k$  is nonbasic in (10.7). Thus, a dual dictionary arising from (10.8) by a single pivot, with  $y_i$  entering and  $y_j$  leaving the basis, will correspond to the primal dictionary arising from (10.7) by a single pivot, with  $x_i$  leaving and  $x_j$  entering the basis. In particular, if (10.8) is one of the feasible dictionaries created by the simplex method, then  $y_i$  and  $y_j$  are determined by the familiar rules: the choice of  $y_i$  is motivated by the desire to increase  $-w$  and the choice of  $y_j$  is dictated by the need to preserve feasibility when  $y_i$  increases. Formally,  $i$  may be any subscript  $i \in B$  with

$$\bar{b}_i < 0 \quad (10.9)$$

and  $j$  must be a subscript  $j \in N$  that has

$$\bar{a}_{ij} < 0 \quad \text{and} \quad \bar{c}_j/\bar{a}_{ij} \leq \bar{c}_s/\bar{a}_{is} \quad \text{for all } s \in N \quad \text{with } \bar{a}_{is} < 0. \quad (10.10)$$

Hence an iteration of the dual simplex method, beginning with a dual-feasible dictionary (10.7), consists of first choosing a subscript  $i \in B$  that satisfies (10.9), then finding a subscript  $j \in N$  that satisfies (10.10), and finally pivoting, with  $x_i$  leaving and  $x_j$  entering the basis.

For illustration, suppose that (10.7) reads

$$\begin{aligned} x_1 &= -4 + 3x_2 - 11x_4 + x_5 \\ x_3 &= 3 - x_2 + 3x_4 - 2x_5 \\ z &= 12 - 4x_2 - x_4 - x_5. \end{aligned}$$

To choose the leaving variable, we examine  $\bar{b}_1 = -4$  and  $\bar{b}_2 = 3$ ; the leaving variable must be  $x_1$ . To find the entering variable, we examine the ratios  $\bar{c}_2/\bar{a}_{12} = 4/3$  and  $\bar{c}_5/\bar{a}_{15} = 1$  (ignoring  $\bar{c}_4/\bar{a}_{14}$  since  $\bar{a}_{14} \geq 0$ ); the entering variable must be  $x_5$ . The pivot with  $x_5$  entering and  $x_1$  leaving the basis yields the dual-feasible dictionary

$$\begin{aligned} x_3 &= -5 - 2x_1 + 5x_2 - 19x_4 \\ x_5 &= 4 + x_1 - 3x_2 + 11x_4 \\ z &= 8 - x_1 - x_2 - 12x_4. \end{aligned}$$

In general, each iteration of the dual simplex method is nothing but a disguised version of an iteration of the simplex method working on the dual problem. This particular example disguises the iteration that leads from

$$\begin{array}{rcl} y_2 &= & 4 - 3y_1 + y_3 \\ y_4 &= & 1 + 11y_1 - 3y_3 \\ y_5 &= & 1 - y_1 + 2y_3 \\ -w &= & -12 + 4y_1 - 3y_3 \end{array} \quad \text{to} \quad \begin{array}{rcl} y_1 &= & 1 + 2y_3 - y_5 \\ y_2 &= & 1 - 5y_3 + 3y_5 \\ y_4 &= & 12 + 19y_3 - 11y_5 \\ -w &= & -8 + 5y_3 - 4y_5. \end{array}$$

If no  $i \in B$  satisfies (10.9), then the computations terminate: since dictionary (10.7) is not only dual-feasible but also feasible, it describes an optimal solution  $x_1^*, x_2^*, \dots, x_{n+m}^*$  by  $x_s^* = 0$  for all  $s \in N$  and  $x_r^* = \bar{b}_r$  for all  $r \in B$ . Similarly, if no  $j \in N$  satisfies (10.10), then the computations terminate: since the dual problem is unbounded, the primal problem is infeasible. The latter conclusion may be also reached directly, without any reference to the dual problem: if no  $j \in N$  satisfies (10.10), then  $\bar{a}_{is} \geq 0$  for all  $s \in N$ , and so the right-hand side of the equation

$$x_i = \bar{b}_i - \sum_{s \in N} \bar{a}_{is} x_s,$$

with  $\bar{b}_i < 0$ , assumes a negative value whenever the values of all  $x_s (s \in N)$  are nonnegative.

In Chapter 7, we saw how the simplex method may be implemented in the revised format without the use of dictionaries. The dual simplex method may be implemented similarly, with only the numbers  $\bar{b}_r (r \in B)$  and  $\bar{c}_s (s \in N)$  stored and updated in each iteration. Working out the details amounts to answering the following four questions:

- How is the leaving variable  $x_i$  found?
- How is the entering variable  $x_j$  found?
- How are the numbers  $\bar{b}_r$  updated?
- How are the numbers  $\bar{c}_s$  updated?

The first question does not present any difficulty since the numbers  $\bar{b}_r (r \in B)$  featured in (10.9) are readily available.

To answer the second question, we need only find a way of recovering the numbers  $\bar{a}_{is} (s \in N)$  featured in (10.10) from the original data. For this purpose, let us recall

that the first  $m$  rows of dictionary (10.7) may be recorded (in the notation of Chapter 7) as

$$\mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b} - \mathbf{B}^{-1}\mathbf{A}_N\mathbf{x}_N. \quad (10.11)$$

If the leaving variable  $x_i$  appears in the  $p$ th position of the basis heading, then it is the  $p$ th equation in (10.11) that expresses  $x_i$  in terms of the nonbasic variables, and so it is the  $p$ th row of the matrix  $\mathbf{B}^{-1}\mathbf{A}_N$  that consists of the desired numbers  $\bar{a}_{is}$ . But the  $p$ th row of  $\mathbf{B}^{-1}\mathbf{A}_N$  equals  $\mathbf{v}\mathbf{A}_N$  with  $\mathbf{v}$  standing for the  $p$ th row of  $\mathbf{B}^{-1}$ ; in turn,  $\mathbf{v}$  itself may be found by solving the system  $\mathbf{v}\mathbf{B} = \mathbf{e}$  with  $\mathbf{e}$  standing for the  $p$ th row of the  $m \times m$  identity matrix. Hence the numbers  $\bar{a}_{is}$  ( $s \in N$ ) may be computed by first solving the system  $\mathbf{v}\mathbf{B} = \mathbf{e}$  and then computing the row vector  $\mathbf{w}_N = \mathbf{v}\mathbf{A}_N$ ; each  $\bar{a}_{is}$  is a component  $w_s$  of this vector  $\mathbf{w}_N$ .

To answer the third question, we need only recall that the numbers  $\bar{b}_r$  ( $r \in B$ ) are nothing but the components  $x_r^*$  ( $r \in B$ ) of the vector  $\mathbf{x}^* = [x_1^*, x_2^*, \dots, x_{n+m}^*]^T$  associated with dictionary (10.7); in fact, (10.11) may be written as

$$\mathbf{x}_B = \mathbf{x}_B^* - \mathbf{B}^{-1}\mathbf{A}_N\mathbf{x}_N.$$

These numbers may be updated as in Chapter 7: having determined the entering variable  $x_j$ , and therefore the entering column  $\mathbf{a}$ , we solve the system  $\mathbf{B}\mathbf{d} = \mathbf{a}$  and replace  $\mathbf{x}_B^*$  by  $\mathbf{x}_B^* - t\mathbf{d}$ , with  $t$  standing for the new value of the entering variable. This value  $t$  equals  $\bar{b}_i/\bar{a}_{ij}$ , which may be written as  $x_i^*/w_j$ .

To answer the fourth question, let us record the last row of (10.7) as

$$z = \bar{d} + \bar{c}_j x_j + \sum_{s \in R} \bar{c}_s x_s$$

with  $s \in R$  if and only if  $s \in N$  and  $s \neq j$ . Since pivoting amounts to substituting for  $x_j$  from

$$x_j = \left( \bar{b}_i - x_i - \sum_{s \in R} \bar{a}_{is} x_s \right) / \bar{a}_{ij}$$

the formula for  $z$  gets updated into

$$z = \bar{d} + \bar{c}_j \left( \bar{b}_i - x_i - \sum_{s \in R} \bar{a}_{is} x_s \right) / \bar{a}_{ij} + \sum_{s \in R} \bar{c}_s x_s.$$

After simplification, and writing  $w_s$  for each  $\bar{a}_{is}$  ( $s \in N$ ), we obtain

$$z = \left( \bar{d} + \bar{b}_i \frac{\bar{c}_j}{w_j} \right) - (\bar{c}_j/w_j)x_i + \sum_{s \in R} \left( \bar{c}_s - w_s \frac{\bar{c}_j}{w_j} \right) x_s.$$

Thus, the new coefficient  $\bar{c}_i$  at  $x_i$  equals  $-\bar{c}_j/w_j$ , and the coefficient  $\bar{c}_s$  at each  $x_s$  with  $s \in R$  gets replaced by  $\bar{c}_s - \bar{c}_j w_s/w_j = \bar{c}_s + \bar{c}_i w_s$ .

These findings are summarized in Box 10.1. Rather than beginning with a dual-feasible dictionary (10.7), the iteration begins just with a vector  $\mathbf{x}_B^*$  whose components  $x_r^*$  ( $r \in B$ ) are the numbers  $\bar{b}_r$ , and with a vector  $\bar{\mathbf{c}}_N$  whose components are the numbers  $\bar{c}_s$  ( $s \in N$ ).

### BOX 10.1 Iteration of the Revised Dual Simplex Method

*Step 1.* If  $\mathbf{x}_B^* \geq 0$  then stop:  $\mathbf{x}^*$  is an optimal solution. Otherwise, choose the leaving variable; this may be any basic variable  $x_i$  with  $x_i^* < 0$ .

*Step 2.* Solve the system  $\mathbf{v}\mathbf{B} = \mathbf{e}$  with  $\mathbf{e}$  standing for the  $p$ th row of the identity matrix and with  $p$  such that  $x_i$  appears in the  $p$ th position of the basis heading. Compute  $\mathbf{w}_N = \mathbf{v}\mathbf{A}_N$ .

*Step 3.* Let  $J$  be the set of those nonbasic variables  $x_j$  for which  $w_j < 0$ . If  $J$  is empty then stop: the problem is infeasible. Otherwise, find the  $x_j$  in  $J$  that minimizes  $\bar{c}_j/w_j$  and let it be the entering variable.

*Step 4.* Solve the system  $\mathbf{B}\mathbf{d} = \mathbf{a}$  with  $\mathbf{a}$  standing for the entering column.

*Step 5.* Set the value  $x_j^*$  of the entering variable at  $t = x_i^*/w_j$  and replace the values  $\mathbf{x}_B^*$  of the basic variables by  $\mathbf{x}_B^* - t\mathbf{d}$ . Replace the leaving column of  $\mathbf{B}$  by the entering column and, in the basis heading, replace the leaving variable by the entering variable. Set  $\bar{c}_i = -\bar{c}_j/w_j$  and add  $\bar{c}_i w_s$  to each  $\bar{c}_s$  with  $s \neq i$ .

For illustration, we shall apply the revised dual simplex method to the problem

$$\text{maximize } \mathbf{c}\mathbf{x} \quad \text{subject to } \mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{x} \geq 0$$

with

$$\mathbf{A} = \begin{bmatrix} -6 & 1 & 2 & 4 & 1 & 0 & 0 \\ 3 & -2 & -1 & -5 & 0 & 1 & 0 \\ -2 & 1 & 0 & 2 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 14 \\ -25 \\ 14 \end{bmatrix},$$

$$\mathbf{c} = [-5 \quad -3 \quad -3 \quad -6 \quad 0 \quad 0 \quad 0].$$

We may initialize by

$$\mathbf{x}_B^* = \begin{bmatrix} x_5^* \\ x_6^* \\ x_7^* \end{bmatrix} = \begin{bmatrix} 14 \\ -25 \\ 14 \end{bmatrix}$$

and  $\bar{\mathbf{c}}_N = [\bar{c}_1, \bar{c}_2, \bar{c}_3, \bar{c}_4] = [-5, -3, -3, -6]$ .

The first iteration begins.

*Step 1.* The leaving variable is  $x_6$ .

*Step 2.* Since  $\mathbf{B} = \mathbf{I}$ , the system  $\mathbf{v}\mathbf{B} = [0, 1, 0]$  reduces to  $\mathbf{v} = [0, 1, 0]$ . We have

$$[w_1, w_2, w_3, w_4] = \mathbf{v} \begin{bmatrix} -6 & 1 & 2 & 4 \\ 3 & -2 & -1 & -5 \\ -2 & 1 & 0 & 2 \end{bmatrix} = [3, -2, -1, -5].$$

Step 3. The set  $J$  consists of  $x_2, x_3, x_4$ . Comparing the ratios  $3/2, 3/1$ , and  $6/5$ , we find that  $x_4$  has to enter the basis.

Step 4. Since  $\mathbf{B} = \mathbf{I}$ , the system

$$\mathbf{B}\mathbf{d} = \begin{bmatrix} 4 \\ -5 \\ 2 \end{bmatrix} \text{ reduces to } \mathbf{d} = \begin{bmatrix} 4 \\ -5 \\ 2 \end{bmatrix}.$$

Step 5. We have  $t = x_6^*/w_4 = 5$ ,

$$\mathbf{x}_B^* = \begin{bmatrix} x_5^* \\ x_4^* \\ x_7^* \end{bmatrix} = \begin{bmatrix} 14 - 4t \\ t \\ 14 - 2t \end{bmatrix} = \begin{bmatrix} -6 \\ 5 \\ 4 \end{bmatrix}, \quad \mathbf{B} = \mathbf{E}_1 \text{ with } \mathbf{E}_1 = \begin{bmatrix} 1 & 4 & \\ & -5 & \\ & 2 & 1 \end{bmatrix},$$

$$\begin{aligned} [\bar{c}_1, \bar{c}_2, \bar{c}_3, \bar{c}_6] &= [-5 + 3\bar{c}_6, -3 - 2\bar{c}_6, -3 - \bar{c}_6, \bar{c}_6] \\ &= [-8.6, -0.6, -1.8, -1.2]. \end{aligned}$$

The second iteration begins.

Step 1. The leaving variable is  $x_5$ .

Step 2. Solving the system  $\mathbf{v}\mathbf{B} = [1, 0, 0]$  we find  $\mathbf{v} = [1, 0.8, 0]$ . Hence

$$[w_1, w_2, w_3, w_6] = \mathbf{v} \begin{bmatrix} -6 & 1 & 2 & 0 \\ 3 & -2 & -1 & 1 \\ -2 & 1 & 0 & 0 \end{bmatrix} = [-3.6, -0.6, 1.2, 0.8].$$

Step 3. The set  $J$  consists of  $x_1$  and  $x_2$ . Comparing the ratios  $8.6/3.6$  and  $0.6/0.6$ , we find that  $x_2$  has to enter the basis.

Step 4. Solving the system

$$\mathbf{B}\mathbf{d} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \text{ we find } \mathbf{d} = \begin{bmatrix} -0.6 \\ 0.4 \\ 0.2 \end{bmatrix}.$$

Step 5. We have  $t = x_5^*/w_2 = 10$ ,

$$\mathbf{x}_B^* = \begin{bmatrix} x_2^* \\ x_4^* \\ x_7^* \end{bmatrix} = \begin{bmatrix} t \\ 5 - 0.4t \\ 4 - 0.2t \end{bmatrix} = \begin{bmatrix} 10 \\ 1 \\ 2 \end{bmatrix}, \quad \mathbf{B} = \mathbf{E}_1\mathbf{E}_2 \text{ with } \mathbf{E}_2 = \begin{bmatrix} -0.6 & & \\ 0.4 & 1 & \\ 0.2 & & 1 \end{bmatrix},$$

$$\begin{aligned} [\bar{c}_1, \bar{c}_3, \bar{c}_5, \bar{c}_6] &= [-8.6 - 3.6\bar{c}_5, -1.8 + 1.2\bar{c}_5, \bar{c}_5, -1.2 + 0.8\bar{c}_5] \\ &= [-5, -3, -1, -2]. \end{aligned}$$

The third iteration begins.

Step 1. Since  $\mathbf{x}_B^* \geq \mathbf{0}$ , the current solution is optimal. (This observation could have been made in step 5 of the previous iteration; the update of  $\bar{c}_N$  was not strictly necessary.)

Extensions to the general setting of problems

$$\text{maximize } \mathbf{c}\mathbf{x} \quad \text{subject to } \mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{l} \leq \mathbf{x} \leq \mathbf{u} \tag{10.12}$$

are as follows. For every basic solution of (10.12), as defined in Chapter 9, we have

$$\mathbf{c}\mathbf{x} = \mathbf{c}_B\mathbf{x}_B + \mathbf{c}_N\mathbf{x}_N = \mathbf{c}_B(\mathbf{B}^{-1}\mathbf{b} - \mathbf{B}^{-1}\mathbf{A}_N\mathbf{x}_N) + \mathbf{c}_N\mathbf{x}_N = \mathbf{c}_B\mathbf{B}^{-1}\mathbf{b} + (\mathbf{c}_N - \mathbf{c}_B\mathbf{B}^{-1}\mathbf{A}_N)\mathbf{x}_N.$$

We shall write  $\bar{\mathbf{c}}_N = \mathbf{c}_N - \mathbf{c}_B\mathbf{B}^{-1}\mathbf{A}_N$ . A basic solution  $\mathbf{x}^*$  is *dual-feasible* if each nonbasic  $x_j$  with  $\bar{c}_j > 0$  has  $x_j^* = u_j$  and if each nonbasic  $x_j$  with  $\bar{c}_j < 0$  has  $x_j^* = l_j$ . Each iteration of the revised dual simplex method, beginning with some dual-feasible basic solution  $\mathbf{x}^*$  and with the corresponding vector  $\mathbf{c}_N$ , is as described in Box 10.2.

**BOX 10.2 An Iteration of Revised Dual Simplex Method Extended**

Step 1. If  $\mathbf{l}_B \leq \mathbf{x}_B^* \leq \mathbf{u}_B$  then stop:  $\mathbf{x}^*$  is an optimal solution. Otherwise, choose the leaving variable: this may be any basic variable  $x_i$  with  $x_i^* < l_i$  or  $x_i^* > u_i$ .

Step 2. Solve the system  $\mathbf{v}\mathbf{B} = \mathbf{e}$  with  $\mathbf{e}$  standing for the  $p$ th row of the identity matrix and with  $p$  such that  $x_i$  appears in the  $p$ th position of the basis heading. Compute  $\mathbf{w}_N = \mathbf{v}\mathbf{A}_N$ .

Step 3. If  $x_i^* < l_i$ , then let  $J$  be the set of those nonbasic variables  $x_j$  for which  $w_j < 0, x_j^* < u_j$  or  $w_j > 0, x_j^* > l_j$ . If  $x_i^* > u_i$ , then let  $J$  be the set of those nonbasic variables  $x_j$  for which  $w_j > 0, x_j^* < u_j$  or  $w_j < 0, x_j^* > l_j$ . If  $J$  is empty then stop: the problem is infeasible. Otherwise, find the  $x_j$  in  $J$  that minimizes  $|\bar{c}_j/w_j|$  and let it be the entering variable.

Step 4. Solve the system  $\mathbf{B}\mathbf{d} = \mathbf{a}$  with  $\mathbf{a}$  standing for the entering column.

Step 5. Set  $t = (x_i^* - l_i)/w_j$  in case  $x_i^* < l_i$  and  $t = (x_i^* - u_i)/w_j$  in case  $x_i^* > u_i$ . Replace the value  $x_j^*$  of  $x_j$  by  $x_j^* + t$  and replace the values  $\mathbf{x}_B^*$  of the basic variables by  $\mathbf{x}_B^* - t\mathbf{d}$ . Replace the leaving column of  $\mathbf{B}$  by the entering column and, in the basis heading, replace the leaving variable by the entering variable. Set  $\bar{c}_i = -\bar{c}_j/w_j$  and add  $\bar{c}_i w_s$  to each  $\bar{c}_s$  with  $s \neq i$ .

We shall not worry about initializing the dual simplex method: whenever an application of this method is called for in sensitivity analysis, a dual-feasible basic solution is readily available.

## SENSITIVITY ANALYSIS

Now we shall explain how the results found by the simplex method in solving a problem

$$\text{maximize } \mathbf{c}\mathbf{x} \quad \text{subject to } \mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{l} \leq \mathbf{x} \leq \mathbf{u} \quad (10.13)$$

can be exploited in solving modified versions of (10.13).

- If the changes leading from (10.13) to its modified version

$$\text{maximize } \tilde{\mathbf{c}}\mathbf{x} \quad \text{subject to } \mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{l} \leq \mathbf{x} \leq \mathbf{u} \quad (10.14)$$

are restricted to the objective function  $\mathbf{c}\mathbf{x}$ , then the previously found optimal basic solution  $\mathbf{x}^*$  of (10.13) remains feasible in (10.14), and so it can be used to initialize the simplex method on (10.14). The intuitive advantage of this strategy over solving (10.14) from scratch is obvious: if  $\tilde{\mathbf{c}}$  differs from  $\mathbf{c}$  only slightly, then an optimal solution of (10.14) is likely to differ from the optimal solution  $\mathbf{x}^*$  of (10.13) only slightly, and the number of simplex iterations leading from  $\mathbf{x}^*$  to an optimal solution of (10.14) is likely to be relatively small. In fact,  $\mathbf{x}^*$  may even be dual-feasible, and therefore optimal, in (10.14).

- If the changes leading from (10.13) to its modified version

$$\text{maximize } \mathbf{c}\mathbf{x} \quad \text{subject to } \mathbf{A}\mathbf{x} = \tilde{\mathbf{b}}, \quad \mathbf{l} \leq \mathbf{x} \leq \mathbf{u} \quad (10.15)$$

are restricted to the right-hand side  $\mathbf{b}$ , then the previously found optimal basic solution  $\mathbf{x}^*$  of (10.13) defines a basic solution  $\bar{\mathbf{x}}$  of (10.15) by  $\bar{\mathbf{x}}_N = \mathbf{x}_N^*$  and  $\mathbf{B}\bar{\mathbf{x}}_B = \tilde{\mathbf{b}} - \mathbf{A}_N\bar{\mathbf{x}}_N$ . It is easy to see that  $\bar{\mathbf{x}}$  is dual-feasible, and so it can be used to initialize the dual simplex method on (10.15). Again, the intuitive advantage of this strategy over solving (10.15) from scratch is obvious: if  $\tilde{\mathbf{b}}$  differs from  $\mathbf{b}$  only slightly, then an optimal solution of (10.15) is likely to differ from  $\mathbf{x}^*$  and  $\bar{\mathbf{x}}$  only slightly, and the number of dual simplex iterations leading from  $\bar{\mathbf{x}}$  to an optimal solution of (10.15) is likely to be relatively small. In fact,  $\bar{\mathbf{x}}$  may even be feasible, and therefore optimal, in (10.15).

- The technique for handling changes in the objective function can handle even more extensive changes as long as a feasible basic solution remains available for initializing the simplex method on the resulting problem. In particular, this is the case when new variables  $x_j$  with  $l_j = 0$  or  $u_j = 0$  are introduced. For example, if  $[x_1^*, x_2^*, \dots, x_6^*]^T$  is an optimal basic solution of the problem

$$\begin{aligned} &\text{maximize} && \sum_{j=1}^6 c_j x_j \\ &\text{subject to} && \sum_{j=1}^6 a_{ij} x_j = b_i \quad (i = 1, 2, 3) \\ &&& l_j \leq x_j \leq u_j \quad (j = 1, 2, \dots, 6) \end{aligned} \quad (10.16)$$

then  $[x_1^*, x_2^*, \dots, x_6^*, 0]^T$  is a feasible basic solution of the problem

$$\begin{aligned} &\text{maximize} && \sum_{j=1}^7 c_j x_j \\ &\text{subject to} && \sum_{j=1}^7 a_{ij} x_j = b_i \quad (i = 1, 2, 3) \\ &&& l_j \leq x_j \leq u_j \quad (j = 1, 2, \dots, 7) \end{aligned}$$

as long as  $l_7 = 0$  or  $u_7 = 0$ . (The new variable  $x_7$  is nonbasic.)

- Similarly, the technique for handling changes in the right-hand side can handle even more extensive changes as long as a dual-feasible solution remains available for initializing the dual simplex method on the resulting problem. In particular, this is the case when new constraints are introduced. For illustration, let us add two new constraints,

$$\sum_{j=1}^6 a_{4j} x_j \leq b_4 \quad \text{and} \quad \sum_{j=1}^6 a_{5j} x_j = b_5$$

to problem (10.16). The resulting problem may be recorded as

$$\begin{aligned} &\text{maximize} && \sum_{j=1}^6 c_j x_j \\ &\text{subject to} && \sum_{j=1}^6 a_{ij} x_j = b_i \quad (i = 1, 2, 3) \\ &&& \sum_{j=1}^6 a_{4j} x_j + x_7 = b_4 \\ &&& \sum_{j=1}^6 a_{5j} x_j + x_8 = b_5 \\ &&& l_j \leq x_j \leq u_j \quad (j = 1, 2, \dots, 6) \\ &&& 0 \leq x_7 \leq +\infty \\ &&& 0 \leq x_8 \leq 0. \end{aligned} \quad (10.17)$$

It is a routine matter to verify that a dual-feasible basic solution  $[x_1^*, x_2^*, \dots, x_8^*]^T$  of (10.17) can be obtained from the previously found optimal solution  $[x_1^*, x_2^*, \dots, x_6^*]^T$  of (10.16) by setting

$$x_7^* = b_4 - \sum_{j=1}^6 a_{4j} x_j^*, \quad x_8^* = b_5 - \sum_{j=1}^6 a_{5j} x_j^*$$

and by letting the new set of five basic variables consist of the three old basic variables along with the two new slack variables  $x_7, x_8$ . The details are left to the reader as an exercise.

More generally, if  $k$  constraints are added to a problem with  $m$  rows and  $n$  columns, then the resulting problem has  $m + k$  rows and  $n + k$  columns. The previously found optimal solution of the original problem extends into a dual-feasible basic solution of the new problem; the new set of  $m + k$  basic variables consists of the old  $m$  basic variables along with the  $k$  new slack variables.

- Now consider a completely general transformation of a problem

$$\text{maximize } \mathbf{c}\mathbf{x} \quad \text{subject to } \mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{l} \leq \mathbf{x} \leq \mathbf{u} \quad (10.18)$$

into a problem

$$\text{maximize } \tilde{\mathbf{c}}\mathbf{x} \quad \text{subject to } \tilde{\mathbf{A}}\mathbf{x} = \tilde{\mathbf{b}}, \quad \tilde{\mathbf{l}} \leq \mathbf{x} \leq \tilde{\mathbf{u}}. \quad (10.19)$$

If the changes leading from (10.18) to (10.19) are only slight, then an optimal solution of (10.19) is likely to differ from the previously found optimal solution  $\mathbf{x}^*$  of (10.18) only slightly. Rather than solving (10.19) from scratch, we may convert  $\mathbf{x}^*$  into a basic solution  $\bar{\mathbf{x}}$  of (10.19) and, if  $\bar{\mathbf{x}}$  turns out to be feasible or dual-feasible, use it to initialize the simplex method or the dual simplex method, respectively. As long as the entries  $a_{ij}$  of  $\mathbf{A}$  remain unchanged for all variables  $x_j$  basic in  $\mathbf{x}^*$ , the vector  $\bar{\mathbf{x}}$  may be obtained by first converting  $\mathbf{x}_N^*$  into  $\bar{\mathbf{x}}_N$  and then solving the system  $\mathbf{B}\bar{\mathbf{x}}_B = \tilde{\mathbf{b}} - \tilde{\mathbf{A}}_N\bar{\mathbf{x}}_N$ . (Changes of  $a_{ij}$  at basic variables  $x_j$  might ruin this plan by making  $\mathbf{B}$  into a singular  $\tilde{\mathbf{B}}$ . A way of bypassing this difficulty is suggested later in this section.)

How do we convert  $\mathbf{x}_N^*$  into  $\bar{\mathbf{x}}_N$ ? Since  $\bar{\mathbf{x}}$  must be a basic solution, each restricted nonbasic variable must be set at one of its bounds. For nonbasic variables with one-sided restrictions, this requirement leaves us no choice; we must have

$$\begin{aligned} \bar{x}_j &= \tilde{l}_j & \text{whenever } \tilde{l}_j \text{ is finite and } \tilde{u}_j &= +\infty \\ \bar{x}_j &= \tilde{u}_j & \text{whenever } \tilde{u}_j \text{ is finite and } \tilde{l}_j &= -\infty. \end{aligned}$$

On the other hand, unrestricted nonbasic variables may assume arbitrary values; to keep  $\bar{\mathbf{x}}_N$  close to  $\mathbf{x}_N^*$ , we may set

$$\bar{x}_j = x_j^* \quad \text{whenever } \tilde{l}_j = -\infty \text{ and } \tilde{u}_j = +\infty.$$

Now there remain only the nonbasic variables that are restricted from both sides in (10.19). For each of these variables  $x_j$ , we must have  $\bar{x}_j = \tilde{l}_j$  or  $\bar{x}_j = \tilde{u}_j$ . It will be convenient to set

$$\begin{aligned} \bar{x}_j &= \tilde{l}_j & \text{if } x_j^* &= l_j \\ \bar{x}_j &= \tilde{u}_j & \text{if } x_j^* &= u_j. \end{aligned}$$

(Actually, these instructions may be self-contradictory if  $l_j = u_j$  but  $\tilde{l}_j < \tilde{u}_j$ . If such is the case, we may choose between  $\bar{x}_j = \tilde{l}_j$  and  $\bar{x}_j = \tilde{u}_j$ . A similar choice can

be made in the only remaining case where  $\tilde{l}_j$  and  $\tilde{u}_j$  are both finite but  $l_j = -\infty$  and  $u_j = +\infty$ .)

In particular, note that  $\bar{\mathbf{x}}$  is dual-feasible in (10.19) if the changes leading from (10.18) to (10.19) are restricted to the lower bounds  $l_j$  and the upper bounds  $u_j$ , with each finite  $l_j$  replaced by a finite  $\tilde{l}_j$  and each finite  $u_j$  replaced by a finite  $\tilde{u}_j$ .

- Even when  $\bar{\mathbf{x}}$  turns out to be neither feasible nor dual-feasible in (10.19), the transformation of (10.18) into (10.19) can still be handled by a combination of the simplex method and the dual simplex method. The trick is to interpose an intermediate problem between (10.18) and (10.19) so that the transformation of (10.18) into the intermediate problem can be handled by the simplex method, and the transformation of the intermediate problem into (10.19) can be handled by the dual simplex method. The problem

$$\text{maximize } \tilde{\mathbf{c}}\mathbf{x} \quad \text{subject to } \tilde{\mathbf{A}}\mathbf{x} = \tilde{\mathbf{b}}, \quad \mathbf{l}_i^* \leq \mathbf{x} \leq \mathbf{u}^* \quad (10.20)$$

with

$$\begin{aligned} l_j^* &= \bar{x}_j, & u_j^* &= \tilde{u}_j & \text{whenever } \bar{x}_j < \tilde{l}_j \\ l_j^* &= \tilde{l}_j, & u_j^* &= \bar{x}_j & \text{whenever } \bar{x}_j > \tilde{u}_j \\ l_j^* &= \tilde{l}_j, & u_j^* &= \tilde{u}_j & \text{whenever } \tilde{l}_j \leq \bar{x}_j \leq \tilde{u}_j \end{aligned}$$

can always serve as the intermediate problem. [In this case,  $\bar{\mathbf{x}}$  is a feasible basic solution of (10.20), and each optimal basic solution of (10.20) can be converted into a dual-feasible basic solution of (10.19) as described earlier.]

- To fit changes of the coefficients  $a_{ij}$  at variables  $x_j$  basic in  $\mathbf{x}^*$  into this scheme, we need only think of each such change as an introduction of a new variable combined with a replacement of  $l_j$  and  $u_j$  by zeros. For instance, consider the problem

$$\text{maximize } \mathbf{c}\mathbf{x} \quad \text{subject to } \mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{l} \leq \mathbf{x} \leq \mathbf{u}$$

with

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} \end{bmatrix} \\ \mathbf{c} &= [c_1 \quad c_2 \quad c_3 \quad c_4 \quad c_5] \\ \mathbf{l} &= [l_1 \quad l_2 \quad l_3 \quad l_4 \quad l_5]^T \\ \mathbf{u} &= [u_1 \quad u_2 \quad u_3 \quad u_4 \quad u_5]^T. \end{aligned}$$

If  $x_2$  is a basic variable in the previously found optimal solution  $\mathbf{x}^*$  and if the coefficients  $a_{12}$ ,  $a_{22}$ ,  $a_{32}$  are replaced by  $\tilde{a}_{12}$ ,  $\tilde{a}_{22}$ ,  $\tilde{a}_{32}$ , respectively, then the resulting problem may be presented as (10.19) with  $\tilde{\mathbf{b}} = \mathbf{b}$  and



$$\tilde{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} & \tilde{a}_{12} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} & \tilde{a}_{22} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} & \tilde{a}_{32} \end{bmatrix}$$

$$\tilde{c} = [c_1 \quad c_2 \quad c_3 \quad c_4 \quad c_5 \quad c_2]$$

$$\tilde{l} = [l_1 \quad 0 \quad l_3 \quad l_5 \quad l_5 \quad l_2]^T$$

$$\tilde{u} = [u_1 \quad 0 \quad u_3 \quad u_4 \quad u_5 \quad u_2]^T.$$

### Parametric Linear Programming

In linear programming problems arising from applications, the coefficients in the objective function may be linear functions of some parameter  $p$  rather than constants. For instance, net profits may depend on a variable price  $p$  of some raw material. In other situations, it may be unclear which of the two objective functions should be maximized. In such cases, a compromise between maximizing  $cx$  and maximizing  $\tilde{c}x$  is to maximize  $pcx + (1 - p)\tilde{c}x = (\tilde{c} + p(c - \tilde{c}))x$  for some fixed value of  $p$  between zero and one. Here, the parameter  $p$  measures the relative importance of  $cx$  as opposed to that of  $\tilde{c}x$ . Since the choice of its actual value may be a highly subjective matter, it may be enlightening to find out how the optimal solution changes with changing  $p$ . Such investigations are known as *parametric linear programming*.

What is called an LP problem with a *parametric objective function* is actually an infinite family of LP problems

$$\text{maximize } (c' + pc'')x \quad \text{subject to } Ax = b, \quad 1 \leq x \leq u \quad (10.21)$$

one for every value of  $p$ . We are going to show that the dependence of the optimal value of (10.21) on the value of the parameter  $p$  always follows the same simple pattern. For illustration, let us consider the problem

$$\begin{aligned} &\text{maximize} && (3 - 2p)x_1 + (-3 + p)x_2 + x_3 \\ &\text{subject to} && x_1 + 2x_2 + 3x_3 \leq 5 \\ & && 2x_1 + x_2 + 4x_3 \leq 7 \\ & && x_1 \geq 0, 0 \leq x_2 \leq 2, x_3 \leq 0 \end{aligned} \quad (10.22)$$

which, after the usual introduction of slack variables, may be presented in the format (10.21) with  $c' = [3, -3, 1, 0, 0]$ ,  $c'' = [-2, 1, 0, 0, 0]$ , and so on. It is a routine, if tedious, matter to verify that (10.22) has precisely six basic feasible solutions,

$$\begin{aligned} x^* &= [3, 1, 0, 0, 0]^T \\ x^* &= [0, 0, 0, 5, 7]^T \\ x^* &= [3.5, 0, 0, 1.5, 0]^T \\ x^* &= [0, 2, 0, 1, 5]^T \\ x^* &= [1, 2, 0, 0, 3]^T \\ x^* &= [5.5, 2, -1.5, 0, 0]^T. \end{aligned}$$

For every fixed value of  $p$ , one of these six solutions must be optimal or else the problem must be unbounded. Hence the optimal value of (10.22), if it exists at all, equals the largest of the six numbers  $6 - 5p$ ,  $0$ ,  $10.5 - 7p$ ,  $-6 + 2p$ ,  $-3$ ,  $9 - 9p$ , specifying the respective values of the objective function at our six solutions. More generally, we have the following result.

**THEOREM 10.1.** If (10.21) has an optimal solution for at least one value of  $p$ , then there are a finite number of linear functions  $r_1 + s_1p$ ,  $r_2 + s_2p$ ,  $\dots$ ,  $r_k + s_kp$  and an interval  $I$  with the following properties:

- (i) If  $p \notin I$  then (10.21) is unbounded.
- (ii) If  $p \in I$  then (10.21) has an optimal solution and its optimal value  $z^* = z^*(p)$  equals the largest of the  $k$  numbers  $r_1 + s_1p$ ,  $r_2 + s_2p$ ,  $\dots$ ,  $r_k + s_kp$ .

**PROOF.** Let  $I$  denote the set of those values of  $p$  for which (10.21) is not unbounded. Since (10.21) has an optimal solution for at least one value of  $p$ , it has a feasible solution for every value of  $p$ , and so it has an optimal solution whenever  $p \in I$ . Showing that  $I$  is an interval amounts to showing that  $p \in I$  whenever  $p_1 < p < p_2$  and  $p_1, p_2 \in I$ . To establish this fact, observe that

$$(c' + pc'')x = \frac{p_2 - p}{p_2 - p_1} (c' + p_1c'')x + \frac{p - p_1}{p_2 - p_1} (c' + p_2c'')x$$

and so every feasible solution  $x$  of (10.21) satisfies

$$(c' + pc'')x \leq \frac{p_2 - p}{p_2 - p_1} z^*(p_1) + \frac{p - p_1}{p_2 - p_1} z^*(p_2).$$

To establish (ii), we rely on the results of Chapter 8. Theorems 8.3 and 8.4 guarantee that (10.21) has a finite number  $k$  of feasible solutions such that for every  $p \in I$ , one of these  $k$  solutions is an optimal solution of (10.21). Hence,  $z^*(p)$  is the largest of the  $k$  numbers  $z_1(p)$ ,  $z_2(p)$ ,  $\dots$ ,  $z_k(p)$ , the  $i$ th of which stands for the value of the objective function at the  $i$ th of our  $k$  solutions. To complete the proof, write  $r_i = c'x^*$  and  $s_i = c''x^*$  for the  $i$ th solution  $x^*$ , and observe that  $z_i(p) = (c' + pc'')x^* = r_i + s_ip$ . ■

Now we turn to the algorithmic problem of solving the infinite family of problems (10.21) in only a finite time. The procedure implicit in our proof of Theorem 10.1 is unsatisfactory on at least two counts: it does not provide a constructive way of finding  $I$  and, in general, it enumerates far too many candidates for optimal solutions of (10.21). To illustrate the second point, we observe that the first and fifth of the six basic feasible solutions in our example are never optimal; in fact,

$$\max(6 - 5p, 0, 10.5 - 7p, -6 + 2p, -3, 9 - 9p) = \begin{cases} 9 - 9p & \text{for } p \leq -0.75 \\ 10.5 - 7p & \text{for } -0.75 \leq p \leq 1.5 \\ 0 & \text{for } 1.5 \leq p \leq 3 \\ -6 + 2p & \text{for } p \geq 3. \end{cases}$$

A procedure designed by S. I. Gass and T. Saaty (1955) bypasses such hopeless candidates automatically by continuously varying the value of  $p$  and monitoring the attendant changes in the optimal solution of (10.21). We are going to illustrate this procedure on our example (10.22).

As a preliminary step, the parameter  $p$  is set at some arbitrarily chosen level, and the resulting LP problem is solved by the simplex method. Setting  $p = 0$  in (10.22), we obtain the problem

$$\begin{aligned} &\text{maximize} && 3x_1 - 3x_2 + x_3 \\ &\text{subject to} && x_1 + 2x_2 + 3x_3 \leq 5 \\ &&& 2x_1 + x_2 + 4x_3 \leq 7 \\ &&& x_1 \geq 0, 0 \leq x_2 \leq 2, x_3 \leq 0. \end{aligned} \quad (10.23)$$

Solving this problem by the simplex method, we arrive at the basic feasible solution  $[3.5, 0, 0, 1.5, 0]^T$ , compute the vector  $y = [0, 1.5]$ , find that  $x_3$  may enter the basis, and discover that there is no candidate for leaving the basis. In short, we discover that (10.23) is unbounded: its solution

$$[3.5 + 2t, 0, -t, 1.5 + t, 0]^T \quad (10.24)$$

is feasible whenever  $t \geq 0$ , and the corresponding value  $10.5 + 5t$  of the objective function can be made arbitrarily large by making  $t$  sufficiently large.

The next step consists of finding all the other values of  $p$  for which this status quo persists. The value of the parametric objective function  $(3 - 2p)x_1 + (-3 + p)x_2 + x_3$  at (10.24) equals  $(10.5 - 7p) + (5 - 4p)t$ . This quantity can be made arbitrarily large as long as  $5 - 4p > 0$ . Thus, we conclude that (10.22) is unbounded whenever  $p < 1.25$ .

To find out what happens when  $p \geq 1.25$ , we raise  $p$  to 1.25 and discover that the basic feasible solution  $[3.5, 0, 0, 1.5, 0]^T$  is optimal in the resulting problem. Again, the next step consists of finding all the other values of  $p$  for which this status quo persists. Toward this end, we first solve the system  $yB = c_B$  with  $c_B = c'_B + pc''_B$  and then compare the vectors  $yA_N$  and  $c_N = c'_N + pc''_N$ . Solving the system

$$y \begin{bmatrix} 1 & 1 \\ 2 & \end{bmatrix} = [3 - 2p, 0] \quad \text{we obtain} \quad y = [0, 1.5 - p].$$

Comparing the vector  $yA_N = [1.5 - p, 6 - 4p, 1.5 - p]$  with  $c_N = [-3 + p, 1, 0]$ , we find that  $[3.5, 0, 0, 1.5, 0]^T$  remains optimal as long as

$$1.5 - p \geq -3 + p, \quad 6 - 4p \leq 1, \quad \text{and} \quad 1.5 - p \geq 0. \quad (10.25)$$

In short,  $[3.5, 0, 0, 1.5, 0]^T$  is an optimal solution of (10.22) whenever  $1.25 \leq p \leq 1.5$ .

To find out what happens when  $p > 1.5$ , we think of  $p$  as being raised just slightly above 1.5 and observe that the inequality  $1.5 - p \geq 0$  in (10.25) becomes violated. Now  $x_5$  becomes eligible for entering the basis; the resulting simplex iteration brings us to the basic feasible solution  $[0, 0, 0, 5, 7]^T$ . For which values of  $p$  is this solution optimal? Solving the system

$$y \begin{bmatrix} 1 & \\ & 1 \end{bmatrix} = [0, 0] \quad \text{we obtain} \quad y = [0, 0].$$

Comparing the vector  $yA_N = [0, 0, 0]$  with  $c_N = [3 - 2p, -3 + p, 1]$  we find that  $[0, 0, 0, 5, 7]^T$  is optimal whenever

$$0 \geq 3 - 2p, \quad 0 \geq -3 + p, \quad \text{and} \quad 0 \leq 1. \quad (10.26)$$

In short,  $[0, 0, 0, 5, 7]^T$  is an optimal solution of (10.22) whenever  $1.5 \leq p \leq 3$ .

To find out what happens when  $p > 3$ , we think of  $p$  as being raised just slightly above 3 and observe that the inequality  $0 \geq -3 + p$  in (10.26) becomes violated. Now  $x_2$  becomes eligible for entering the basis; the resulting simplex iteration brings us to the basic feasible solution

$[0, 2, 0, 1, 5]^T$ . For which values of  $p$  is this solution optimal? Comparing the vector  $yA_N = [0, 0, 0]$  with  $c_N = [3 - 2p, -3 + p, 1]$  we find that the answer is affirmative whenever

$$0 \geq 3 - 2p, \quad 0 \leq -3 + p, \quad \text{and} \quad 0 \leq 1.$$

In short,  $[0, 2, 0, 1, 5]^T$  is an optimal solution of (10.22) whenever  $p \geq 3$ . Now (10.22) is solved completely; note that the useless basic feasible solutions  $[3, 1, 0, 0, 0]^T$ ,  $[1, 2, 0, 0, 3]^T$ , and  $[5.5, 2, -1.5, 0, 0]^T$  have never even come under consideration.

Incidentally, even this procedure may take a considerable time on small but sufficiently nasty examples such as

$$\begin{aligned} &\text{maximize} && \sum_{j=1}^n (p - 2^{n-j})x_j \\ &\text{subject to} && x_i + 2 \sum_{j=i+1}^n x_j \leq 4^{n-i} \quad (i = 1, 2, \dots, n) \\ &&& x_j \geq 0 \quad (j = 1, 2, \dots, n). \end{aligned} \quad (10.27)$$

These examples, constructed by K. G. Murty (1980), are closely related to the Klee–Minty examples of Chapter 4. They have  $2^n$  basic feasible solutions, each of which is a unique optimal solution of (10.27) for a suitably chosen value of  $p$ . It follows that any procedure for solving (10.27) must enumerate  $2^n$  basic feasible solutions in this case; already when  $n$  is as small as ten, there are more than a thousand basic feasible solutions to be enumerated.

Before leaving the subject of parametric objective function, let us note that Theorem 10.1 applies just as well to minimization problems

$$\text{minimize} \quad (c' + pc'')x \quad \text{subject to} \quad Ax = b, \quad l \leq x \leq u$$

as soon as “the largest” in (ii) is replaced by “the smallest.” After this observation, we are ready to tackle LP problems with a *parametric right-hand side*,

$$\text{maximize} \quad cx \quad \text{subject to} \quad Ax = b' + pb'', \quad l \leq x \leq u$$

and, more generally, parametric LP problems

$$\text{maximize} \quad cx \quad \text{subject to} \quad Ax = b' + pb'', \quad l' + pl'' \leq x \leq u' + pu''. \quad (10.28)$$

**THEOREM 10.2.** If (10.28) has an optimal solution for at least one value of  $p$ , then there are a finite number of linear functions  $r_1 + s_1p, r_2 + s_2p, \dots, r_k + s_kp$  and an interval  $I$  with the following properties:

- (i) If  $p \notin I$  then (10.28) is infeasible.
- (ii) If  $p \in I$  then (10.28) has an optimal solution and its optimal value  $z^* = z^*(p)$  equals the smallest of the  $k$  numbers  $r_1 + s_1p, r_2 + s_2p, \dots, r_k + s_kp$ .

**PROOF.** Note that the dual of (10.28) is a minimization problem with a parametric objective function. Now the desired conclusion follows directly from the observation made just before the statement of this theorem, combined with the duality theorem: if the dual is unbounded, then the primal is infeasible, and if the dual has an optimal value  $z^*$ , then  $z^*$  is also the optimal value of the primal. ■

Every infinite family (10.28) of LP problems can be solved in only a finite time by a procedure analogous to the procedure suggested for solving (10.21), except that now the optimal solutions vary continuously with  $p$  and simplex iterations are replaced by dual simplex iterations. For illustration, let us consider the parametric problem

$$\begin{aligned} &\text{maximize} && x_1 + 4x_2 + x_3 \\ &\text{subject to} && 2x_1 + 5x_2 + x_3 \leq 7 - 3p \\ &&& x_1 + 3x_2 + x_3 \leq 5 - 2p \\ &&& x_1 \geq p, x_2 \geq 0, 0 \leq x_3 \leq 3 + 2p. \end{aligned} \quad (10.29)$$

Setting  $p$  at zero and applying the simplex method to the resulting problem, we find the optimal solution  $[0, 1, 2, 0, 0]^T$ . Even though this vector ceases to be a solution of (10.29) as soon as  $p \neq 0$ , it points out a family of basic solutions

$$[p, 1 - p, 2, 0, 0]^T \quad (10.30)$$

by specifying that  $x_2, x_3$  are basic and that  $x_1, x_4, x_5$  are at their lower bounds. In fact, each of these basic solutions is dual-feasible regardless of the actual value of  $p$ ; in addition, it is feasible as long as  $1 - p \geq 0$  and  $2 \leq 3 + 2p$ . Thus we conclude that (10.30) is an optimal solution of (10.29) whenever  $-0.5 \leq p \leq 1$ .

To find out what happens for the remaining values of  $p$ , we first think of  $p$  as being raised just slightly above 1. Now  $x_2$  becomes eligible for leaving the basis in the dual-feasible basic solution (10.30); the resulting dual simplex iteration brings us to the basic solution

$$[p, 0, 7 - 5p, 0, -2 + 2p]^T. \quad (10.31)$$

Since this dual-feasible solution remains feasible as long as  $0 \leq 7 - 5p \leq 3 + 2p$  and  $-2 + 2p \geq 0$ , we conclude that (10.31) is an optimal solution of (10.29) whenever  $1 \leq p \leq 1.4$ .

Next, we think of  $p$  as being raised above 1.4. Now  $x_3$  becomes eligible for leaving the basis in the dual-feasible basic solution (10.31); as there are no candidates for entering the basis, we conclude that (10.29) is infeasible whenever  $p > 1.4$ .

Finally, we return to our starting point  $-0.5 \leq p \leq 1$  and think of  $p$  as being lowered below  $-0.5$ . Now  $x_3$  becomes eligible for leaving the basis in the dual-feasible basic solution (10.30); the resulting dual simplex iteration brings us to the basic solution

$$[p, 0.8 - 1.4p, 3 + 2p, 0, -0.4 - 0.8p]^T. \quad (10.32)$$

Since this dual-feasible solution remains feasible as long as  $0.8 - 1.4p \geq 0$  and  $-0.4 - 0.8p \geq 0$ , we conclude that (10.32) is an optimal solution of (10.29) whenever  $p \leq -0.5$ .

## PROBLEMS

### 10.1 Writing

$$\begin{aligned} c_j = 0 \quad \text{and} \quad a_{ij} &= \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad \text{for } j = 1, 2, \dots, m \\ b_i = 0 \quad \text{and} \quad a_{ij} &= \begin{cases} -1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad \text{for } i = m + 1, m + 2, \dots, m + n \\ \bar{c}_s = 0 \quad \text{and} \quad \bar{a}_{rs} &= \begin{cases} 1 & \text{if } r = s \\ 0 & \text{if } r \neq s \end{cases} \quad \text{for } s \in B \\ \bar{b}_r = 0 \quad \text{and} \quad \bar{a}_{rs} &= \begin{cases} -1 & \text{if } r = s \\ 0 & \text{if } r \neq s \end{cases} \quad \text{for } r \notin B \end{aligned}$$

prove the following:

(i) If every solution  $x_1, x_2, \dots, x_{m+n}, z$  of (10.4) satisfies (10.7), then

$$\sum_{i=1}^{m+n} \bar{a}_{ri} a_{ij} = 0, \quad \sum_{i=1}^{m+n} \bar{a}_{ri} b_i = b_r, \quad \sum_{i=1}^{m+n} \bar{c}_i a_{ij} = -c_j, \quad \sum_{i=1}^{m+n} \bar{c}_i b_i = -\bar{d} \quad (10.33)$$

for all  $r \in B$  and all  $j = m + 1, m + 2, \dots, m + n$ .

(ii) If every solution  $x_1, x_2, \dots, x_{m+n}, z$  of (10.7) satisfies (10.4), then

$$\sum_{r=1}^{m+n} a_{ir} \bar{a}_{rs} = 0, \quad \sum_{r=1}^{m+n} a_{ir} \bar{b}_r = b_i, \quad \sum_{r=1}^{m+n} c_r \bar{a}_{rs} = -\bar{c}_s, \quad \sum_{r=1}^{m+n} c_r \bar{b}_r = \bar{d} \quad (10.34)$$

for all  $s \in N$  and all  $i = 1, 2, \dots, m$ .

(iii) If (10.34) holds for all  $s \in N$  and all  $i = 1, 2, \dots, m$ , then every solution  $y_1, y_2, \dots, y_{m+n}, w$  of (10.6) is a solution of (10.8).

(iv) If (10.33) holds for all  $r \in B$  and all  $j = m + 1, m + 2, \dots, m + n$ , then every solution  $y_1, y_2, \dots, y_{m+n}, w$  of (10.8) satisfies (10.6).

△10.2 The optimal solution of the illustrative problem of Chapter 7 calls for 39 bookcases, 48 chairs, and 30 bedframes to be produced per day. Solve the following variations:

- The net profit brought in by each desk increases from \$13 to \$15.
- The availability of metal increases from 117 to 125 units per day.
- The company may also produce coffee tables, each of which requires three hours of work, one unit of metal, two units of wood, and brings in a net profit of \$14.
- The number of chairs produced must be at most five times the numbers of desks.
- The demand for wood per chair increases from three to four units.

10.3 The optimal solution of the problem

$$\text{maximize } \mathbf{c}\mathbf{x} \quad \text{subject to } \mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0}$$

with

$$\mathbf{A} = \begin{bmatrix} 8 & 5 & 1 & 0 & 0 \\ 8 & 6 & 0 & 1 & 0 \\ 8 & 7 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 32 \\ 33 \\ 35 \end{bmatrix},$$

$$\mathbf{c} = [1 \quad 1 \quad 0 \quad 0 \quad 0]$$